

# **MATH3701 Statistical Modelling**

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# Weekly schedule

## i Provisional Weekly Lecture Schedule<sup>a</sup>

<sup>a</sup>Some sections will be left as *directed reading*, but please note that all material is examinable.

|        |     |                |                                                        |
|--------|-----|----------------|--------------------------------------------------------|
| Week 1 | L1  | Chapter 1      | Introduction to Statistical Modelling                  |
|        | L2  | Chapter 2      | Normal Linear Models: Structure and Matrix Form        |
|        | L3  | Chapter 2      | Normal Linear Models: Fitting in R and Diagnostics     |
| Week 2 | L4  | Chapter 3      | GLM Theory: The Exponential Family                     |
|        | L5  | Chapter 3      | GLM Theory: Moments and Variance Functions             |
|        | L6  | Chapter 3      | GLM Theory: Link Functions                             |
|        | L7  | Chapter 4      | GLM Estimation: Maximum Likelihood                     |
| Week 3 | L8  | Chapter 4      | GLM Estimation: IRLS                                   |
|        | L9  | Chapter 4      | GLM Estimation: Deviance, Residuals and R              |
|        | L10 | Chapter 5      | Modelling Proportions: Logistic Regression             |
|        | L11 | Chapter 5      | Modelling Proportions: Overdispersion and Odds Ratios  |
| Week 4 | L12 | Chapter 5      | Modelling Proportions: Dose-Response                   |
|        | L13 | Chapter 6      | Loglinear Models                                       |
|        | L14 | Chapter 7      | Extensions to Loglinear Models                         |
|        | L15 | Chapter 8      | GLM Consolidation                                      |
|        | L16 | <i>Part I</i>  | <b>Part I Review and Exercises</b>                     |
| Week 5 | L17 | Chapter 9      | Beyond Linear Predictors: Non-parametric Motivation    |
|        | L18 | Chapter 10     | Introducing Splines                                    |
|        | L19 | Chapter 11     | Interpolating Splines: Natural Splines and Optimality  |
| Week 6 | L20 | Chapter 11     | Interpolating Splines: Matrix Derivation               |
|        | L21 | Chapter 12     | Smoothing Splines: Theory                              |
|        | L22 | Chapter 13     | Smoothing Splines: Practice and Cross-Validation       |
|        | L23 | Chapter 13     | GCV, Smoothing Matrix and Effective Degrees of Freedom |
| Week 7 | L24 | Chapter 14     | Generalised Additive Models                            |
|        | L25 | <i>Part II</i> | <b>Part II Review and Exercises</b>                    |

|         |     |                 |                                                       |
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|         | L26 | Chapter 15      | Introduction to Survival Analysis                     |
|         | L27 | Chapter 16      | Survival, Hazard and Cumulative Hazard Functions      |
| Week 8  | L28 | Chapter 17      | Kaplan–Meier Estimator                                |
|         | L29 | Chapter 18      | Parametric Families: Exponential Distribution         |
|         | L30 | Chapter 18      | Parametric Families: Weibull and Graphical Assessment |
|         | L31 | Chapter 19      | MLE with Censored Data                                |
| Week 9  | L32 | Chapter 20      | Parametric Two-Sample Comparison                      |
|         | L33 | Chapter 21      | Proportional Hazards Property                         |
|         | L34 | Chapter 22      | PH Regression: Simple Case                            |
|         | L35 | Chapter 22      | PH Regression: Multiple Covariates                    |
| Week 10 | L36 | Chapter 23      | The Cox Model: Motivation and Partial Likelihood      |
|         | L37 | Chapter 23      | The Cox Model: Estimation and Inference               |
|         | L38 | Chapter 23      | The Cox Model in R                                    |
|         | L39 | Chapter 24      | Cox Model Diagnostics                                 |
|         | L40 | Chapter 25      | Model Selection                                       |
| Week 11 | L41 | Chapter 26      | Extended Case Study                                   |
|         | L42 | <i>Part III</i> | <b>Part III Review and Exercises</b>                  |
|         | L43 |                 | Course Review and Synthesis                           |

# Overview

## Preface

These lecture notes are produced for the University of Leeds module *MATH3701 - Statistical Modelling* for the academic year 2026-27. They draw on material from several sources, and I am grateful to previous contributors whose work has informed these notes. A PDF version will be made available on the University of Leeds Minerva system.

RP Mann, Leeds, 2026

## Generative AI usage within this module

The assessments for this module fall in the red and amber category for using Generative AI. For the exam you will not be able to use Generative AI. For the coursework component of the assessment you may use AI to help draft the structure of your report and for assistance with language and grammar. You should not use AI to perform mathematical or computational analyses. It is essential that you understand everything you have written that is submitted.

If you have used Gen AI to prepare or produce part or all of your work, you must acknowledge your use of AI.

### Warning

#### **Statistical ethics and sensitive data**

Please note that from time to time we will be using data sets from situations which some might perceive as sensitive. All such data sets will, however, be derived from real-world studies which appear in textbooks or in scientific journals. The daily work of many statisticians involves applying their professional skills in a wide variety of situations and as such it is important to include a range of commonly encountered examples in this module. Whenever possible, sensitive topics will be signposted in advance. If you feel that any examples may be personally upsetting then, if possible,

please contact the module lecturer in advance. If you are significantly affected by any of these situations, then you can seek support from the [Student Counselling and Wellbeing service](#).

# Official Module Description

## Module summary

Statistical models allow us to understand complex data and make principled predictions. This module develops three powerful and widely-used classes of model. Generalised linear models extend ordinary linear regression to handle non-normal responses such as counts and proportions. Generalised additive models relax the linearity assumption, allowing flexible non-parametric smoothing whilst retaining interpretability. Survival analysis addresses data where the response is a time-to-event, subject to censoring, and develops both parametric and semi-parametric approaches including the Cox proportional hazards model.

## Learning outcomes

On successful completion of the module students will be able to:

1. carry out regression analysis with generalised linear models including the use of link functions;
2. understand the use of deviance in model selection;
3. appreciate the problems caused by overdispersion;
4. fit and interpret the special cases of log linear models and logistic regression;
5. compare methods for scatterplot smoothing suitable for use in a generalised additive model;
6. interpret a fitted generalised additive model;
7. describe the characteristic features of survival data including censoring;
8. use a range of parametric and semi-parametric regression models to assess the effects of covariates on a survival distribution; and
9. use a statistical package with real data to fit these models to data and to write a report giving and interpreting the results.

## Syllabus

**Part I — Generalised Linear Models:** Normal linear models (revision); exponential family distributions; GLM structure and link functions; maximum likelihood estimation and IRLS; model deviance and residuals; logistic regression; log-linear models.

**Part II — Generalised Additive Models:** Limitations of parametric models; splines; interpolating and smoothing splines; cross-validation and effective degrees of freedom; generalised additive models.

**Part III — Survival Analysis:** Survival and hazard functions; censoring; Kaplan-Meier estimation; parametric survival distributions; maximum likelihood with censored data; proportional hazards models; the Cox model; model diagnostics and selection.

## **Part I**

# **Part I: Generalised Linear Models**

# 1 Introduction

## 1.1 Overview

In previous modules you have studied linear models with a normally distributed error term, such as simple linear regression, multiple linear regression and ANOVA for normally distributed observations. In this module we will study three major extensions of this framework.

Outline of the module:

1. **Part I: Generalised Linear Models.** Extending linear regression to non-normal responses — counts, proportions, and more.
2. **Part II: Generalised Additive Models.** Relaxing the linearity assumption using smooth, non-parametric functions of the predictors.
3. **Part III: Survival Analysis.** Modelling time-to-event data, where observations may be censored, using parametric and semi-parametric models.

### ! Important

This module will make extensive use of **R** and hence it is very important that you are comfortable with its use. If you need some revision, then material is available on the website for the 1st year module MATH1710: <https://mpaldrige.github.io/math1710/R.html>

The purpose of a generalised linear model is to describe the dependence of a *response* variable  $y$  on a set of  $p$  *explanatory* variables  $x = (x_1, x_2, \dots, x_p)$  where, conditionally on  $x$ , observation  $y$  has a distribution which is **not necessarily** normal. Note that the normal distribution situation is a special case of the general framework and we will study that in the next Chapter.

Please be aware that in this learning material we may use lowercase letters, for example  $y$  or  $y_i$ , to denote both observed values or random variables; which is being considered should be clear from the context.

**! Important**

This module will make extensive use of many basic ideas from statistics. If you need some revision, please refer to your notes from previous statistics modules.

## 1.2 Motivating example

Table 1.1 shows data<sup>1</sup> on the number of beetles killed by five hours of exposure to 8 different concentrations of gaseous carbon disulphide.

Table 1.1: Numbers of beetles killed by five hours of exposure to 8 different concentrations of gaseous carbon disulphide

| Dose   | No. of beetle | No. killed |
|--------|---------------|------------|
| $x_i$  | $m_i$         | $y_i$      |
| 1.6907 | 59            | 6          |
| 1.7242 | 60            | 13         |
| 1.7552 | 62            | 18         |
| 1.7842 | 56            | 28         |
| 1.8113 | 63            | 52         |
| 1.8369 | 59            | 53         |
| 1.8610 | 62            | 61         |
| 1.8839 | 60            | 60         |

Figure 1.1a shows the same data with a linear regression line superimposed. Although this line goes close to the plotted points, we can see some fluctuations around it. More seriously, this is a stupid model: it would predict a mortality rate of greater than 100% at a dose of 1.9 units, and a negative mortality rate at 1.65 units!

A more sensible dose-response relationship for the beetle mortality data might be based on the *logistic* function (to be defined later), as plotted in Figure 1.1b. The resulting curve is a closer, more-sensible, fit. Later in this module we will see how this curve was fitted using maximum likelihood estimation for an appropriate generalised linear model.

This is an example of a dose-response experiment which are widely used in medical and pharmaceutical situations.

**! Warning**

**Warning of potentially sensitive material.** For further information on dose-response experiments see, for example, [www.britannica.com/science/dose-response-relationship](http://www.britannica.com/science/dose-response-relationship).

<sup>1</sup>Dobson and Barnett, 3rd edn, p.127

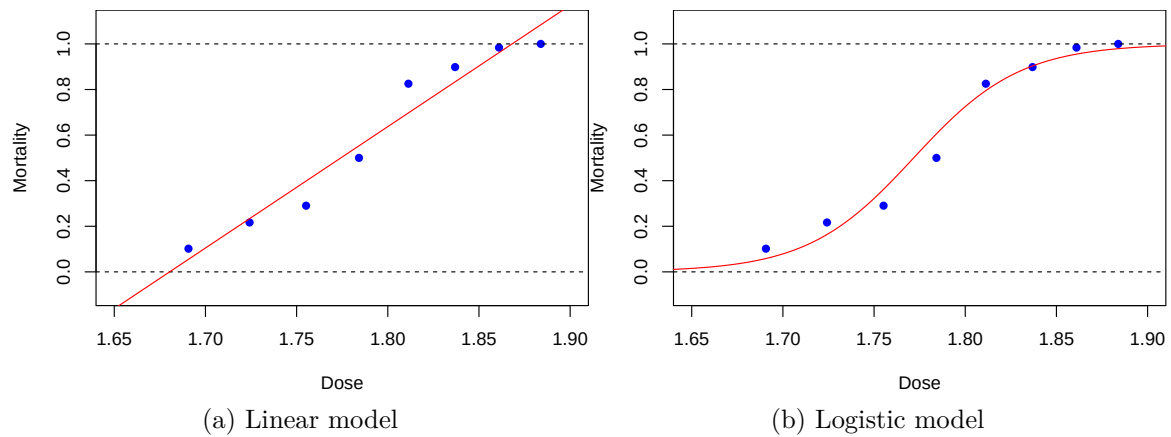


Figure 1.1: Beetle mortality rates with fitted dose-response curves.

## Focus on correlation quiz

Test your knowledge recall and comprehension to reinforce idea of linear relationships and correlation.

For each situation, choose one of the following statements which you think is **most likely** to apply.

1. The diastolic blood pressure and the weight of patients attending a heart health clinic at the Leeds General Infirmary.
  - (A) positive correlation
  - (B) uncorrelated
  - (C) negative correlation
  - (D) other
2. The daily stock market closing prices of British Telecom and Virgin Media shares on the London Stock Exchange.
  - (A) positive correlation
  - (B) uncorrelated
  - (C) negative correlation
  - (D) other
3. The daily rainfall and hours of sunshine collected at a weather monitoring station in the Pennines of Yorkshire.

- (A) positive correlation
  - (B) uncorrelated
  - (C) negative correlation
  - (D) other
4. The number of road accidents occurring at a busy roundabout and the UK Retail Prices Index.
- (A) positive correlation
  - (B) uncorrelated
  - (C) negative correlation
  - (D) other

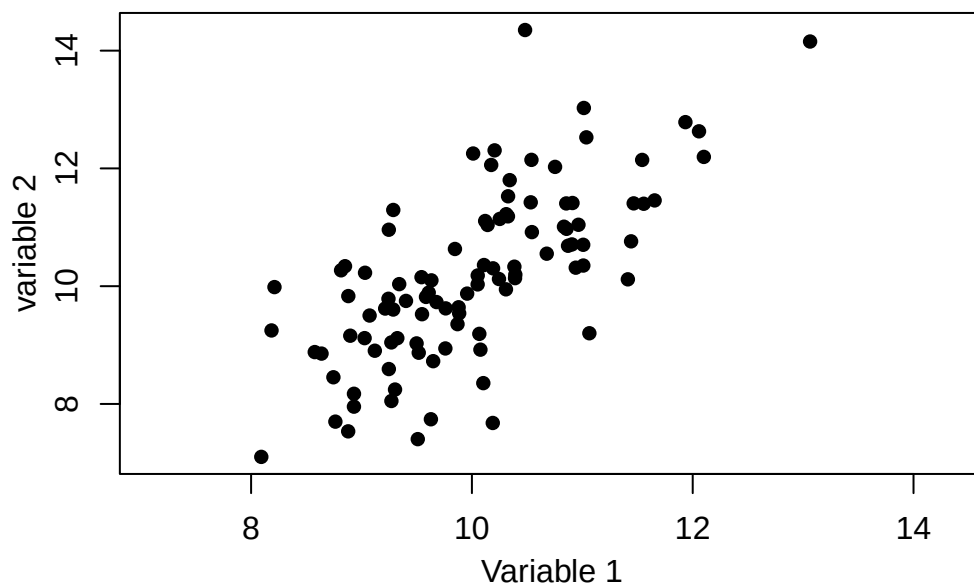
[Click here to see explanations](#)

1. It is well documented that people who are over-weight (or at least high BMI) are more likely to have high blood pressure. This is true for systolic and diastolic blood pressure as well as for men and women. It is not certain if the relationship will be linear, but it will lead to a strong positive correlation value. This is likely to be a causal relationship, excess weight causes high blood pressure.
2. Although these two companies are in the same business sectors, technology and entertainment, it is unlikely that direct competition will be the main factor in the relative behaviour – if it were, then we might expect a negative correlation, that is one does well if the other does badly. Instead, they are both likely to be driven by the same economic and social trends. This is not a causal relationship but both are being driven by an (unseen) third variable. It does, however, lead to a correlation.
3. Although both weather features, rainfall and sun, can happen at the same time – perhaps leading to a rainbow – and there are very many cases when neither occurs – for example a cloudy but dry day – there is a general pattern that it will not rain when it is sunny and it will not be sunny when it rains. This leads to a negative correlation and a moderate value is likely even if the relationship is not linear. Again, this is not a causal relationship but is being driven, perhaps, by the presence of clouds.
4. It is hard to imagine that there will be a relationship between the number of road accidents and an inflation measure, hence a value of correlation close to zero – though do not expect to ever get a value of exactly zero. It is not completely inconceivable that the number of road accidents will be higher when the economy is active, but this is unlikely to lead to a substantial correlation value.

From each of the following scatter plots, choose from the drop down list which correlation value is **most likely**.

5. What is the most likely correlation for the data below?

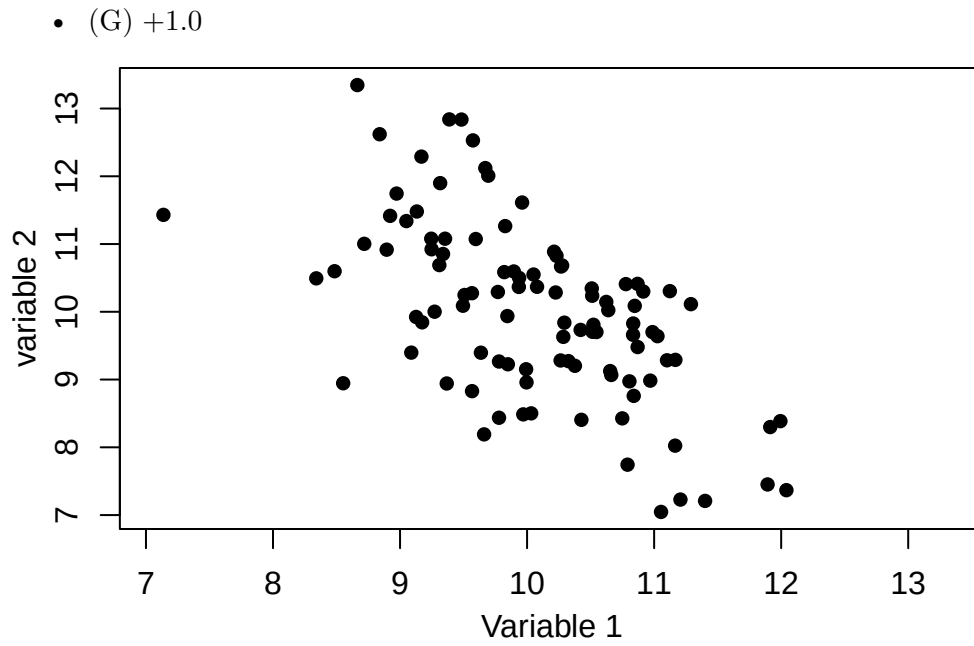
- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7
- (G) +1.0



most likely correlation for the data below?

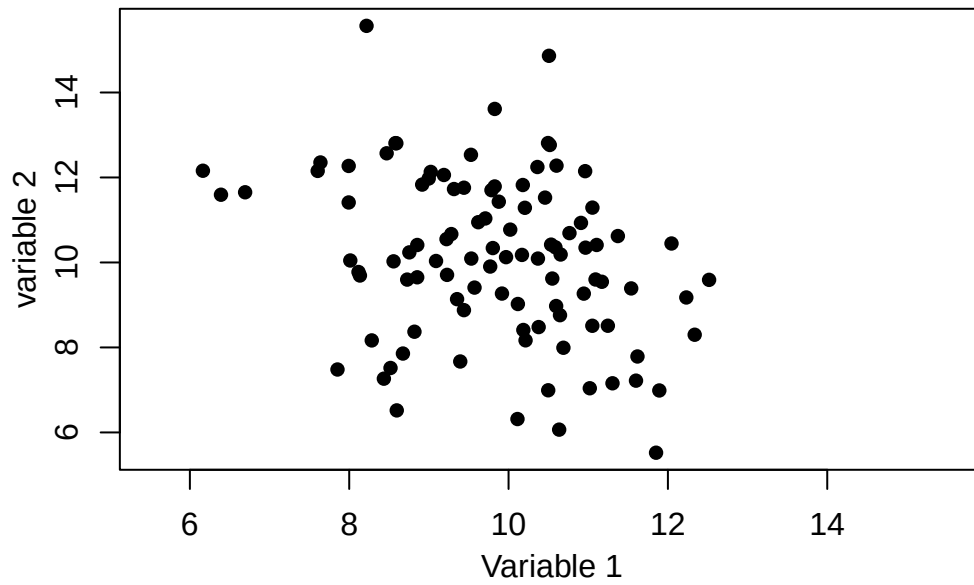
- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7

6. What is the



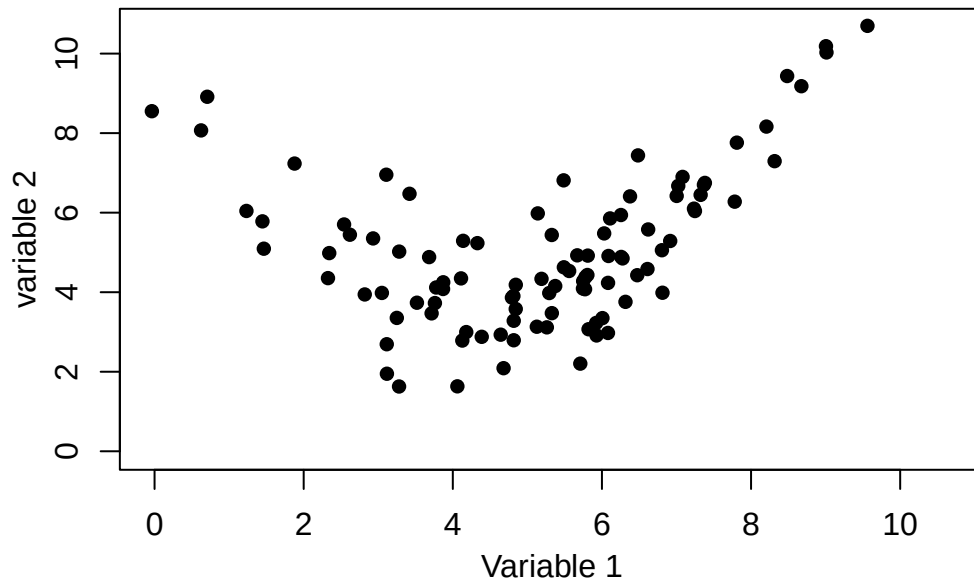
7. What is the most likely correlation for the data below?

- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7
- (G) +1.0



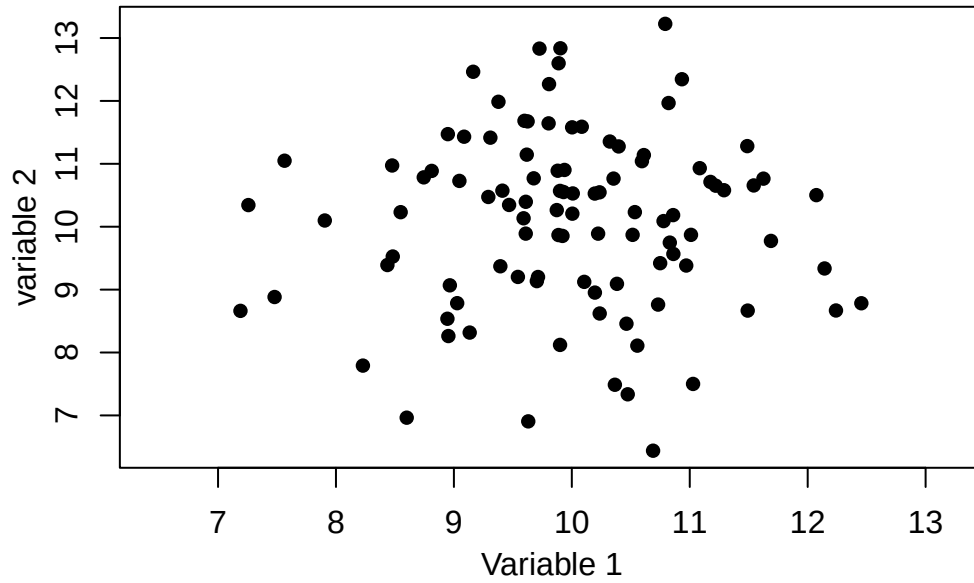
8. What is the most likely correlation for the data below?

- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7
- (G) +1.0



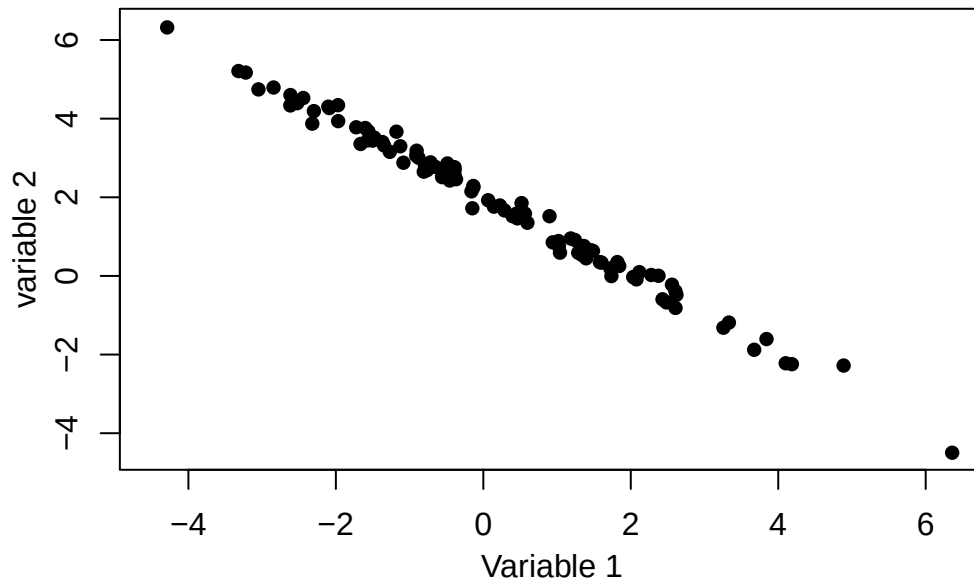
9. What is the most likely correlation for the data below?

- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7
- (G) +1.0



10. What is the most likely correlation for the data below?

- (A) -1.0
- (B) -0.7
- (C) -0.3
- (D) 0.0
- (E) +0.3
- (F) +0.7
- (G) +1.0



[Click here to see explanations](#)

5. Imagine dividing the plot by horizontal and vertical lines at the respective mean values. There would be a majority of points in the top-right and bottom-left indicating a positive correlation, but there are still some in the other quadrants. Hence,  $+1$  is too high and  $0.3$  is too low. For information, the exact value is  $0.70$ .
6. Again, imagine dividing the plot by horizontal and vertical lines at the respective mean values. This time the majority of points would be in the top-left and bottom-right quadrants, and there is moderate spread. A value  $-1$  is too extreme and  $-0.3$  is too close to zero. For information, the exact value is  $-0.60$ .
7. A similar situation to Question 6, but there is noticeably more spread. Although the majority of points are in the top-left and bottom-right quadrants there are substantial numbers in the other quadrants. It would be inaccurate to say that this shows uncorrelated variables as there is a definite negative slope to the pattern. For information, the exact value is  $-0.30$ .
8. This is a difficult one as there is a clear relationship, but it is quadratic rather than linear. For information, the exact value is  $0.30$ . Perhaps without the accompanying graph, this correlation value would be misleading.
9. Dividing the plot by horizontal and vertical lines at the respective mean values leaves very similar numbers of points in all four quadrants. This indicates that the correlation will be close to zero – there is no linear relationship. For information, the exact value is  $0.01$ .
10. Almost all of the points are in the top-left and bottom-right quadrants indicating a negative correlation. The points are very close to the line and hence a value close to  $-1$  is likely – such extreme cases are rare. For information, the exact value is  $-0.995$ .

### 1.3 Revision of least-squares estimation

Suppose that we have  $n$  paired data values  $(x_1, y_1), \dots, (x_n, y_n)$  and that we believe these are related by a linear model

$$y_i = \alpha + \beta x_i + \epsilon_i$$

for all  $i \in \{1, 2, \dots, n\}$ , where  $\epsilon_1, \dots, \epsilon_n$  are independent and identically distributed (iid) with  $E[\epsilon_i] = 0$  and  $\text{Var}[\epsilon_i] = \sigma^2$ . The aim will be to find values of the model parameters,  $\alpha, \beta$  and  $\sigma^2$  using the data. Specifically, we will estimate  $\alpha$  and  $\beta$  using the values which minimize the residual sum of squares (RSS)

$$RSS(\alpha, \beta) = \sum_{i=1}^n (y_i - (\alpha + \beta x_i))^2. \quad (1.1)$$

This measures how close the data points are around the regression line and hence the resulting estimates,  $\hat{\alpha}$  and  $\hat{\beta}$ , will give us a fitted regression line which is *closest* to the data.

Figure 1.2 illustrates this process. The data points are fixed but we are free to choose values for  $\alpha$  and  $\beta$ , that is to move the line up and down and to rotate it as needed. In general, the data points do not sit exactly on any line. For any values of  $\alpha$  and  $\beta$  the value on the line  $y = \alpha + \beta x$  can be calculated and the discrepancy, as measured in the vertical direction, is defined as the residual,  $r_i = y_i - (\alpha + \beta x_i)$ . Then, the residual sum of squares is formed as the sum of the squares of these individual residuals.

It can be shown that Equation 1.1 takes its minimum when the parameters are given by

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \quad \text{and} \quad \hat{\beta} = \frac{s_{xy}}{s_x^2} \quad (1.2)$$

where  $\bar{x}$  and  $\bar{y}$  are the sample means,

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

is the sample covariance and

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

is the sample variance of the  $x$  values. It can be shown that these estimators are unbiased, that is  $E[\hat{\alpha}] = \alpha$  and  $E[\hat{\beta}] = \beta$  – see Section 1.5.

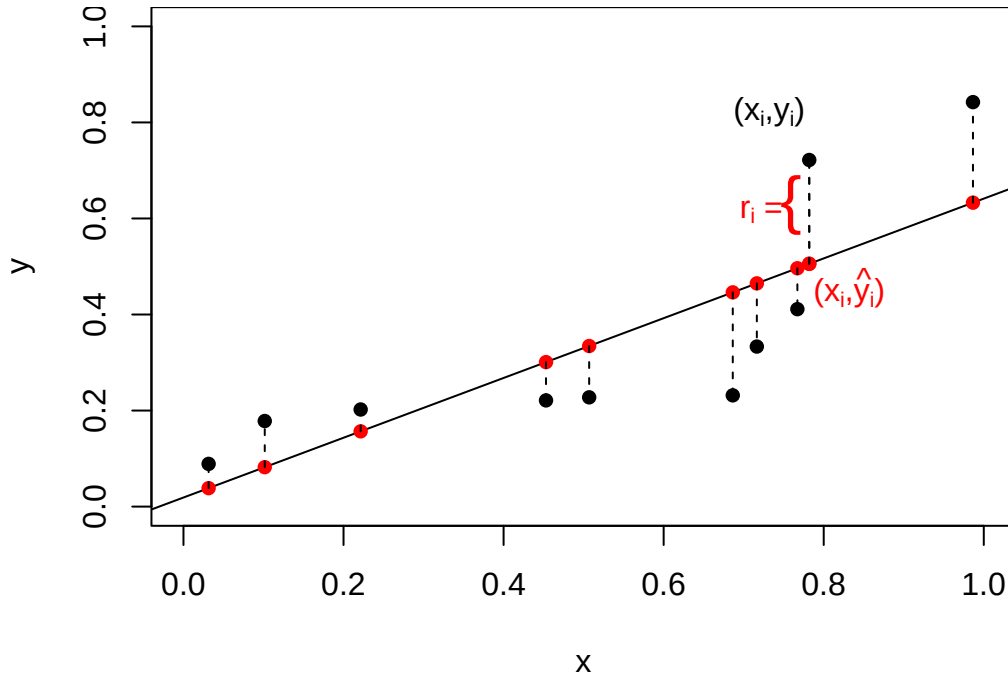


Figure 1.2: Diagram showing linear regression method

The fitted regression line is then given by  $\hat{y} = \hat{\alpha} + \hat{\beta}x$ , the fitted values by  $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ , and the model residuals by  $r_i = \hat{\epsilon}_i = y_i - \hat{y}_i$  for all  $i \in \{1, \dots, n\}$ .

To complete the model fitting, we also estimate the error variance,  $\sigma^2$ , using

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n r_i^2. \quad (1.3)$$

Note that, by construction,  $\bar{r} = 0$  and, further, it can be shown that  $\hat{\sigma}^2$  is an unbiased estimator of  $\sigma^2$ , that is  $E[\hat{\sigma}^2] = \sigma^2$ .

Returning to the beetle data example above, we have  $\hat{\alpha} = -8.947843$ ,  $\hat{\beta} = 5.324937$ , and  $\hat{\sigma}^2 = 0.0075151$ .

We will interpret the output later, but in *R*, the fitting can be done with a single command with corresponding fitting output from a second command:

Call:

```
lm(formula = mortality ~ dose)
```

Residuals:

|  | Min      | 1Q       | Median  | 3Q      | Max     |
|--|----------|----------|---------|---------|---------|
|  | -0.10816 | -0.06063 | 0.00263 | 0.05119 | 0.12818 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -8.9478  | 0.8717     | -10.27  | 4.99e-05 *** |
| dose        | 5.3249   | 0.4857     | 10.96   | 3.42e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08669 on 6 degrees of freedom

Multiple R-squared: 0.9524, Adjusted R-squared: 0.9445

F-statistic: 120.2 on 1 and 6 DF, p-value: 3.422e-05

## 1.4 Types of variables

The way a variable enters a model will depend on its type. The most common five types of variable are:

### 1. Quantitative

- Continuous: for example, height; weight; duration. Real valued. Note that although recorded data is rounded it is still usually best regarded as continuous.
- Count (discrete): for example, number of children in a family; accidents at a road junction; number of items sold. Non-negative and integer-valued.

### 2. Qualitative

- Ordered categorical (ordinal): for example, severity of illness (Mild/ Moderate/Severe); degree classification (first/ upper-second/ lower-second/ third).
- Unordered categorical (nominal):
  - Dichotomous (binary): two categories: for example sex (M/ F); agreement (Yes/ No); coin toss (Head/ Tail).
  - Polytomous (also known as polychotomous): more than two categories: for example blood group (A/ B/ O); eye colour (Brown/ Blue/ Green).

Note that although dichotomous is clearly a special case of polytomous, making the distinction is usually worthwhile as it often leads to a simplified modelling and testing approach.

## Focus on data type quiz

Test your knowledge recall and comprehension to reinforce ideas ready for later in the module

For each of the following situations what is the **most appropriate** data type: nominal, ordinal, discrete, or continuous?

- The eye colour of 100 patients visiting the Yorkshire Cancer Research Centre, for example, grey, green, brown, blue....

- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
2. The nationality of students at the University of Leeds, for example, British, Chinese, Greek, Indian....
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
3. The five-star ratings submitted by 50 customers on TripAdvisor for the Leeds Queens Hotel, for example 1 star, 2 star,... 5 star.
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
4. The diastolic blood pressure of 20 male and 20 female patients attending a heart health clinic at the Leeds General Infirmary in a study to investigate differences between men and women, for example, 80 mm Hg, 130 mm Hg,...
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
5. The daily stock market closing price of British Telecom shares on the London Stock Exchange over a year to study the change over time, for example, 114.75p, 115.10p,...
- (A) nominal

- (B) ordinal
  - (C) discrete
  - (D) continuous
6. The January monthly rainfall collected since 1961 at a weather monitoring station in the Pennines of Yorkshire, for example, 8 mm , 12 mm,...
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
7. The level of satisfaction of 100 randomly chosen voters with the policies of a political party, for example, agree, fully agree, neither agree nor disagree, disagree, and fully disagree.
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
8. The number of new people following the *TheRoyalFamily* twitter page per day over a year, for example 459, 700,... to study the change due to a royal wedding.
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous
9. The number of road accidents occurring per month, at a busy roundabout over a 10-year period to study the change over time, for example, 0, 1, 2,...
- (A) nominal
  - (B) ordinal

- (C) discrete
  - (D) continuous
10. The number of Scottish strawberries in 50 randomly selected boxes bought from ASDA supermarket, for example, 50, 58, 68,...
- (A) nominal
  - (B) ordinal
  - (C) discrete
  - (D) continuous

[Click here to see explanations](#)

1. Eye colour is qualitative and can take any one of an unordered set of categories. Although the eye colours are categories, there is no clear ordering to the colours.
2. Nationality is qualitative and can take any one of an unordered set of categories. Although the nationalities are categories, there is no clear ordering to the countries.
3. The rating is qualitative and can take any one of a set of categories but the categories are clearly ordered, 5 star is better than 4 star etc. Although the ratings are represented by integers, there is no reason why the difference between 1 and 2 stars has the same interpretation as between 4 and 5 stars and hence it cannot be discrete.
4. Although the recorded values might take only integer values, blood pressure is a measurement and could take any real number.
5. Share price is a measurement and could be any real number, even though in practice it will be rounded.
6. Rainfall is a measurement and although the recorded values might take only integer values, rainfall could be any real number.
7. Satisfaction score is qualitative and can take any one of a set of categories but the categories are clearly ordered, fully agree is better than agree etc. Although the scores could be represented by numerical values, e.g. 1,2,3,4,5, there is no reason why the difference between 1 and 2 has the same interpretation as between 4 and 5 and hence it is not discrete.
8. The number of people is a quantitative count which is limited to the non-negative integers. The variable is discrete.
9. The number of accidents is a quantitative count, being limited to the non-negative integers and hence is discrete.
10. The number of strawberries is a quantitative count which is limited to the non-negative integers. The variable is discrete.

## 1.5 Exercises

### ! Important

Data files for this module are available in the <https://www.richardpmann.com/MATH3701/Datasets/> directory of the course materials. For example:

```
beetle = read.table("https://www.richardpmann.com/MATH3701/Datasets/beetle.txt",
header=T)
```

1.1 Consider again the beetle data in Table 1.1. Perform the calculations by hand and then check the answers using *R*. Finally plot the fitted regression line on a scatter plot of the data.

[Click here to see hints.](#)

See the code chunk used to produce Figure 1.1.

1.2 Consider the following synthetic data:

|       | $i = 1$ | $i = 2$ | $i = 3$ | $i = 4$ | $i = 5$ | $i = 6$ | $i = 7$ | $i = 8$ |
|-------|---------|---------|---------|---------|---------|---------|---------|---------|
| $x_i$ | -1      | 0       | 1       | 2       | 2.5     | 3       | 4       | 6       |
| $y_i$ | -2.8    | -1.1    | 7.2     | 8.0     | 8.9     | 9.2     | 14.8    | 24.7    |

Plot the data to check that a linear model is suitable and then fit a linear regression model. Do you think that the fitted model can be reliably used to predict the values of  $y$  when  $x = 5$  and  $x = 10$ ? Justify your answers.

[Click here to see hints.](#)

Which is more reliable: prediction for a value within the range of the data (interpolation) or outside the range of the data (extrapolation)?

1.3 Starting from Equation 1.1, derive the estimation equations given in Equation 1.2. Show that  $\hat{\alpha}$  and  $\hat{\beta}$  are unbiased estimators of  $\alpha$  and  $\beta$ . What can be said about  $\hat{\sigma}^2$  as an estimator of  $\sigma^2$ ?

[Click here to see hints.](#)

For unbiasedness of intercept and slope, differentiate the RSS with respect to  $\alpha$  and  $\beta$  and set to zero. For the variance estimator, an intuitive explanation of why we divide by  $n - 2$  rather than  $n$  is sufficient.

1.4 The *Brownlee's Stack Loss Plant Data*<sup>2</sup> is already available in **R**, with background details on the help page, `?stackloss`.

<sup>2</sup>Brownlee, K. A. (1960, 2nd ed. 1965) *Statistical Theory and Methodology in Science and Engineering*. New York: Wiley. pp. 491–500.

After plotting all pairs of variables, which of `Air.Flow`, `Water.Temp` and `Acid.Conc` do you think could be used to model `stack.loss` using a linear regression? Justify your answer.

Perform a simple linear regression using `stack.loss` as the response variable and your chosen variable as the explanatory variable. Add the fitted regression line to a scatter plot of the data and comment.

[Click here to see hints.](#)

Use `pairs()` to plot all variable pairs simultaneously, then `lm()` to fit the chosen model.

1.5 In an experiment conducted by de Silva et al. in 2020<sup>3</sup> data was obtained to investigate falling objects and gravity, as first considered by Galileo and Newton. A copy of the data is available in the file `physics_from_data.csv`.

Suppose that we wish to develop a method to predict the maximum Reynolds number from a single explanatory variable. Which of the variables do you think helps explain Reynolds number the best? Why do you think this?

[Click here to see hints.](#)

Read the data into `R` and perform a simple linear regression of the maximum Reynolds number as the response variable and, in turn, each of the other variables as the explanatory variable. Plot the data and add the corresponding fitted linear models.

---

<sup>3</sup>de Silva BM, Higdon DM, Brunton SL, Kutz JN. Discovery of Physics From Data: Universal Laws and Discrepancies. *Front Artif Intell.* 2020 Apr 28;3:25. doi: 10.3389/frai.2020.00025. PMID: 33733144; PMCID: PMC7861345.

## 2 Essentials of Normal Linear Models

### 2.1 Overview

In many fields of application, we might assume the response variable is normally distributed. For example: heights, weights, log prices, etc.

The data<sup>1</sup> in Table 2.1 record the birth weights of 12 girls and 12 boys and their gestational ages (time from conception to birth).

A key question is: can we predict the birth weight of a baby born at a given gestational age? Figure 2.1 shows initial exploratory plots.

### Focus on modelling quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the module.

For each situation, choose one of the following statements which you think is **most likely** to apply.

1. What is the most useful graphical summary for identifying a potential relationship between two variables?
  - (A) boxplot
  - (B) scatter plot
  - (C) kernel density plot
  - (D) histogram
  - (E) other
2. What is the most useful numerical summary for identifying a potential linear relationship between two variables?
  - (A) variances

---

<sup>1</sup>Dobson and Barnett, 3rd edition, Table 2.3.

Table 2.1: Gestational ages (in weeks) and birth weights (in grams) for 24 babies (12 girls and 12 boys).

Table 2.2: Girls

| Gestational Age | Birth weight |
|-----------------|--------------|
| 40              | 3317         |
| 36              | 2729         |
| 40              | 2935         |
| 37              | 2754         |
| 42              | 3210         |
| 39              | 2817         |
| 40              | 3126         |
| 37              | 2539         |
| 36              | 2412         |
| 38              | 2991         |
| 39              | 2875         |
| 40              | 3231         |

Table 2.3: Boys

| Gestational Age | Birth weight |
|-----------------|--------------|
| 40              | 2968         |
| 38              | 2795         |
| 40              | 3163         |
| 35              | 2925         |
| 36              | 2625         |
| 37              | 2847         |
| 41              | 3292         |
| 40              | 3473         |
| 37              | 2628         |
| 38              | 3176         |
| 40              | 3421         |
| 38              | 3975         |

- (B) correlation
  - (C) skew
  - (D) sample means
  - (E) other
3. Which of the following is a true statement about the correlation coefficient? (Choose any that apply.)
- (A) Can be positive or negative
  - (B) A value close to -1 means uncorrelated and close to +1 indicates a high correlation
  - (C) Is always a positive number
  - (D) Can take any real number
  - (E) Is always between -1 and +1
4. Which of the following is a true statement about regression? (Choose any that apply.)
- (A) All regression models describe linear relationships

- (B) A linear regression model will fit well if the correlation is close to zero
- (C) A linear regression can sometimes be used to approximate a non-linear relationship
- (D) After fitting a regression model we should always consider residuals
- (E) A linear model can be fitted to any data set

5. Which of the following is a true statement about statistical modelling? (Choose any that apply.)

- (A) Only use a linear model
- (B) Only use a model that is 100% correct
- (C) Models only involve a statistical distribution
- (D) Models are part deterministic and part random
- (E) All models are approximations

The histogram shows a spread around 2800–3000 g; the boxplot indicates slightly higher birth weights for boys; and the scatter plot shows an increasing relationship with gestational age. Together, these suggest that both gestational age and sex are likely to be useful predictors of birth weight.

## 2.2 Linear models

We consider four nested candidate models:

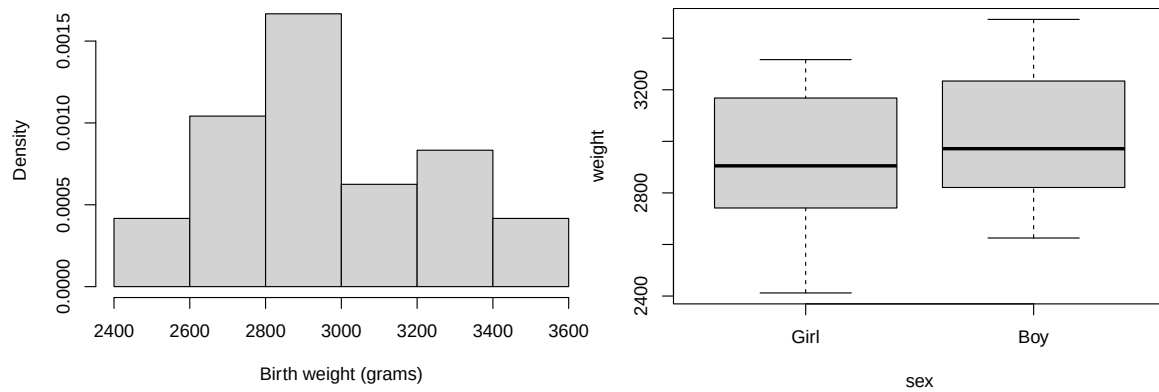
---

|           |                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------|
| Model 0 : | $\text{Weight} = \alpha$                                                                                               |
| Model 1 : | $\text{Weight} = \alpha + \beta \cdot \text{Age}$                                                                      |
| Model 2 : | $\text{Weight} = \alpha + \beta \cdot \text{Age} + \gamma \cdot \text{Sex}$                                            |
| Model 3 : | $\text{Weight} = \alpha + \beta \cdot \text{Age} + \gamma \cdot \text{Sex} + \delta \cdot \text{Age} \cdot \text{Sex}$ |

---

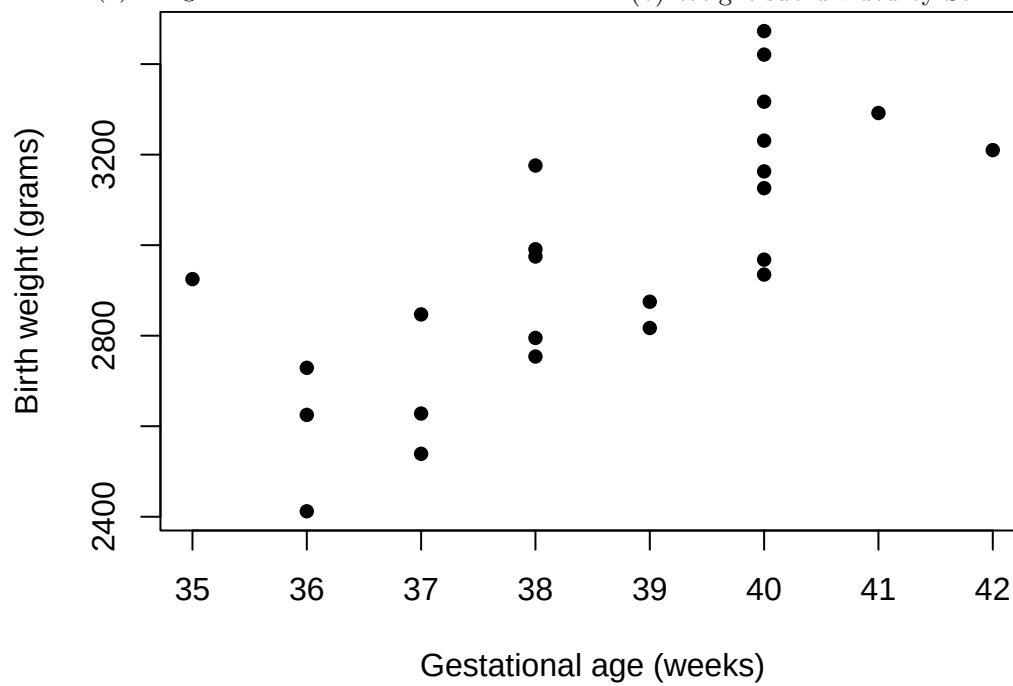
Here, **Weight** is the *response* variable and **Age** (continuous) and **Sex** (a *dummy* variable: 0 for girls, 1 for boys) are the *explanatory* variables. Each model is a *special case* of the next — such models are called *nested*.

$\gamma$  is the main effect of **Sex**: the difference in birth weight between boys and girls at the same gestational age.  $\delta$  is the *interaction* between **Age** and **Sex**: it captures whether the effect of gestational age on birth weight differs between boys and girls.



(a) Weight distribution

(b) Weight sub-divided by Sex



(c) Relationship between variables

Figure 2.1: Birthweight and gestational age for 24 babies.

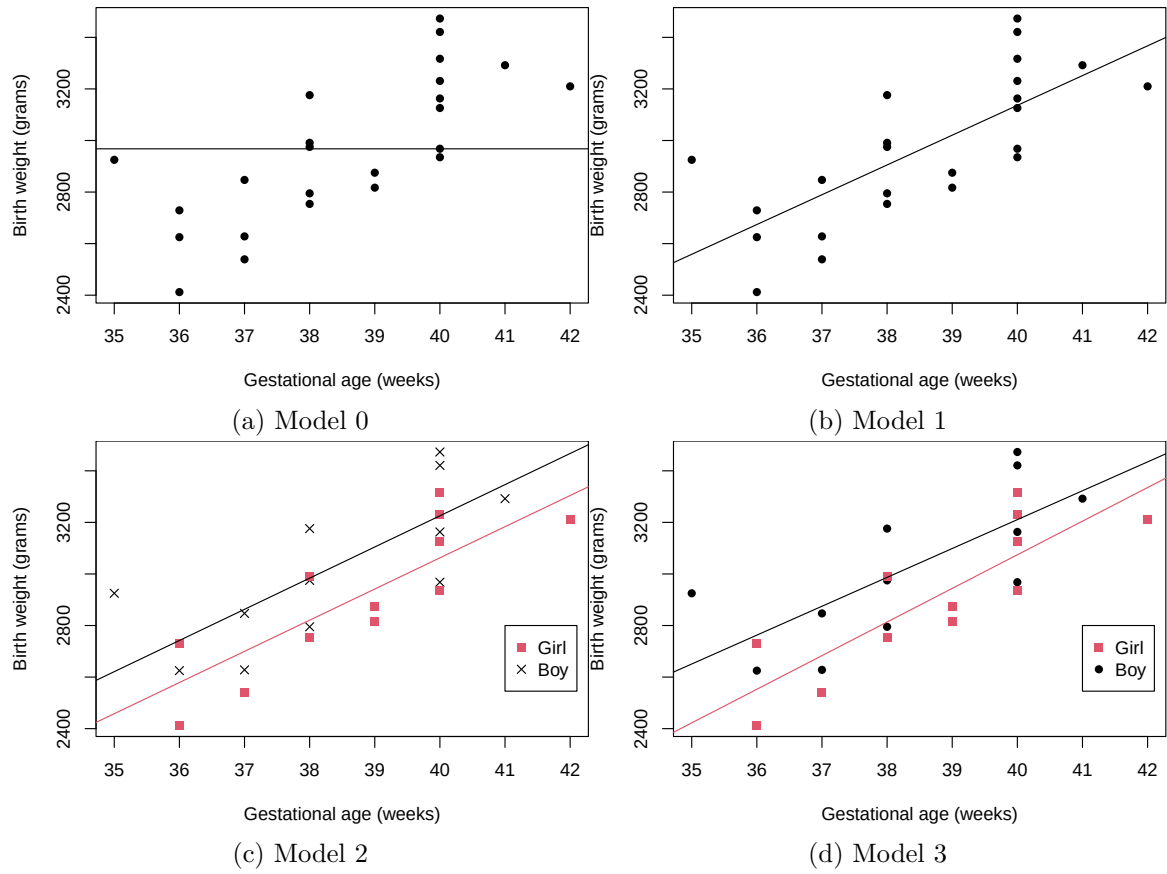


Figure 2.2: Birthweight data with fitted regression lines from competing models.

To choose between nested models we use  $F$ -tests. Let  $R_k$  denote the residual sum of squares (RSS) for Model  $k$  with  $r_k$  residual degrees of freedom. The test statistic for comparing Model  $k + 1$  against the simpler Model  $k$  is

$$F = \frac{(R_k - R_{k+1})/(r_k - r_{k+1})}{R_{k+1}/r_{k+1}},$$

which under  $H_0$  (Model  $k$  is adequate) follows an  $F_{r_k - r_{k+1}, r_{k+1}}$  distribution. A large  $F$  (small  $p$ -value) is evidence that the more complex model is needed. In **R**, `anova()` computes this table directly.

Fitting Model 1 in **R**:

```
(Intercept)      age
-1484.9846      115.5283
```

Analysis of Variance Table

```
Response: weight
      Df Sum Sq Mean Sq F value    Pr(>F)
age      1 1013799 1013799   27.33 3.04e-05 ***
Residuals 22  816074   37094
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA table gives  $F = 27.3$  ( $p < 0.001$ ): strong evidence that gestational age is a significant predictor. Fitting Model 2:

```
(Intercept)      age      sexM
-1773.3218      120.8943      163.0393
```

Analysis of Variance Table

```
Response: weight
      Df Sum Sq Mean Sq F value    Pr(>F)
age      1 1013799 1013799 32.3174 1.213e-05 ***
sex      1  157304  157304   5.0145  0.03609 *
Residuals 21  658771   31370
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The estimate  $\hat{\gamma} = 163$  g is the additional birth weight for boys after adjusting for gestational age, and is statistically significant ( $F = 5.01$ ,  $p = 0.036$ ). Comparing Models 2 and 3 is left as an exercise.

## Focus on regression quiz

1. Which of the following is a true statement about correlation and linear regression? (Choose any that apply.)
  - (A) The correlation and the slope parameter will have the opposite sign
  - (B) There is no relationship between correlation and linear regression
  - (C) The correlation and the regression slope parameter will have the same sign
  - (D) If the correlation is away from zero then a linear regression should always be used
  - (E) If the correlation is close to zero then a linear regression will not be useful
2. Which of the following is NOT a true statement about model residuals? (Choose any that apply.)
  - (A) All should be zero if the model is a good fit
  - (B) Can help to identify unusual data points
  - (C) Should always be plotted as part of a data analysis
  - (D) Will always sum to zero
  - (E) Can be used to judge how well the model fits the data
3. Which of the following is a true statement about prediction using linear regression? (Choose any that apply.)
  - (A) We should be careful about extrapolating
  - (B) Interpolation and extrapolation are both types of prediction
  - (C) Interpolation should only be used if we know the true relationship is linear
  - (D) Interpolation is performed using the fitted model
  - (E) Prediction is always reliable
4. Which of the following is NOT an important part of regression model fitting? (Choose any that apply.)
  - (A) A scatter plot of the residuals

- (B) A scatter plot of the data
- (C) A histogram of the data
- (D) The command `lm` to fit a linear model
- (E) The command `cor` to check for a relationship between variables

## 2.3 Types of normal linear model

The dependent variable  $y$  is modelled as a linear combination of  $p$  explanatory variables  $\mathbf{x} = (x_1, x_2, \dots, x_p)$  plus a random error  $\epsilon \sim N(0, \sigma^2)$ . Table 2.5 summarises common special cases.

Table 2.5: Types of normal linear model and their explanatory variable types, where indicator function  $I(x = j) = 1$  if  $x = j$  and 0 otherwise.

| $p$ | Explanatory variables       | Model                                                                            |
|-----|-----------------------------|----------------------------------------------------------------------------------|
| 1   | Quantitative                | Simple linear regression $y = \alpha + \beta x + \epsilon$                       |
| >1  | Quantitative                | Multiple linear regression<br>$y = \alpha + \sum_{i=1}^p \beta_i x_i + \epsilon$ |
| 1   | Dichotomous ( $x = 1$ or 2) | Two-sample $t$ -test<br>$y = \alpha + \delta I(x = 2) + \epsilon$                |
| 1   | Polytomous, $k$ levels      | One-way ANOVA<br>$y = \alpha + \sum_{j=1}^k \delta_j I(x = j) + \epsilon$        |
| >1  | Qualitative                 | $p$ -way ANOVA                                                                   |

For the two-sample  $t$ -test model, observations in the two groups have means  $\alpha$  and  $\alpha + \beta_2$  under the *corner* constraint  $\beta_1 = 0$  — so  $\beta_2$  is the difference in means relative to a baseline. For one-way ANOVA with  $k$  groups, the corner constraint  $\delta_1 = 0$  means  $\delta_j$  is the mean difference between group  $j$  and group 1.

## 2.4 Matrix representation of linear models

All models in Table 2.5 can be written as

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon, \quad (2.1)$$

where

- $\mathbf{Y}$  is an  $n \times 1$  vector of observed responses;

- $\mathbf{X}$  is an  $n \times p$  *design* matrix;
- $\beta$  is a  $p \times 1$  vector of parameters;
- $\epsilon$  is an  $n \times 1$  vector of IID  $N(0, \sigma^2)$  errors.

**Constructing the design matrix.** Start with a column of ones (the intercept). For each quantitative variable add one column of values. For each qualitative variable with  $k$  levels, add  $k$  indicator columns and then delete one to avoid singularity (the *corner constraint*).

**Example: Simple linear regression.** For  $y_i = \alpha + \beta x_i + \epsilon_i$ :

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

**Example: One-way ANOVA with  $k$  groups.** If observation  $i$  is in group  $g_i$ , the design matrix has a column of ones plus  $k - 1$  indicator columns (after the corner constraint). For a two-group ( $k = 2$ ) example with  $n = 5$  observations:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad \beta = \begin{bmatrix} \alpha \\ \delta_2 \end{bmatrix},$$

where  $\delta_2$  is the mean difference between group 2 and group 1.

**OLS estimation.** The ordinary least-squares (OLS) estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}, \quad (2.2)$$

with fitted values  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{H}\mathbf{Y}$ , where  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is the *hat matrix*. The residual vector is  $\mathbf{r} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$ .

## Focus on matrix representations quiz

1. What is the dimension of the design matrix?

- (A)  $n$  by 1
- (B) 2
- (C)  $p$  by 1
- (D)  $X$

- (E)  $n$  by  $p$
2. What is the distribution of the error term?
- (A) mean zero
  - (B) normal
  - (C) IID
  - (D)  $n$  by 1
  - (E) random variable
3. Which quantity represents the model parameters?
- (A)  $n$
  - (B)  $X$
  - (C) epsilon
  - (D) beta
  - (E)  $Y$
4. Which two terms in the model have the same dimensions?
- (A)  $Y$  and epsilon
  - (B) beta and epsilon
  - (C)  $Y$  and beta
  - (D)  $Y$  and  $X$
  - (E)  $X$  and beta
5. Which of the following is a potential problem when using qualitative variables?
- (A)  $X$  is not square
  - (B)  $X$  is singular
  - (C)  $X$  can be negative
  - (D)  $X$  can be zero

- (E) X is full rank

## 2.5 Model shorthand notation

In R, a qualitative variable is called a *factor* and its categories are called *levels*. A convenient compact notation specifies models without writing out the full design matrix. Below,  $E$ ,  $F$  denote generic variables.

Table 2.6: Model shorthand notation

| Notation               | Meaning                                                             |
|------------------------|---------------------------------------------------------------------|
| $E$                    | Include $E$ (one column if quantitative; $k - 1$ columns if factor) |
| $E + F$                | Include $E$ and $F$ (no interaction)                                |
| $E : F$                | Interaction of $E$ and $F$                                          |
| $E * F$                | Shorthand for $1 + E + F + E:F$                                     |
| $E/F$                  | Shorthand for $1 + E + E:F$ (nested model)                          |
| $\text{poly}(E, \ell)$ | Orthogonal polynomial in $E$ up to degree $\ell$                    |
| $-E$                   | Remove term $E$ from the model                                      |
| $I(\cdot)$             | Arithmetic expression, e.g. $I(E * F)$ multiplies two variables     |

The notation uses  $\sim$  as shorthand for “is modelled by”. Examples:

- Weight regressed on age and sex, no interaction:  
`Weight ~ Age + Sex`
- Wellbeing regressed on age and income with interaction:  
`Wellbeing ~ Age * Income`
- Degree class regressed on school and subject nested within school:  
`DegreeClass ~ School/Subject`
- Yield regressed on variety and a quadratic in rainfall:  
`Yield ~ Variety + poly(Rainfall, 2)`

## Focus on model notation quiz

1. What **R** command can be used to convert numerical values into a nominal variable?

- (A) `as.nominal`
- (B) `factorial`
- (C) `as.factor`
- (D) `as.numerical`

- (E) as.ordinal
2. Which of the following defines a model where variable Y is regressed on variables V1 and V2, but without an interaction?
- (A)  $Y \sim V1:V2$
  - (B)  $Y \sim V1*V2$
  - (C)  $Y \sim V1 + V2 + V1:V2$
  - (D)  $Y \sim V1 + V2$
  - (E)  $V1 + V2 \sim Y$
3. Which of the following defines a model where variable Y is regressed on variables V1 and V2, including a constant and an interaction?
- (A)  $Y \sim V1*V2$
  - (B)  $Y \sim 1 + V1 + V2$
  - (C)  $Y \sim 1+V1/V2$
  - (D)  $Y \sim 1+V1:V2$
  - (E)  $Y \sim V1 + V2 + V1:V2 -1$
4. Which of the following defines a model regressing Y on the product of V1 and V2?
- (A)  $Y \sim V1:V2$
  - (B)  $Y \sim I(V1 + V2)$
  - (C)  $Y \sim V1.V2$
  - (D)  $Y \sim \text{poly}(V1,V2)$
  - (E)  $Y \sim I(V1*V2)$
5. Which of the following defines a model where variable Y is regressed on a second-order polynomial in V1?
- (A)  $Y \sim V1*V1$
  - (B)  $Y \sim \text{poly}(V1,2)$

- (C)  $Y \sim I(V1)$
- (D)  $Y \sim 1+V1$
- (E)  $Y \sim V1^2$

## 2.6 Fitting linear models in R

The primary command is `lm(formula)`. Key functions for working with a fitted model object (here called `my.lm`):

| Command                              | Output                                      |
|--------------------------------------|---------------------------------------------|
| <code>my.lm = lm(y ~ x + a*b)</code> | Fit the model                               |
| <code>summary(my.lm)</code>          | Parameter estimates, SEs, $t$ -tests, $R^2$ |
| <code>anova(my.lm)</code>            | ANOVA table with $F$ -tests                 |
| <code>coefficients(my.lm)</code>     | Parameter estimates                         |
| <code>fitted.values(my.lm)</code>    | Fitted values $\hat{Y}$                     |
| <code>residuals(my.lm)</code>        | Residuals $\mathbf{r}$                      |
| <code>df.residual(my.lm)</code>      | Residual degrees of freedom                 |
| <code>model.matrix(y ~ a*b)</code>   | Design matrix                               |
| <code>predict(my.lm, newdata)</code> | Predicted values at new $x$ values          |

Always check residual plots after fitting. Residuals should show no systematic pattern when plotted against fitted values, should be approximately symmetric, and a normal QQ plot should be approximately linear.

Call:

```
lm(formula = y ~ x)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-0.38790 -0.12026 -0.01578  0.12751  0.44184
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.941552   0.062524   31.05  <2e-16 ***
x             0.204299   0.008609   23.73  <2e-16 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.1972 on 48 degrees of freedom

Multiple R-squared: 0.9215, Adjusted R-squared: 0.9198

F-statistic: 563.1 on 1 and 48 DF, p-value:  $< 2.2e-16$

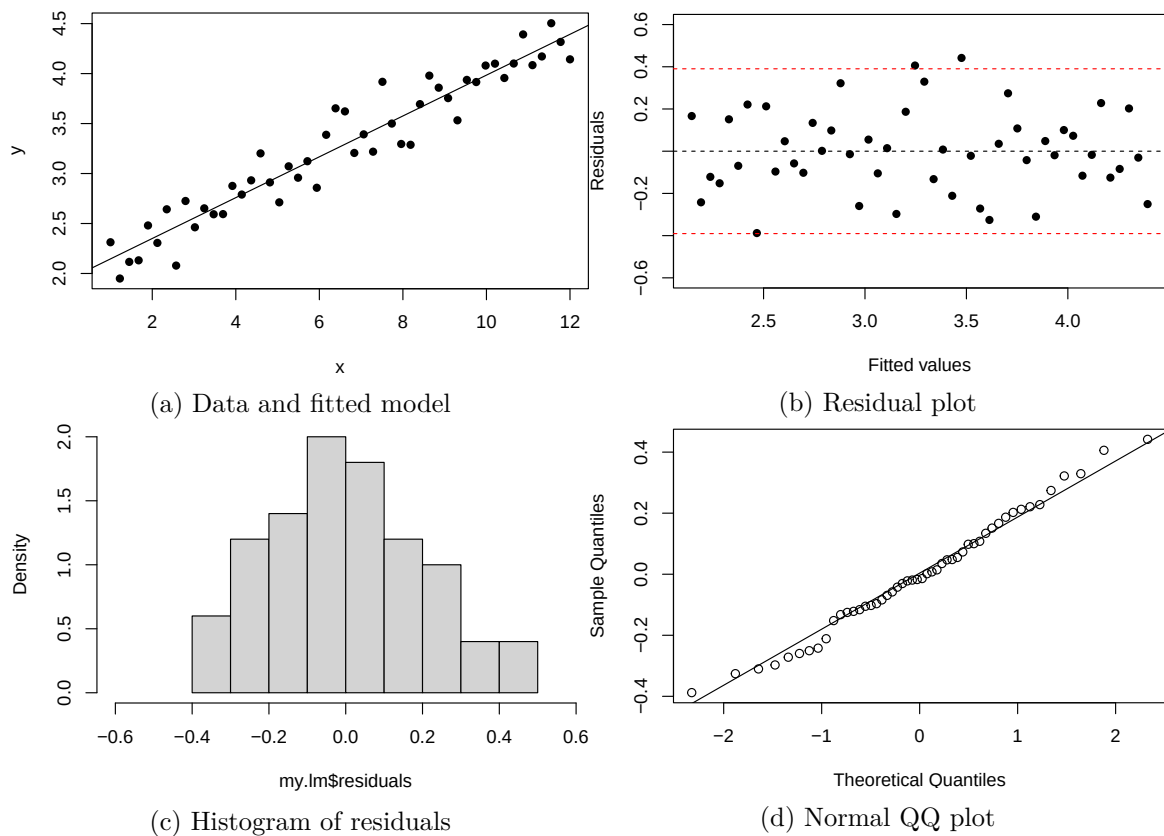


Figure 2.3: Model fitting on a toy example.

The residual plot shows no systematic pattern and all residuals are within two standard deviations of zero. The histogram and QQ plot support normality. To predict at new values of  $x$ :

|          |          |
|----------|----------|
| 1        | 2        |
| 2.758747 | 3.167345 |

## 2.7 Ethics in statistics and data science

### **i** Note

#### **Directed reading.**

In previous modules you will have seen that professional and ethical considerations are important in statistics and data science: how we choose which graph to present, how we handle missing data, and how we deal with suspected outliers. It is important for statisticians and data scientists to be aware of the ethical dimensions of their work.

Roughly speaking, ethics concerns what we ought to do and the kind of person we ought to be. Questions relevant to data science include:

- How should we collect data in a way that respects participants?
- Why is privacy important? How should data scientists protect privacy?
- How can algorithmic bias wrong members of the public?
- To what extent are data scientists responsible for the impact of their work?
- When and how should data scientists challenge authority in the workplace?

Please keep these issues in mind throughout any data analysis and modelling work. For further discussion, the Royal Statistical Society and the Alan Turing Institute both publish guidance on data ethics.

## 2.8 Exercises

2.1 The following situations each involve two variables. For each: explore the data with a suitable plot, calculate the correlation, and decide whether a linear model would be appropriate.

- a. Use the data in `schoolstudy.csv` to investigate the relationship between calorie intake and BMI in 11-year-old children.
- b. Use the data in `smartmeter.csv` to investigate the relationship between daily electricity and gas consumption in a typical house.
- c. Use the data in `housing.csv` (2022–23 UK Housing Survey) to investigate the relationship between number of rooms and household income.

[Click here to see hints.](#)

For each data set, use `read.csv()` to load the data, `plot()` for a scatter plot, `cor()` for the correlation, and `lm()` to fit a linear model. Does a straight line seem reasonable?

2.2 An extra model for the birthweight data would be one where **Weight** differs by sex but does not depend on gestational age. Write down the model equation, fit it in **R**, and test whether sex is statistically significant.

[Click here to see hints.](#)

The model is  $\text{Weight} = \alpha + \gamma \cdot \text{Sex}$ . Use `lm()` and `anova()`.

2.3. For each given situation, consider the description and then investigate the suitability of a linear model.

[Click here to see hints.](#)

For each given data set, produce an appropriate graph within RStudio, fit a linear regression model and add the fitted model to the graph. Comment on the quality of fit.

- a. Continuing the childhood obesity example, use the data in file `schoolstudy.csv` to model the relationship between BMI and calorie intake in 11-year old children.
- b. To study the profitability of several iron ore (hematite) extraction quarries, small samples are taken from lorries arriving at an iron purification site. The lorries are open-topped with most travelling less than 20 Km but one quarry is more than 100 Km away. A chemical analysis provides a percentage of pure iron in the sample. Quarries with iron content less than 30% are not considered economically viable. The data file `iron.csv` contains measurements of percentage pure iron arriving at an iron purification site recorded over a 50 year period. Model the relationship between iron purity and time.
- c. A study aims to investigate osteoporosis in women before and after menopause. The X-rays of a randomly selected sample of patients taking routine mammograms are analysed. The age of the patient and their menopause status are recorded, along with a measure of bone density calculated from the X-ray. Use the data in the file `bmd.csv` to model the relationship between age and Tscore, noting that a value of below -2.5 indicates osteoporosis, between -2.5 and -1.0 indicates osteopenia whereas above -1.0 is normal.
- d. A primary school head teaching wishes to investigate the relationship between social skills of children and the ages of their brothers and sisters. The hypothesis is that those with older siblings will be better able to deal with social interaction. The file `skills.csv` contains data on the age of the eldest sibling of a class of 6-year old school children along with a *social skills* score for each child assessed during the school lunch break. Model the relationship between sibling age and social skills.

2.4 In an experiment to investigate Ohm's Law,  $V = IR$ , the following data<sup>2</sup> were recorded:

Table 2.8: Experimental verification of Ohm's Law

|                 |    |    |    |    |    |    |      |    |
|-----------------|----|----|----|----|----|----|------|----|
| Voltage (Volts) | 4  | 8  | 10 | 12 | 14 | 18 | 20   | 24 |
| Current (mAmps) | 11 | 24 | 30 | 36 | 40 | 53 | 58.5 | 70 |

Does this data support Ohm's Law? What is the resistance of the material?

[Click here to see hints.](#)

Note that Ohm's Law is a linear function through the origin. To fit a model without intercept, use `lm(y ~ x - 1)`. The resistance is the reciprocal of the estimated slope.

2.5 In an investigation<sup>3</sup> into the effect of eating on pulse rate, 6 men and 6 women were tested before and after a meal:

Table 2.9: Pulse rates before and after a meal

|     |        |     |    |    |     |    |    |
|-----|--------|-----|----|----|-----|----|----|
| Men | before | 105 | 79 | 79 | 103 | 87 | 97 |
|-----|--------|-----|----|----|-----|----|----|

<sup>2</sup>Aykroyd, P.J. (1956). Unpublished.

<sup>3</sup>Source unknown.

|       |        |     |    |    |     |     |     |
|-------|--------|-----|----|----|-----|-----|-----|
|       | after  | 109 | 87 | 86 | 109 | 100 | 101 |
| Women | before | 74  | 73 | 82 | 78  | 86  | 77  |
|       | after  | 82  | 80 | 90 | 90  | 93  | 81  |

---

Suggest a suitable model and write down the corresponding design matrix. Test whether the change in pulse rate due to the meal differs between men and women.

[Click here to see hints.](#)

The explanatory variables are **Sex** (Men/Women) and **Time** (before/after), both categorical. Declare them as factors. An interaction between Sex and Time captures whether the before-to-after change differs by sex. Use `lm()` and `anova()`.

2.6 A laboratory experiment<sup>4</sup> investigated the effect of seasonal floods on the height of barley seedlings. Three barley varieties (Goldmarker, Midas, Igri) were grown under two watering conditions (Normal and Waterlogged) and on four shelf positions (Top, Second, Third, Bottom). Data are in `barley.csv`.

Fit appropriate models to test the importance of: (a) watering condition, (b) barley variety, and (c) shelf position. Do not include interactions involving shelf position. If you find a significant interaction between watering condition and variety, interpret the parameter estimates carefully.

[Click here to see hints.](#)

Declare all three explanatory variables as factors with `as.factor()`. Fit a model including main effects and the watering-by-variety interaction with `lm()`, then use `anova()` to assess significance.

---

<sup>4</sup>Source unknown.

## 3 GLM Theory

### 3.1 Motivating examples

We cannot always assume that the dependent variable  $Y$  is normally distributed. For example, for the beetle mortality data in Table 1.1, suppose each beetle subjected to a dose  $x_i$  has a probability  $p_i$  of being killed. Then, the number of beetles killed  $Y_i$  out of a total number  $m_i$  at dose-level  $x_i$  will have a  $\text{Bin}(m_i, p_i)$  distribution with probability mass function

$$\Pr(y_i; p_i, m_i) = \binom{m_i}{y_i} p_i^{y_i} (1 - p_i)^{m_i - y_i} \quad (3.1)$$

where  $y_i$  takes values in  $\{0, 1, \dots, m_i\}$ .

Table 12.1 contains seasonal data on tropical cyclones for 13 seasons. Suppose that, within season  $i$ , there is a constant probability  $\lambda_i dt$  of a cyclone occurring in any short time-interval  $dt$ . Then the total number of cyclones  $Y_i$  during season  $i$  will have a Poisson distribution with mean  $\lambda_i$ , that is  $Y_i \sim \text{Po}(\lambda_i)$ , with probability mass function

$$\Pr(y_i; \lambda_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} \quad (3.2)$$

where  $y_i$  takes values in  $\{0, 1, 2, \dots\}$ .

Table 3.1: Numbers of tropical cyclones in  $n = 13$  successive seasons<sup>1</sup>

|                   |   |   |   |   |   |   |    |   |   |    |    |    |    |
|-------------------|---|---|---|---|---|---|----|---|---|----|----|----|----|
| Season            | 1 | 2 | 3 | 4 | 5 | 6 | 7  | 8 | 9 | 10 | 11 | 12 | 13 |
| No of<br>cyclones | 6 | 5 | 4 | 6 | 6 | 3 | 12 | 7 | 4 | 2  | 6  | 7  | 4  |

In these two examples, we have non-normal data and would like to know whether and how the dependent variable  $Y_i$  depends on the covariate dose or season.

Generalised linear models provide a modelling framework for data analysis in the non-normal setting. We will revisit the beetle mortality and cyclone data sets after describing the structure of a generalised linear model.

<sup>1</sup>Dobson and Barnett, 3rd edn, Table 1.2

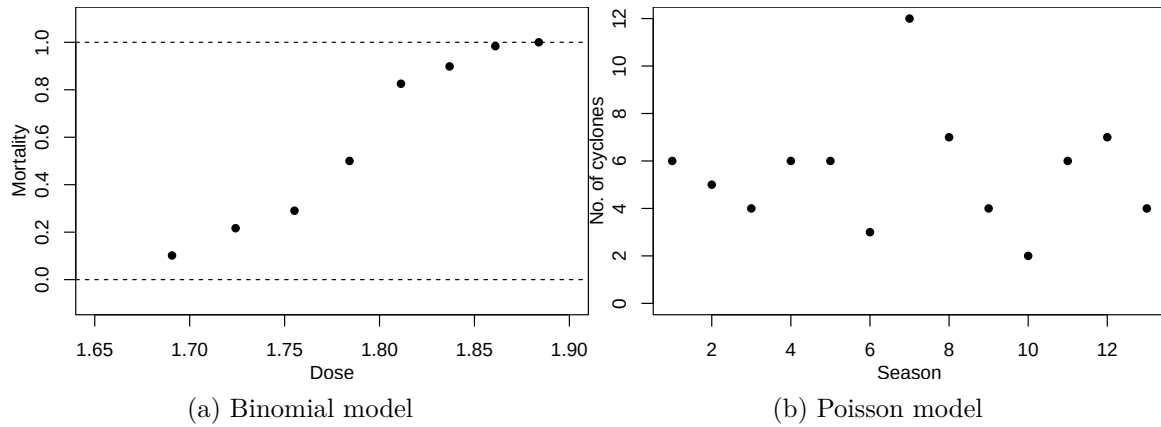


Figure 3.1: Examples of non-normally distributed data.

## Focus on distributions quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the Chapter.

- Which of the following best describes the sample space of a binomial random variable?
  - (A) All real numbers
  - (B) Any real number between 0 and  $m$
  - (C) An integer value between 0 and  $m$
  - (D) Positive real numbers
  - (E) Non-negative integers
  - (F) None of the above
- Which of the following best describes the sample space of a Poisson random variable?
  - (A) Non-negative integers
  - (B) An integer value between 0 and  $m$
  - (C) Any real number between 0 and  $m$
  - (D) Positive real numbers
  - (E) All real numbers
  - (F) None of the above

3. Which of the following best describes the sample space of a normal random variable?
- (A) An integer value between 0 and  $m$
  - (B) Any real number between 0 and  $m$
  - (C) Positive real numbers
  - (D) All real numbers
  - (E) Non-negative integers
  - (F) None of the above
4. For the binomial distribution, which of the following best describes the range of values possible for  $\log(p/(1-p))$ ?
- (A) An integer value between 0 and 1
  - (B) Positive real numbers
  - (C) All real numbers
  - (D) Any real number between 0 and 1
  - (E) Non-negative integers
  - (F) None of the above
5. For the Poisson distribution, which of the following best describes the range of values possible for  $\log(\ )$ ?
- (A) Non-negative integers
  - (B) An integer value between 0 and 1
  - (C) All real numbers
  - (D) Any real number between 0 and 1
  - (E) Positive real numbers
  - (F) None of the above
6. For the normal distribution, which of the following best describes the range of values possible for  $\mu$ ?

- (A) All real numbers
- (B) An integer value between 0 and 1
- (C) Any real number between 0 and 1
- (D) Positive real numbers
- (E) Non-negative integers
- (F) None of the above

## 3.2 The GLM structure

A *generalised linear model* relates a continuous or discrete response variable  $Y$  to a set of explanatory variables  $\mathbf{x} = (x_1, \dots, x_p)$ . The model contains three parts:

**Random part:** The probability (mass or density) function of  $Y$  is assumed to belong to the *two-parameter exponential family* of distributions with parameters  $\theta$  and  $\phi$  :

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}, \quad (3.3)$$

where  $\phi > 0$ . Here,  $\theta$  is called the *canonical* or *natural* parameter of the distribution and  $\phi$  is called the *scale* parameter. We show below that the mean  $E[Y]$  depends only on  $\theta$ , and  $\text{Var}[Y]$  depends on  $\phi$  and possibly also  $\theta$ . Various choices for functions  $b(\cdot)$  and  $c(\cdot)$  produce a wide variety of familiar distributions (see below). Sometimes we may set  $\phi = 1$ ; then Equation 3.3 is called the *one-parameter exponential family*.

Most of the common statistical distributions are members of the regular exponential family, such as binomial, Poisson, geometric, gamma, normal, but not the Student's  $t$  and uniform with unknown bounds.

In this module, we will generally assume that each observation  $Y_i$ ,  $i = 1, \dots, n$ , is *independently* drawn from an exponential family where  $\theta$  depends on the covariates. Thus we write

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}.$$

Note the subscripts on both  $y$  and  $\theta$ , and hence the observations may not be identically distributed.

**Systematic part:** This is a *linear predictor*:

$$\eta = \sum_{j=1}^p \beta_j x_j. \quad (3.4)$$

Note that the symbol  $\eta$  is pronounced *eta*.

**Link function:** This is a one-to-one function providing the link between the linear predictor  $\eta$  and the mean  $\mu = E[Y]$ :

$$\eta = g(\mu), \quad \text{and} \quad \mu = g^{-1}(\eta) = h(\eta). \quad (3.5)$$

Here,  $g(\mu)$  is called the *link function*, and  $h(\eta)$  is called the *inverse link function*.

We will now discuss each of these parts in more detail.

## Focus on GLM structure quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the Chapter.

1. What symbols are usually used for the parameters in the definition of the two-parameter exponential family?
  - (A)  $\eta$  and  $\mu$
  - (B)  $y$  and  $x$
  - (C)  $y$  and  $\mu$
  - (D)  $y$ ,  $\eta$  and  $\mu$
  - (E)  $b$  and  $c$
  - (F) None of the above
2. Which of the following is NOT a member of the two-parameter exponential family?
  - (A) binomial
  - (B) Bernoulli
  - (C) Poisson
  - (D) Student's  $t$
  - (E) normal
  - (F) None of the above
3. Which of the following is NOT part of the usual definition of a generalised linear model?

- (A) Random part
  - (B) Design matrix
  - (C) Link function
  - (D) Systematic part
  - (E) None of the above
4. In the definition of a GLM, which of the following best describes an assumption about the distribution of the response variable?
- (A) Linear function of the explanatory variables
  - (B) Systematic
  - (C) Member of the exponential family
  - (D) Link function
  - (E) None of the above
5. Which part of the GLM definition involves the mean of the response variable and the explanatory variables?
- (A) Linear predictor
  - (B) Expectation of  $Y$
  - (C) Regression parameters,
  - (D) Link function
  - (E) None of the above

### 3.3 The random part of a GLM

We begin with some examples of exponential family members.

**Example: Poisson distribution**

If  $Y$  has a Poisson distribution with parameter  $\lambda$ , that is  $Y \sim \text{Po}(\lambda)$ , then  $Y$  takes values in  $\{0, 1, 2, \dots\}$  and has probability mass function:

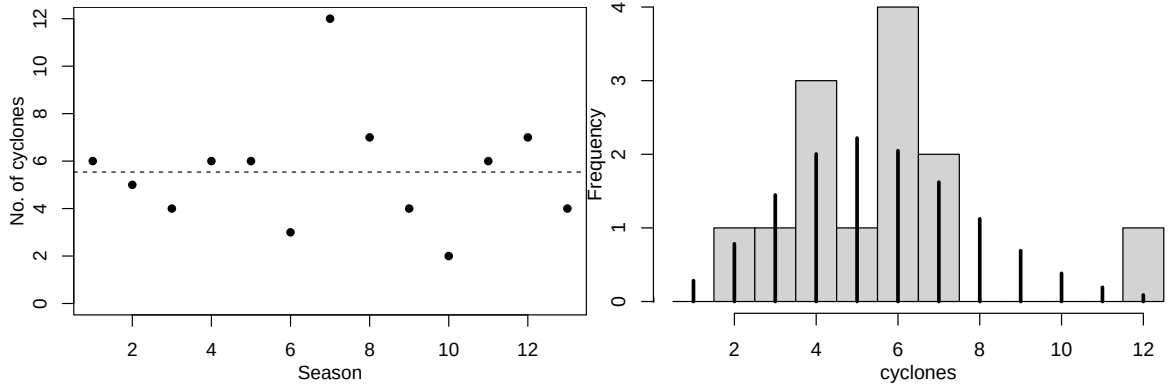
$$f(y) = \frac{e^{-\lambda} \lambda^y}{y!} = \exp \{y \log \lambda - \lambda - \log y!\}, \quad (3.6)$$

which has the form of Equation 3.3 with components as in Table 3.2.

Table 3.2: Exponential model components for the Poisson

| $\theta$       | $\phi$ | $b(\theta)$          | $c(y, \phi)$ |
|----------------|--------|----------------------|--------------|
| $\log \lambda$ | 1      | $\lambda = e^\theta$ | $-\log y!$   |

For example, to model the cyclone data in Table 12.1, we might simply assume that the number of cyclones in each season has a Poisson distribution with a constant rate  $\lambda$  across all seasons  $i$ , that is  $Y_i \sim \text{Po}(\lambda)$ . The parameter is simply estimated by the sample mean,  $\hat{\lambda} = \bar{y} = 5.5384615$ .



(a) No. of cyclones against season with mean (b) Fitted Poisson model assuming constant rate

Figure 3.2: Poisson model fitted to cyclone data.

### Example: Binomial distribution

Let  $Y$  have a Binomial distribution, that is  $Y \sim \text{Bin}(m, p)$ , with  $m$  fixed. Then  $Y$  is discrete, taking values in  $\{0, 1, \dots, m\}$ , and has probability mass function:

$$f(y) = \binom{m}{y} p^y (1-p)^{m-y} = \binom{m}{y} \left( \frac{p}{1-p} \right)^y (1-p)^m$$

which can be re-written as

$$f(y) = \exp \left\{ y \logit p + m \log(1-p) + \log \binom{m}{y} \right\}, \quad (3.7)$$

which has the form of Equation 3.3 with

$$\theta = \logit p = \log \left( \frac{p}{1-p} \right),$$

and with components as in Table 3.3.

Table 3.3: Exponential model components for the Binomial

| $\theta$  | $\phi$ | $b(\theta)$            | $c(y, \phi)$        |
|-----------|--------|------------------------|---------------------|
| logit $p$ | 1      | $m \log(1 + e^\theta)$ | $\log \binom{m}{y}$ |

Note that it can be shown that  $-m \log(1 - p) = m \log(1 + e^\theta)$  – see Exercises.

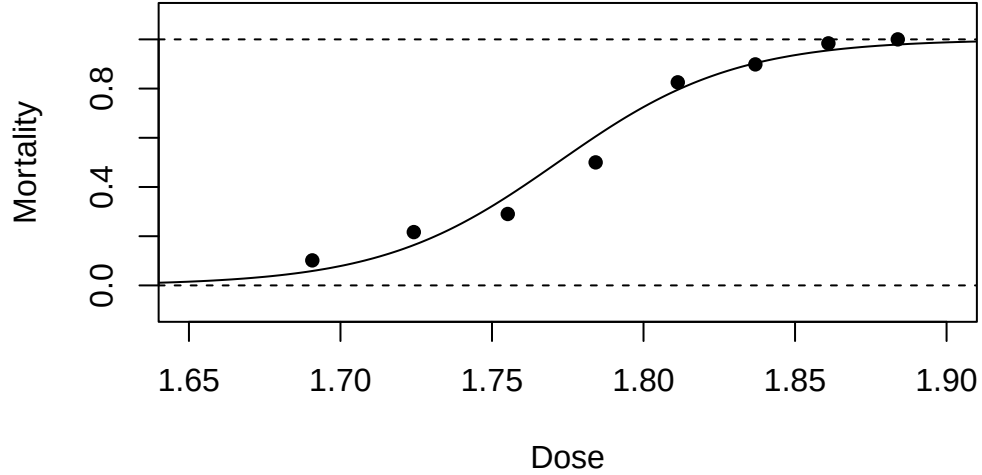


Figure 3.3: Binomial model fitted to beetle data.

### Example: Normal distribution

Let  $Y$  have a Normal distribution with mean  $\mu$  and variance  $\sigma^2$ , that is  $Y \sim N(\mu, \sigma^2)$ . Then  $Y$  takes values on the whole real line and has probability density function

$$\begin{aligned}
 f(y; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ \frac{-1}{2\sigma^2} (y - \mu)^2 \right\}, \\
 &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{y^2}{2\sigma^2} + \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} \right\} \\
 &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} + \left[ \frac{-y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right] \right\},
 \end{aligned}$$

which has the form of Equation 3.3 with components as in Table 3.4.

Table 3.4: Exponential model components for the Gaussian

| $\theta$ | $\phi$     | $b(\theta)$  | $c(y, \phi)$                                      |
|----------|------------|--------------|---------------------------------------------------|
| $\mu$    | $\sigma^2$ | $\theta^2/2$ | $-\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi)$ |

From the usual regression point of view, we write  $y = \alpha + \beta x + \epsilon$ , with  $\epsilon \sim N(0, \sigma^2)$ . From the point of view of a generalised linear model, we write  $Y \sim N(\mu, \sigma^2)$  where  $\mu(x) = \alpha + \beta x$ .

### 3.4 Moments of exponential-family distributions

It is straightforward to find the mean and variance of  $Y$  in terms of  $b(\theta)$  and  $\phi$ . Since we want to explore the dependence of  $E[Y]$  on explanatory variables, this property makes the exponential family very convenient.

**Proposition 3.1.** *For random variables in the exponential family:*

$$E[Y] = b'(\theta), \quad \text{and} \quad \text{Var}[Y] = b''(\theta)\phi. \quad (3.8)$$

**Proof** We give the proof for a continuous random variable. For the discrete case, replace all integrals by sums – see Exercises.

Starting with the simple property that all probability density functions integrate to 1, we have

$$1 = \int \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\} dy$$

and then differentiating both sides with respect to  $\theta$  gives

$$0 = \int \left[ \frac{y - b'(\theta)}{\phi} \right] \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\} dy. \quad (3.9)$$

Next, using the definition of the exponential family to simplify the equation gives

$$0 = \int \left[ \frac{y - b'(\theta)}{\phi} \right] f(y; \theta) dy$$

and expanding the brackets leads to

$$0 = \frac{1}{\phi} \left( \int y f(y; \theta) dy - b'(\theta) \int f(y; \theta) dy \right).$$

The first integral is simply the expectation of  $Y$  and the second is the integral of the probability density function of  $Y$ , and hence

$$0 = \frac{1}{\phi} (E[Y] - b'(\theta))$$

which implies that

$$E[Y] = b'(\theta), \quad (3.10)$$

which proves the first part of the proposition.

Differentiating Equation 3.9 by parts and then using the definition of the exponential family to simplify again yields

$$0 = \int \left\{ -\frac{b''(\theta)}{\phi} + \left[ \frac{y - b'(\theta)}{\phi} \right]^2 \right\} f(y; \theta) dy$$

and using Equation 3.10 gives,

$$\begin{aligned} 0 &= -\frac{b''(\theta)}{\phi} + \int \left[ \frac{y - E[Y]}{\phi} \right]^2 f(y; \theta) dy \\ 0 &= -\frac{b''(\theta)}{\phi} + \frac{\text{Var}[Y]}{\phi^2} \end{aligned}$$

which implies that

$$\text{Var}[Y] = \phi b''(\theta),$$

which proves the second part of the proposition.

Together, these two results allow us to write down the expectation and variance for any random variable once we have shown that it is a member of the exponential family.

### Example: Poisson and normal distribution moments

Table 3.5: Summary of moment calculations via exponential family properties

|                               | $\theta$       | $b(\theta)$  | $\phi$     | $E[Y] = b'(\theta)$  | $b''(\theta)$ | $\text{Var}[Y] = b''(\theta)\phi$ |
|-------------------------------|----------------|--------------|------------|----------------------|---------------|-----------------------------------|
| Poisson,<br>$Po(\lambda)$     | $\log \lambda$ | $e^\theta$   | 1          | $e^\theta = \lambda$ | $e^\theta$    | $e^\theta \times 1 = \lambda$     |
| Normal,<br>$N(\mu, \sigma^2)$ | $\mu$          | $\theta^2/2$ | $\sigma^2$ | $\theta = \mu$       | 1             | $1 \times \sigma^2 = \sigma^2$    |

## 3.5 The systematic part of the model

The second part of the generalised linear model, the linear predictor, is given as  $\eta = \sum_{j=1}^p \beta_j x_j$ , where  $x_j$  is the  $j$ th explanatory variable (with  $x_1 = 1$  for the intercept). For each observation  $y_i$ ,  $i = 1, \dots, n$ , the explanatory variables may differ. To make explicit this dependence on  $i$ , we write:

$$\eta_i = \sum_{j=1}^p \beta_j x_{ij}, \quad (3.11)$$

where  $x_{ij}$  is the value of the  $j$ th explanatory variable on individual  $i$  (with  $x_{i1} = 1$ ). Rewriting in matrix notation:

$$\eta = \mathbf{X}\beta, \quad (3.12)$$

where  $\eta = (\eta_1, \dots, \eta_n)$  is a vector of linear predictor values,  $\beta = (\beta_1, \dots, \beta_p)$  is a vector of regression parameters, and  $\mathbf{X}$  is the  $n \times p$  design matrix (as described in Section 2.4).

Recall from Section 1.4 and Section 2.3 that we are concerned with two kinds of explanatory variable:

- Quantitative — for example,  $x \in (-\infty, \infty)$  etc.
- Qualitative — for example,  $x \in \{A, B, C\}$  etc.

Each quantitative variable is represented in  $\mathbf{X}$  by a single column. Each qualitative variable with  $k + 1$  levels is represented by  $k$  dummy columns (one column dropped to avoid identification problems).

## 3.6 The link function

In Section 3.2 we saw that the random part of an observation,  $y$ , might be described by a member of the exponential family. We also saw that the systematic part of  $y$  might be described using a linear predictor,  $\eta$ , of the explanatory variables. The *link function*  $\eta = g(\mu)$  brings these two parts together.

Occasionally, the choice of link function  $g(\mu)$  is motivated by theory underlying the data at hand. When there is no compelling underlying theory, we typically choose a link function that transforms the restricted range of the dependent variable onto the whole real line. For example, when observations are typically positive, so we have  $\mu > 0$ , we might choose the logarithmic link:

$$g(\mu) = \log(\mu). \quad (3.13)$$

When observations are binomial counts from  $B(m, p)$ ,  $0 < p < 1$ , with mean  $\mu = mp$ , we might choose the *logit* link

$$\eta = g(\mu) = \text{logit}(\mu/m) = \text{logit}(p) = \log\{p/(1-p)\} \quad (3.14)$$

or the *probit* link

$$\eta = g(\mu) = \Phi^{-1}(\mu/m) = \Phi^{-1}(p), \quad (3.15)$$

or the *complementary log-log* (*cloglog*) link:

$$\eta = g(\mu) = \log(-\log(1 - \mu/m)) = \log(-\log(1 - p)), \quad (3.16)$$

or the *cauchit* link:

$$\eta = g(\mu) = \tan(\pi(\mu/m - \frac{1}{2})) = \tan(\pi(p - \frac{1}{2})). \quad (3.17)$$

Figure 3.4 shows these link functions fitted to the beetle mortality data. The logit and probit links are very similar; the complementary log-log fits slightly better in the extremes; the cauchit link fits these data quite poorly in the extremes.

A mathematically and computationally convenient choice of link function  $g(\mu)$  can be constructed by setting  $\theta = \eta$ ,  $\{\#eq-g3\}$  where  $\theta$  is the canonical parameter of the exponential family as defined in Equation 3.3. Then Equation 3.8 shows that the mean  $\mu$  is a function

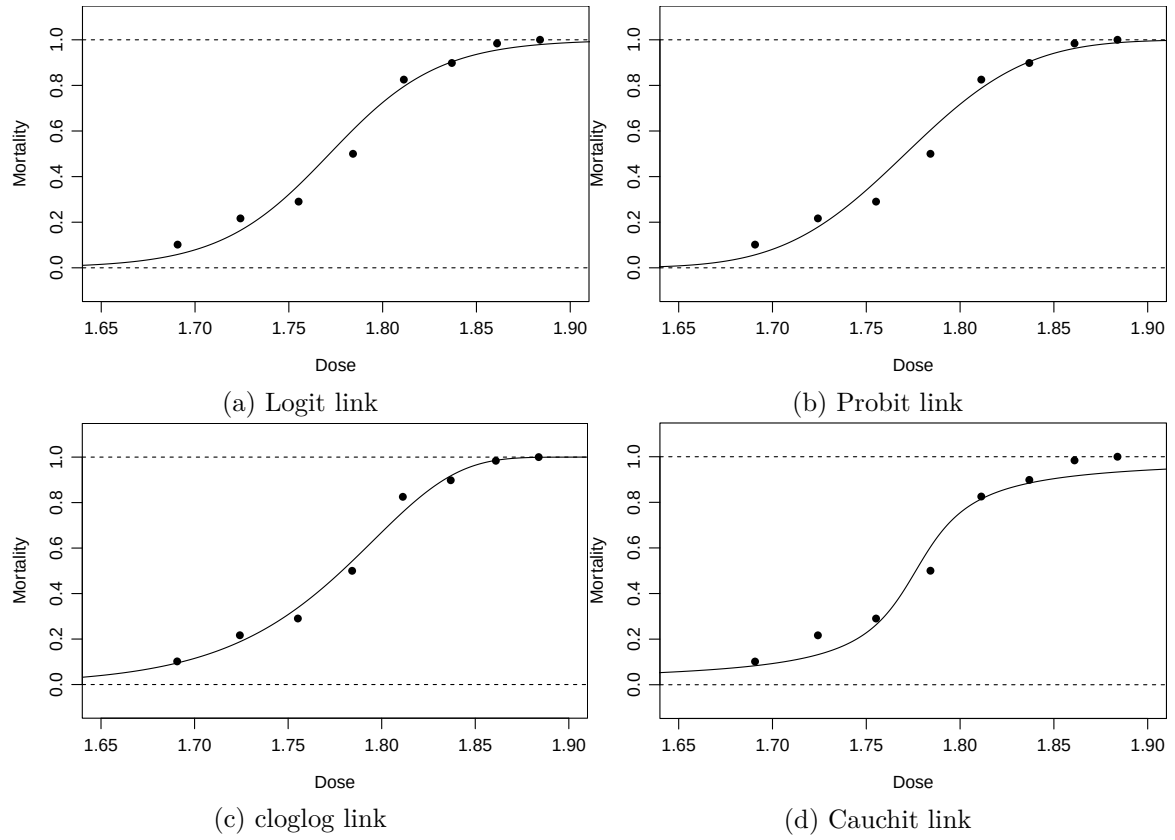


Figure 3.4: Dose-response curves fitted to the beetle mortality data with different choices of link function.

of  $\theta$  and therefore this indirectly provides a link between  $\mu$  and  $\eta$ . Since  $\mu = b'(\theta)$ , provided  $b'(\cdot)$  has an inverse, we may write

$$g(\mu) = \theta = (b')^{-1}(\mu). \quad (3.18)$$

The function produced from Equation 3.18 is called the *canonical* link function.

**Proposition 3.2.** *For the canonical link function,*

$$g'(\mu) = 1/b''(\theta).$$

**Proof:** From Proposition 3.1,  $\mu = b'(\theta)$ , so  $d\mu/d\theta = b''(\theta)$ . From Equation 3.18,  $d\theta/d\mu = g'(\mu)$ . Since  $d\theta/d\mu = (d\mu/d\theta)^{-1}$ , we obtain  $g'(\mu) = 1/b''(\theta)$ .

**Example: Poisson canonical link function**

For the Poisson distribution  $\text{Po}(\lambda)$ , from Table 3.2,  $b(\theta) = e^\theta$  so  $b'(\theta) = e^\theta$ . Applying Equation 3.18:  $g(\mu) = \log(\mu)$ . Thus the canonical link for the Poisson is  $\log$ .

**Example: Normal canonical link function**

For the Normal distribution  $N(\mu, \sigma^2)$ , from Table 3.4,  $b(\theta) = \theta^2/2$  so  $b'(\theta) = \theta$ . Applying Equation 3.18:  $g(\mu) = \mu$ . Thus the canonical link for the Normal distribution is the identity function.

Table 3.6 summarises the canonical link functions for several distributions, showing that the canonical link maps the restricted range of  $\mu$  onto the whole real line for Normal, Poisson, and Binomial, but not for the Gamma.

Table 3.6: Canonical link functions and their ranges ( $\dagger$ binomial distribution with index  $m$  and mean  $\mu$ ;  $\ddagger$ gamma distribution with mean  $\mu$  — see Exercises for details).

| $f(y)$             | Range of $\mu$      | $b(\theta)$            | $\mu = b'(\theta)$    | Canonical link, $g(\mu)$ | Range of $\eta$     |
|--------------------|---------------------|------------------------|-----------------------|--------------------------|---------------------|
| Normal             | $(-\infty, \infty)$ | $\frac{1}{2}\theta^2$  | $\theta$              | $\mu$                    | $(-\infty, \infty)$ |
| Poisson            | $(0, \infty)$       | $e^\theta$             | $e^\theta$            | $\log \mu$               | $(-\infty, \infty)$ |
| Binomial $\dagger$ | $(0, m)$            | $m \log(1 + e^\theta)$ | $m/(1 + e^{-\theta})$ | $\text{logit}(\mu/m)$    | $(-\infty, \infty)$ |
| Gamma $\ddagger$   | $(0, \infty)$       | $-\log(-\theta)$       | $-\theta^{-1}$        | $-\mu^{-1}$              | $(-\infty, 0)$      |

Using the canonical link in the exponential family formula Equation 3.3 and taking  $n$  independent observations, the log-likelihood simplifies to

$$\log f(\mathbf{y}; \mathbf{X}, \beta, \phi) = \frac{\sum_j \beta_j S_j - \sum_i b\left(\sum_j \beta_j x_{ij}\right)}{\phi} + \sum_i c(y_i, \phi) \quad (3.19)$$

where  $S_j = \sum_{i=1}^n y_i x_{ij}$  are called *sufficient statistics*. Their appearance confers both theoretical and practical advantages.

## Focus on link functions quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the module.

1. If the response variable  $Y$  can take only positive values, then which of the following transformations would produce an unrestricted range?
  - (A)  $\exp(Y)$
  - (B)  $\log(Y)$
  - (C)  $1/Y$
  - (D)  $\text{logit}(Y)$
  - (E) None of the above
2. If the range of the response variable  $Y$  is restricted to the interval  $(0,1)$ , then which of the following transformations would produce an unrestricted range?
  - (A)  $\text{logit}(Y)$
  - (B)  $\exp(Y)$
  - (C)  $\log(Y)$
  - (D)  $1/Y$
  - (E) None of the above
3. If response variable  $Y$  follows a normal distribution, then which of the following transformations would produce an unrestricted range?
  - (A)  $\sqrt{Y}$
  - (B)  $\exp(Y)$
  - (C)  $\log(Y)$
  - (D)  $\text{logit}(Y)$
  - (E) None of the above
4. If the range of the response variable  $Y$  is a percentage, then which of the following transformations would produce an unrestricted range?

- (A)  $1/Y$
  - (B)  $\exp(Y)$
  - (C)  $\log(Y)$
  - (D)  $\text{logit}(Y/100)$
  - (E) None of the above
5. If the response variable  $Y$  can take any real value, then which of the following transformations would produce an unrestricted range?
- (A) Identity
  - (B) Logistic
  - (C) Square-root
  - (D) Log
  - (E) None of the above

### 3.7 Exercises

3.1 Consider the beetle data again (see Table 1.1), but suppose that we had only been given the  $y$  values, that is the number killed, and misinformed that each came from a sample of size 62. Further, suppose that we did not know that different doses had been used. That is, we were given data:  $\mathbf{y} = \{6, 13, 18, 28, 52, 53, 61, 60\}$  and led to believe that the model  $Y \sim \text{Bin}(62, p)$  was appropriate. Use the given data to estimate  $p$ . Then, calculate the fitted probabilities and superimpose them on a histogram of the data.

[Click here to see hints.](#)

See the code chunk used to create Figure 3.2.

3.2 Verify that, in general, if  $q = 1/(1 + e^{-x})$  then  $x = \log(q/(1 - q))$  and then for the binomial distribution,  $Y \sim \text{Bin}(m, p)$ , show that

$$-m \log(1 - p) = m \log(1 + e^\theta)$$

where  $\theta = \text{logit } p = \log(p/(1 - p))$ .

[Click here to see hints.](#)

Each derivation requires algebraic rearrangement and simplification only.

3.3 Suppose that  $Y$  has a gamma distribution with parameters  $\alpha$  and  $\lambda$ , that is  $Y \sim \text{Gamma}(\alpha, \lambda)$ , with probability density function

$$f(y; \alpha, \lambda) = \frac{\lambda^\alpha y^{\alpha-1} e^{-\lambda y}}{\Gamma(\alpha)} \quad y > 0; \alpha, \lambda > 0.$$

Write this in the form of the exponential family and clearly identify  $\theta$ ,  $\phi$ ,  $b(\theta)$  and  $c(y, \phi)$  – as was done for the Poisson and binomial in Table 3.2 and Table 3.3.

[Click here to see hints.](#)

Consider treating  $1/\alpha$  as a scale parameter and let  $\theta$  be a suitable function of  $\lambda$  and  $\alpha$ .

3.4 Express the geometric distribution,  $Y \sim \text{Geom}(p)$ ,  $0 < p < 1$ , with probability mass function

$$f(y) = (1 - p)^{y-1} p; \quad y = 1, 2, 3 \dots$$

as an exponential family distribution.

[Click here to see hints.](#)

Follow the same process as for the Poisson and binomial.

3.5 Prove Proposition 3.1 assuming that  $Y$  follows a discrete distribution. Use the proposition, starting from the given  $b(\theta)$ , to verify the results for expectation and variance of the Poisson in Table 3.5 and then derive similar results for the binomial,  $Y \sim \text{Bin}(m, p)$ .

[Click here to see hints.](#)

Replace the integration in the continuous case by summation for the discrete case. For the binomial, find  $b(\theta)$  from Table 3.3 and then compute its first and second derivatives.

3.6 Use the properties of exponential families to find the expectation and variance of each of the geometric and gamma distributions considered above.

[Click here to see hints.](#)

Use the results from Exercises 3.3 and 3.4, and apply Proposition 3.1.

3.7 Using Equation 3.18, verify the canonical link function,  $g(\mu)$ , for the binomial and gamma distributions shown in Table 3.6.

[Click here to see hints.](#)

Use the appropriate  $b'(\theta)$  and, using  $\mu = b'(\theta)$  and Equation 3.18, rearrange to find  $\theta$  and hence  $g(\mu)$ .

## 4 GLM Estimation

Throughout this module we use the principle of maximum likelihood estimation (MLE) to estimate model parameters and will consider two cases.

### 4.1 The identically distributed case

#### 4.1.1 Maximum likelihood estimation

Suppose we have  $n$  independent and identically distributed (i.i.d.) observations  $\mathbf{y} = \{y_i, i = 1, \dots, n\}$ , where each  $y_i$  is sampled from the same exponential family density

$$f(y_i; \theta, \phi) = \exp \left\{ \frac{y_i \theta - b(\theta)}{\phi} + c(y_i, \phi) \right\}, \quad (4.1)$$

for  $i = 1, \dots, n$ . In this case, the canonical parameter  $\theta$  does not depend on  $i$  as the observations are identically distributed and hence must have the same parameter.

By independence, the joint distribution of all the observations  $\mathbf{y}$  is:

$$f(\mathbf{y}; \theta, \phi) = \prod_{i=1}^n f(y_i; \theta, \phi).$$

So, taking logs and then substituting for the probability function using the exponential family form, Equation 3.3, gives

$$\log f(\mathbf{y}; \theta, \phi) = \sum_{i=1}^n \log f(y_i; \theta, \phi) = \sum_{i=1}^n \left[ \frac{y_i \theta - b(\theta)}{\phi} + c(y_i, \phi) \right].$$

Regarding the observations  $\mathbf{y}$  as constants (which they are, once we have data) and the scale parameter  $\phi$  as a fixed *nuisance* parameter (whose value we may not know), the log-likelihood as a function of the parameter of interest  $\theta$  is:

$$l(\theta; \mathbf{y}, \phi) = n \left( \frac{\bar{y} \theta - b(\theta)}{\phi} \right) + \text{constant}, \quad (4.2)$$

where  $\bar{y} = \sum y_i / n$ .

We estimate  $\theta$  by maximizing the log likelihood – i.e. given the data  $\mathbf{y}$ , we estimate the value of  $\theta$  to be that value for which the likelihood, and hence the log-likelihood, is greatest.

We maximize the log-likelihood by differentiating it and setting it to zero:

$$\frac{dl(\theta; \mathbf{y}, \phi)}{d\theta} = n \left( \frac{\bar{y} - b'(\theta)}{\phi} \right)$$

and hence the MLE for  $\theta$ , which we denote  $\hat{\theta}$ , satisfies

$$b'(\hat{\theta}) = \bar{y} \quad (4.3)$$

and hence

$$\hat{\theta} = (b')^{-1}(\bar{y}). \quad (4.4)$$

Further, we showed in Proposition 3.1 that  $E[Y] = \mu = b'(\theta)$  and if we let  $\hat{\mu}$  denote the MLE of  $\mu$ , then  $\hat{\mu} = b'(\hat{\theta})$ <sup>1</sup>, hence we have  $\hat{\mu} = \bar{y}$ . So we find that  $\hat{\theta}$  is the value of  $\theta$  for which the theoretical mean  $\hat{\mu} = b'(\hat{\theta})$  matches the sample mean  $\bar{y}$ .

### Example: MLE of the Poisson distribution

For the Poisson distribution,  $\text{Po}(\lambda)$ , we have found that  $b(\theta) = e^\theta$  and therefore  $b'(\theta) = e^\theta$ . Hence, the MLE of natural parameter  $\theta$  is found as the solution of  $b'(\hat{\theta}) = e^{\hat{\theta}} = \bar{y}$ , that is  $\hat{\theta} = \log(\bar{y})$ .

## 4.1.2 Estimation accuracy

For our i.i.d. sample  $\mathbf{y} = \{y_i, i = 1, \dots, n\}$ , we have  $b'(\hat{\theta}) = \hat{\mu} = \bar{y}$ . Let  $\theta_0$  be the true value of  $\theta$  with corresponding mean  $\mu_0$ , i.e.

$$b'(\theta_0) = \mu_0. \quad (4.5)$$

How accurate is  $\hat{\theta}$ ? We know that

$$E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \mu_0 = b'(\theta_0), \quad (4.6)$$

using Equation 4.5. Also,

$$\text{Var}[\bar{Y}] = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}[Y_i]$$

because the observations are independent. Then, using the result Equation 3.8 gives

$$\text{Var}[\bar{Y}] = \frac{1}{n} b''(\theta_0) \phi. \quad (4.7)$$

We can use Taylor's theorem to expand  $b'(\hat{\theta})$  about  $\theta_0$ :

$$\bar{y} = b'(\hat{\theta}) \approx b'(\theta_0) + (\hat{\theta} - \theta_0)b''(\theta_0),$$

---

<sup>1</sup>Using the result that the MLE of any function of a parameter is given by the same function applied to the MLE of the parameter.

which implies that

$$(\hat{\theta} - \theta_0) \approx b''(\theta_0)^{-1} \{b'(\hat{\theta}) - b'(\theta_0)\} = b''(\theta_0)^{-1}(\bar{Y} - \mu_0), \quad (4.8)$$

using Equation 4.3 and Equation 4.5. We can use Equation 4.8 to get approximations to the mean and variance of  $\hat{\theta}$ :

$$E[\hat{\theta} - \theta_0] \approx b''(\theta_0)^{-1} E[\bar{Y} - \mu_0] = 0,$$

using Equation 4.6, so

$$E[\hat{\theta}] \approx \theta_0, \quad (4.9)$$

and

$$\text{Var}(\hat{\theta}) \approx E[(\hat{\theta} - \theta_0)^2]$$

using Equation 4.9,

$$\text{Var}(\hat{\theta}) \approx E\left[\left(b''(\theta_0)^{-1}(\bar{Y} - \mu_0)\right)^2\right]$$

using Equation 4.8,

$$\text{Var}(\hat{\theta}) \approx (b''(\theta_0))^{-2} \text{Var}[\bar{Y}]$$

using Equation 4.6,

$$\text{Var}(\hat{\theta}) = \frac{\phi}{n b''(\theta_0)} \quad (4.10)$$

using Equation 4.7.

Thus we see that the first two derivatives of  $b(\theta)$  play a key role in inference.

### Example: Accuracy for the Poisson distribution

For the Poisson distribution,  $\text{Po}(\lambda)$ , we have found that  $\hat{\theta} = \log(\bar{Y})$ . Using Equation 4.9 we know that  $\hat{\theta}$  is, at least, approximately unbiased. Then, using that  $\phi = 1$  (Table 3.2),  $b'(\theta) = e^\theta$  (Table 3.6) hence  $b''(\theta) = e^\theta$ , and using Equation 4.10, leads to the result

$$\text{Var}(\hat{\theta}) = \frac{\phi}{n b''(\theta_0)} = \frac{1}{ne^{\theta_0}}.$$

## Focus on maximum likelihood quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the Chapter

1. Which of the following statements best describes the purpose of maximum likelihood estimation?
  - (A) To estimate probabilities of unknown events
  - (B) To calculate the mean of a distribution

- (C) To summarise a data set
  - (D) To find good values of unknown model parameters
  - (E) To find the true values of model parameters
  - (F) None of the above
2. Which of the following mathematical ideas is NOT used in the above maximum likelihood estimation?
- (A) By independence
  - (B) The parameter  $\theta$  depends on  $i$
  - (C) Identically distributed
  - (D) Substituting for the probability function
  - (E) Taking logs
  - (F) None of the above
3. Which of the following terms is used to refer to the fixed parameter  $\Phi$ ?
- (A) Nuisance
  - (B) Annoyance
  - (C) Normally distributed
  - (D) Independent
  - (E) Canonical
  - (F) None of the above
4. Which of the following is NOT a good property of a parameter estimator?
- (A) Small variance
  - (B) Has fixed variance for all sample sizes
  - (C) Has expectation equal to the true value
  - (D) Variance decreases as sample size increases

- (E) Unbiased
- (F) None of the above

5. Which of the following does NOT influence the variance of the estimator of ?

- (A)  $b(\cdot)$
- (B)  $\Phi$
- (C) The true parameter value
- (D)  $c(y, \Phi)$
- (E) The sample size
- (F) None of the above

## 4.2 The general case

### 4.2.1 MLE Estimation

Suppose that now the  $n$  independent observations  $\mathbf{y} = \{y_i, i = 1, \dots, n\}$  are not identically distributed. They are, however, sampled from the same exponential family density but with differing parameters, that is

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\},$$

for  $i = 1, \dots, n$ . In this case, the canonical parameter does depend on  $i$  – but we assume that the scale parameter  $\phi$  does not – and let  $\theta = \{\theta_i, i = 1, \dots, n\}$ .

In most applications, we are not interested in estimation of  $\theta$  but instead we are interested in the linear predictor parameters  $\beta = \{\beta_1, \dots, \beta_p\}$ . Note, however, that each  $\theta_i$  will depend on all  $\beta_1, \dots, \beta_p$ . This is most obvious for the canonical parameter case where a convenient choice of link function is obtained using  $\theta = \eta = X\beta$ , hence  $\theta_i = \mathbf{x}_i^T \beta = \sum_{j=1}^p x_{ij} \beta_j$  and each  $\theta_i$  clearly depends on all  $\beta_1, \dots, \beta_p$ .

The principle of maximum likelihood will be used to estimate the model parameters  $\beta$ . Using the independence of  $\mathbf{y} = \{y_i, i = 1, \dots, n\}$ , given the parameters  $\beta$ , the likelihood function is:

$$L(\beta; \mathbf{y}, \phi) = f(\mathbf{y}; X, \beta, \phi) = \prod_{i=1}^n f(y_i; \mathbf{x}_i, \beta, \phi) \quad (4.11)$$

and the log-likelihood by

$$l(\beta; \mathbf{y}, \phi) = \log L(\beta; \mathbf{y}, \phi) = \sum_{i=1}^n \log f(y_i; \mathbf{x}_i, \beta, \phi). \quad (4.12)$$

Then we wish to find the value of  $\beta$  which maximizes the log-likelihood function,

$$\hat{\beta} = \max_{\beta} l(\beta; \mathbf{y}, \phi).$$

For generalised linear models there is usually no closed-form expression for the MLE  $\hat{\beta}$ . Instead, an iterative approach based on Newton's Method is often used.

For the normal linear regression model, however, that is where the exponential family is Gaussian and the link function  $g(\mu)$  is the identity function, we have the familiar closed-form expression

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}. \quad (4.13)$$

#### 4.2.2 The score function and Fisher information

We define the *score function*:

$$U(\beta) = \frac{\partial l(\beta; \mathbf{y}, \phi)}{\partial \beta}, \quad (4.14)$$

which is a  $p \times 1$  vector. We define the *observed Fisher information*:

$$I(\beta) = -\frac{\partial U(\beta)}{\partial \beta^T} = -\frac{\partial^2 l(\beta; \mathbf{y}, \phi)}{\partial \beta \partial \beta^T}, \quad (4.15)$$

which is a  $p \times p$  matrix whose  $(j, k)$ th element is:  $-\frac{\partial^2 l(\beta; \mathbf{y}, \phi)}{\partial \beta_j \partial \beta_k}$ . We also define the *expected Fisher information*:

$$J(\beta) = E \left[ -\frac{\partial^2 l(\beta; \mathbf{y}, \phi)}{\partial \beta \partial \beta^T} \right], \quad (4.16)$$

which is also a  $p \times p$  matrix.

Newton's Method then involves choosing an initial value for the parameter vector,  $\hat{\beta}_0$  say, and then repeatedly updating, until convergence, the value using  $\hat{\beta}_t = \hat{\beta}_{t-1} - I^{-1}(\hat{\beta}_{t-1}) U(\hat{\beta}_{t-1})$ . A related method, called Fisher Scoring, replaces  $I$  in the iterative step by  $J$  – calculating the expected Fisher information is more work but does lead to better numerical properties.

**Proposition 4.1.** *With definitions Equation 4.14, Equation 4.15, Equation 4.16 above,*

- $E[U(\beta)] = 0$ ,
- $J(\beta) = E[U(\beta)U^T(\beta)]$ .

**Proof** We give the proof for continuous random variables. For the discrete case, replace integration by sums – see Exercises.

To start the proof, notice that we can re-write the joint density of the data given the parameters as

$$f(\mathbf{y}; X, \beta, \phi) = L(\beta; \mathbf{y}, \phi) = \exp \{l(\beta; \mathbf{y}, \phi)\}$$

and then

$$1 = \int f(\mathbf{y}; X\beta, \phi) d\mathbf{y} = \int \exp\{l(\beta; \mathbf{y}, \phi)\} d\mathbf{y},$$

where  $d\mathbf{y} = dy_1 \cdots dy_n$ . Differentiating this with respect to  $\beta = (\beta_1, \dots, \beta_p)^T$  gives:

$$\begin{aligned} 0 &= \int \frac{\partial l(\beta; \mathbf{y}, \phi)}{\partial \beta} \exp\{l(\beta; \mathbf{y}, \phi)\} d\mathbf{y} \\ &= \int U(\beta) f(\mathbf{y}; X, \beta, \phi) d\mathbf{y} \\ &= E[U(\beta)]. \end{aligned} \tag{*}$$

Proving the first part.

Next, differentiating (\*) by parts with respect to  $\beta^T$ ,

$$\begin{aligned} 0 &= \int \frac{\partial U(\beta)}{\partial \beta^T} f(\mathbf{y}; X, \beta, \phi) + \frac{\partial l(\beta; \mathbf{y}, \phi)}{\partial \beta} \frac{\partial l(\beta; \mathbf{y}, \phi)}{\partial \beta^T} f(\mathbf{y}; X, \beta, \phi) d\mathbf{y} \\ &= \int -I(\beta) f(\mathbf{y}; X, \beta, \phi) d\mathbf{y} + \int U(\beta) U^T(\beta) f(\mathbf{y}; X, \beta, \phi) d\mathbf{y} \\ &= -J(\beta) + E[U(\beta) U^T(\beta)]. \end{aligned}$$

proving the second part.

**Proposition 4.2.** *Under some regularity conditions, the MLE  $\hat{\beta}$  of  $\beta$  has the following asymptotic properties:*

- $E(\hat{\beta}) = \beta$ ; i.e.  $\hat{\beta}$  is unbiased for  $\beta$ .
- $\text{Var}(\hat{\beta}) = J^{-1}(\beta)$ .
- $\hat{\beta}$  follows a  $p$ -dimensional Normal distribution.

Combining these three statements we have that asymptotically

$$\hat{\beta} \sim N_p(\beta, J^{-1}(\beta)). \tag{4.17}$$

**Proof:** Omitted. Similar to the proof using Taylor's theorem in Section 4.1.

From Equation 4.17, variances  $\text{Var}(\hat{\beta}_k)$ , standard errors  $\text{se}(\hat{\beta}_k)$  and correlations between parameter estimates  $\text{Corr}(\hat{\beta}_k, \hat{\beta}_h)$  can be estimated.

### 4.2.3 The saturated case

Again we assume the observations  $y_i$ ,  $i = 1, \dots, n$  are independent but now we assume that  $y_i$  is sampled from an exponential family probability function with canonical parameter  $\theta_i$ ,

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\},$$

for  $i = 1, \dots, n$ . We can form the log-likelihood in the usual way to give

$$l(\theta; \mathbf{y}, \phi) = \sum_{i=1}^n \log f(y_i; \theta_i, \phi) = \sum_{i=1}^n \left[ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right]$$

and so we find the the MLE of  $\theta$  using

$$\hat{\theta}_i = (b')^{-1}(y_i), \quad i = 1, \dots, n.$$

Note that each parameter is only a function of the corresponding observation.

Further, from Proposition 3.1,  $E[Y_i] = \mu_i = b'(\theta_i)$  and if we again let  $\hat{\mu}_i$  denote the MLE of  $\mu_i$ , then  $\hat{\mu}_i = b'(\hat{\theta}_i)$ , hence we have  $\hat{\mu}_i = y_i$ . Thus we see that, under the saturated model, the mean of the distribution of  $y_i$  is estimated to be equal to  $y_i$  itself. That is, the data are fitted exactly by the model. Of course this model is quite useless for explanation or prediction, since it misinterprets random variation as systematic variation. Nevertheless, the saturated model is useful as a benchmark for comparing models, as we will see later.

It is worth noting that the same situation can occur even when modelling in terms of the regression parameters  $\beta_j$ ,  $j = 1, \dots, p$ . When  $p \geq n$ , and if the covariates are linearly independent, each  $\theta_i$  can take on any value independently of the others and so estimating the  $\beta_j$ 's is equivalent to estimating the  $\theta_i$ 's. That is we are also considering the *saturated* or *full* model. This highlights the danger of putting too many covariates into the model. There is a big literature on how to deal with more parameters than data using techniques of *regularised regression*.

## Focus on general maximum likelihood quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the Chapter

1. Which of the following statements best describes the purpose of maximum likelihood estimation?
  - (A) To find the true values of model parameters
  - (B) To estimate probabilities of unknown events
  - (C) To summarise a data set
  - (D) To calculate the mean of a distribution
  - (E) To find good values of unknown model parameters
  - (F) None of the above

2. Which of the following mathematical ideas is NOT used in the general case of maximum likelihood estimation?
  - (A) The log-likelihood
  - (B) Usually no closed-form expression
  - (C) Not identically distributed
  - (D) Using the independence
  - (E) The parameter  $\theta$  does not depend on  $i$
  - (F) None of the above
3. Which of the following terms is NOT used in maximum likelihood theory?
  - (A) The moment generating function
  - (B) The likelihood function
  - (C) Fisher information
  - (D) The log-likelihood function
  - (E) Score function
  - (F) None of the above
4. Which of the following is a true statement about the saturated case?
  - (A) The parameters  $\theta_i$  are unrelated to the values of  $y_i$
  - (B) The model fits the data exactly
  - (C) The values of  $\theta_i$  will be exactly the same as the values of  $y_i$
  - (D) It should never be considered
  - (E) Has fewer parameters than data observations
  - (F) None of the above
5. Which of the following is NOT a true statement about the asymptotic properties of the maximum likelihood estimator of  $\theta$ ?
  - (A) The asymptotic properties are when  $n$  tends to infinity

- (B) Follows a normal distribution
- (C) Is unbiased
- (D) Will always be equal to the true value.
- (E) The variance can be found from the expected Fisher information
- (F) None of the above

### 4.3 Model deviance

The *deviance* is a quantity we use to assess the fit of a model to the data. Let  $M$  be a model of interest with fitted parameters  $\hat{\theta}$  and corresponding fitted values  $\hat{\mu}$ . Also consider the saturated model with fitted parameters  $\tilde{\theta}$  and fitted values  $\tilde{\mu}$ .

The *deviance* of model  $M$  is defined as twice the difference between the log-likelihood of the saturated model,  $l(\tilde{\theta}; \mathbf{y}, \phi)$ , and the log-likelihood of model  $M$ ,  $l(\hat{\theta}; \mathbf{y}, \phi)$ , multiplied by  $\phi$ ,

$$\begin{aligned}
 D &= 2\phi \{l(\tilde{\theta}; \mathbf{y}, \phi) - l(\hat{\theta}; \mathbf{y}, \phi)\} \\
 &= \begin{cases} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2 = \text{Residual sum of squares} & \text{Normal} \\ 2 \sum_{i=1}^n \left\{ y_i \log \left( \frac{y_i}{\hat{\mu}_i} \right) + (m_i - y_i) \log \left( \frac{m_i - y_i}{m_i - \hat{\mu}_i} \right) \right\} & \text{Binomial} \\ 2 \sum_{i=1}^n \left\{ y_i \log \left( \frac{y_i}{\hat{\mu}_i} \right) - y_i + \hat{\mu}_i \right\} & \text{Poisson} \end{cases}
 \end{aligned} \tag{4.18}$$

Note that Dobson, and others, call  $D^* = D/\phi = 2\{l(\tilde{\theta}; y, \phi) - l(\hat{\theta}; y, \phi)\}$  the *scaled deviance*.

We now consider two situations:

**Scale parameter  $\phi$  known.** For some data-types (e.g. Poisson, Binomial), we know  $\phi = 1$ . Consider two nested models  $M_1$  and  $M_2$  with  $r_1$  and  $r_2$  parameters respectively where the parameters in  $M_1$  are a subset of those in  $M_2$  and hence  $r_1 < r_2$ . Further, let  $D_1$  and  $D_2$  be the deviances of model  $M_1$  and  $M_2$  respectively.

Then, asymptotically,

- the log likelihood-ratio statistic  $D_1 - D_2 \sim \chi_{r_2 - r_1}^2$  can be used to test the importance of the extra parameters in  $M_2$  not included in  $M_1$ ;
- a goodness-of-fit test for  $M_2$  can be done based on  $D_2 \sim \chi_{n - r_2}^2$ .

The quality of the approximations involved depends on there being a large amount of *information*, for example, large counts for Binomial and Poisson data, or a large sample size for Normal data.

**Scale parameter  $\phi$  unknown.** For some data-types (e.g. Normal, Gamma),  $\phi$  is not known (typically  $\phi = \sigma^2$ ). We must find a model  $M_3$  *big* enough to be believed, then estimate  $\phi$  by the residual mean square:

$$\hat{\phi} = \frac{D_3}{n - r_3}. \quad (4.19)$$

Then test  $M_1$  against  $M_2$  using

$$F = \frac{(D_1 - D_2)/(r_2 - r_1)}{\hat{\phi}} = \frac{(D_1 - D_2)/(r_2 - r_1)}{D_3/(n - r_3)} \quad (4.20)$$

with

$$F \sim F_{r_2 - r_1, n - r_3}. \quad (4.21)$$

So if the observed value of the statistic Equation 4.20 was within the upper (say 5%) tail of the  $F$ -distribution Equation 4.21, we would infer that Model  $M_2$  is better than Model  $M_1$ .

## 4.4 Model residuals

Consider a generalised linear model with observed values  $y_i, i = 1, \dots, n$  and fitted values  $\hat{\mu}_i$ . Then the *raw* or *response* residuals are defined by

$$e_i^{\text{raw}} = y_i - \hat{\mu}_i.$$

More useful are the *standardised* or *Pearson* residuals defined by

$$e_i^{\text{std}} = e_i^{\text{P}} = \frac{y_i - \hat{\mu}_i}{\sqrt{b''(\theta_i)}}.$$

Recall from Equation 3.8 that  $\text{Var}(Y_i) = \phi b''(\theta_i)$ .

*Deviance residuals* are defined so that the sum of squared deviance residuals equals the total deviance. Thus we set

$$e_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i},$$

where  $d_i$  is the contribution of observation  $i$  to the deviance,  $D$ . For example, when  $y_i$  has a Poisson distribution with estimated mean  $\hat{\mu}_i$ , we have

$$e_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{2 \left[ y_i \log \left( \frac{y_i}{\hat{\mu}_i} \right) - y_i + \hat{\mu}_i \right]}.$$

Residuals are useful for assessing the overall fit of a model to the data, and for identifying where the model might need to be improved.

## 4.5 Fitting generalised linear models in R

### 4.5.1 GLM-related R commands

The function used to fit a generalised linear model in **R** is

```
glm(formula, family).
```

Let  $x, y, z, a, b, c, \dots$  be a set of vectors all of the same length  $n$  (perhaps read in from a data file using the `read.table` and `attach` commands). If  $a, b, c$  are qualitative variables, then they first need to be declared as factors by `a = as.factor(a)`, etc.

The `formula` argument of `glm` specifies the required model in compact notation, e.g.  $y \sim x*a$  or  $y \sim x + z*a$  where  $\sim, +, *$  have the same meaning as in Section 2.5.

The `family` argument specifies which exponential family is to be used. We shall use `gaussian`, `poisson` and `binomial`; `gaussian` is the default. Other options are available; see `help(family)` for further information.

Along with the family, a link function can be specified. The possible choices are:

- `gaussian` – "identity" (default)
- `poisson` – "log" (default), "sqrt", "identity"
- `binomial` – "logit" (default), "probit", "cloglog".

**R** assumes the default options unless we state otherwise. For example,

```
glm(y ~ a+b) # Gaussian errors, identity link
glm(y ~ a + b, poisson) # Poisson errors,
log link
glm(y ~ a+b, poisson("sqrt")) # Poisson errors, sqrt link
```

**Note that for the binomial case**, the response variable should be an  $n \times 2$  matrix `ym`, say, not a vector, where the first column contains the numbers of successes and the second column the numbers of failures, for example:

```
glm(ym ~ a+b, binomial) # binomial errors, logit link
```

To extract information about a fitted generalised linear model, it is best to store the result of `glm` as a variable and then to use the following functions:

- To fit a GLM and store the result in `y.glm` (for example): `y.glm = glm(y ~ a*b, poisson("sqrt"))`
- To print various pieces of information including deviance residuals, parameter estimates and standard errors, deviances, and (if specified) correlations of parameter estimates: `summary(y.glm, correlation=T)`
- To print the anova table of the fitted model: `anova(y.glm)`
- To print the deviance of the fitted model: `deviance(y.glm)`
- To print the residual degrees of freedom of the fitted model: `df.residual(y.glm)`
- To print the vector of fitted values under the fitted model: `fitted.values(y.glm)`
- To print the residuals from the fitted model: `residuals(y.glm, type)` Note: `type` should be "deviance" (default), "pearson", or "response"

- To print the parameter estimates from the fitted model: `coefficients(y.glm)`
- To print the design matrix for a specified model formula: `model.matrix(y ~ a*b)`

The functions `summary` and `anova` are the most useful for printing out information about the fitted model. The results of the other functions can be saved as variables for further computation, if desired.

#### 4.5.2 Example of fitting Poisson GLM in R

Here is a toy example of **R** commands for modelling a response in terms of two qualitative explanatory variables (that is *factors*). The model assumes the data are Poisson-distributed and uses the logarithmic link function.

Call:

```
glm(formula = y ~ a + b, family = poisson)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 1.1408   | 0.3351     | 3.404   | 0.000663 | *** |
| a2          | -0.3054  | 0.3522     | -0.867  | 0.385934 |     |
| a3          | 0.1466   | 0.3132     | 0.468   | 0.639712 |     |
| a4          | 0.4568   | 0.2932     | 1.558   | 0.119269 |     |
| b2          | 0.9163   | 0.3162     | 2.898   | 0.003761 | **  |
| b3          | 0.9445   | 0.3150     | 2.999   | 0.002712 | **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 31.725 on 11 degrees of freedom  
 Residual deviance: 13.150 on 6 degrees of freedom  
 AIC: 68.227

Number of Fisher Scoring iterations: 5

Correlation of Coefficients:

|    | (Intercept) | a2   | a3   | a4   | b2   |
|----|-------------|------|------|------|------|
| a2 | -0.45       |      |      |      |      |
| a3 | -0.50       | 0.48 |      |      |      |
| a4 | -0.54       | 0.51 | 0.57 |      |      |
| b2 | -0.67       | 0.00 | 0.00 | 0.00 |      |
| b3 | -0.68       | 0.00 | 0.00 | 0.00 | 0.72 |

## Analysis of Deviance Table

Model: poisson, link: log

Response: y

Terms added sequentially (first to last)

|      | Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)    |
|------|----|----------|-----------|------------|-------------|
| NULL |    |          | 11        | 31.725     |             |
| a    | 3  | 6.2793   | 8         | 25.445     | 0.098784 .  |
| b    | 2  | 12.2947  | 6         | 13.150     | 0.002139 ** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

[1] 13.15047

[1] 6

|          |          |          |           |          |          |          |           |
|----------|----------|----------|-----------|----------|----------|----------|-----------|
| 1        | 2        | 3        | 4         | 5        | 6        | 7        | 8         |
| 3.129412 | 2.305882 | 3.623529 | 4.941176  | 7.823529 | 5.764706 | 9.058824 | 12.352941 |
| 9        | 10       | 11       | 12        |          |          |          |           |
| 8.047059 | 5.929412 | 9.317647 | 12.705882 |          |          |          |           |

## Focus on GLM fitting in R quiz

Test your knowledge recall and application to reinforce basic ideas and prepare for similar concepts later in the Chapter

1. Which of the following is the main R command used to fit generalised linear models?

- (A) fitglm
- (B) glm
- (C) formula
- (D) lm
- (E) family
- (F) None of the above

2. Which of the following is NOT an option for the model family?

- (A) binomial
- (B) gaussian
- (C) poisson
- (D) Gamma
- (E) logit
- (F) None of the above

3. Which of the following is NOT a link option in a GLM?

- (A) log
- (B) identity
- (C) logit
- (D) constant
- (E) probit
- (F) None of the above

4. Which of the following is NOT a link option with the binomial family?

- (A) logit
- (B) cauchit
- (C) inverse
- (D) cloglog
- (E) log
- (F) None of the above

5. Which of the following is NOT an R command used to check model fit?

- (A) anova
- (B) deviance

- (C) residuals
- (D) summary
- (E) glm
- (F) None of the above

## 4.6 Exercises

4.1 Use Equation 4.4 to obtain estimation equations for the natural parameter  $\theta$ , based on a sample  $\mathbf{y} = \{y_1, \dots, y_n\}$ , for each of the following situations:

- (a) the binomial,  $Y \sim \text{Bin}(m, p)$ ,
- (b) the geometric,  $Y \sim \text{Ge}(p)$ ,
- (c) the exponential,  $Y \sim \text{Exp}(\lambda)$ .

Assuming that  $n$  is very large, use Equation 4.9 to comment on possible bias of the estimator for large samples. What is the corresponding variance of the estimator?

[Click here to see hints.](#)

For each situation equate the theoretical expectation, in terms of the derivative of  $b$ , to the sample mean – that is start from Equation 4.3 and solve. For bias and variance you should consider the asymptotic results.

4.2 For a sample of size  $n$  from the normal distribution,  $Y \sim N(\mu, \sigma^2)$ , how do the results produced using Equation 4.9 and Equation 4.10 compare with the familiar results  $\hat{\mu} = \bar{Y}$ ,  $E[\hat{\mu}] = \mu$ , and  $\text{Var}[\hat{\mu}] = \sigma^2/n$ ?

[Click here to see hints.](#)

Identify,  $\theta$ ,  $\phi$  and  $b''(\theta)$  then substitute into the equations.

4.3 For the normal linear regression model,  $\mathbf{Y} = X\beta + \epsilon$  where  $\epsilon \sim N_n(0, \sigma^2 I_n)$  use the principle of maximum likelihood to show that the MLE has the closed form given in Equation 4.13.

[Click here to see hints.](#)

Write out the log-likelihood, expand the quadratic form  $(\mathbf{y} - X\beta)^T(\mathbf{y} - X\beta)$ , then apply standard results on matrix differentiation:  $\frac{\partial}{\partial \beta}(\mathbf{a}^T \beta) = \mathbf{a}$  and  $\frac{\partial}{\partial \beta}(\beta^T A \beta) = 2A\beta$  for symmetric  $A$ .

4.4 Check that you can derive the formulas given for deviance of normal, binomial, and Poisson models at the start of Section 4.3.

[Click here to see hints.](#)

Form the likelihoods of the saturated model, recalling that the fitted values are exactly equal to the observed  $y$  values, and substitute them into the general deviance equation. Then, simplify.

4.5 How do the deviance tests defined in Section 4.3 relate to the  $F$ -tests used in normal linear model comparison?

[Click here to see hints.](#)

Start by comparing the notation. Then, replace the RSS terms in the  $F$ -statistic with deviances and adjust the notation for degrees of freedom. You should then get very similar expressions. The only difference being from which model  $\sigma^2$  is estimated.

4.6 Consider a clinical trial into the effectiveness of Amoxicillin (a widely used antibiotic) against bacterial chest infections in children. A sample of 200 children with chest infections were randomly assigned to one of five groups with Amoxicillin dose levels of 20-100 mg/kg per day. Four weeks later it was recorded whether the child had required a re-treatment with a further dose of antibiotic, or not, with results summarised in Table 4.1

Table 4.1: Effectiveness of Amoxicillin against bacterial chest infections in children

|              | 20mg | 40mg | 60mg | 80mg | 100mg |
|--------------|------|------|------|------|-------|
| Group size   | 23   | 18   | 14   | 23   | 22    |
| Re-treatment | 20   | 14   | 5    | 6    | 3     |

First fit a normal linear model and then fit a generalised linear model assuming binomial errors and a logistic link. Superimpose both fitted models onto a scatter plot of the data. Which do you think is the better model? Justify your answer.

[Click here to see hints.](#)

The linear model part would be straightforward, look at earlier examples if not. For the glm case, use `glm()` with `family = binomial` and compare the fit by eye.

# 5 Modelling Proportions

## 5.1 Introduction

In this chapter we will focus on applications of generalised linear modelling where the response variable follows a binomial distribution. This can arise when the outcome is binary, that is it can take one of only two possible values, or is the sum of a set of such binary outcomes. These two outcomes record whether some event of interest has occurred or not. In the simplest case, the response variable,  $B$  say, is defined as

$$B = \begin{cases} 1 & \text{if the event has occurred} \\ 0 & \text{if the event has not occurred} \end{cases} \quad (5.1)$$

and we set  $Pr(B = 1) = p$ , and hence  $Pr(B = 0) = 1 - p$ . This is, of course, the definition of a Bernoulli trial leading to a Bernoulli random variables,  $B \sim \text{Bernoulli}(p)$ .

Suppose now that there are  $m$  similar binary outcomes,  $B_i, i = 1, \dots, m$  with  $Pr(B_i = 1) = p$ , and that the total number of times that the event occurred is recorded as the response  $Y$ . Assuming that  $m$  is known before the trials start, that  $p$  is fixed and that the individual Bernoulli trials are independent then  $Y$  follows a binomial distribution,  $Y \sim B(m, p)$  with probability mass function

$$f(y) = \binom{m}{y} p^y (1 - p)^{m-y}. \quad (5.2)$$

Note that the binomial random variable can be thought of as the sum of i.i.d Bernoulli random variables,  $Y = B_1 + \dots + B_m$ , if that is helpful.

The Binomial distribution  $B(m, p)$  is often described in terms of *success* and *failure* and a binomial distribution in terms of the number of successes in  $m$  independent trials, where  $p$  is the probability of success in each trial. The term *success* need not correspond to a favourable outcome; it is merely the language traditionally used in connection with this model. For example, success might correspond to death.

Of course, the special case with  $m = 1$  reduces to the Bernoulli distribution. Further, if  $Y_1 \sim B(m_1, p)$  and independently  $Y_2 \sim B(m_2, p)$ , then  $Y_1 + Y_2$  also follows a binomial distribution,  $B(m_1 + m_2, p)$  – note that is only valid when  $p$  is common.

Recall that the binomial can be re-written in the exponential family form of Equation 3.3 with

$$f(y) = \exp \left\{ y \operatorname{logit} p + m \log(1 - p) + \log \binom{m}{y} \right\},$$

with natural or canonical parameter  $\theta = \operatorname{logit} p = \log(p(1 - p))$ , scale parameter  $\phi = 1$ ,  $b(\theta) = m \log(1 + e^\theta)$  and  $c(y, \phi) = \log \binom{m}{y}$  as in Table 3.3.

## 5.2 The logistic model

Throughout this module we have assumed that a response variable,  $Y$ , depends on a set of explanatory variables,  $\mathbf{x} = \{x_1, \dots, x_p\}$ . In particular, for binomial counts from  $B(m, p)$ ,  $0 < p < 1$  with mean  $\mu = mp$  we set the link function,  $g(\mu)$ , equal to the linear predictor  $\eta = \sum \beta_j x_j$  and hence have

$$g(\mu) = \sum_{j=1}^p \beta_j x_j = \mathbf{x}^T \boldsymbol{\beta}$$

where  $\boldsymbol{\beta} = \{\beta_1, \dots, \beta_p\}$  are the linear predictor parameters. Recall that the canonical logit link Equation 3.14 for the binomial leads to the systematic part of the model

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \mathbf{x}^T \boldsymbol{\beta} \quad (5.3)$$

but that other links function are possible, such as the probit, Equation 3.15, the complementary log-log, Equation 3.16, and the cauchit Equation 3.17.

This model can alternatively be written

$$Y \sim B(m, p), \quad \text{where } p = \frac{\exp\{\mathbf{x}^T \boldsymbol{\beta}\}}{1 + \exp\{\mathbf{x}^T \boldsymbol{\beta}\}} \quad (5.4)$$

which makes the dependence of  $Y$  on  $\mathbf{x}$ , and  $\boldsymbol{\beta}$ , more explicit.

In general, we might want to consider  $n$  independent binomial random variables representing subgroups of the sample with  $Y_1 \sim B(m_1, p_1), \dots, Y_n \sim B(m_n, p_n)$ , see Table 5.1 and hence

$$\mathbf{Y} \sim B(\mathbf{m}, \mathbf{p}), \quad \text{and } \mathbf{p} = \frac{\exp\{X\boldsymbol{\beta}\}}{1 + \exp\{X\boldsymbol{\beta}\}},$$

with response  $\mathbf{Y} = \{Y_1, \dots, Y_n\}$ ,  $\mathbf{m} = \{m_1, \dots, m_n\}$ ,  $\mathbf{p} = \{p_1, \dots, p_n\}$ ,  $X$  being the  $n \times p$  design matrix and  $\boldsymbol{\beta} = \{\beta_1, \dots, \beta_p\}$  the linear predictor parameters.

Table 5.1: Linear logistic model

|              | Group 1     | Group 2     | ... | Group $n$   |
|--------------|-------------|-------------|-----|-------------|
| Successes    | $Y_1$       | $Y_2$       | ... | $Y_n$       |
| Failures     | $m_1 - Y_1$ | $m_2 - Y_2$ | ... | $m_n - Y_n$ |
| <b>Total</b> | $m_1$       | $m_2$       | ... | $m_n$       |

Then we can write down the log likelihood of  $\boldsymbol{\beta}$  using Equation 4.12 and Equation 5.2 as

$$l(\boldsymbol{\beta}; \mathbf{y}) = \sum_{i=1}^n \left\{ y_i \log(p_i) + (m_i - y_i) \log(1 - p_i) + \log\binom{m_i}{y_i} \right\} \quad (5.5)$$

where

$$p_i = \frac{\exp\{\mathbf{x}_i^T \boldsymbol{\beta}\}}{1 + \exp\{\mathbf{x}_i^T \boldsymbol{\beta}\}}.$$

We would then use the principle of maximum likelihood to estimate  $\beta$

$$\hat{\beta} = \arg \max_{\beta} l(\beta; \mathbf{y}).$$

Then, the fitted probability are given by

$$\hat{p}_i = \frac{\exp\{\mathbf{x}_i^T \hat{\beta}\}}{1 + \exp\{\mathbf{x}_i^T \hat{\beta}\}}$$

and corresponding fitted values

$$\hat{y}_i = m_i \hat{p}_i.$$

The Pearson residuals for binomial data take the form

$$e_i^P = \frac{y_i - m_i \hat{p}_i}{\sqrt{m_i \hat{p}_i (1 - \hat{p}_i)}}.$$

From the general result  $\text{Var}(Y_i) = \phi b''(\theta)$  in Proposition 3.1, it can be shown that  $\text{Var}(Y_i) = m_i p_i (1 - p_i)$ . Then, using the estimate of  $p_i$  leads to the denominator above.

For large  $m_i$ , the usual Normal approximation to the Binomial means that the Pearson residuals are approximately  $N(0, 1)$  distributed.

It can be shown that the deviance for the fitted model is

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \left( \frac{y_i}{m_i \hat{p}_i} \right) + (m_i - y_i) \log \left( \frac{m_i - y_i}{m_i (1 - \hat{p}_i)} \right) \right\}, \quad (5.6)$$

which is approximately  $\chi_{n-r}^2$  distributed if the model is correct, where  $r$  is the number of degrees of freedom in the model (that is the number of parameters or columns of the design matrix).

This formula can be shown to be equivalent to

$$D = 2 \sum_{j=1}^2 \sum_{i=1}^n O_{ji} \log \frac{O_{ji}}{E_{ji}}$$

where  $O_{ji}$  denotes the observed value and  $E_{ji}$  denotes the expected value in cell  $(j, i)$  of the  $2 \times n$  table of successes and failures:

Table 5.2: Observed and expected frequencies in the linear logistic model

|                    | Group 1  | Group 2  | ... | Group $n$ |
|--------------------|----------|----------|-----|-----------|
| Successes, $j = 1$ | $O_{11}$ | $O_{12}$ | ... | $O_{1n}$  |
|                    | $E_{11}$ | $E_{12}$ | ... | $E_{1n}$  |
| Failures, $j = 2$  | $O_{21}$ | $O_{22}$ | ... | $O_{2n}$  |
|                    | $E_{21}$ | $E_{22}$ | ... | $E_{2n}$  |

Another goodness-of-fit statistic is the Pearson chi-squared statistic:

$$X^2 = \sum_{j=1}^2 \sum_{i=1}^n \frac{(O_{ji} - E_{ji})^2}{E_{ji}}.$$

This is asymptotically equivalent to the deviance Equation 5.6 (proof is by Taylor series expansion; omitted). Thus, asymptotically,  $X^2$  is also approximately  $\chi_{n-r}^2$  distributed. Both approximations can be poor if the expected frequencies are small, but  $X^2$  copes slightly better with this problem. See Dobson, p.136 for more details.

## Focus on model choice quiz

Test knowledge recall and comprehension to reinforce ideas of linear and non-linear models.

1. Which of the following is NOT a true statements about modelling?
  - (A) Some situations might suggest more than one model to try.
  - (B) The context of the problem and description of the data can be ignored – it's only the data that matter
  - (C) Goodness of fit can be assessed using the model residuals.
  - (D) It is important to plot the fitted model on the same graph as the data.
  - (E) All model assumptions should be checked.
2. Which of the following is a true statements about linear regression?
  - (A) It is always possible to use the fitted model for extrapolation.
  - (B) A linear model should always be the first model considered.
  - (C) When there is a moderate correlation then the relationship between variables is always linear.
  - (D) A linear model will fit best when the correlation is close to zero.
  - (E) It is always important to check model residuals.
3. Which of the following is a true statements about logistic regression?
  - (A) A logisitc model should always be used if the correlation is close to zero.

- (B) The explanatory variables follow a binomial distribution
- (C) It is always possible to use the fitted model for extrapolation.
- (D) Should only be used if a linear model is not a good fit to the data.
- (E) The response variable follows a binomial distribution.

Click here to see explanations

1. The starting point should always to learn about the background to a data set as this often suggests models to consider, appropriate distributions and any potential problems. Hence saying that it is only the data that matters is inappropriate.
2. Of course there are many important things to check, but looking at residuals is one of the most important.
3. Logistic regression assumes that the data follow a binomial distribution.

### 5.3 Overdispersion

Examination of residuals and deviances may indicate that a model is not an adequate fit to the data. One possible reason is *overdispersion*. In particular, it is advisable to compare the residual deviance of the proposed model with the corresponding degrees of freedom. If the ratio, deviance divided by degrees of freedom, is substantially greater than 1, then there is evidence for overdispersion – substantially less than 1 indicates under dispersion but that is very rare. Overdispersion can occur for any error distribution where the variance is linked to the mean — e.g. Binomial, Poisson. In the binomial case, overdispersion is called *extra-Binomial* variation.

Recall that if  $Y_i \sim \text{Bin}(m_i, p_i)$ ,  $\text{Var}(Y_i) = m_i p_i (1 - p_i)$ . Overdispersion occurs if observations which have been modelled by a  $\text{Bin}(m_i, \hat{p}_i)$  distribution have substantially greater variation than  $m_i \hat{p}_i (1 - \hat{p}_i)$ . This will lead to a value of  $D$  substantially greater than the expected value of  $n - r$ . This can occur if the model is missing appropriate explanatory variables or has the wrong link function, or if the  $Y_i$  are not independent.

One solution is to include an extra parameter  $\tau$  in the model so that  $\text{Var}(Y_i) = \tau \times m_i p_i (1 - p_i)$ . For more details, see Section 7.7 of Dobson or Chapter 6 of Collett (1991) *Modelling Binary data*, Chapman & Hall.

The `glm` function in **R** allows for *extra-Binomial* variation by setting `family=quasibinomial()` with the usual link functions available.

## 5.4 Odds ratios for 2x2 tables

In the special case of the data forming a  $2 \times 2$  table, then a commonly calculated summary statistic is the *odds ratio*. This is a measure of association between the presence or absence of some property and an outcome. It represents the odds that an outcome will occur given presence, compared to the odds of the outcome occurring in the absence of that property.

The following  $2 \times 2$  table summarizes the different possibilities.

|         | Success | Failure     |
|---------|---------|-------------|
| Present | $y_1$   | $m_1 - y_1$ |
| Absent  | $y_2$   | $m_2 - y_2$ |

Similarly, by dividing by the corresponding row total, the probabilities are summaries as follows.

|         | Success | Failure     |
|---------|---------|-------------|
| Present | $\pi_1$ | $1 - \pi_1$ |
| Absent  | $\pi_2$ | $1 - \pi_2$ |

There are conditional probabilities, for example  $\Pr(\text{Success}|\text{Present})$  etc., and form a *stochastic matrix* where sum of each row is now 1.

For each row, the *odds* of success is defined by

$$O_i = \pi_i / (1 - \pi_i), \quad i = 1, 2,$$

and a comparison between these two probabilities is given by the *odds ratio*

$$\phi = \frac{O_1}{O_2} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}.$$

If  $\phi = 1$  then the property does not affect the odds of success,  $\phi > 1$  indicates that the property is associated with higher odds of success, and  $\phi < 1$  indicates the property is associated with lower odds of success.

Note that the *log-odds*,  $\log O_i = \log(\pi_i / (1 - \pi_i))$ , which is the  $\text{logit}(\pi_i)$ , and hence is the natural parameter in the regular exponential family form of the binomial distribution.

Finally, an approximate  $100(1-\alpha)\%$  confidence interval,  $(\phi_{\alpha/2}, \phi_{1-\alpha/2})$ , for the odds ratio is calculated using the formulas:

$$\phi_{\alpha/2} = \phi \times \exp \left( -z_{1-\alpha/2} \sqrt{\frac{1}{y_1} + \frac{1}{(m_1 - y_1)} + \frac{1}{y_2} + \frac{1}{(m_2 - y_2)}} \right)$$

and

$$\phi_{1-\alpha/2} = \phi \times \exp \left( +z_{1-\alpha/2} \sqrt{\frac{1}{y_1} + \frac{1}{(m_1 - y_1)} + \frac{1}{y_2} + \frac{1}{(m_2 - y_2)}} \right)$$

where  $z_{1-\alpha/2}$  is such that  $\Pr(Z \leq z_{1-\alpha/2}) = 1 - \alpha/2$ . In particular, for a 95% confidence interval  $z_{1-\alpha/2} = z_{0.975} = 1.96$ .

If the confidence interval for the odds ratio includes the value 1 then the calculated odds ratio would not be considered statistically significant.

## 5.5 Application to dose–response experiments

A variable dose of some reagent is administered to each study subject, and the occurrence of a specific response is recorded. This is a *dose–response* experiment, one of the first uses of regression models for Bernoulli (or Binomial) responses.

For example, the Table 5.5 (including the information from Table 1.1) gives the number of beetles killed  $y_i$  and the number not killed  $(m_i - y_i)$  out of a total number  $m_i$  that were exposed to a dose  $x_i$  of gaseous carbon disulphide, for  $n = 8$  dose levels  $i = 1, \dots, 8$  (Dobson: pp.109 in 1st edn; pp.119 in 2nd edn; pp.127 in 3rd edn). The proportion killed  $p_i = y_i/m_i$  at each dose level  $i$  is also shown in Table 5.5 and plotted in Figure 5.1.

Table 5.5: Numbers of beetles killed by five hours of exposure to 8 different concentrations of gaseous carbon disulphide

| Dose<br>$x_i$ | No. of beetle<br>$m_i$ | No. killed<br>$y_i$ | No. not killed<br>$m_i - y_i$ | Proportion<br>$p_i = y_i/m_i$ |
|---------------|------------------------|---------------------|-------------------------------|-------------------------------|
| 1.6907        | 59                     | 6                   | 53                            | 0.10                          |
| 1.7242        | 60                     | 13                  | 47                            | 0.22                          |
| 1.7552        | 62                     | 18                  | 44                            | 0.29                          |
| 1.7842        | 56                     | 28                  | 28                            | 0.50                          |
| 1.8113        | 63                     | 52                  | 11                            | 0.83                          |
| 1.8369        | 59                     | 53                  | 6                             | 0.90                          |
| 1.8610        | 62                     | 61                  | 1                             | 0.98                          |
| 1.8839        | 60                     | 60                  | 0                             | 1.00                          |

Now Equation 5.4 motivates modelling the beetle data as

$$Y_i \sim B(m_i, p_i), \quad \text{for } i = 1, \dots, n = 8$$

where

$$\eta_i = \alpha + \beta x_i$$

and so

$$p_i = \frac{\exp\{\alpha + \beta x_i\}}{(1 + \exp\{\alpha + \beta x_i\})}.$$

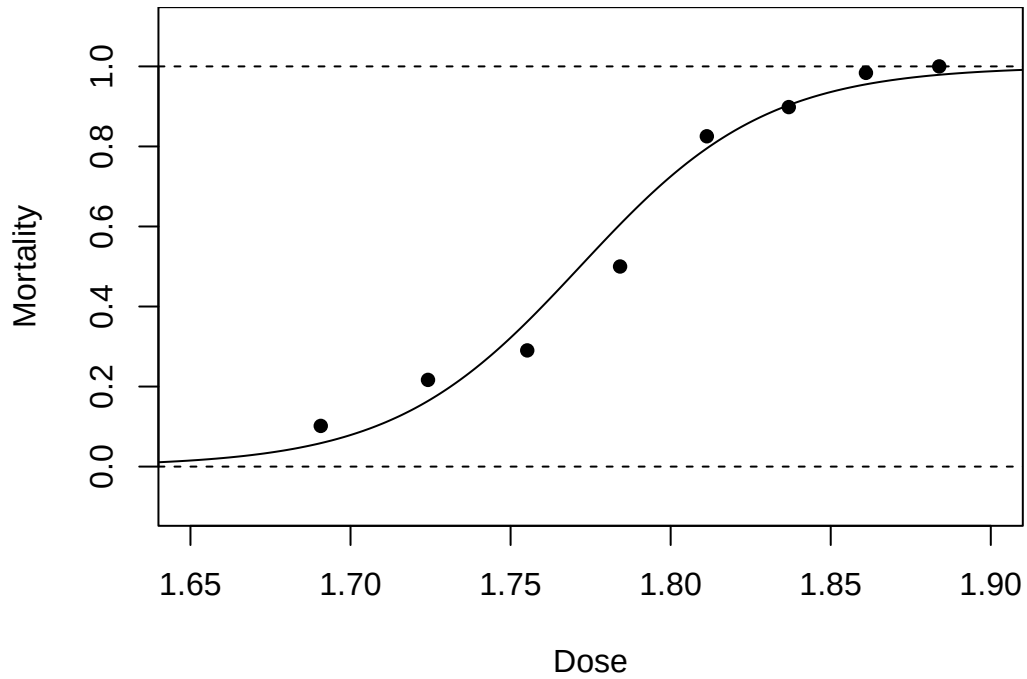


Figure 5.1: Beetle mortality rates with fitted dose-response curves.

To fit this model in **R**, a matrix with columns containing the numbers killed  $y_i$  and the numbers not killed  $m_i - y_i$  is first calculated

|      |      |      |
|------|------|------|
|      | [,1] | [,2] |
| [1,] | 6    | 53   |
| [2,] | 13   | 47   |
| [3,] | 18   | 44   |
| [4,] | 28   | 28   |
| [5,] | 52   | 11   |
| [6,] | 53   | 6    |

which is then used in the **glm** command

The model parameter estimates are given by

|             |          |
|-------------|----------|
| (Intercept) | dose     |
| -60.71745   | 34.27033 |

and hence

$$\hat{p}_i = \frac{\exp\{-60.7 + 34.3x_i\}}{(1 + \exp\{-60.7 + 34.3x_i\})}. \quad (5.7)$$

Further, the deviance is given by

```
[1] 11.23223
```

or more helpfully a full summary, which contains parameter estimates and deviance values using

```
Call:
```

```
glm(formula = y ~ dose, family = binomial)
```

```
Coefficients:
```

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -60.717  | 5.181      | -11.72  | <2e-16 *** |
| dose        | 34.270   | 2.912      | 11.77   | <2e-16 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 284.202 on 7 degrees of freedom  
Residual deviance: 11.232 on 6 degrees of freedom  
AIC: 41.43
```

```
Number of Fisher Scoring iterations: 4
```

or the anova table

```
Analysis of Deviance Table
```

```
Model: binomial, link: logit
```

```
Response: y
```

```
Terms added sequentially (first to last)
```

|        | Df     | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|--------|--------|----------|-----------|------------|---------------|
| NULL   |        |          | 7         | 284.202    |               |
| dose 1 | 272.97 |          | 6         | 11.232     | < 2.2e-16 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Clearly, many of the results are repeated in the various **R** output.

When considering testing model goodness of fit, it is important to distinguish between deviance values which compare a given model with the saturated model and deviance values

which compare two competing models. For example, in the Analysis of Deviance Table, the right hand values compare the Null model with the saturated and the model including dose with the saturated. Whereas the left hand value is relevant to the comparison of Null model with the model including dose.

### Summary of results:

From the output, we can first test if the Null model, which contains only a constant term, is a good fit to the data using the hypotheses:

$$H_0 : \text{Null model is true against } H_1 : \text{Null model is false.}$$

From the output, note that the Null deviance is  $D_0 = 284.202$  and that this model has  $r_0 = 1$  parameters. Therefore,  $D_0$  follows a  $\chi^2$  distribution with  $n - r_0 = 8 - 1 = 7$  degrees of freedom and hence the p-value is:

[1] 1.425247e-57

This means that we reject  $H_0$  as the observed value of  $D$  is in the upper tail of the  $\chi^2$  distribution.

If we had obtained a non-significant result, then we would stop the analysis and conclude that the Null model is an adequate fit.

Next, consider testing the model which includes the explanatory variable where the hypotheses are

$$H_0 : \text{Null model is true against } H_1 : \text{Model with dose is true.}$$

From the output, the deviance of the proposed model is  $D_1 = 11.232$  with  $r_1 = 2$  parameters. The test statistics is then  $D_0 - D_1 = 284.202 - 11.232 = 272.97$  which follows a  $\chi^2$  distribution with  $r_1 - r_0 = 2 - 1 = 1$  degrees of freedom. The p-value is then

[1] 2.556369e-61

and we reject  $H_0$  in favour of  $H_1$  and conclude that dose is important.

We can finish the testing by checking if this model is a good fit to the data. That is,

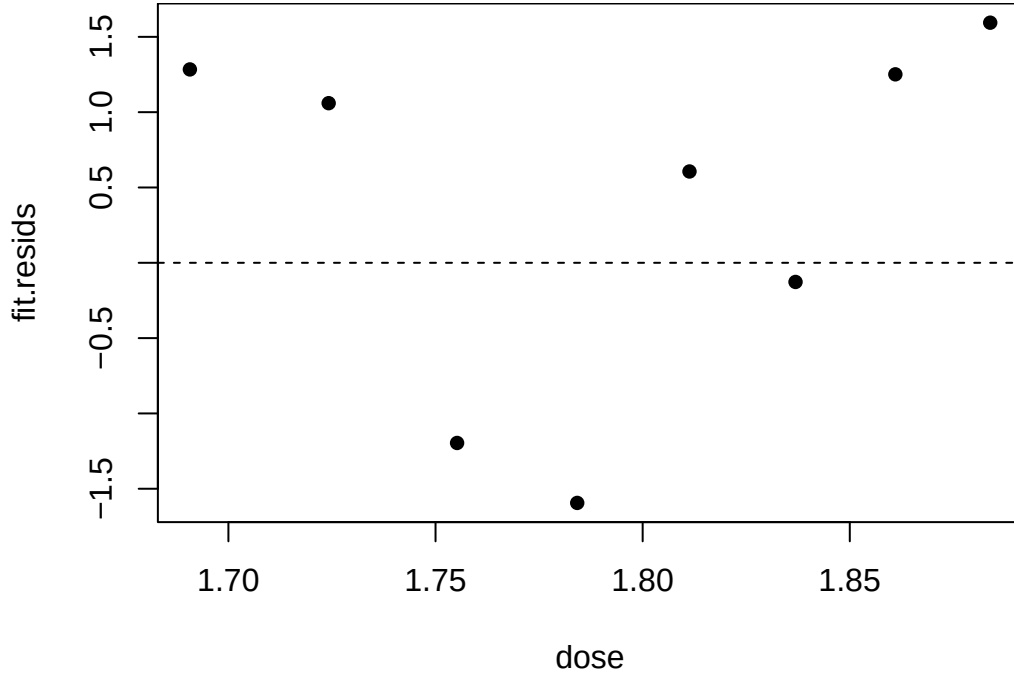
$$H_0 : \text{The model with dose is true against } H_1 : \text{The model with dose is false}$$

The deviance of this is  $D_1 = 11.232$  with  $r_1 = 2$  parameters which follows a  $\chi^2$  distribution with  $n - r_1 = 8 - 2 = 6$  degrees of freedom and has p-value

[1] 0.08146544

which means that we accept  $H_0$  at the 5% level and conclude that this model is an adequate fit to the data.

Finally, we can look at the residuals:



These show a very slight *u-shaped* pattern and hence it would be worth exploring fitting using the other link functions. For this data set, the complementary log-log (cloglog) link has a slightly better fit, see Figure 3.4, but a slight pattern remains.

Before finishing, suppose that we wish to predict, by hand, the probability for dose level  $x_0 = 1.85$ , say, and the expected number of beetles killed when  $m_0 = 60$  beetles, say, are exposed.

The probability is predicted as

$$\hat{p}_i = \frac{\exp\{-60.7 + 34.3x_0\}}{(1 + \exp\{-60.7 + 34.3x_0\})} = 0.936$$

and the expected number of beetles

$$m_0 \hat{p}_i = 56.2$$

Finally, suppose that we wish to find the minimum dose which kills 95% of the beetles, that is  $p_0 = 0.95$  which requires us to invert the logistic equation.

In general, suppose we wish to find  $x_p$  which corresponds to proportion  $p$  then

$$x_p = \frac{1}{\beta} \left\{ \log \left( \frac{p}{1-p} \right) - \alpha \right\}$$

and hence here  $\hat{x}_p = 1.856$ .

## 5.6 Exercises

5.1. Which of the following descriptions suggest that a logistic regression should be considered. Justify your answers.

- a. A pharmaceutical company perform clinical trials where volunteers are given varying doses of an experimental drug to study the chances of potential negative side effects. From historical data 4 previous volunteers out of a total of 120 have had negative side effects.
- b. A credit card company monitors sales to identify fraudulent transactions. On the previous five days, 26, 20, 18, 19 and 21 regular transactions occurred before the first fraudulent transaction was identified.
- c. A national internet service provider monitors the frequency of helpline phone calls regarding internet failures. From records, there are an average of 0.27 calls per day.
- d. The University Student Education Services monitor the success of registered students every year. In particular, they record whether students pass their examinations, or not, and what *application score* they were given when they applied for a place at University.

[Click here to see hints.](#)

For each, consider what is response, is it binomial and what is the explanatory variable. In cases which are not binomial, suggest an alternative distribution.

5.2. For each of the situation produce appropriate graphs within RStudio. Describe the type of relationship, if any, shown in the plots. Is a linear, a logistic, or some other model appropriate? If linear or logistic, go ahead with model fitting and perform a residual analysis. Do you think that prediction from the model would be reliable? Justify each step of your analysis.

- a. A UK loan company requires a detailed questionnaire to be completed for any loan application including such information as income, regular expenditure, previous loan details and other information from data analytics and consumer credit reporting companies. All the information is combined into a single *credit score* from 0 to 1000. A large sample of credit applicants were contacted by an independent market research company to discover the outcome of their application. The data file *loans.csv* gives the number of successful application, number declined and the credit scores.
- b. A small estate agent company is interested in studying the changes in sales of houses due to targeted advertising campaigns and in particular to determine if additional spending is worthwhile. The company record the weekly sales before each advertising campaign and then again for one week a full month later, along with the cost of each campaign. The datafile *advertising.csv* contains the advertising budget values in pounds and corresponding changes in sale, also in pounds.

- c. The yield of arable crops is sensitive to the nutrient levels in the soil. A key fertilizer used to boost production is the mineral phosphorus which is applied as phosphorus pentoxide,  $P_2O_5$ . At a wheat large farm, various amounts from 0 kg/ha to 80 kg/ha were applied to investigate the effect on wheat yield, in kg/ha. Use the data in file *wheat.csv* to investigate this situation.
- d. Cardiac troponin (cTn) is a human protein which is commonly measured in the diagnosis of heart attacks. A small sample of blood is taken from patients suspected of having heart attacks and the level of cardiac troponin is recorded. The cardiac troponin levels, in ng/mL, of 100 patients arriving at a hospital Accident & Emergency department with significant chest discomfort along with the later confirmed diagnosis (0=No and 1=Yes) are stored in the file *heart.csv*.

[Click here to see hints.](#)

For each, consider what is response, is it binomial and what is the explanatory variable. If the plot reveals a relationship which is either linear or logistic, then follow the appropriate steps previously discussed. For the residual analysis, plot a graph of residuals against fitted values and check for a lack of pattern. For the linear cases, also consider the assumption of normality, for example using a histogram.

5.3 In the beetle example, use the fitted logistic regression model, Equation 5.7, to predict what dose of gaseous carbon disulphide would kill 90% of beetles. Then, fit alternative link functions to the data and plot the results. Do you think that the choice of link function makes a difference to your conclusions? Justify your answer.

[Click here to see hints.](#)

For the prediction, rearrange the usual expression for  $p$  in terms of  $x$  and the model parameters to give an expression for  $x$ . To fit alternative link functions, just change the argument in the *family* option.

5.4 (Based on Dobson & Barnett, pp 144–145). Suppose there are 2 groups of people: the first group is exposed to some pollutant and the second group is not. In a prospective study, each group is followed for several years and categorised according to the presence or absence of some disease. Let  $\pi_i$  denote the probability that a person in group  $i$  contracts the disease,  $i = 1, 2$ . The following  $2 \times 2$  table summarizes the different possibilities.

|             | Diseased | Not diseased |
|-------------|----------|--------------|
| Exposed     | $\pi_1$  | $1 - \pi_1$  |
| Not exposed | $\pi_2$  | $1 - \pi_2$  |

Note that the sum of each row is 1. For each  $i = 1, 2$ , the *odds* of contracting the disease is defined by

$$O_i = \pi_i / (1 - \pi_i),$$

and a comparison between these two probabilities is given by the *odds ratio*

$$\phi = \frac{O_1}{O_2} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}.$$

- Show that  $\phi = 1$  if and only if there is no difference between the control and exposed groups. What does it mean if  $\phi > 1$ ?
- Consider now  $m$  tables where each is of this form, with probabilities  $\pi_{ij}$  represented by a logistic model

$$\text{logit}(\pi_{ij}) = \alpha_i + \beta_i x_j, \quad i = 1, 2, \quad j = 1, \dots, m,$$

where  $x_j$  is some specified quantitative explanatory variable. Interpret the parameters  $\alpha_i$  and  $\beta_j$ , and give their effect on the log odds ratio  $\log \phi_j$ , say, for each table. Show that  $\log \phi_j$  is constant across the  $m$  tables if  $\beta_1 = \beta_2$ .

- Give a practical example where such a model might be appropriate.
- How would you express this model in the R computer language?

[Click here to see hints.](#)

In (a), one way is easy. For the other way, rearrange the odds ratio and consider the ratios of the event occurring and the ratio that the events not occurring. These are only equal if the probabilities are the same. In (b), note that the log odds ratio is exactly the logistic transform. Substitute in and simplify. An example should be easy, and check earlier examples for the R command.

5.5. Consider again the beetle example, but suppose that rather than the actual dose concentration had been measured, instead only whether a *Low dose* or *High dose* was used recorded. This would lead to the following table:

|           | Died | Not died | Total |
|-----------|------|----------|-------|
| Low dose  | 65   | 172      | 237   |
| High dose | 226  | 18       | 244   |

What are the respective probability of death values in the two dose groups? The *relative risk* is defined as the ratio of these two values. Calculate the risk of death due to High dose exposure relative to Low dose for this example. How could this value be used to measure the association between dose and death?

What are the respective odds of death in each of the two groups? Calculate the *odds ratio*, defined as the ratio of these two values. How could this value be used to interpret the data?

[Click here to see hints.](#)

If there is no difference then the relative risk should be 1 and a value substantially different indicates an association. The odds and the odds ratio have a similar interpretation. All measures do the same thing, but each is standard in particular application areas.

# 6 Loglinear Models

## 6.1 Overview

In this chapter, we deal with data sets in which the response variable  $y$  is a count and the explanatory variables are all factors – i.e. qualitative variables. Initially we assume  $y$  has a Poisson distribution.

See Chapter 9 of Dobson and Barnett (2008). See also Agresti (1996) *An introduction to categorical data analysis*.

We assume a generalised linear model with

- responses (counts) having independent Poisson distributions;
- a logarithmic link function (hence the name *log-linear model*).

Consider, for example, the two-way contingency table in Table 6.1:

Table 6.1: A two-way contingency table with  $k_1$  rows and  $k_2$  columns, where each entry  $y_{ij}$  is a count.

|       | 1           | 2           | ... | $j$         | ... | $k_2$         | Total       |
|-------|-------------|-------------|-----|-------------|-----|---------------|-------------|
| 1     | $y_{11}$    | $y_{12}$    | ... | $y_{1j}$    | ... | $y_{1k_2}$    | $y_{1+}$    |
| ...   | ...         | ...         | ... | ...         | ... | ...           | ...         |
| $i$   | $y_{i1}$    | $y_{i2}$    | ... | $y_{ij}$    | ... | $y_{ik_2}$    | $y_{i+}$    |
| ...   | ...         | ...         | ... | ...         | ... | ...           | ...         |
| $k_1$ | $y_{k_1 1}$ | $y_{k_1 2}$ | ... | $y_{k_1 j}$ | ... | $y_{k_1 k_2}$ | $y_{k_1 +}$ |
| Total | $y_{+1}$    | $y_{+2}$    | ... | $y_{+j}$    | ... | $y_{+k_2}$    | $y_{++}$    |

In row  $i$  and column  $j$  of Table 6.1 we assume

$$Y_{ij} \sim \text{Po}(\lambda_{ij})$$

where

$$\log \lambda_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}. \quad (6.1)$$

Here  $\mu$  is called the *main effect*;  $\alpha_i$  is a *row effect*;  $\beta_j$  is a *column effect*; and  $(\alpha\beta)_{ij}$  denotes an *interaction effect* parameter. Some or all of these effects must be present in the model, with constraints to ensure that the model is identifiable – this is sometimes referred to as the *identifiability* or *aliasing problem*. Generally, **R** function `glm` automatically sets the first level of each effect to zero to achieve model identification. Thus, for the two-way model,

$$\alpha_1 = 0, \quad \beta_1 = 0, \quad (\alpha\beta)_{11} = (\alpha\beta)_{1j} = (\alpha\beta)_{k_1} = 0.$$

## 6.2 Motivating examples

### 6.2.1 Malignant melanoma

The data in Table 6.2 are from a study of 400 patients with malignant melanoma, a particular form of skin cancer (see Dobson and Barnett, p.172). For each tumour, its type and site were recorded. The data in Table 6.2 comprise the numbers of tumours  $y$  in each combination of site and tumour-type. This is a two-way *contingency table*. We want to know how melanoma frequency depends on site and type.

Table 6.2: Melanoma counts by type and site.

| Type                           | Head | Trunk | Extremities | Total |
|--------------------------------|------|-------|-------------|-------|
| Hutchinson's melanotic freckle | 22   | 2     | 19          | 34    |
| Superficial spreading melanoma | 16   | 54    | 115         | 185   |
| Nodular                        | 19   | 33    | 73          | 125   |
| Indeterminate                  | 11   | 17    | 28          | 56    |
| Total                          | 68   | 106   | 226         | 400   |

Table 6.3: Percentages across columns within rows for melanoma data.

| Type                           | Head | Trunk | Extremities | Total |
|--------------------------------|------|-------|-------------|-------|
| Hutchinson's melanotic freckle | 64.7 | 5.9   | 29.4        | 100   |
| Superficial spreading melanoma | 8.6  | 29.2  | 62.2        | 100   |
| Nodular                        | 15.2 | 26.4  | 58.4        | 100   |
| Indeterminate                  | 19.6 | 30.4  | 50.0        | 100   |
| Total                          | 17.0 | 26.5  | 56.5        | 100   |

Table 6.4: Percentages across rows within columns for melanoma data.

| Type                           | Head  | Trunk | Extremities | Total |
|--------------------------------|-------|-------|-------------|-------|
| Hutchinson's melanotic freckle | 32.4  | 1.9   | 4.4         | 8.5   |
| Superficial spreading melanoma | 23.5  | 50.9  | 50.9        | 46.3  |
| Nodular                        | 27.9  | 31.1  | 32.3        | 31.3  |
| Indeterminate                  | 16.2  | 16.0  | 12.4        | 14.0  |
| Total                          | 100.0 | 99.9  | 100.0       | 100.0 |

Although we have two factors, **Type** and **Site**, that we may use as predictors, standard ANOVA regression methods are inappropriate here as the dependent variable is not continuous but is instead a count. We will use *log-linear regression*, a type of generalised linear model, to analyse these data.

Table 6.3 shows row and Table 6.4 column percentages for these data. For example, 15.2% of nodular melanomas occurred in the head and neck, 26.4% in the trunk, and 58.4% in the extremities. Compare this to the equivalent figures for Hutchinson’s melanotic freckles: 64.7%, 5.9%, and 29.4% - strikingly different. So different types of melanomas are more likely to occur in different locations.

### 6.2.2 Flu vaccine

The data in Table 6.5 are from a randomised controlled trial in which 73 patients were randomised into two groups (see Dobson and Barnett, 2008, p.173). The treatment group was given a flu vaccine, while the control group was given a placebo. Levels of an antibody (HIA) were measured after 6 weeks and classified into three groups: Low, Moderate, and High.

Table 6.5: Antibody responses to flu vaccine from a randomised controlled trial.

| Group   | Low | Moderate | High |
|---------|-----|----------|------|
| Placebo | 25  | 8        | 5    |
| Vaccine | 6   | 18       | 11   |
| Total   | 31  | 26       | 16   |

Is the pattern of response the same for each treatment group? The percentages in Table 6.6 suggest not - row percentages indicate lower responses in the placebo group.

Table 6.6: Percentages across rows within columns for antibody responses to flu vaccine.

| Group   | Low  | Moderate | High | Total |
|---------|------|----------|------|-------|
| Placebo | 65.8 | 21.1     | 13.2 | 100   |
| Vaccine | 17.1 | 51.4     | 31.4 | 100   |
| Total   | 42.4 | 35.6     | 22.0 | 100   |

Table 6.7: Percentages across rows within columns for antibody responses to flu vaccine.

| Group   | Low  | Moderate | High |
|---------|------|----------|------|
| Placebo | 80.6 | 30.8     | 31.2 |
| Vaccine | 19.4 | 69.2     | 68.8 |
| Total   | 100  | 100      | 100  |

### 6.3 Maximum likelihood estimation

Recall that for each cell  $Y_{ij} \sim \text{Po}(\lambda_{ij})$  so  $E[Y_{ij}] = \lambda_{ij}$ . However, we estimate  $\hat{\lambda}_{ij} = y_{ij}$  only for the *saturated model* given by Equation 6.1. In general, for non-saturated models, the estimate  $\hat{\lambda}_{ij} \neq y_{ij}$ .

Consider the independence model for a 2-way table, that is where  $(\alpha\beta)_{ij} = 0$  for all  $i, j$ . Here,

$$\log \lambda_{ij} = \mu + \alpha_i + \beta_j \quad (6.2)$$

and  $y_{ij}$  is the observed count for cell  $(i, j)$ , so the likelihood is given by

$$L(\lambda; y) = \prod_{i,j} \frac{e^{-\lambda_{ij}} \lambda_{ij}^{y_{ij}}}{y_{ij}!}$$

and the log-likelihood is

$$l(\lambda; y) = \sum_{i,j} \{y_{ij} \log \lambda_{ij} - \lambda_{ij} - \log y_{ij}!\}.$$

Next, using Equation 6.2, gives

$$l(\lambda; y) = \sum_{i,j} \{y_{ij}(\mu + \alpha_i + \beta_j) - \exp(\mu + \alpha_i + \beta_j) - \log y_{ij}!\}$$

which can be re-written as

$$l(\lambda; y) = \mu y_{++} + \sum_i \alpha_i y_{i+} + \sum_j \beta_j y_{+j} - e^\mu \left( \sum_i e^{\alpha_i} \right) \left( \sum_j e^{\beta_j} \right) - \sum_{ij} \log y_{ij}!. \quad (6.3)$$

The maximum likelihood estimates of the model parameters are obtained in the usual way. Differentiating Equation 6.3 with respect to  $\mu$  and setting the result to zero gives, at the MLE,

$$y_{++} = e^{\hat{\mu}} \left( \sum_i e^{\hat{\alpha}_i} \right) \left( \sum_j e^{\hat{\beta}_j} \right). \quad (6.4)$$

Differentiating Equation 6.3 with respect to  $\alpha_i$  (where  $i \neq 1$  because  $\alpha_1 = 0$  to avoid an identifiability problem) and setting the result to zero gives, at the MLE,

$$y_{i+} = e^{\hat{\mu}} e^{\hat{\alpha}_i} \left( \sum_j e^{\hat{\beta}_j} \right). \quad (6.5)$$

Differentiating Equation 6.3 with respect to  $\beta_j$  (where  $j \neq 1$  because  $\beta_1 = 0$ ) and setting the result to zero gives, at the MLE,

$$y_{+j} = e^{\hat{\mu}} e^{\hat{\beta}_j} \left( \sum_i e^{\hat{\alpha}_i} \right). \quad (6.6)$$

Then

$$\frac{y_{i+} y_{+j}}{y_{++}} = e^{\widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j} = \widehat{\lambda}_{ij}. \quad (6.7)$$

It follows that

$$\widehat{\lambda}_{i+} = y_{i+} \quad \widehat{\lambda}_{+j} = y_{+j} \quad \widehat{\lambda}_{++} = y_{++}.$$

Thus, the total fitted count in row  $i$  is identical to the total observed count in row  $i$ . Further, the total fitted count in column  $j$  is equal to the total observed count in column  $j$  and the total fitted count equals the total observed count.

## 6.4 Model fitting in R

### 6.4.1 Malignant melanoma

Consider an analysis of Melanoma data introduced in Section 6.2.1.

To test the independence of **Type** and **Site**, we fit the model

$$\text{Count} \sim \text{Type} + \text{Site}$$

assuming Poisson counts and a logarithmic link function, using the commands:

Call:

```
glm(formula = mcount ~ type.F + site.F, family = "poisson")
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 1.7544   | 0.2040     | 8.600   | < 2e-16 ***  |
| type.F2     | 1.6940   | 0.1866     | 9.079   | < 2e-16 ***  |
| type.F3     | 1.3020   | 0.1934     | 6.731   | 1.68e-11 *** |
| type.F4     | 0.4990   | 0.2174     | 2.295   | 0.02173 *    |
| site.F2     | 0.4439   | 0.1554     | 2.857   | 0.00427 **   |
| site.F3     | 1.2010   | 0.1383     | 8.683   | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 295.203 on 11 degrees of freedom  
 Residual deviance: 51.795 on 6 degrees of freedom  
 AIC: 122.91

Number of Fisher Scoring iterations: 5

We see that the residual deviance for this model is 51.795 on 6 degrees of freedom. If the model is true, the residual deviance will have an approximate  $\chi^2$  distribution on 6 degrees of freedom. If the model is not true, the residual deviance will probably be too large to correspond to this distribution. Thus we calculate the  $p$ -value in the upper tail of the  $\chi^2_6$  distribution, which can be computed using the command:

```
[1] 2.050465e-09
```

This strongly indicates that the independence model is inadequate. Therefore, unless we can spot alternative simplifications, we will have to use the saturated model.

We can look at residuals from the model see where the departures from independence occur. The largest residual is for Hutchinson's freckle on the head and neck, which occurs more often than would be expected under independence.

For the saturated model,  $\hat{\lambda}_{ij} = y_{ij}$ . In this example,  $\sum_{ij} \hat{\lambda}_{ij} = \sum_{ij} y_{ij} = 400$ , so the probability of a tumour being in category  $(i, j)$  is  $y_{ij}/400$  — just the observed proportions.

Note that in Table 6.2 we have a total of  $y_{++} = 400$  observations. If the data were truly Poisson, this would be a suspiciously round number. In reality, this total was fixed by design, so we should take into account the fact that  $y_{++} = 400$  and fit a more suitable model, such as the *multinomial* model which we will meet in the next chapter.

Overdispersion can occur for the Poisson model, just as in the binomial case — see Section 5.3. This is called *extra-Poisson* variation. The `glm` function in **R** can take this into account by including an extra parameter  $\tau$  in the model using `family=quasipoisson()`.

### 6.4.2 Flu vaccine

There is no saved data file for this example — recall it is very small — and so define R variables for the `count`, `response` (perhaps not a good name as it may be confusing but it seems the correct description from the context) and `drug` used. Don't forget to declare the explanatory variables as factors.

Now fit the first model, here consider the saturated model with main effects and interaction.

Call:

```
glm(formula = count ~ response.F * drug.F, family = poisson)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 1.60944  | 0.44721    | 3.599   | 0.00032  | *** |
| response.FL | 1.60944  | 0.48990    | 3.285   | 0.00102  | **  |
| response.FM | 0.47000  | 0.57009    | 0.824   | 0.40969  |     |

```

drug.FV          0.78846    0.53936    1.462    0.14379
response.FL:drug.FV -2.21557    0.70539   -3.141    0.00168 **
response.FM:drug.FV  0.02247    0.68663    0.033    0.97389
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 2.3807e+01 on 5 degrees of freedom
Residual deviance: 1.7764e-15 on 0 degrees of freedom
AIC: 37.128

```

Number of Fisher Scoring iterations: 3

Even though the model is a perfect fit – the fitted values are exactly the data values (check next) we see that `response.F="M"` is not significantly different to the baseline `response.F="H"` and also that `drug.F="V"` is not significantly different to `drug.F="P"`.

Let's check the fitted values:

```

 1  2  3  4  5  6
25  8  5  6 18 11

```

where we see that, indeed, they are the same as the data.

The Null deviance is 23.807 and follows a  $\chi^2$  distribution on 5 degrees of freedom (under the Null hypothesis). This has a p-value of  $p < 0.001$  which is highly significant and hence the null model is not adequate.

We cannot test the saturated model as it fits perfectly.

Next let's try a reduced model with the interaction removed:

Call:

```
glm(formula = count ~ response.F + drug.F, family = poisson)
```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.11972    0.27408   7.734 1.04e-14 ***
response.FL   0.66140    0.30783   2.149  0.0317 *
response.FM   0.48551    0.31774   1.528  0.1265
drug.FV      -0.08224    0.23428  -0.351  0.7256
---

```

```

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 23.807  on 5  degrees of freedom
Residual deviance: 18.643  on 2  degrees of freedom
AIC: 51.771
```

Number of Fisher Scoring iterations: 5

The `response.F="M"` is again not significantly different to the baseline `response.F="H"` and `drug.F="V"` is not significantly different to `drug.F="P"`.

The model deviance is 18.643 following a  $\chi^2_2$  distribution with corresponding p-value of  $p < 0.001$ . This is highly significant suggesting that this and any model without an interaction term is unlikely to be adequate.

Let us return to a model with interaction, and let's try merging response `response.F="M"` and `"H"` – calling them `"HM"` (I chose this name so that alphabetically it would be first just as `"H"` was first):

Call:

```
glm(formula = count ~ response2.F * drug.F, family = poisson)
```

Coefficients:

|                      | Estimate | Std. Error | z value | Pr(> z )     |
|----------------------|----------|------------|---------|--------------|
| (Intercept)          | 1.8718   | 0.2774     | 6.749   | 1.49e-11 *** |
| response2.FL         | 1.3471   | 0.3419     | 3.940   | 8.17e-05 *** |
| drug.FV              | 0.8023   | 0.3338     | 2.404   | 0.0162 *     |
| response2.FL:drug.FV | -2.2295  | 0.5640     | -3.953  | 7.71e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 23.807  on 5  degrees of freedom
Residual deviance:  2.405  on 2  degrees of freedom
AIC: 35.533
```

Number of Fisher Scoring iterations: 4

Interestingly all terms in this model are significant but let's check the overall goodness of fit.

The model deviance is 2.405 following a  $\chi^2_2$  distribution with corresponding p-value of  $p = 0.3$ . This is non-significant suggesting that this model is a good fit to the data.

The fitted values are:

|      |     |     |     |      |      |
|------|-----|-----|-----|------|------|
| 1    | 2   | 3   | 4   | 5    | 6    |
| 25.0 | 6.5 | 6.5 | 6.0 | 14.5 | 14.5 |

and the residuals (which I have rounded for display purposes):

|     |     |      |     |     |      |
|-----|-----|------|-----|-----|------|
| 1   | 2   | 3    | 4   | 5   | 6    |
| 0.0 | 1.5 | -1.5 | 0.0 | 3.5 | -3.5 |

The fitted values are a good fit to the data and hence, of course, the residuals are not too large.

The overall conclusion is that the model with response and drug, and their interaction, but with the merging of response levels H and M is an appropriate model. This indicates that the vaccine does have an effect on the antibody response but that it does not influence whether that is a medium or high response.

## 6.5 Multi-way contingency tables

We can generalize the model notation introduced in Equation 6.1 to accommodate multi-way contingency tables; that is tables of counts indexed by multiple factors. For example, for a 3-way table of cell counts  $y_{ijk}$ , the saturated model would be written:

$$Y_{ijk} \sim \text{Po}(\lambda_{ijk})$$

where

$$\log \lambda_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk}.$$

Here,  $\alpha_i$ ,  $\beta_j$  and  $\gamma_k$  are *main effects*;  $(\alpha\beta)_{ij}$ ,  $(\alpha\gamma)_{ik}$ , and  $(\beta\gamma)_{jk}$  are two-way *interaction effects*; and  $(\alpha\beta\gamma)_{ijk}$  is a *three-way interaction*.

This approach is easily extended to tables of any number of factors. Note, however, that not all terms need be present in a model, though 3-way or higher-order interactions might sometimes be required.

A *hierarchical* model is one in which each term in the model is accompanied by all lower-order terms. For example, including the term  $(\alpha\beta\gamma)_{ijk}$  would require inclusion of each of the terms  $\alpha_i, \beta_j, \gamma_k, (\alpha\beta)_{ij}, (\alpha\gamma)_{ik}, (\beta\gamma)_{jk}$ . We will be concerned only with *hierarchical* contingency table models. Non-hierarchical models are sometimes appropriate, depending on the nature of the factors involved, but hierarchical models are more generally applicable and easier to interpret.

## 6.6 Exercises

6.1 Overdispersion is an occasional problem when fitting generalised linear models with a known scale parameter in situations where there is unexplained variation. In this exercise we illustrate the problem in Poisson regression. The starting point is the observation that if  $Y \sim P(\lambda)$ , then  $E[Y] = \lambda$  and  $\text{Var}[Y] = \lambda$ .

- a. Consider joint random variables  $(X, Y)$  where  $X$  takes two possible values with equal probabilities,  $Pr(X = 1) = Pr(X = 2) = 1/2$ .

Suppose the conditional distribution of  $Y$  given  $X$  is Poisson,

$$Y|(X = 1) \sim P(\lambda_1), \quad Y|(X = 2) \sim P(\lambda_2), \quad (*)$$

where  $\lambda_1 < \lambda_2$ . Let  $\lambda = (\lambda_1 + \lambda_2)/2$  denote the average value. Thus the marginal distribution of  $Y$  is a mixture of two Poisson distributions. Show that

$$E[Y] = \lambda, \quad \text{Var}[Y] = \lambda + (\lambda_1 - \lambda_2)^2/4,$$

that is, although the mean of  $Y$  is the same under the mixture (or conditional Poisson) model as under the Poisson model, the variance is larger.

[Click here to see hints.](#)

Start with the definition of expectation as a weighted average of the conditional expectations. Then, for the variance, apply a similar result to find the expectation of the square.

- b. This phenomenon might be observed in data as follows. Let  $n = 60$  and let an explanatory variable  $x_i$  take the value  $x_i = 1$  for  $i = 1, \dots, 30$  and  $x_i = 2$  for  $i = 31, \dots, 60$ . Suppose that the observations  $y_i|x_i$  come from the above conditional Poisson model (\*).

Consider fitting the following two models in R with Poisson errors and a log link function:

$$(i) \quad y \sim 1, \quad (ii) \quad y \sim x.$$

Since model (ii) is the correct model, it should yield a good fit to the data. But if the experimenter does not know about the variable  $x$ , it will only be feasible to fit model (i). Let  $\bar{Y}$  and  $S^2$  denote the sample mean and variance of the  $Y_i$ ,  $i = 1, \dots, 60$ . Show that

$$E[\bar{Y}] = \lambda, \quad E[S^2] = \lambda + \frac{60}{59}(\lambda_1 - \lambda_2)^2/4.$$

Hence show that the Pearson  $\chi^2$  goodness of fit statistic for model (i) will indicate a poorly fitting model if  $\lambda_1$  and  $\lambda_2$  are far apart.

[Click here to see hints.](#)

For the expectation of the sample variance, first find the expectation of the square of the sample mean. Next, note that the Pearson goodness of fit statistics depends

on the ratio of the sample variance divided by the sample mean. Under the conditional Poisson model the variance would be larger but the mean unchanged, hence the statistics gets bigger and so will be further into the upper tail.

- c. This example is very simple, but overdispersion can occur much more widely. Why is overdispersion not a problem for generalised linear models in which the response distribution includes a scale parameter?

[Click here to see hints.](#)

In the Poisson, and the Binomial, the problem is caused by the mean and variance being defined by the same model parameter.

6.2 (From Dobson & Barnett, p 163) This question should be done in a computer package such as R. You should think carefully about which variables, if any, to condition on in your analysis: HOME = 1,2,3, CONTACT = 1,2, or SATISFACTION = 1,2,3.

The data relate to an investigation into satisfaction with housing conditions in Copenhagen. Residents of selected areas living in rented houses built between 1960 and 1968 were questioned about their satisfaction and their degree of contact with other residents. The data were tabulated by type of housing. Investigate the associations between satisfaction, contact with other residents and type of housing.

#### Low Contact:

| Satisfaction: | Low | Medium | High |
|---------------|-----|--------|------|
| Tower blocks  | 65  | 54     | 100  |
| Apartments    | 130 | 76     | 111  |
| Houses        | 67  | 48     | 62   |

#### High Contact:

| Satisfaction: | Low | Medium | High |
|---------------|-----|--------|------|
| Tower blocks  | 34  | 47     | 100  |
| Apartments    | 141 | 116    | 191  |
| Houses        | 130 | 105    | 104  |

- a. Produce appropriate tables of percentages to gain initial insights into the data; for example, percentages in each contact level by type of housing and level of satisfaction, or percentages in each level of satisfaction by contact and type of housing.

[Click here to see hints.](#)

Consider many 2x2 tables for separate levels of the third variable. Each time re-scaling to create rows or columns, as appropriate, summing to 100%.

- b. Using e.g. R, fit various log-linear models to investigate interactions between the variables.

[Click here to see hints.](#)

Define your own variables and make sure the explanatory variables are converted to factor. Including the three-way interaction would lead to the saturated model – why? – and hence only consider pair-wise two-way interactions (as well as the variables separately).

- c. For some model that fits (at least moderately) well, calculate the Pearson residuals and use them to find where the largest discrepancies are between the observed and expected values.

[Click here to see hints.](#)

Use the residuals command with type=“pearson”

# 7 Extensions to Loglinear models

## 7.1 Overview

In this chapter we consider two extensions to the log-linear models studied in the previous chapter. Recall that the log-linear model assumes that each observation follows a Poisson distribution, but in many cases these will not be all independent. Instead, row totals or the grand total will be fixed. This structure means that the observed counts will no longer follow the Poisson distribution and hence modification to the theory and calculations are needed.

## 7.2 Contingency tables with fixed marginals

### 7.2.1 Introduction

In the melanoma example, see Table 6.2, the overall total number of observations was exactly 400. If the data were truly Poisson, this would be a suspiciously round number. In reality, this total was fixed by design, so we should take into account the fact that  $y_{++} = 400$  and fit a more appropriate model. Also, recall the flu vaccine example of Table 6.5. Suppose that some of the marginal totals are fixed by design. For example, that the number of patients in each arm of the trial (38 in the Placebo group, 35 in the Vaccine group) was fixed before data collection started. In this case, we should also take into account the fact that  $y_{1+} = 38$  and  $y_{2+} = 35$  and fit a more suitable model.

Before we can re-consider these data sets, we first need to establish some theoretical results to deal with such *conditional distribution* cases.

### 7.2.2 Modelling

Consider, again, a general two-way contingency table such as that in Table 7.1. The fixed marginals are the column sums:  $y_{+j}$ , the row sums:  $y_{i+}$ , and the overall sum:  $y_{++}$ .

Table 7.1: A two-way contingency table with  $k_1$  rows and  $k_2$  columns, where each entry  $y_{ij}$  is a count.

|   | 1        | 2        | ... | $j$      | ... | $k_2$      | Total    |
|---|----------|----------|-----|----------|-----|------------|----------|
| 1 | $y_{11}$ | $y_{12}$ | ... | $y_{1j}$ | ... | $y_{1k_2}$ | $y_{1+}$ |

|              | 1           | 2           | ... | $j$         | ... | $k_2$         | Total       |
|--------------|-------------|-------------|-----|-------------|-----|---------------|-------------|
| ...          | ...         | ...         | ... | ...         | ... | ...           | ...         |
| $i$          | $y_{i1}$    | $y_{i2}$    | ... | $y_{ij}$    | ... | $y_{ik_2}$    | $y_{i+}$    |
| ...          | ...         | ...         | ... | ...         | ... | ...           | ...         |
| $k_1$        | $y_{k_1 1}$ | $y_{k_1 2}$ | ... | $y_{k_1 j}$ | ... | $y_{k_1 k_2}$ | $y_{k_1 +}$ |
| <b>Total</b> | $y_{+1}$    | $y_{+2}$    | ... | $y_{+j}$    | ... | $y_{+k_2}$    | $y_{++}$    |

Suppose that we have fixed row totals  $y_{i+}$ , or a fixed overall total  $y_{++}$ , how would such constraints affect the distribution of the random variables  $Y_{ij}$ ?

To answer this question, we will need the following results.

**Proposition 7.1.** *Let  $Y_{ij}$ , for  $i = 1, \dots, k_1$  and  $j = 1, \dots, k_2$ , be the responses in a two-way contingency table, with  $k_1$  rows and  $k_2$  columns. Further, assume that the counts follow independent Poisson distributions with corresponding means  $\lambda_{ij}$ , that is  $Y_{ij} \sim \text{Po}(\lambda_{ij})$ . Then, the distributions of the row sums are,*

$$Y_{i+} \sim \text{Po}(\lambda_{i+}),$$

$$\text{where } Y_{i+} = \sum_{j=1}^{k_2} Y_{ij} \text{ and } \lambda_{i+} = \sum_{j=1}^{k_2} \lambda_{ij}.$$

**Notes:**

- Here we are not yet assuming that the marginal totals are fixed.
- Similar results hold when considering column sums and the overall sum.

**Proof:** This follows from the moment generating function of the Poisson distribution. If  $Y_1, \dots, Y_k$  are independent with  $Y_j \sim \text{Po}(\lambda_j)$ , then the MGF of their sum is  $\prod_j e^{\lambda_j(e^t-1)} = e^{(\sum_j \lambda_j)(e^t-1)}$ , which is the MGF of a  $\text{Po}(\sum_j \lambda_j)$  distribution.

**Proposition 7.2.** *Let  $Y_{ij} \sim \text{Po}(\lambda_{ij})$  for  $i = 1, \dots, k_1$  and  $j = 1, \dots, k_2$  then the conditional distribution of each count in a row given that the row sum is fixed is given by*

$$(Y_{ij} | Y_{i+} = m) \sim \text{Bin}\left(m, \frac{\lambda_{ij}}{\lambda_{i+}}\right).$$

*That is, conditioning on the sum of independent Poisson random variables induces a Binomial distribution.*

**Proof:** First define  $Y_{i-j} = Y_{i+} - Y_{ij}$  with correspondingly  $y_{i-j} = m - y_{ij}$  and also  $\lambda_{i-j} = \lambda_{i+} - \lambda_{ij}$ .

Consider the conditional probability mass function

$$\begin{aligned}
p(Y_{ij} = y_{ij} | Y_{i+} = m) &= \frac{p(Y_{ij} = y_{ij}, Y_{i+} = m)}{p(Y_{i+} = m)} \quad \text{by conditional probability definition} \\
&= \frac{p(Y_{ij} = y_{ij}, Y_{i-j} = m - y_{ij})}{p(Y_{i+} = m)} \quad \text{rearranging} \\
&= \frac{p(Y_{ij} = y_{ij})p(Y_{i-j} = m - y_{ij})}{p(Y_{i+} = m)} \quad \text{by independence.}
\end{aligned}$$

Then, each of these follows a Poisson distribution:  $Y_{ij} \sim \text{Po}(\lambda_{ij})$ ,  $Y_{i-j} \sim \text{Po}(\lambda_{i-j})$ , and  $Y_{i+} \sim \text{Po}(\lambda_{i+})$ . Therefore,

$$\begin{aligned}
p(Y_{ij} = y_{ij} | Y_{i+} = m) &= \frac{\lambda_{ij}^{y_{ij}} e^{-\lambda_{ij}}}{y_{ij}!} \frac{\lambda_{i-j}^{m-y_{ij}} e^{-\lambda_{i-j}}}{m-y_{ij}!} \bigg/ \frac{\lambda_{i+}^m e^{-\lambda_{i+}}}{m!} \\
&= \binom{m}{y_{ij}} \left( \frac{\lambda_{ij}}{\lambda_{i+}} \right)^{y_{ij}} \left( \frac{\lambda_{i-j}}{\lambda_{i+}} \right)^{m-y_{ij}} \\
&= \binom{m}{y_{ij}} \pi^{y_{ij}} (1 - \pi)^{m-y_{ij}}
\end{aligned}$$

which is the probability mass function of a  $\text{Bin}(m, \pi)$  random variable, where  $\pi = \lambda_{ij}/\lambda_{i+}$ .

**Proposition 7.3.** *Let  $Y_{ij} \sim \text{Po}(\lambda_{ij})$  for  $i = 1, \dots, k_1$  and  $j = 1, \dots, k_2$  then the **joint** conditional distribution of the counts in a row given that the row sum is fixed is given by*

$$p(Y_{i1}, \dots, Y_{ik_2} | Y_{i+} = m) = \frac{m!}{y_{i1}! \dots y_{ik_2}!} \pi_{i1}^{y_{i1}} \dots \pi_{ik_2}^{y_{ik_2}}$$

with

$$\pi_{ij} = \lambda_{ij}/\lambda_{i+} \quad \text{and} \quad \sum_{j=1}^{k_2} \pi_{ij} = 1.$$

This distribution is called the Multinomial distribution, with index  $m$  and parameters  $\pi_{ij}$ ,  $j = 1, \dots, k_2$ , which can be denoted

$$(Y_{i1}, \dots, Y_{ik_2} | Y_{i+} = m) \sim \text{Mult}(m, \pi_{i1}, \dots, \pi_{ik_2}).$$

**Proof:** Omitted.

**Proposition 7.4.** *Let  $Y_{ij} \sim \text{Po}(\lambda_{ij})$  for  $i = 1, \dots, k_1$  and  $j = 1, \dots, k_2$  then the conditional distribution of the counts given that the overall sum is fixed is*

$$(Y_{11}, \dots, Y_{k_1 k_2} | Y_{++} = m) \sim \text{Mult}(m, \pi_{11}, \dots, \pi_{k_1 k_2}).$$

with probability mass function given by

$$p(Y_{11}, \dots, Y_{k_1 k_2} | Y_{++} = m) = \frac{m!}{y_{11}! \dots y_{k_1 k_2}!} \pi_{11}^{y_{11}} \dots \pi_{k_1 k_2}^{y_{k_1 k_2}}$$

where  $\pi_{ij} = \lambda_{ij}/\lambda_{++}$ .

**Proof:** Omitted.

### 7.3 Product-multinomial models

The Poisson contingency table model assumes that each cell count (and therefore row and column total) is random – specifically, Poisson.

However, in many examples, some of the marginal totals are fixed by design.

For example, as discussed in the Flu-vaccine example of Section 6.2.2, the numbers in the placebo (38) and vaccine (35) groups are not random, but fixed by the experimenter. Further, we are not interested in modelling these row totals but instead the response to treatment (placebo or vaccine). Thus, in the Flu vaccine example, we should condition the Poisson model on the row totals.

From Equation 6.1 the saturated model for a two-factor table is

$$\lambda_{ij} = \exp \{ \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} \}.$$

Then, using the result in Proposition 7.3, we see that

$$\pi_{ij} = \frac{\lambda_{ij}}{\lambda_{i+}} = \frac{\exp \{ \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} \}}{\sum_j \exp \{ \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} \}}$$

which simplifies to give

$$\pi_{ij} = \frac{\exp \{ \beta_j + (\alpha\beta)_{ij} \}}{\sum_j \exp \{ \beta_j + (\alpha\beta)_{ij} \}}. \quad (7.1)$$

Thus, for each row, that is each  $i$ , this multinomial distribution depends on  $\beta_j$  and  $(\alpha\beta)_{ij}$  for  $j = 1, \dots, k_2$ , but not on  $\mu$  and  $\alpha_i$ . This means that the data in each row of the table are independent and hence, overall, the model likelihood is given by

$$L(\beta; \mathbf{y}) = \prod_{i=1}^{k_1} p(Y_{i1}, \dots, Y_{ik_2} | Y_{i+} = y_{i+}),$$

with  $\beta = \{ \mu, \alpha_1, \dots, \alpha_{k_1}, \beta_1, \dots, \beta_{k_2}, (\alpha\beta)_{11}, \dots, (\alpha\beta)_{k_1 k_2} \}$  and where

$$p(Y_{i1}, \dots, Y_{ik_2} | Y_{i+} = y_{i+}) = \frac{y_{i+}!}{y_{i1}! \dots y_{ik_2}!} \pi_{i1}^{y_{i1}} \dots \pi_{ik_2}^{y_{ik_2}}.$$

Further, using Equation 7.1 the log-likelihood is given by

$$l(\beta; \mathbf{y}) = \text{const} + \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} y_{ij} \left\{ \beta_j + (\alpha\beta)_{ij} - \log \sum_j \exp \{ \beta_j + (\alpha\beta)_{ij} \} \right\}.$$

Recall that the saturated model will have the fitted values equal to the data values in all cases. Therefore, we concentrate on the product multinomial models with interactions set to zero, that is  $(\alpha\beta)_{ij} = 0$ , then

$$l(\beta; \mathbf{y}) = \text{const} + \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} y_{ij} \left\{ \beta_j - \log \sum_j \exp \{ \beta_j \} \right\}$$

which simplifies to

$$l(\beta; \mathbf{y}) = \text{const} + \sum_{j=1}^{k_2} y_{+j} \beta_j - y_{++} \log \sum_j \exp \{ \beta_j \}.$$

Differentiating this with respect to single parameter  $\beta_k$  and setting to zero gives

$$\frac{y_{+k}}{y_{++}} = e^{\hat{\beta}_k} / \sum_j e^{\hat{\beta}_j} \quad (7.2)$$

which is identical to the parameter estimate for the previous independence Poisson model. Thus, in this case, if we want to fit the product multinomial model (which is hard) we can instead fit the Poisson model (which is easy). Note, however, that these estimates only coincide when  $\alpha_1, \dots, \alpha_{k_1}$  are included in the Poisson independence model, even though we see that these are eliminated from the multinomial model. In fact, these parameter estimates reflect the fixed marginal totals in the multinomial situation.

Although this derivation has only considered the cases of the independence Poisson model:  $\log \lambda_{ij} = \mu + \alpha_i + \beta_j$  and the product multinomial model with  $(\alpha\beta)_{ij} = 0$ , the following result gives general guidance.

**Theorem 7.1.** *Tables with fixed margin sums can be analyzed using a multinomial or product-multinomial model as though they were independent Poisson models, with log-linear predictor, provided terms corresponding to the fixed margins are included in the model. Then the MLEs for the two models yield the same values for the parameters of interest and have the same deviances.*

For a 2-way table:

| Multinomial conditioning on | Poisson model must include |
|-----------------------------|----------------------------|
| $y_{++}$                    | $\mu$                      |
| $y_{i+}$                    | $\mu, \alpha_i$            |

For a 3-way table:

| Multinomial conditioning on | Poisson model must include                                                 |
|-----------------------------|----------------------------------------------------------------------------|
| $y_{+++}$                   | $\mu$                                                                      |
| $y_{i++}$                   | $\mu, \alpha_i$                                                            |
| $y_{ij+}$                   | $\mu, \alpha_i, \beta_j, (\alpha\beta)_{ij}$                               |
| $y_{ij+}, y_{+jk}$          | $\mu, \alpha_i, \beta_j, \gamma_k, (\alpha\beta)_{ij}, (\beta\gamma)_{jk}$ |

**Proof:** Omitted. For details see Birch, 1963, JRSS(B), 220–233. The proof is not especially difficult but is a bit messy. It depends on the property of the Poisson model with log link function that observed totals equals fitted totals for the effects present in the Poisson model.

## 7.4 Model fitting in R

Consider again the flu vaccine data described in Section 6.2.2 with data repeated below. Recall that 38 people were recruited to the Placebo group and 35 to the Vaccine group and that we assume these were fixed before data collection started.

Table 7.4: Antibody responses to flu vaccine from a randomised controlled trial.

| Group   | Low | Moderate | High | Total     |
|---------|-----|----------|------|-----------|
| Placebo | 25  | 8        | 5    | <b>38</b> |
| Vaccine | 6   | 18       | 11   | <b>35</b> |
| Total   | 31  | 26       | 16   | 73        |

The fixed row sums mean that a product-multinomial model is appropriate but this can be fitted using the usual same command as the Independence Poisson model as long as we include the term for the row effect, that is for Group.

Call:

```
glm(formula = count ~ antibodyF + groupF, family = "poisson")
```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.78111    0.21184  13.129  <2e-16 ***
antibodyF2   -0.17589    0.26593  -0.661   0.5083
antibodyF3   -0.66140    0.30783  -2.149   0.0317 *
groupF2      -0.08224    0.23428  -0.351   0.7256
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for poisson family taken to be 1)
```

```

Null deviance: 23.807  on 5  degrees of freedom
Residual deviance: 18.643  on 2  degrees of freedom
AIC: 51.771

```

```
Number of Fisher Scoring iterations: 5
```

Hence, our proposed model has a residual deviance of 18.643 on 2 degrees of freedom. The chi-square test gives a  $p$ -value of

$$p = \Pr(\chi_2^2 > 18.643) = 8.95 \times 10^{-5},$$

so we reject the independence model in favour of the saturated model.

For the independence model

$$\hat{\pi}_{ij} = e^{\hat{\beta}_j} / \sum_j e^{\hat{\beta}_j}$$

whose right-hand side does not depend on  $i$  and hence is the same for all rows. The corresponding fitted values are then given by  $\hat{y}_{ij} = y_{i+} \hat{\pi}_{ij}$ .

## 7.5 Exercises

7.1 Consider a  $2 \times m$  contingency table with entries  $y_{ij}$ ,  $i = 1, 2$ ,  $j = 1, \dots, m$ . The rows label STATUS represents *alive* (STATUS = 1) or *dead* (STATUS=2) and the columns label AGE represents  $m$  age groups (AGE = 1, ...,  $m$ ) and not the exact age.

Suppose that a Poisson model  $P(\lambda_{ij})$  with

$$\log(\lambda_{ij}) = \delta + \alpha_i + \beta_j + \gamma_i AGE_j$$

is considered appropriate. Note that  $AGE_j = j$  is treated as a quantitative variable in the last term. What additional constraints on the parameters should be included to make estimation unique – that is to remove the aliasing problem.

Suppose that the column totals,  $Y_{+j}$ , are fixed. Explain why a product binomial model  $B(y_{+j}, \pi_j)$ ,  $j = 1, \dots, m$ , is suitable in this case and find the form of  $\pi_j$ . Which parameters can be estimated under this model?

7.2 In the product-multinomial model for a table with fixed row totals, model parameter estimates are given as the solution of Equation 7.2 and it was claimed that this had the same form as for the previous independence Poisson model (without fixed row totals). Use Equation 6.4 and Equation 6.6 to show that indeed the estimation equations are identical and hence that the parameter estimates will be the same.

# 8 GLM Consolidation and Outlook

## 8.1 Overview

We have now covered all the core elements of generalised linear models. This chapter consolidates those ideas by revisiting the three-component structure of a GLM, summarising the key results from earlier chapters, and working through a complete analysis from start to finish. We then look forward to Part II, where we will relax the linearity assumption on the systematic component.

## 8.2 The three-component structure revisited

Every generalised linear model has exactly three components:

1. **Random component.** The response variable  $Y_i$  follows a distribution from the exponential family:

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}.$$

The mean and variance are determined by  $b(\theta_i)$ :

$$E[Y_i] = b'(\theta_i) = \mu_i, \quad \text{Var}[Y_i] = \phi b''(\theta_i).$$

2. **Systematic component.** A linear predictor in the covariates:

$$\eta_i = \mathbf{x}_i^T \beta = \sum_{j=1}^p \beta_j x_{ij}.$$

3. **Link function.** A monotone differentiable function  $g$  connecting mean to linear predictor:

$$g(\mu_i) = \eta_i.$$

When  $g = (b')^{-1}$ , the link is *canonical* and simplifies the score equations.

The following table summarises the choices for the three distributions we have studied in detail.

Table 8.1: Summary of GLM components for the three standard exponential family distributions.

| Distribution | Canonical link                     | Variance function       | Typical use          |
|--------------|------------------------------------|-------------------------|----------------------|
| Normal       | Identity: $\eta = \mu$             | $b''(\theta) = 1$       | Continuous response  |
| Binomial     | Logit: $\eta = \log \frac{p}{1-p}$ | $b''(\theta) = mp(1-p)$ | Counts / proportions |
| Poisson      | Log: $\eta = \log \lambda$         | $b''(\theta) = \lambda$ | Count data           |

### 8.3 Fitting, inference, and model comparison

**Estimation.** Parameters are estimated by maximum likelihood using the iteratively reweighted least squares (IRLS) algorithm (see supplementary notes for the full derivation). The MLE satisfies asymptotically:

$$\hat{\beta} \sim N_p(\beta, \mathcal{J}^{-1}(\beta)).$$

**Deviance.** The deviance measures how much a fitted model  $M$  falls short of the saturated model:

$$D = 2\phi \{l(\tilde{\theta}; \mathbf{y}, \phi) - l(\hat{\theta}; \mathbf{y}, \phi)\}.$$

For the three standard families,  $D$  reduces to:

- *Normal*: residual sum of squares  $\sum_i (y_i - \hat{\mu}_i)^2$ .
- *Binomial*:  $2 \sum_i \left[ y_i \log \frac{y_i}{m_i \hat{p}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i (1 - \hat{p}_i)} \right]$ .
- *Poisson*:  $2 \sum_i \left[ y_i \log \frac{y_i}{\hat{\mu}_i} - y_i + \hat{\mu}_i \right]$ .

**Model comparison.** For two nested models  $M_1 \subset M_2$  with  $r_1 < r_2$  parameters:

- If  $\phi$  is known:  $(D_1 - D_2) \sim \chi^2_{r_2 - r_1}$  asymptotically.
- If  $\phi$  is unknown: use the  $F$ -statistic Equation 4.20.

**Residuals.** Deviance residuals  $e_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$  and Pearson residuals  $e_i^P = (y_i - \hat{\mu}_i) / \sqrt{b''(\hat{\theta}_i)}$  are the main diagnostic tools.

### 8.4 Worked case study: Coronary heart disease

This example illustrates the complete GLM workflow: data exploration, model formulation, fitting, inference, and model criticism.

The data (Dobson and Barnett, 2008, p.127) relate to a study of coronary heart disease (CHD). A total of 1,329 men were cross-classified by cholesterol level (4 ordered groups: 1 = lowest, 4 = highest) and blood pressure (4 ordered groups: 1

= lowest, 4 = highest). For each of the  $4 \times 4 = 16$  cells, the number of men with CHD ( $y_{ij}$ ) and the total number of men ( $m_{ij}$ ) are recorded. The data are in <https://www.richardpmann.com/MATH3701/Datasets/chd.txt>.

### 8.4.1 Step 1: Load and explore the data

|    | y  | m   | chol | bp |
|----|----|-----|------|----|
| 1  | 2  | 119 | 1    | 1  |
| 2  | 3  | 124 | 1    | 2  |
| 3  | 3  | 50  | 1    | 3  |
| 4  | 4  | 26  | 1    | 4  |
| 5  | 3  | 88  | 2    | 1  |
| 6  | 2  | 100 | 2    | 2  |
| 7  | 0  | 43  | 2    | 3  |
| 8  | 3  | 23  | 2    | 4  |
| 9  | 8  | 127 | 3    | 1  |
| 10 | 11 | 220 | 3    | 2  |
| 11 | 6  | 74  | 3    | 3  |
| 12 | 6  | 49  | 3    | 4  |
| 13 | 7  | 74  | 4    | 1  |
| 14 | 12 | 111 | 4    | 2  |
| 15 | 11 | 57  | 4    | 3  |
| 16 | 11 | 44  | 4    | 4  |

The proportion with CHD in each group:

Both increasing cholesterol and increasing blood pressure appear associated with higher CHD proportions.

### 8.4.2 Step 2: Model formulation

Each  $Y_{ij}$  is a count of CHD cases out of  $m_{ij}$  men, so a Binomial GLM with logit link is appropriate:

$$Y_{ij} \sim B(m_{ij}, p_{ij}), \quad \text{logit}(p_{ij}) = \mu + \alpha_i + \beta_j,$$

where  $\alpha_i$  is the effect of cholesterol group  $i$  and  $\beta_j$  is the effect of blood pressure group  $j$ . We will also consider a saturated model (with an interaction term) and the null model.

Since both factors are ordered, we will first treat them as factors (to allow flexible effect estimates) and then consider whether a simpler model with numeric scores is adequate.

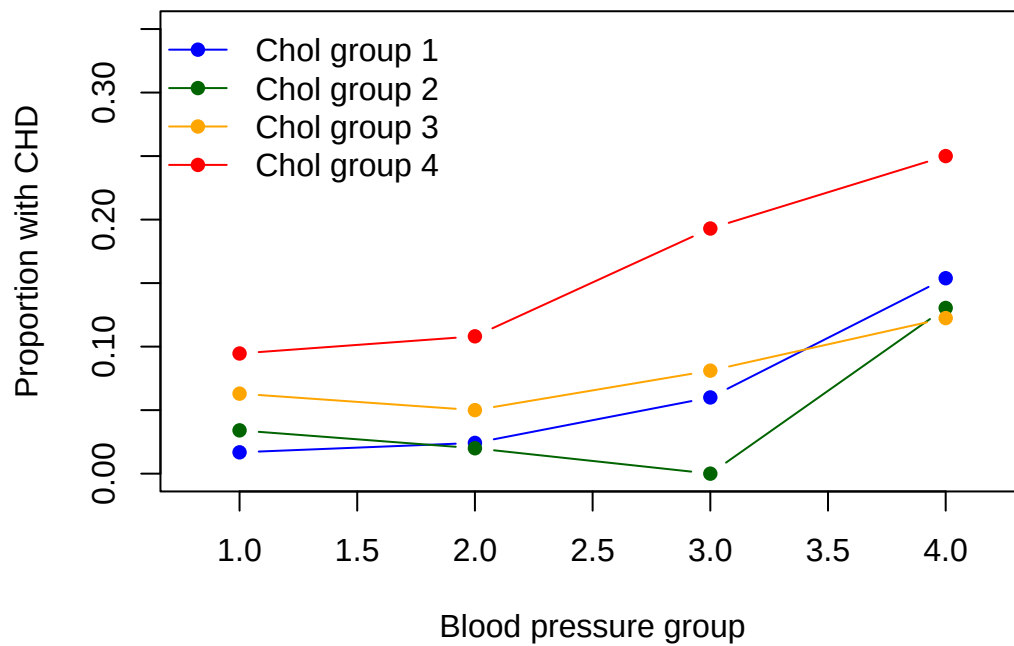


Figure 8.1: Proportion with CHD by cholesterol and blood pressure group. Lines connect groups at the same cholesterol level.

### 8.4.3 Step 3: Fit and compare models

Null deviance: 58.73 on 15 df

Main effects deviance: 8.08 on 9 df

Saturated deviance: 0 on 0 df

### 8.4.4 Step 4: Inference

Test main effects vs null:

[1] 3.48186e-09

Both cholesterol and blood pressure are highly significant.

Test interaction vs main effects:

[1] 0.5264905

The interaction is not significant, so the main-effects model is adequate.

Goodness of fit for main-effects model:

```
[1] 0.5264905
```

The main-effects model fits the data well ( $p > 0.05$ ).

#### 8.4.5 Step 5: Interpret the fitted model

Call:

```
glm(formula = ym ~ cholF + bpF, family = binomial, data = chd)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -3.48194 | 0.34865    | -9.987  | < 2e-16 ***  |
| cholF2      | -0.20798 | 0.46641    | -0.446  | 0.65566      |
| cholF3      | 0.56223  | 0.35080    | 1.603   | 0.10900      |
| cholF4      | 1.34412  | 0.34297    | 3.919   | 8.89e-05 *** |
| bpF2        | -0.04146 | 0.30365    | -0.137  | 0.89139      |
| bpF3        | 0.53236  | 0.33240    | 1.602   | 0.10925      |
| bpF4        | 1.20042  | 0.32689    | 3.672   | 0.00024 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 58.7262 on 15 degrees of freedom  
Residual deviance: 8.0762 on 9 degrees of freedom  
AIC: 73.18

Number of Fisher Scoring iterations: 4

The estimated log-odds ratios for cholesterol and blood pressure groups (relative to group 1 in each case) show a clear increasing trend: higher cholesterol and higher blood pressure each increase the log-odds of CHD.

#### 8.4.6 Step 6: Residual analysis

No strong pattern is apparent, consistent with a well-fitting model.

### 8.5 Looking ahead: limitations of the linear predictor

The GLM framework is powerful, but the linearity constraint  $\eta_i = \mathbf{x}_i^T \beta$  can be restrictive.

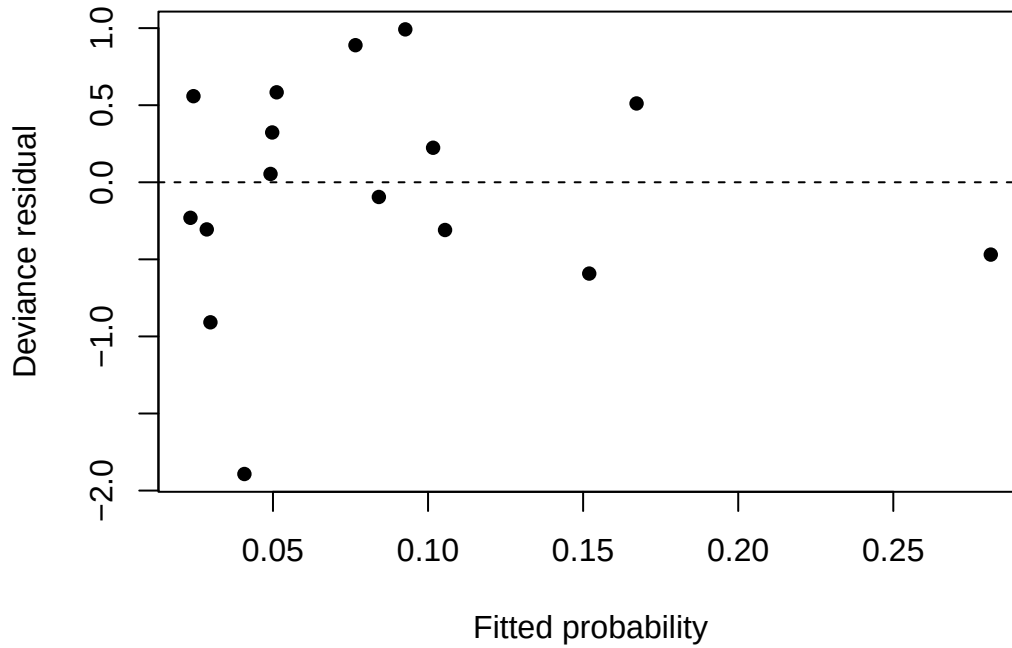


Figure 8.2: Deviance residuals from the main-effects logistic regression model.

Consider a situation where the true relationship between a *continuous* covariate  $x$  and the response is smooth but nonlinear. We have two broad options:

1. **Polynomial regression:** add  $x^2, x^3, \dots$  to the linear predictor. This is easy but can produce unstable fits and the polynomial degree must be chosen in advance.
2. **Non-parametric smoothing:** replace  $\beta x$  with a smooth function  $f(x)$  estimated from the data. This forms the basis of **Generalised Additive Models (GAMs)**, studied in Part II.

The GAM model is written:

$$g(\mu_i) = \alpha + f_1(x_{i1}) + f_2(x_{i2}) + \dots + f_p(x_{ip}),$$

where each  $f_j$  is a smooth function (a *spline*) chosen to fit the data while penalising excessive wiggleness.

We will see that GAMs retain the interpretability of GLMs while gaining considerable flexibility — and that the estimation and inference methods are natural generalisations of what we have developed in Part I.

## 8.6 Summary

### Tip

#### Key points from Part I

- A GLM is defined by its random component (exponential family), systematic component (linear predictor), and link function.
- Maximum likelihood estimation is carried out by IRLS.
- Model fit is assessed via deviance; nested models are compared by likelihood ratio tests ( $\chi^2$  or  $F$ ).
- Logistic regression (binomial/logit) is used for binary or proportion responses.
- Log-linear models (Poisson/log) model count data; product-multinomial models arise when marginal totals are fixed.
- Overdispersion (variance exceeding the theoretical value) can be addressed via quasi-likelihood.

## **Part II**

# **Part II: Generalised Additive Models**

# 9 Non-parametric Modelling

## 9.1 Introduction

In Part I of these notes, we extended simple linear regression to the generalised linear model, which can cope with non-normally distributed response variables, in particular data following Binomial and Poisson distributions. However, we still used only linear functions of the predictor variables. A further extension of the linear model is the *generalised additive model*. Here, we no longer insist that the predictor variables affect the response via a linear function of the predictors, but allow the response to depend on a more general smooth function of each predictor. In Part II of these notes, we study splines and their use in interpolating and smoothing the effects of explanatory variables, culminating in the generalised additive model.

Outline of Part II:

1. Interpolating and smoothing splines.
2. Cross-validation and fitting splines to data.
3. The generalised additive model.

## 9.2 Motivation

Table 9.1 reports on the depth of a coal seam determined by drilling bore holes at regular intervals along a line. The depth  $y$  at location  $x = 6$  is missing: could we estimate it?

Table 9.1: Coal-seam depths (in metres) below the land surface at intervals of 1 km along a linear transect.

|               |     |     |      |      |      |     |    |      |      |      |     |
|---------------|-----|-----|------|------|------|-----|----|------|------|------|-----|
| Location, $x$ | 0   | 1   | 2    | 3    | 4    | 5   | 6  | 7    | 8    | 9    | 10  |
| Depth, $y$    | -90 | -95 | -140 | -120 | -100 | -75 | NA | -130 | -110 | -105 | -50 |

Figure 9.1 plots these data, superimposed with predictions from several polynomial regression models.

Each of these models would predict a different value for the missing observation  $y_6$ . We do not know the accuracy of the depth measurements, so in principle any of these curves could be correct. Clearly, the residual variance is largest for the constant-depth model in Figure 9.1a, and smallest for the cubic polynomial in Figure 9.1d – the residual sums of

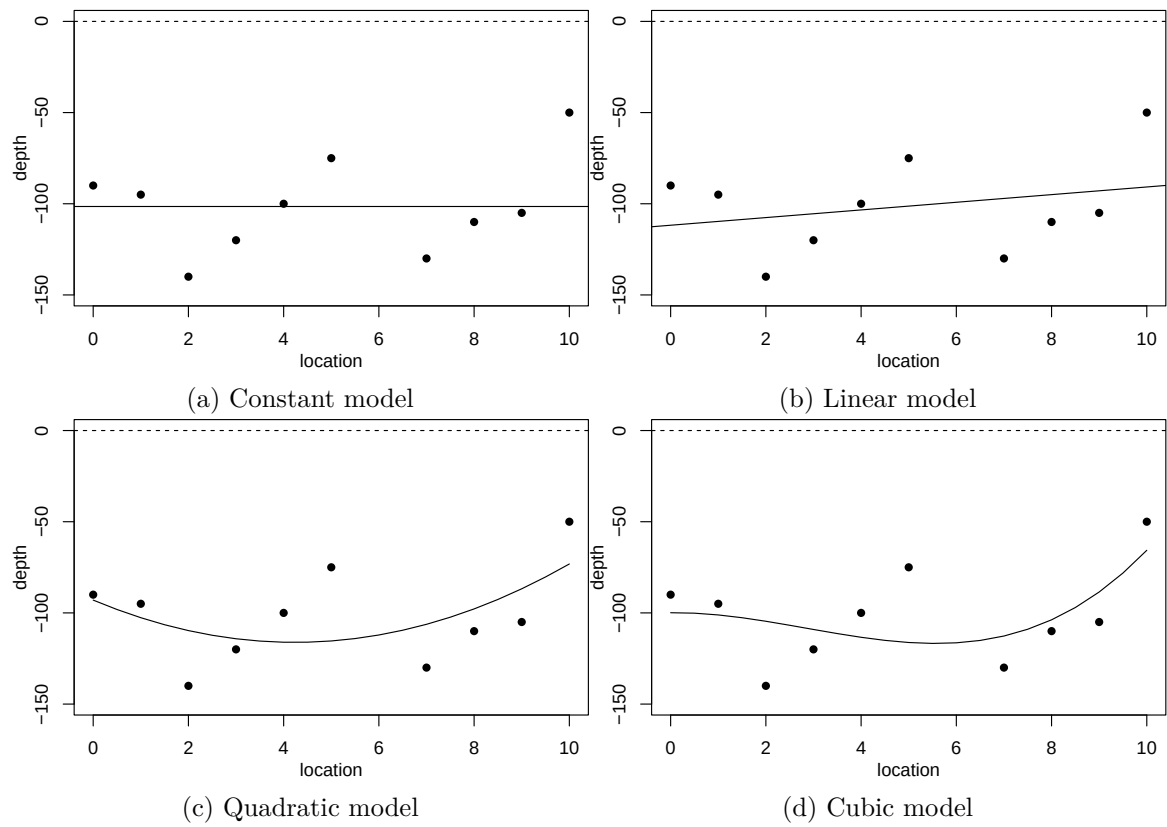


Figure 9.1: The coal-seam data superimposed with predictions from polynomial regression models.

squares are: 6252.5, 5773.05, 4489.84, 4242.89. However, none of these models produces a convincingly good fit. Moreover, these models are not particularly believable, since we know that geological pressures exerted over very long periods of time cause the landscape and its underlying layers of rock to undulate and fracture. This suggests we need a different strategy.

Next, consider the simulated example in Figure 9.2. At first look we might be happy with the fitted curves in Figure 9.2a or Figure 9.2b. The data, however, are created with a *change-point* at  $x = 0.67$  where the relationship changes from linear with slope 0.6 to a constant value of 0.75. This description is completely lost with these two models.

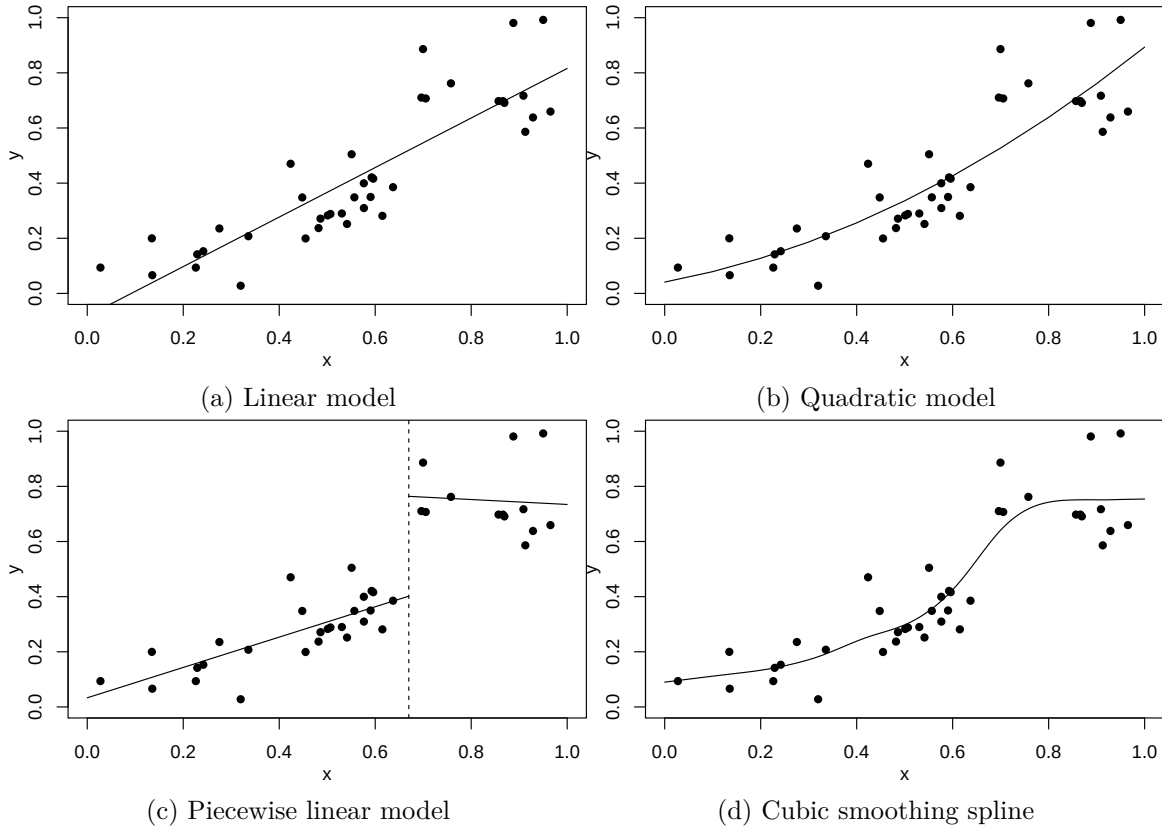


Figure 9.2: Simulated data superimposed with predictions from various models.

Figure 9.2c shows the result of fitting one linear function to the data below 0.67 and a second linear function above. Clearly, this fits well but it has assumed that the change-point location is known – which is unrealistic. Finally, Figure 9.2d shows a fitted *cubic smoothing spline* to the data – we will study these models later. This shows an excellent fit and leads to appropriate conclusions. That is, the relationship is approximately linear for small values, then there is a rapid increase, and finally a near constant value for high values. Of course, this is not exactly as the true relationship with a discontinuity at  $x = 0.67$  but it would definitely suggest something extreme occurs between about 0.6 to 0.7. Full details will follow later, but the cubic spline fits local cubic polynomials which are constrained to create a continuous curve.

Now returning to the coal seam data. Figure 9.3 shows the data again, superimposed with predictions from methods which are not constrained to produce such smooth curves.

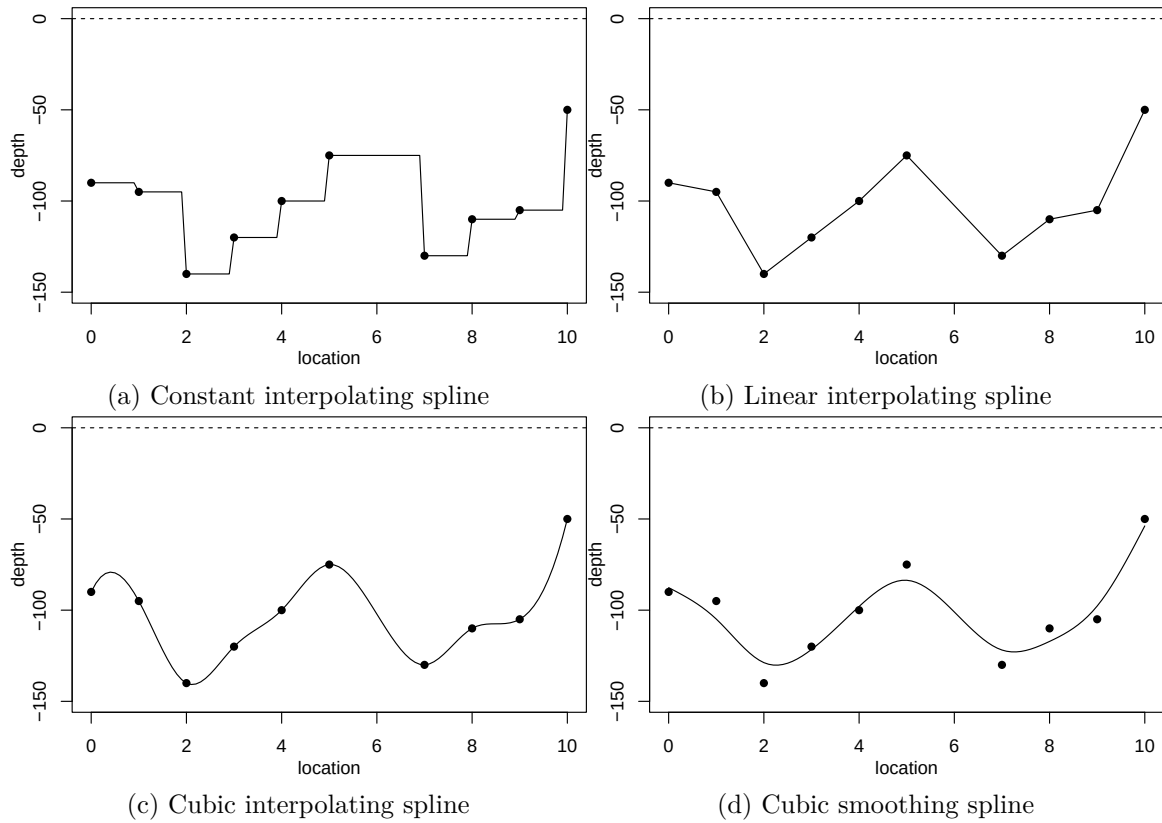


Figure 9.3: The coal-seam data superimposed with predictions from various spline models.

The simplest method, *constant-spline interpolation*, assumes that the dependent variable remains constant between successive observations, with the result shown in Figure 9.3a. However, the discontinuities in this model make it quite unreliable. A better method, whose results are shown in Figure 9.3b, is *linear-spline interpolation*, which fits a straight line between successive observations. Even so, this method produces discontinuities in the *gradient* at each data point. A better method still, shown in Figure 9.3c, is *cubic spline interpolation*, which fits a cubic polynomial between successive data points such that both the gradient and the curvature at each data point is continuous.

A feature of all these interpolation methods is that they fit the data exactly. Is this a good thing? The final method assumes that there may be some measurement error in the observations, which justifies fitting a smoother cubic spline than the cubic interpolating spline, but as we see in Figure 9.3d which does not reproduce the data points exactly. Is this a bad thing? We will see during this module how to construct and evaluate these curves. Here, the results are presented only for motivation.

## Focus on polynomials quiz

Test your knowledge recall and comprehension to reinforce basic ideas about continuity and differentiability.

1. Which of the predicted curves in Figure 9.3 are continuous?
  - (A) None of the models
  - (B) Model (a) only
  - (C) Models (c) and (d) only
  - (D) All models
2. Which of the predicted curves in Figure 9.3 have a continuous first derivative?
  - (A) None of the models
  - (B) Model (b) only
  - (C) Models (c) and (d) only
  - (D) All models
3. Which of the predicted curves in Figure 9.3 have a continuous second derivative?
  - (A) None of the models
  - (B) Model (b) only
  - (C) Models (c) and (d) only
  - (D) All models
4. Which of the predicted curves in Figure 9.3 have a continuous third derivative?
  - (A) None of the models
  - (B) Model (b) only
  - (C) Models (c) and (d) only
  - (D) All models
5. Which of the predicted curves in Figure 9.3 has the highest residual sum of squares?

- (A) None of the models
- (B) Model (a)
- (C) Model (b)
- (D) Model (c)
- (E) Model (d)
- (F) All models

### 9.3 General modelling approaches

We wish to model the dependence of a response variable  $y$  on an explanatory variable  $x$ , where  $y$  and  $x$  are both continuous. We observe  $y_i$  at each time  $x_i$ , for  $i = 1, \dots, n$ , where the observation locations are ordered:  $x_1 < x_2 < \dots < x_n$ . We imagine that the  $y$ 's are noisy versions of a smooth function of  $x$ , say  $f(x)$ . That is,

$$y_i = f(x_i) + \epsilon_i, \quad (9.1)$$

where the  $\{\epsilon_i\}$  are i.i.d:

$$\epsilon_i \sim N(0, \sigma^2). \quad (9.2)$$

We suppose we do not know the correct form of function  $f$ : how can we estimate it?

It is useful to divide modelling approaches into two broad types: parametric and non-parametric.

#### Parametric models

By far the most common parametric model is simple linear regression, for example,  $f(x) = \alpha + \beta x$ , where parameters  $\alpha$  and  $\beta$  are to be estimated. This is, of course, the simplest example of the polynomial model family,  $f(x) = \alpha + \beta x + \gamma x^2 + \dots + \omega x^p$ , where  $p$  is the *order* of the polynomial and where all of  $\alpha, \beta, \gamma, \dots, \omega$  are to be estimated. This has as special cases: quadratic, cubic, quartic, and quintic polynomial models. Also common are exponential models, for example  $f(x) = \alpha e^{-\beta x}$ , where  $\alpha, \beta$  are to be estimated – do not confuse this with the exponential probability density function.

Note that the polynomial models are all linear functions *of the parameters*; this is the parametric family that underpins the GLMs studied in Part I. The exponential model, however, is an example of a model which is non-linear in the parameters – it is an example of a *non-linear regression model*.

Although very many parametric models exist, they are all somewhat inflexible in their description of  $f$ . They cannot accommodate arbitrary fluctuations in  $f(x)$  over  $x$  because they contain only a small number of parameters (degrees-of-freedom).

## Non-parametric models

In such models,  $f$  is assumed to be a smooth function of  $x$ , but otherwise we do not know what  $f$  looks like. A *smooth function*  $f$  is such that  $f(x_i)$  is close to  $f(x_j)$  whenever  $x_i$  is close to  $x_j$ . To characterise and fit  $f$  we will use an approach based on *splines*. In practice, different approaches to characterizing and fitting smooth  $f$  lead to similar fits to the data. The spline approach fits neatly with normal and generalised linear models, but so do other approaches (for example, kernel smoothing and wavelets).

## Piecewise polynomial models

A common problem with low-order polynomials is that they can often fit well for part of the data but have unappealing features elsewhere. For example, although none of the models in Figure 9.1 fit the data at all well, we might imagine that three short linear segments might be a good fit to the coal-seam data. Also, the piecewise linear model was a good description of the data in Figure 9.2c. This suggests that local polynomial models might be useful. In some situation, for example when we know that the function  $f$  is continuous, jumps in the fitted model, as in Figure 9.2c, are unacceptable. Alternatively, we may require differentiability of  $f$ . Such technical issues lead to the use of *splines*, which is introduced in the next chapter.

## 9.4 Exercises

9.1 Consider the first three models fitted in Figure 9.1 and let the data be denoted,  $\{(x_i, y_i) : i = 1, 2, \dots, n\}$ . These three models can be written

- (a)  $y = \alpha + \epsilon$
- (b)  $y = \alpha + \beta x + \epsilon$
- (c)  $y = \alpha + \beta x + \gamma x^2 + \epsilon$

where  $\epsilon$  represents normally distributed random error. Use the principle of least squares, or otherwise, to obtain estimates of the model parameters.

[Click here to see hints.](#)

For each, start by defining the residual sum of squares (RSS),  $RSS = \sum (y_i - \hat{y}_i)^2$  where, in turn (a)  $\hat{y}_i = \alpha$ , (b)  $\hat{y}_i = \alpha + \beta x_i$ , (c)  $\hat{y}_i = \alpha + \beta x_i + \gamma x_i^2$ . Then, find the parameter values which minimise the RSS by (possibly partial) differentiation.

9.2 Discuss possible approaches to fitting an exponential model,  $y = \alpha e^{\beta x}$ , to data. Note that no actual algebraic derivation, nor numerical coded algorithm is expected.

[Click here to see hints.](#)

There is more than one approach. Can least squares be used? Could a simple transformation of the data and the fitted model make solving the problem easier? Is there an algebraic solution? Is there a purely numerical solution?

9.3 In Figure 9.2c, discuss how you might fit a two-part linear model for the case where the change-point is *unknown*. Note that no actual algebraic derivation, nor numerical coded algorithm is expected.

[Click here to see hints.](#)

With many similar problems, imagine breaking the problem down into steps. If you know the location of the change-point then what should you do? Can you then try different possible change-point locations?

9.4 Discuss the four fitted models in Figure 9.3. Can you give positive and negative properties of each model? Which do you think is best and which worst? Do you think which is best/worst depends on the data? Justify your answers.

[Click here to see hints.](#)

Don't get too stuck on the data used here, but think of general issues: ease of use, reliability of the data, is there error with the data or is it very reliable? What if the response is discrete?

# 10 Introducing Splines

## 10.1 Basic definitions

Let  $t_1 < t_2 < \dots < t_m$  be a fixed set of *sites* or *knots* which need not correspond to observation locations, as in Figure 10.1.

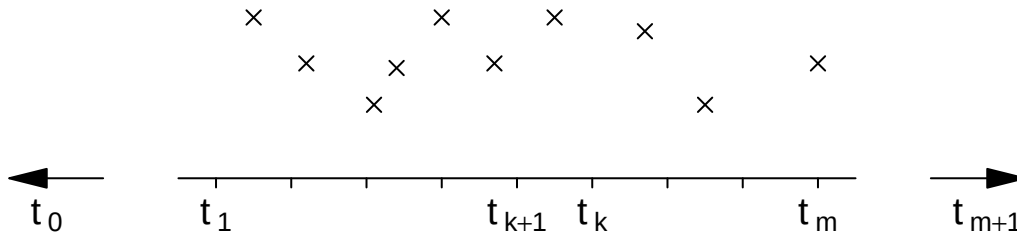


Figure 10.1: Diagram of knots and data points.

Note that we use the symbol  $t$ , rather than  $x$ , so that we do not confuse knots and observation locations.

A spline of order  $p \geq 1$  is a piecewise-polynomial of order  $p$  which is  $(p - 1)$  times differentiable at the knots. Thus there are coefficients  $\{a_{k\ell}, k = 0, \dots, m, \ell = 0, \dots, p\}$  such that

$$f(t) = \sum_{\ell=0}^p a_{k\ell} t^\ell, \quad \text{for } t_k \leq t < t_{k+1}, \quad (10.1)$$

where we take  $t_0 = -\infty$  and  $t_{m+1} = +\infty$ .

If we are using cubic polynomials, ( $p = 3$ ), then  $f$  is given by the following equations:

$$f(t) = a_{00} + a_{01}t + a_{02}t^2 + a_{03}t^3, \quad t_0 \leq t < t_1$$

to the left of the first knot,

$$f(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3, \quad t_1 \leq t < t_2$$

between the first and second knots, and so on until

$$f(t) = a_{m0} + a_{m1}t + a_{m2}t^2 + a_{m3}t^3, \quad t_m \leq t < t_{m+1}$$

to the right of the final knot. This is illustrated in Figure 10.2a with  $m = 2$ .

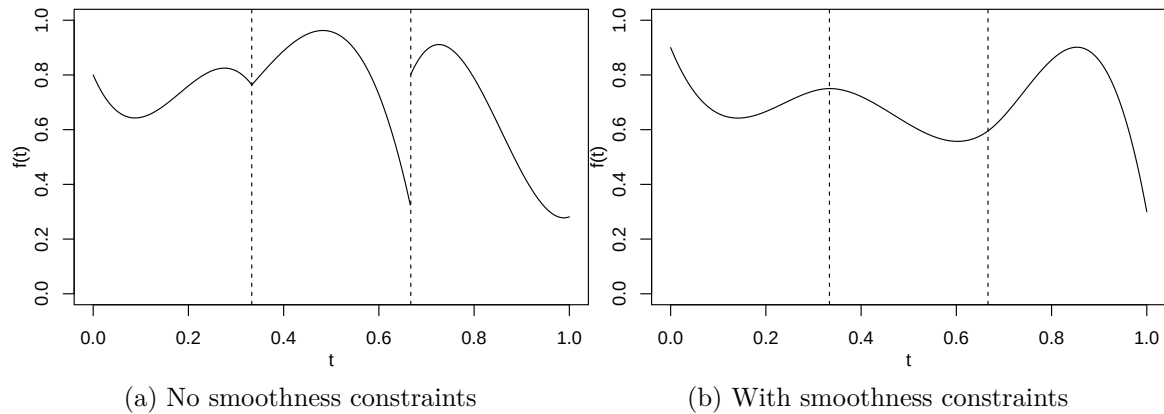


Figure 10.2: Piecewise-cubic functions in three intervals with knot positions indicated with vertical lines.

## Focus on splines quiz

Test your knowledge recall and comprehension to reinforce basic ideas about splines.

- Which of the following best describes the relationship between knots and data locations.
  - (A) Knots are the response and data locations are the explanatory variable
  - (B) There must be fewer knots than data locations
  - (C) Knots and data locations are both marked on the x-axis
  - (D) Knots and data points are exactly the same
- How many polynomial equations are needed to define a cubic spline with four knots?
  - (A) 3
  - (B) 4
  - (C) 5
  - (D) 12
  - (E) None of the above
- What is the highest power possible in a cubic spline?
  - (A) 0

- (B) 1
  - (C) 2
  - (D) 3
  - (E) None of the above
4. How many times can a spline of order 5 be differentiated at the knots?
- (A) Cannot be differentiated
  - (B) Twice differentiable
  - (C) Four times differentiable
  - (D) Infinitely differentiable
5. Which of the following is NOT a property of cubic splines?
- (A) Piecewise cubic polynomial
  - (B) Twice differentiable
  - (C) May contain change-points
  - (D) Continuous at the knots

## 10.2 Imposing smoothness

Because of the use of polynomials,  $f$  is smooth *between* each successive pair of knots. At the knots, however,  $f$  might not be continuous and it might not be differentiable – in such cases we would say that the function is not smooth.

To ensure that  $f$  is also smooth at each of the knots, we impose smoothness constraints which control continuity of the function and its derivatives at the knots.

Let  $f^{(\ell)}$  be the  $\ell$ -th order derivative, with  $f^{(0)} = f$  being the function itself,  $f^{(1)} = f'$  is the first derivative and  $f^{(2)} = f''$  the second derivative. Further, let  $f^{(\ell)}(t - \epsilon)$  and  $f^{(\ell)}(t + \epsilon)$ , for  $\epsilon \geq 0$ , denote evaluation of the function or its derivative at points just below and just above  $t$  – we will be interested in their relative values as  $\epsilon \rightarrow 0$ .

To impose smoothness, we require that

$$\lim_{\epsilon \rightarrow 0} f^{(\ell)}(t_k - \epsilon) = \lim_{\epsilon \rightarrow 0} f^{(\ell)}(t_k + \epsilon), \quad (10.2)$$

for all  $k = 1, \dots, m$  and for  $\ell = 0, \dots, (p - 1)$ .

In other words we say that  $f$  is smooth if the limits, from below and from above, of the function and its  $(p - 1)$  derivatives exist and are equal.

The meaning of these smoothness constraints is illustrated in Figure 10.2. In Figure 10.2a, a piecewise cubic function with two knots has been plotted. The first derivative,  $f'$ , is discontinuous at the first knot and the function itself,  $f$ , is discontinuous at the second knot. Figure 10.2b shows a similar shaped cubic spline with two knots. This time, the function  $f$  and its first two derivatives are continuous at both knots.

The smoothness conditions in Equation 10.2 induce constraints on the coefficients  $\{a_{k\ell}\}$ . A polynomial of order  $p$  has  $p + 1$  coefficients, and there are  $m + 1$  intervals when we have  $m$  knots. This leads to  $(p + 1) \times (m + 1)$  coefficients but there are  $p$  constraints at each of the  $m$  knots. Thus the total *degrees of freedom* of the system is

$$\text{df}_{\text{spline}} = (p + 1)(m + 1) - pm = m + p + 1. \quad (10.3)$$

These degrees of freedom provide the necessary flexibility in the spline.

Note that  $f$  is infinitely differentiable everywhere, except at the knots where it is  $p - 1$  times differentiable. In particular, for  $p = 1$ ,  $f$  is a linear spline comprising linear pieces constrained to be continuous at the knots, although the slope of  $f$  is discontinuous at the knots. Also, for  $p = 3$ ,  $f$  is a cubic spline comprising cubic polynomial pieces continuous at the knots; where the first and second derivatives of  $f$  are also continuous, but the third derivative is discontinuous at the knots.

## Focus on smoothness quiz

Test your knowledge recall and comprehension to reinforce basic ideas about smoothness.

1. Which of the following best describes the motivation for using smooth fitted models?
  - (A) Makes model fitting easier
  - (B) Calculations are easy to do in R
  - (C) It produces nice graphs
  - (D) Reduces the effect of measurement error
  - (E) None of the above
2. Which of the following best describes the motivation for using piecewise polynomial components?
  - (A) Can involve change-points

- (B) Well understood and easy to use
  - (C) Can model jumps well
  - (D) They lead to normally distributed errors
  - (E) None of the above
3. Is the following a true statement? ‘The higher the order of the polynomial components the smoother the spline’ TRUE / FALSE
4. If a model fitted to a particular data set has zero degrees of freedom, then which of the following statements about the solution is most likely to be true?
- (A) Does not fit the data well
  - (B) No solution
  - (C) Unique solution
  - (D) Multiple solutions
5. Which of the following best describes spline modeling?
- (A) The only non-parametric method available
  - (B) It is a parametric approach
  - (C) Requires high-level coding
  - (D) A flexible non-parametric approach

## 10.3 Exercises

10.1 Why is it not sensible to define a smooth function made up of constant components? Similarly, why is it not sensible to create a differentiable function from linear splines?

[Click here to see hints.](#)

For each case, think about the number of parameters for each component and the implications of any constraints.

10.2 In the situation illustrated in Figure 10.2b, where  $p = 3$  and  $m = 2$ , clearly identify the  $(p + 1) \times (m + 1) = 12$  model parameters and the  $pm = 6$  smoothness constraints in terms of the cubic polynomials and their derivatives.

[Click here to see hints.](#)

Define a cubic polynomial for each interval and consider continuity and differentiability.

10.3 Further consider the situation illustrated in Figure 10.2b. Suppose now that we require the splines to pass through specified coordinates  $(t_1, f(t_1))$  and  $(t_2, f(t_2))$ . What is the degrees of freedom for this model? How many such cubic splines would satisfy these constraints? Discuss potential additional constraints which would lead to a unique fitted model. Do you think having a unique solution is a positive or negative property?

[Click here to see hints.](#)

Think about the degrees of freedom, that is the total number of parameters and the number of constraints, including forcing the spine to pass through two points. Think about the implications of having zero and non-zero degrees of freedom. There are very many (infinitely many?) potential additional constraints, but suggest one or two which sound a good idea.

10.4 For a general problem, what would be the effect of requiring additional constraints of the form of Equation 10.2 but with  $\ell = p$ ? Would this lead to an acceptable fitted cubic spline model? Justify your answer.

[Click here to see hints.](#)

Think about the implication of this on the curvature of neighbouring components, and hence on overall curvature.

# 11 Interpolating Splines

## 11.1 Overview

Chapter 9 considered general limitations of parametric models, and polynomial regression in particular (see Figure 9.1), which motivated the use of the more flexible spline models (see Figure 9.3) – though at that stage no mathematical details were presented. In Chapter 10, basic spline definitions were given, including the notation of smoothness constraints, and these ideas were further explored in the Exercises. This chapter will now give mathematical details of the interpolating spline problem and consider application to data. A feature of all these interpolation methods is that they fit the data exactly and that the fitted functions are smooth. Figure 11.1 shows *cubic spline interpolation*, which fits a cubic polynomial between successive data points such that the function, gradient and the curvature are all continuous at each data point. The solid line shows the fitted values within the range of the data, whereas the dashed line shows the fitted values outside the range of the data – *extrapolation*.

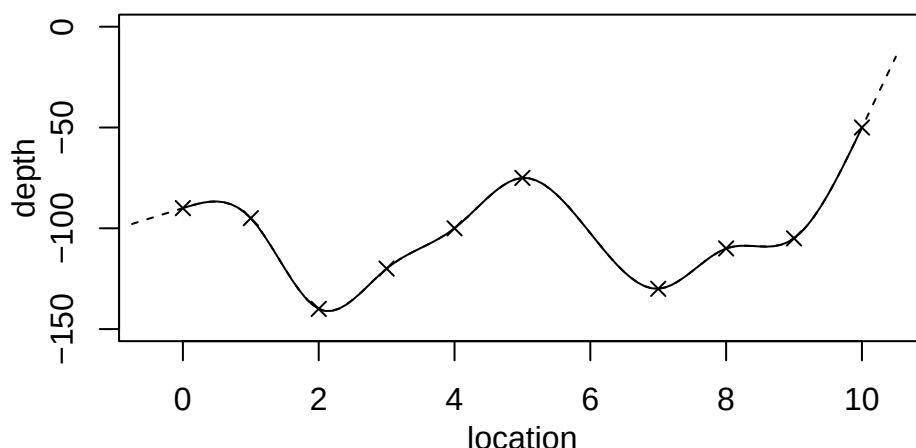


Figure 11.1: A cubic interpolating spline fitted to the coal-seam data, with the dashed line showing extrapolation.

## 11.2 Natural splines

Suppose we have  $n$  observations  $\{y_1, \dots, y_m\}$  at locations  $\{t_1, \dots, t_m\}$ . We can construct a cubic spline (that is with  $p = 3$ ) to pass through (interpolate) all the points  $(t_i, y_i)$ ,  $i = 1, \dots, m$ . In fact, for any given set of points, there is an infinite number of cubic splines

which interpolate them, see Figure 11.2 for examples. Exactly one of these splines has the property that, in the leftmost and rightmost intervals, it is a straight line. Such a spline is called a *natural* cubic spline.

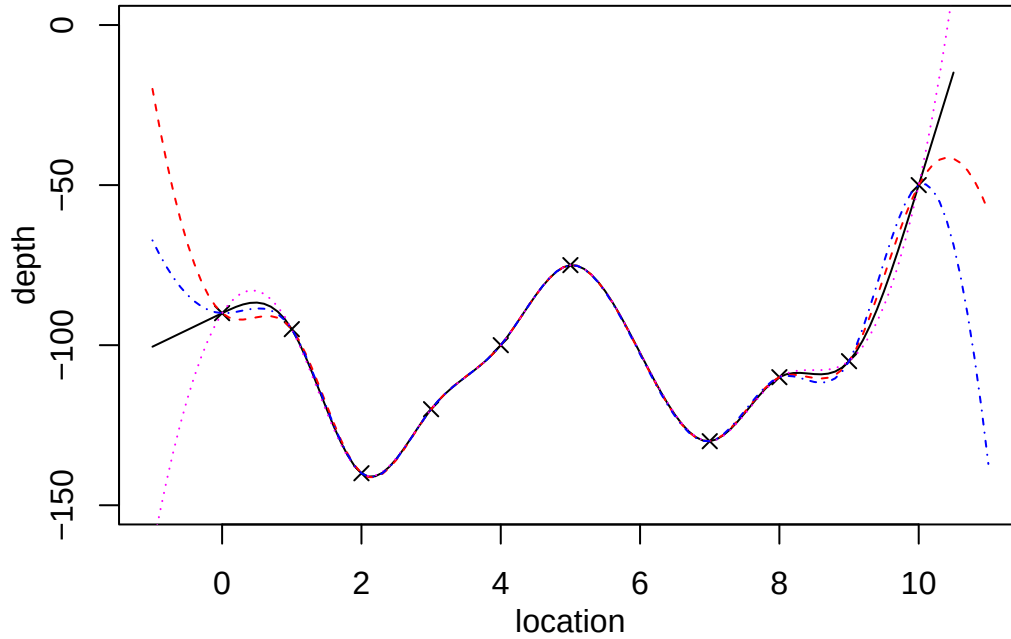


Figure 11.2: Cubic interpolating splines fitted to the coal-seam data, with the dashed lines showing extrapolation – the natural spline is shown in solid black.

Note that natural splines are not the only choice for the spline method – see the R help page for other options – with perhaps the most useful other being `periodic` which can be considered if we expect the unknown function to also be periodic. Figure 11.3 shows the fitted natural and periodic spline fitted to a simulated *cosine* function. The only noticeable difference is outside the range of the data, that is for extrapolation. Great care should be used, however, as imposing such additional restrictions on the fitted function can lead to unforeseen modelling errors when we do not observe the full range of  $\mathbf{x}$  values.

### 11.3 Properties of natural splines

*Natural* splines are a special case of polynomial splines of *odd* order  $p$ . Thus we have natural linear splines ( $p = 1$ ), natural cubic splines ( $p = 3$ ), etc. A spline is said to be *natural* if, beyond the boundary knots  $t_1$  and  $t_m$ , its  $(p + 1)/2$  higher-order derivatives are zero:

$$f^{(j)}(t) = 0, \quad (11.1)$$

for  $j = (p + 1)/2, \dots, p$  and either  $t \leq t_1$  or  $t \geq t_m$ .

Thus a natural spline of order  $p$  has the following  $p + 1$  constraints, in addition to those of Equation 10.2 :

$$\lim_{\epsilon \rightarrow 0} f^{(\ell)}(t_1 + \epsilon) = \lim_{\epsilon \rightarrow 0} f^{(\ell)}(t_m - \epsilon) = 0, \quad (11.2)$$

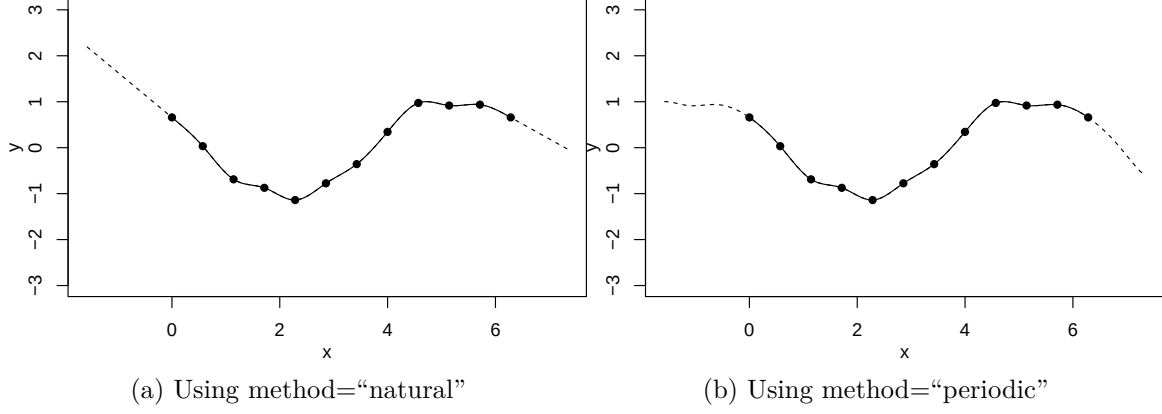


Figure 11.3: Example of different interpolating splines.

for  $\ell = (p + 1)/2, \dots, p$ .

In particular,

- a natural *linear* spline has  $p + 1 = 2$  additional constraints:

$$\lim_{\epsilon \rightarrow 0} f^{(1)}(t_1 + \epsilon) = \lim_{\epsilon \rightarrow 0} f^{(1)}(t_m - \epsilon) = 0, \quad (11.3)$$

implying that  $f(t)$  is constant in the outer intervals of a natural linear spline,

- a natural *cubic* spline has  $p + 1 = 4$  additional constraints:

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} f^{(2)}(t_1 + \epsilon) &= \lim_{\epsilon \rightarrow 0} f^{(2)}(t_m - \epsilon) = 0, \\ \lim_{\epsilon \rightarrow 0} f^{(3)}(t_1 + \epsilon) &= \lim_{\epsilon \rightarrow 0} f^{(3)}(t_m - \epsilon) = 0, \end{aligned} \quad (11.4)$$

implying that  $f(t)$  is linear in the outer intervals of a natural cubic spline.

The total degrees of freedom of a natural spline is, starting from Equation 10.3, but taking into account the additional  $p + 1$  constraints is

$$\text{df}_{\text{nat.spline}} = m + p + 1 - (p + 1) = m. \quad (11.5)$$

That is the degrees of freedom for natural splines equals  $m$  whatever the value of  $p$ .

**Proposition 11.1.** *Linear and cubic natural splines have the following representations:*

- *Linear natural splines:*

$$f(t) = a_0 + \sum_{i=1}^m b_i |t - t_i|; \quad \sum_{i=1}^m b_i = 0 \quad (11.6)$$

- *Cubic natural splines:*

$$f(t) = a_0 + a_1 t + \sum_{i=1}^m b_i |t - t_i|^3; \quad \sum_{i=1}^m b_i = \sum_{i=1}^m b_i t_i = 0. \quad (11.7)$$

**Proof:** See the supplementary notes `Supplementary/proof_natural_spline_optimality.pdf`.

## Focus on natural splines quiz

Test your knowledge recall and comprehension to reinforce basic ideas about natural splines.

1. Which of the following is a key property of every spline?
  - (A) Has jump discontinuities
  - (B) Is a smooth non-parametric function
  - (C) Is an example of a parametric model
  - (D) Always interpolate data
  - (E) Always extrapolate data
2. Which of the following is NOT a key property of a cubic spline function?
  - (A) Is always continuous
  - (B) Is always infinitely differentiable
  - (C) Is a smooth function
  - (D) Is a piecewise polynomial
  - (E) Has extra constraints at knots
3. Which of the following statements is true?
  - (A) Natural splines are a special case of polynomial splines
  - (B) Polynomial splines are a special case of natural splines
  - (C) Polynomial splines are identical to natural splines
  - (D) Natural splines have fewer constraints than polynomial splines
  - (E) Natural splines are smoother than polynomial splines
4. Which of the following statements is true?
  - (A) Natural splines have fewer constraints in the outer intervals
  - (B) Natural splines have additional constraints in the internal intervals

- (C) Natural splines have additional constraints in the outer intervals
  - (D) Natural splines are not smooth functions
  - (E) Natural splines have fewer constraints in the internal intervals
5. What determines the degrees of freedom of a natural spline?
- (A) Both the number of knots and the polynomial order
  - (B) The sample size
  - (C) The number of knots
  - (D) Both the number of knots and the sample size
  - (E) The order of the polynomial used

## 11.4 Roughness penalties

An aim of spline models is to describe an unknown function using piecewise-polynomials which are smooth. In the previous section, smoothness was imposed by explicitly constraining specified high-order derivatives. An alternative approach is to measure and control the degree of smoothness of the splines. In practice the *roughness* of the spline is usually measured and one definition of roughness is:

$$J_\nu(f) = \int_{-\infty}^{\infty} [f^{(\nu)}(t)]^2 dt \quad (11.8)$$

where  $\nu \geq 1$  is an integer and  $f^{(\nu)}$  denotes the  $\nu$ th derivative of  $f$ . Thus  $f^{(1)}(t)$  denotes the first derivative and  $f^{(2)}(t)$  denotes the second derivative of  $f$ .

Intuitively, roughness measures the “wiggleness” of a function.

An alternative approach to that in previous sections is to find the function  $f$  which minimizes Equation 11.8 and satisfies  $f(t_i) = y_i$  for  $i = 1, \dots, m$ . We refer to the solutions of this problem as the *optimal interpolating function*.

It turns out that there is a very close link between  $J_\nu(\cdot)$  and  $p$ th-order natural splines, where  $p = 2\nu - 1$  (so  $p$  is odd). Important special cases are:  $\nu = 1$  and  $p = 1$ , and  $\nu = 2$  and  $p = 3$ . This relationship is defined in the following proposition.

**Proposition 11.2.** *The optimal interpolating function is a  $p$ th-order natural spline, where  $p = 2\nu - 1$ . That is, the natural spline  $f$  is the unique minimizer of  $J_\nu(f)$ .*

**Proof:** See the supplementary notes `Supplementary/proof_natural_spline_optimality.pdf`.

**Comments**

- Linear and cubic interpolating splines are also of interest in numerical analysis, for example to interpolate tables of numbers.
- The linear interpolating spline is simply the piecewise-linear path connecting the data points.
- Of course, in the linear spline case, knot points are clearly visible as kinks in the interpolating function. But, in the cubic spline case, knot points are invisible to the naked eye. Hence, in general, there is little motivation to use higher-order splines.
- Numerical considerations: the interpolating spline solutions involve matrix inversion. The inversion of an  $n \times n$  matrix involves  $O(n^3)$  operations – hence it is time consuming if  $n$  is large (for example,  $n = 1000$  or  $10000$ ). Fortunately there are tricks to reduce the computation to  $O(n)$ .

## 11.5 Fitting interpolating splines in R

There are two main functions within **R** for fitting interpolating splines to data: **spline** which outputs fitted values for specified points, or **splinefun** which returns an **R** *function* which can be used directly by other commands, such as **curve**. The following illustrates the two approaches.

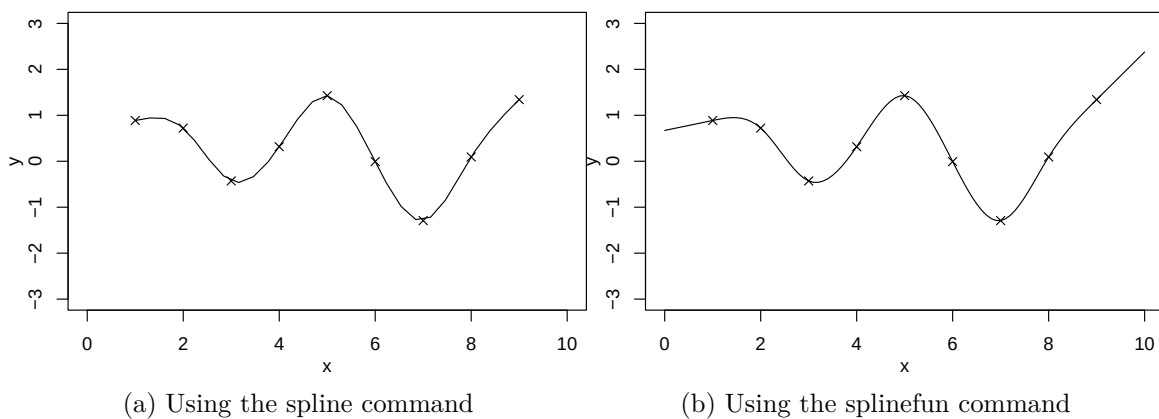


Figure 11.4: R code for cubic interpolating splines.

The following illustrates the different ways to draw the spline and to calculate fitted values.

```
$x
[1] 2.5 7.5

$y
[1] 0.08785792 -0.78655273

[1] 0.08785792 -0.78655273
```

Before finishing, let us consider the derivatives of the fitted spline function. Figure 11.5 shows (a) the fitted natural spline, along with its first three derivatives in (b)-(d). Note that the function and the first two derivatives are continuous everywhere, but that the third derivative is not continuous and has jumps at the knot locations. Note also that the first derivative, and higher derivatives, are all constant outside the range of the interior knots.

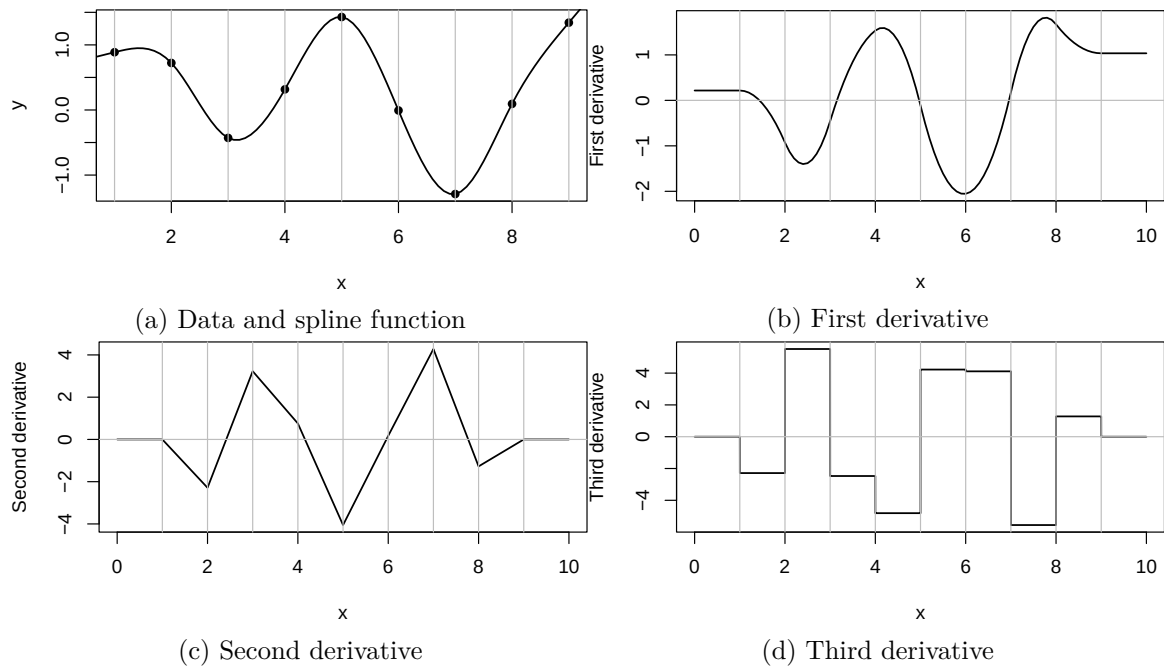


Figure 11.5: Derivatives of cubic interpolating splines.

## Focus on fitting splines quiz

Test your knowledge recall and comprehension to reinforce basic ideas about R commands for splines.

1. Which of the following should be first in every data analysis?
  - (A) Fit a spline model
  - (B) Calculate the correlation
  - (C) Add the fitted curve to a plot of the data
  - (D) Fit a linear model

- (E) Draw a scatter plot
2. Which of the following is NOT part of a data analysis using interpolating splines?
- (A) Read the data description
  - (B) Examine model residuals
  - (C) Add the fitted curve to a plot of the data
  - (D) Fit a spline model
  - (E) Draw a scatter plot of the data
3. What is the main difference between the R commands `spline` and `splinefun`?
- (A) `splinefun` produces a function
  - (B) `spline` interpolates whereas `splinefun` smooths
  - (C) `spline` is more boring
  - (D) Only `spline` has knots
  - (E) They are identical, only the name is different
4. Which of the following is NOT an optional method for the R command `spline`?
- (A) `fm`
  - (B) `periodic`
  - (C) `natural`
  - (D) `hyman`
  - (E) `smooth`
5. When using the output from `splinefun`, which of the following CANNOT be plotted?
- (A) Fourth derivative
  - (B) Function value
  - (C) Second derivative
  - (D) First derivative

- (E) Third derivative

## 11.6 Exercises

11.1 For the situation shown in Figure 10.2, but taking  $p = 1$ , write down the linear functions for the three intervals and clearly identify all the 6 model parameters. Next, write down the constraints required to make the functions pass through the  $m = 2$  data points, and the two constraints which impose continuity of function. What additional constraints are needed to fix the first derivative at zero for the outer two intervals?

[Click here to see hints.](#)

You might get two redundant constraints but think carefully about which can be removed.

11.2 Continuing the problem described in Exercise 11.1, write the constraints as a system of 6 linear equations in the 6 unknown model parameters. How might you solve this system to give the parameter values which solve the interpolation problem?

[Click here to see hints.](#)

You need to write the six equations as a set of simultaneous equations. A simple matrix inversion is all that is needed for the solution – but don't try to actually do the inversion!

11.3 Continuing the linear system described in Exercise 11.2, create a synthetic problem by choosing two data response values. Then solve the system in **R**, or otherwise, and plot the fitted spline interpolating function.

[Click here to see hints.](#)

The choice of the two data point locations (x-values) and function value (y-value) is completely your choice. Then, define the two vectors and the matrix defined in the previous question. R has a function “solve” which will invert the design matrix – look at “help(solve)”

11.4 Again, considering the situation shown in Figure 10.2, but taking  $p = 1$ . Using the alternative representation in Equation 11.6, write down two constraints involving the data points and the additional constraint on the  $b_i$  parameters. Write this linear system of 3 equations in three unknowns in matrix form.

[Click here to see hints.](#)

The first part only requires checking the definition of the alternative form in the notes. Then, write it in vector/matrix form.

11.5 Continuing the linear system described in Exercise 11.4, using the same points created in Exercise 11.3, calculate the parameter values in this new parameterization. Check that your two fitted interpolating splines give the same answers. Which approach do you prefer? Justify your answer.

[Click here to see hints.](#)

Again, in R, define the vectors and matrix, and use “solve” to find the parameter estimates. You can check for equality by any approach, for example just plot both solutions and see if they match. The choice of which is preferable is yours, no wrong answer, but it’s the justification that matters.

11.6 Create your own version of the R code used to produce Figure 11.4 and experiment with the two alternative spline fitting commands. Remove the `set.seed(15342)` command so that you produce different data each time and comment on the similarities and differences when using different data sets.

[Click here to see hints.](#)

Simply copy the code from the referred to figure and remove the `set.seed` command. Then, each time you run the code you will get a different answer.

11.7 Let

$$f(t) = 3 + 2t + 4|t|^3 + |t - 1|^3.$$

Write  $f$  as a cubic polynomial in each of the intervals  $(-\infty, 0)$ ,  $(0, 1)$  and  $(1, \infty)$ . Verify that  $f$  and its first two derivatives are continuous at the knots.

Is  $f$  a spline? Is  $f$  a natural spline?

[Click here to see hints.](#)

To show this is a spline, you need to re-write the equations as polynomials. That is consider what happens to the absolute values in each interval. Then, substitute in the knot locations (0 and 1) into the equations for function, derivative and second derivative and make sure all matches. There is also a solution based on the equation of the alternative form – see if you can argue it this way also.

11.8 Let

$$f(t) = 3 + |t| - |t - 2|.$$

Show by direct calculation that

$$\int_{-\infty}^{\infty} \{f'(t)\}^2 dt = 8.$$

Show that this integral can also be written in the form  $-2\mathbf{b}^T K \mathbf{b}$ , where you should define  $\mathbf{b}$  and  $K$ .

[Click here to see hints.](#)

Use a similar approach as in the previous exercise to “remove” the absolute value. You should then see that the integral is trivial. For the matrix form, go back to the definitions of each part. Multiplying out will then give the same answer as the integral.

# 12 Smoothing Splines

## 12.1 Overview

We observe  $y_i$  at each location  $x_i$ , for  $i = 1, \dots, n$ . We imagine that the  $y$ 's are noisy versions of a smooth function, say  $f(\cdot)$ , where the errors follow a normal distribution with constant variance. That is

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

for  $i = 1, \dots, n$ , where  $f$  is smooth, the  $\epsilon_i$  are i.i.d., and  $f$  and  $\sigma^2$  are unknown. The log-likelihood for this situation is:

$$l(f; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f(x_i))^2 - n \log \sigma \quad (12.1)$$

and we wish to estimate  $f$  for a given data set  $\mathbf{y} = \{y_1, \dots, y_n\}$ . With no constraints on  $f$ , the log-likelihood would be maximised by setting  $f(x_i) = y_i$  for all  $i$ , and we would estimate the noise variance as  $\hat{\sigma}^2 = 0$ . This takes no account of randomness in the data and  $f$  would in general need to be quite wiggly to achieve this fit.

Figure 12.1a shows such an interpolation of noisy data. This would be of little use for explanation, interpolation or prediction.

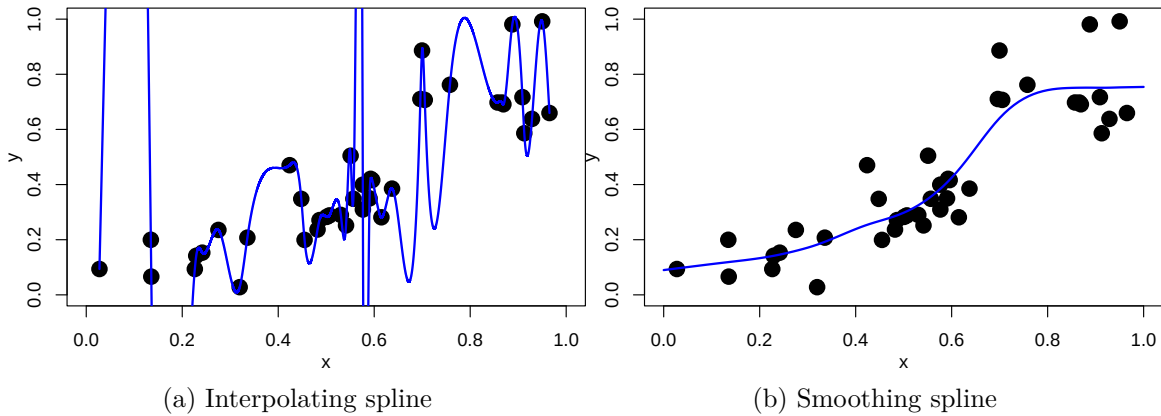


Figure 12.1: Comparison of interpolating and smoothing methods applied to a noisy data set.

We do not expect, or even want, the fitted function  $f$  to pass exactly through the data points  $\{(x_i, y_i), i = 1, \dots, n\}$ , but merely to lie close to them. We would rather trade-off goodness-of-fit against smoothness of  $f$ . Figure 12.1b shows a smoothing spline fit to the

same data. This is much better as there is a clear explanation of the relationship, it could be used reasonably well for interpolation and prediction.

## 12.2 The penalised least-squares criterion

Noting that maximizing Equation 12.1 is equivalent to minimizing the residual sum of squares  $\sum_{i=1}^n (y_i - f(x_i))^2$ , we can achieve this trade-off by minimizing a *penalised* sum of squared residuals:

$$R_\nu(f, \lambda) = \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda J_\nu(f), \quad (12.2)$$

where  $J_\nu(f)$ , as first defined in Equation 11.8, penalizes the roughness of  $f$ . The *smoothing parameter*  $\lambda \geq 0$  controls the severity of this penalty. For now we will assume  $\lambda$ , which absorbs the  $\sigma^2$  in Equation 12.1, is known.

Figure 12.2 shows example smoothing spline fits using a range of smoothing parameters,  $\lambda$ . In Figure 12.2a the fit is essentially a straight line; perhaps Figure 12.2b and Figure 12.2c show acceptable fits. Figure 12.2d, with a very small  $\lambda$  value, is close to an interpolating spline fit and is clearly unacceptable.

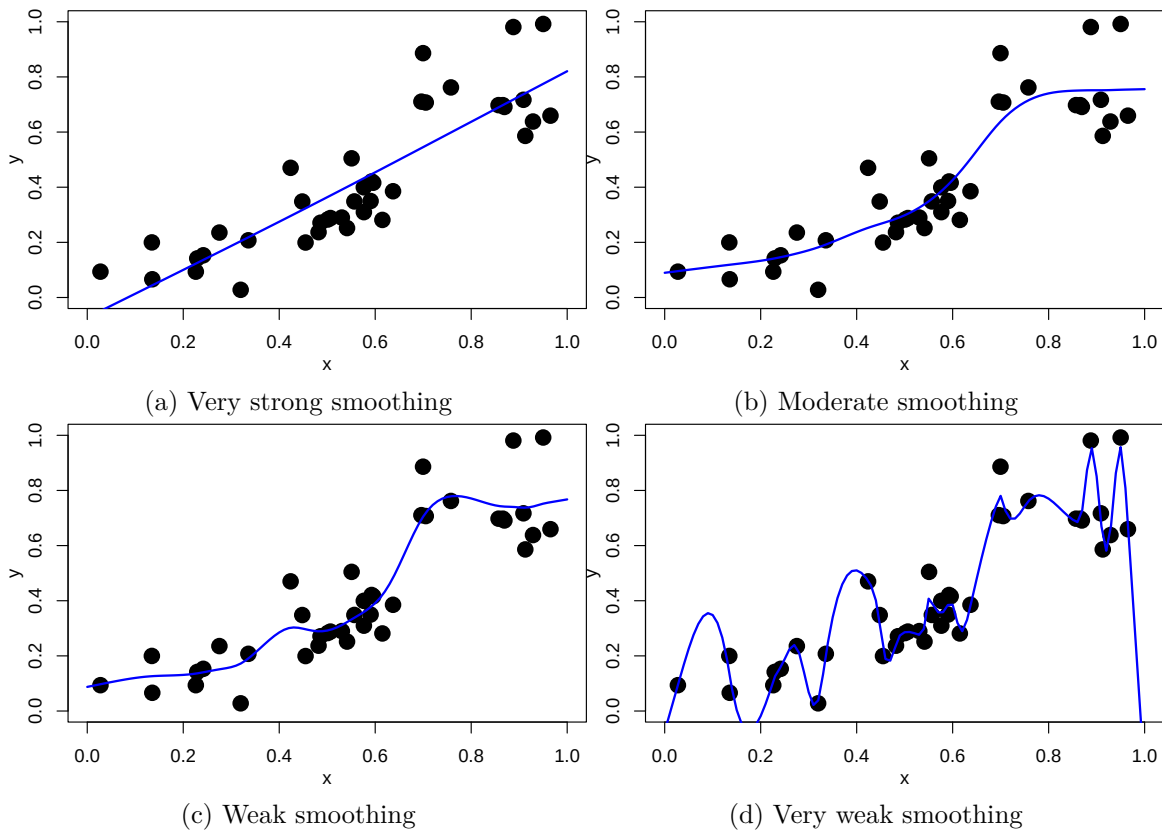


Figure 12.2: Smoothing spline fits using a range of smoothing parameters.

As the smoothing parameter  $\lambda$  increases, the optimal  $f$  becomes smoother. In particular, it can be shown that as  $\lambda \rightarrow \infty$ , the vector of coefficients  $\mathbf{b} \rightarrow \mathbf{0}$ , and  $\mathbf{a}$  tends to the usual least-squares estimate of the regression parameters. Thus, the smoothing spline converges to the sample mean  $f(x) = \bar{y}$  when  $\nu = 1$ , and to the ordinary least squares fitted line,  $f(x) = \hat{\alpha} + \hat{\beta}x$ , when  $\nu = 2$ . In the other direction, as  $\lambda \rightarrow 0$  the smoothing solution converges to the interpolating spline.

## Focus on smoothing splines quiz

Test your knowledge recall and comprehension to reinforce basic ideas about smoothing splines.

1. Which of the following statements is NOT true?
  - (A) The smoothing spline parameter  $\lambda$  controls the amount of smoothing
  - (B) Interpolating splines pass through all data points
  - (C) Smoothing splines should be used for all problems
  - (D) Smoothing splines are better for noisy data
  - (E) Both interpolating and smoothing splines are piecewise polynomial functions
2. Which of the following statements is true about splines?
  - (A) The roughness measures the wiggleness of a function
  - (B) The roughness can be integrated to give the smoothness
  - (C) The roughness defines the likelihood function
  - (D) The roughness is a measure of goodness of fit
  - (E) The roughness measures the lack of fit
3. Which of the following best describes the result of using a very large value of  $\lambda$  in a cubic smoothing spline?
  - (A) The spline is approximately quadratic
  - (B) The spline is approximately constant
  - (C) The spline is approximately linear

- (D) The spline is approximately cubic
  - (E) The spline interpolates the data
4. Which of the following best describes the result of using a very small value of  $\lambda$  in a linear smoothing spline?
- (A) The spline is approximately quadratic
  - (B) The spline is approximately cubic
  - (C) The spline is approximately linear
  - (D) The spline is approximately constant
  - (E) The spline interpolates the data
5. Which of the following statements best describes how to choose the value of  $\lambda$ ?
- (A) To balance closeness to data and roughness
  - (B) To minimise the roughness
  - (C) To maximise the likelihood
  - (D) To maximise the roughness
  - (E) To better fit the data with a high roughness

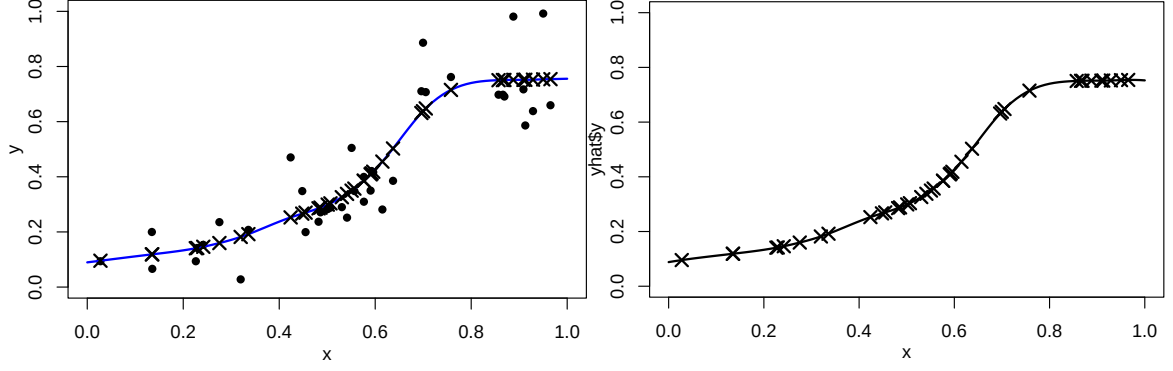
## 12.3 Relation to interpolating splines

We show in the following proposition that the function  $\hat{f}$  which minimizes Equation 12.2 is the interpolating spline of its fitted values.

**Proposition 12.1.** *Suppose  $\hat{f}$  minimizes  $R_\nu(f, \lambda)$  and let  $\hat{y}_i = \hat{f}(x_i), i = 1, \dots, n$  denote the corresponding fitted values. Then,  $\hat{f}$  solves the interpolation problem for the artificial data set  $(x_i, \hat{y}_i), i = 1, \dots, n$ . That is,  $\hat{f}$  minimizes  $J_\nu(f)$  over functions  $f$  satisfying  $\hat{f}(x_i) = \hat{y}_i, i = 1, \dots, n$ . Consequently,  $\hat{f}$  is a  $p^{\text{th}}$ -order natural spline, where  $p = 2\nu - 1$ . This means that, when solving the smoothing problem, we only need consider spline functions given by the representations in Proposition 11.1.*

**Proof:** Suppose the assertion is not true. In this case, we must be able to find a function  $\hat{f}^*$ , say, which also interpolates the artificial data  $(x_i, \hat{y}_i), i = 1, \dots, n$ , but which has a smaller roughness penalty. That is

$$J_\nu(\hat{f}^*) < J_\nu(\hat{f}) \text{ with } \hat{f}^*(x_i) = \hat{y}_i, \quad i = 1, \dots, n.$$



(a) Smoothing spline of data (dots) with fitted values (crosses) (b) Interpolating spline of the fitted values from the smoothing problem

Figure 12.3: Illustration of the proposition showing that the solution of the smoothing problem is a natural interpolating spline.

Note that the fitted values from the function  $\hat{f}^*$  are also equal to  $\hat{y}_i, i = 1, \dots, n$  as it interpolates the same artificial data as  $\hat{f}$ .

Now, from Equation 12.2,

$$\begin{aligned} R_\nu(\hat{f}^*, \lambda) &= \sum_i (y_i - \hat{y}_i)^2 + \lambda J_\nu(\hat{f}^*) \\ &< \sum_i (y_i - \hat{y}_i)^2 + \lambda J_\nu(\hat{f}) = R_\nu(\hat{f}, \lambda). \end{aligned}$$

Hence  $R_\nu(\hat{f}^*, \lambda) < R_\nu(\hat{f}, \lambda)$ . But, by construction  $\hat{f}$  minimizes  $R_\nu(f, \lambda)$ , which is a contradiction. Hence, it must not be possible to find a function  $\hat{f}^*$  which also interpolates the artificial data but which has a smaller roughness penalty.

We have shown that  $\hat{f}$  is the optimal interpolant of the fitted values  $\hat{y}_i, i = 1, \dots, n$ , so it follows from Proposition 11.1 that  $\hat{f}$  is a natural spline of order  $p = 2\nu - 1$ .

## 12.4 The smoothing problem in matrix notation

We have just proved that the function  $\hat{f}$  that minimizes  $R_\nu(f, \lambda)$  must be a natural spline (linear if  $\nu = 1$ , cubic if  $\nu = 2$ ) with knots at  $\{x_i, i = 1, \dots, n\}$ . That is, as in Proposition 11.1, we can write:

$$\hat{f}(t) = \sum_{i=1}^n b_i |t - x_i|^p + \begin{cases} a_0, & \nu = 1 \\ a_0 + a_1 t, & \nu = 2, \end{cases} \quad (12.3)$$

where  $p = 2\nu - 1$  and constraints

$$\sum_{i=1}^n b_i = 0 \text{ for } \nu = 1 \quad \sum_{i=1}^n b_i = \sum_{i=1}^n b_i x_i = 0 \text{ for } \nu = 2. \quad (12.4)$$

However, we have not yet figured out how to calculate the parameter values  $\hat{a}_0, \dots, \hat{a}_{\nu-1}, \hat{b}_1, \dots, \hat{b}_n$  in Equation 12.3 which optimally fit the data  $y_1, \dots, y_n$ . For this, it is convenient to re-express the penalised sum of squared residuals Equation 12.2 in matrix notation.

**Proposition 12.2.** *The roughness of a natural linear spline ( $\nu = 1$ , i.e.  $p = 1$ ) is*

$$J_1(f) = -2 \sum_{i=1}^n \sum_{k=1}^n b_i b_k |x_i - x_k| = c_1 \mathbf{b}^T K_1 \mathbf{b}, \quad (12.5)$$

and the roughness of a natural cubic spline ( $\nu = 2$ , i.e.  $p = 3$ ) is

$$J_2(f) = 12 \sum_{i=1}^n \sum_{k=1}^n b_i b_k |x_i - x_k|^3 = c_2 \mathbf{b}^T K_2 \mathbf{b}, \quad (12.6)$$

where the constants are given by

$$c_1 = -2, \quad c_2 = 12. \quad (12.7)$$

Here  $\mathbf{b} = (b_1, \dots, b_n)^T$  is the vector of spline coefficients and  $K_\nu$  is the  $n \times n$  matrix whose  $(i, k)$ th element is  $|x_i - x_k|^p$ , where  $p = 2\nu - 1$ , that is

$$K_\nu = \begin{bmatrix} |x_1 - x_1|^p & |x_1 - x_2|^p & \dots & |x_1 - x_n|^p \\ |x_2 - x_1|^p & |x_2 - x_2|^p & \dots & |x_2 - x_n|^p \\ \vdots & \vdots & \ddots & \vdots \\ |x_n - x_1|^p & |x_n - x_2|^p & \dots & |x_n - x_n|^p \end{bmatrix}$$

**Proof:** See the supplementary notes `Supplementary/proof_smoothing_spline_solution.pdf`.

From Proposition 12.2 the roughness penalty satisfies:

$$J_\nu(\hat{f}) = c_\nu \mathbf{b}^T K_\nu \mathbf{b}, \quad (12.8)$$

where  $c_1 = -2$ ;  $c_2 = 12$ ;  $\mathbf{b} = (b_1, \dots, b_n)^T$ ; and  $K_\nu$  is the  $n \times n$  matrix whose  $(i, k)$ th element is  $|x_i - x_k|^p$ .

From Equation 12.3, the value of  $\hat{f}$  at data location  $x_k$  is

$$\hat{f}(x_k) = \sum_{i=1}^n b_i |x_k - x_i|^p + \begin{cases} a_0, & \nu = 1 \\ a_0 + a_1 x_k, & \nu = 2 \end{cases}. \quad (12.9)$$

In matrix form, this is

$$\hat{\mathbf{f}} = \begin{bmatrix} \hat{f}(x_1) \\ \vdots \\ \hat{f}(x_n) \end{bmatrix} = K_\nu \mathbf{b} + L_\nu \mathbf{a}_\nu \quad (12.10)$$

where

$$L_1 = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{a}_1 = a_0, \quad \text{and} \quad L_2 = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}.$$

Thus, from Equation 12.4 – Equation 12.10, the penalised least-squares criterion Equation 12.2 reduces to a quadratic function of the parameters  $\mathbf{a}$  and  $\mathbf{b}$ :

$$R_\nu(f, \lambda) = (\mathbf{y} - K_\nu \mathbf{b} - L_\nu \mathbf{a})^T (\mathbf{y} - K_\nu \mathbf{b} - L_\nu \mathbf{a}) + \lambda c_\nu \mathbf{b}^T K_\nu \mathbf{b} \quad (12.11)$$

subject to

$$L_\nu^T \mathbf{b} = 0, \quad (12.12)$$

where  $\mathbf{y} = (y_1, \dots, y_n)^T$ .

To find the explicit values for  $\mathbf{b}$  and  $\mathbf{a}$ , we must minimize the quadratic function Equation 12.11 subject to the linear constraints Equation 12.12.

**Proposition 12.3.** *The solution to the smoothing spline problem is given by*

$$\begin{bmatrix} \hat{\mathbf{a}} \\ \hat{\mathbf{b}} \end{bmatrix} = \begin{bmatrix} 0 & L_\nu \\ L_\nu^T & K_\nu + \lambda^* I_n \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} \\ \mathbf{y} \end{bmatrix} \quad (12.13)$$

and  $\lambda^* = c_\nu \lambda$ .

**Proof:** See the supplementary notes `Supplementary/proof_smoothing_spline_solution.pdf`.

## 12.5 Smoothing splines in R

Fitting a smoothing spline to data involves estimating values of  $\mathbf{a}$  and  $\mathbf{b}$  to minimize the penalised roughness measure in Equation 12.11 subject to the constraints in Equation 12.12, which yields the explicit solution in Equation 12.13.

Of course, it is possible to code the solution of this matrix system, but it is usual instead to use in-built commands in software such as **R**.

In **R** the basic command is:

where  $\mathbf{x}$  and  $\mathbf{y}$  are vectors of data coordinates, and `lambda` specifies the value of the smoothing parameter to use – note that the effect of a given value of  $\lambda$  depends on the problem considered.

For example, consider the synthetic data set containing a break-point at  $x = 2/3$  first seen in Figure 9.2. Fitted spline curves with  $\lambda = 0.00001$  and  $\lambda = 1$ , shown in Figure 12.4, are based on the following commands:

If no value of the smoothing parameter is specified, the optimal value is calculated using generalised cross-validation as discussed in the next chapter.

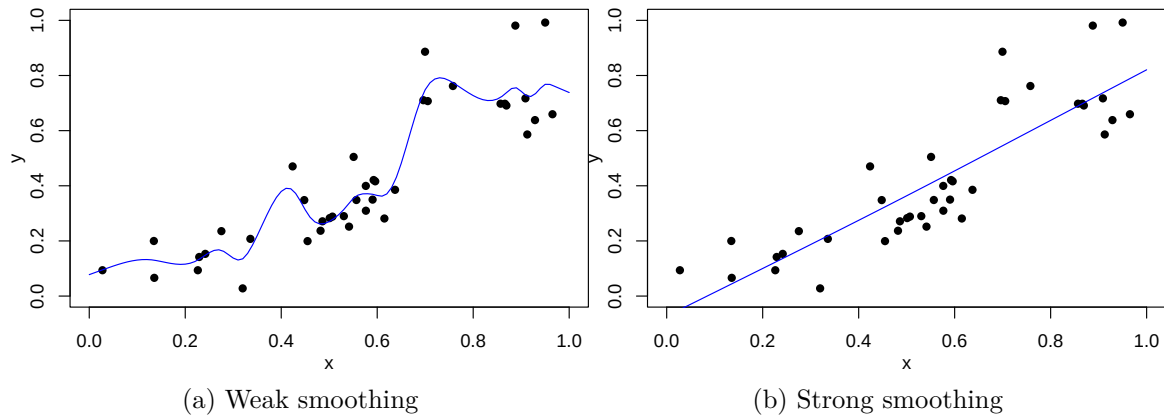


Figure 12.4: Simulated data superimposed with fitted smoothing splines.

## 12.6 Exercises

12.1 The following data record the number of tropical cyclones in each of 13 seasons (Dobson and Barnett, 3rd edn, Table 1.2):

Table 12.1: Numbers of tropical cyclones in  $n = 13$  successive seasons.

| Season         | 1 | 2 | 3 | 4 | 5 | 6 | 7  | 8 | 9 | 10 | 11 | 12 | 13 |
|----------------|---|---|---|---|---|---|----|---|---|----|----|----|----|
| No of cyclones | 6 | 5 | 4 | 6 | 6 | 3 | 12 | 7 | 4 | 2  | 6  | 7  | 4  |

It is reasonable to allow the average number of cyclones to vary with season, rather than assuming a constant rate. Fit a cubic smoothing spline to these data, using a range of values for the smoothing parameter  $\lambda$  to estimate the time-varying cyclone rate. By eye, suggest a suitable value for  $\lambda$  and justify your choice. Compare the smoothed estimate of rate with the value obtained by assuming a constant rate.

[Click here to see hints.](#)

Follow the R code used in Section 12.5 but don't worry about finding an exact value for the smoothing parameter, only the order of magnitude.

12.2 Consider the *Old Faithful* data set on geyser eruptions available in R. Use the following commands to learn more about and visualize the data:

```
data(faithful)
help(faithful)
plot(faithful)
```

Fit a cubic smoothing spline to these data, using a range of values for the smoothing parameter  $\lambda$ . By eye, suggest a suitable value for  $\lambda$ .

[Click here to see hints.](#)

Taking `waiting` as the explanatory variable and `eruptions` as the response, follow the R code used in Section 12.5 but don't worry about finding an exact value for the smoothing parameter, only the order of magnitude.

# 13 Choosing the Smoothing Parameter

## 13.1 Overview

Suppose we are given data  $D = \{(x_i, y_i), i = 1, \dots, n\}$  and that our model is:

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2) \quad (13.1)$$

where the  $\epsilon_i$  are i.i.d. and  $f(x)$  is assumed to be smooth. Given knot positions  $\{x_i, i = 1, \dots, n\}$ , we can estimate  $f(x)$  with a smoothing spline  $\hat{f}_\lambda(x)$ .

How then should we choose the value of the smoothing parameter  $\lambda$ ? By setting  $\lambda \rightarrow 0$ , we obtain exactly the interpolating spline  $\hat{f}_0(x)$  and a perfect fit to the data. However, this tends to *overfit* the data: applying it to a new sample of data where model Equation 13.1 still applies would produce a poor fit. Conversely, by setting  $\lambda \rightarrow \infty$ , we get:

$$f_\infty(x) = \begin{cases} \hat{a}_0, & \nu = 1, p = 1 \\ \hat{a}_0 + \hat{a}_1 x, & \nu = 2, p = 3. \end{cases}$$

Here,  $\hat{a}_0 = \bar{y}$ , for the  $\nu = 1, p = 1$  case, and  $\{\hat{a}_0, \hat{a}_1\}$ , for the  $\nu = 2, p = 3$  case, are the OLS linear regression parameters. If the true  $f(x)$  were constant or linear, this solution would be reasonable, but often we are interested in less regular functions.

## 13.2 Training/test approach

One way to approach estimation of  $\lambda$  is to partition the set of indices  $I = \{1, \dots, n\}$  into two subsets  $I_1$  and  $I_2$ , where  $I_1 \cup I_2 = I$  and  $I_1 \cap I_2 = \emptyset$ . Thus we obtain two datasets:

- Training dataset:  $D_1 = \{(x_i, y_i), i \in I_1\}$ ,
- Test dataset:  $D_2 = \{(x_i, y_i), i \in I_2\}$ .

We fit a smoothing spline  $\hat{f}_{\lambda, I_1}(x)$  to the training dataset, and judge the quality of the fit using the test dataset:

$$Q_{I_1: I_2}(\lambda) = \sum_{i \in I_2} (y_i - \hat{f}_{\lambda, I_1}(x_i))^2. \quad (13.2)$$

We choose  $\lambda$  to minimise  $Q_{I_1: I_2}(\lambda)$ . Many algorithms exist for such minimization, for example through evaluation on a fine grid of  $\lambda$  values, although many more computationally efficient algorithms exist.

### 13.3 Cross-validation or *leave-one-out*

This is an extreme form of the above principle. The test dataset  $D_2$  comprises a single observation,  $(x_j, y_j)$ , for a given value of  $j$ . The training set  $D_1$  is then  $D_{-j} = \{(x_i, y_i), i \in I_{-j}\}$ , where  $I_{-j}$  denotes the full set  $I$  excluding  $j$ . Then in a slightly amended notation we can write

$$Q_{-j;j}(\lambda) = (y_j - \hat{f}_{\lambda,-j}(x_j))^2$$

to assess the quality of fit. Of course,  $j$  is arbitrary, so we repeat this process for each  $j \in \{1, \dots, n\}$  then average the assessments to form the *ordinary cross-validation criterion*:

$$Q_{OCV}(\lambda) = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{f}_{\lambda,-j}(x_j))^2. \quad (13.3)$$

We then choose the value  $\hat{\lambda}$  which minimises  $Q_{OCV}(\lambda)$ . Hopefully, a plot of  $Q_{OCV}(\lambda)$  will appear as in Figure 13.1, but there is no theoretical guarantee that this curve will have a unique turning point, making it difficult to locate the minimum.

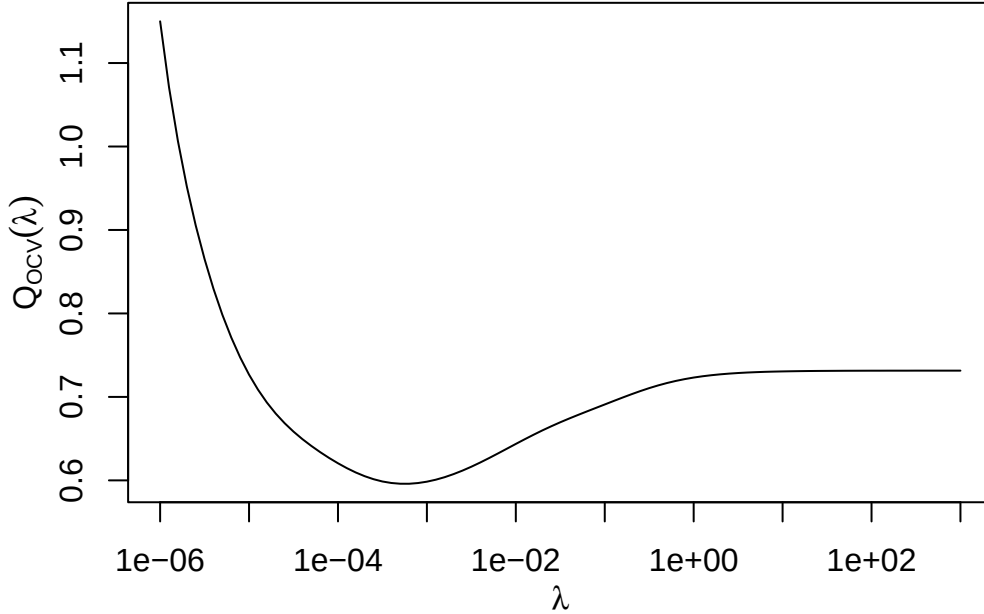


Figure 13.1: Ordinary cross-validation plot.

At first sight, evaluation of  $Q_{OCV}(\lambda)$  for a given  $\lambda$  appears computationally intensive: we must compute  $n$  different smoothing solutions, each corresponding to one of the *left-out* data points. Fortunately, there is a computational trick which enables us to compute  $Q_{OCV}(\lambda)$  directly from the smoothing spline solution constructed from the whole dataset.

## 13.4 The smoothing matrix

Here we show, for a given value of the smoothing parameter  $\lambda$  and the index  $\nu \geq 1$ , that the fitted value  $\hat{f}_\lambda(x_k)$  at each data location  $x_k$  may be written as a linear combination of the observations,  $y_1, \dots, y_n$ .

Recall from Proposition 12.3 that the smoothing spline  $\hat{f}_\lambda(x)$ , which minimises the penalised sum of squares criterion Equation 12.2 for a given value of the smoothing parameter  $\lambda$ , has coefficients  $\hat{\mathbf{a}}, \hat{\mathbf{b}}$  where

$$\begin{bmatrix} \hat{\mathbf{a}} \\ \hat{\mathbf{b}} \end{bmatrix} = M_\lambda^{-1} \begin{bmatrix} \mathbf{0} \\ \mathbf{y} \end{bmatrix} \quad \text{where} \quad M_\lambda = \begin{bmatrix} 0 & L_\nu \\ L_\nu^T & K_\nu + \lambda^* I_n \end{bmatrix}.$$

The fitted values of the smoothing spline at the data locations can be represented in matrix form as

$$\hat{\mathbf{f}} = \begin{bmatrix} \hat{f}(x_1) \\ \vdots \\ \hat{f}(x_n) \end{bmatrix} = K_\nu \hat{\mathbf{b}} + L_\nu \hat{\mathbf{a}} = \begin{bmatrix} K_\nu & L_\nu \end{bmatrix} \begin{bmatrix} M_\lambda^{12} \\ M_\lambda^{22} \end{bmatrix} \mathbf{y},$$

where  $M_\lambda^{-1}$  has been partitioned in the form

$$M_\lambda^{-1} = \begin{bmatrix} M_\lambda^{11} & M_\lambda^{12} \\ M_\lambda^{21} & M_\lambda^{22} \end{bmatrix},$$

where  $M_\lambda^{12}$  is  $\nu \times n$  and  $M_\lambda^{22}$  is  $n \times n$ .

It can be shown that the matrix

$$S_\lambda = \begin{bmatrix} K_\nu & L_\nu \end{bmatrix} \begin{bmatrix} M_\lambda^{12} \\ M_\lambda^{22} \end{bmatrix}$$

is a symmetric positive definite matrix for  $\lambda > 0$ , called the *smoothing matrix*. Then  $S_\lambda$  connects the data  $\mathbf{y}$  to the fitted values through

$$\hat{\mathbf{f}} = S_\lambda \mathbf{y}. \quad (13.4)$$

That is, the fitted values are a simple linear function of the data.

## 13.5 Effective degrees of freedom

How many degrees of freedom are there in the smoothing spline? There are altogether  $n + \nu$  parameters,  $\nu$  in  $\mathbf{a}$  and  $n$  in  $\mathbf{b}$ , but these are not completely free.

We showed earlier that as the smoothing parameter  $\lambda \rightarrow \infty$ , the smoothing spline  $\hat{f}(x)$  becomes the least-squares regression solution for model formula  $y \sim 1$  when  $\nu = 1$ , or model formula  $y \sim 1 + x$  when  $\nu = 2$ . Thus, when  $\lambda = \infty$ , the degrees of freedom in the spline is  $\nu$ . We also showed that when  $\lambda = 0$ , the smoothing spline  $\hat{f}(x)$  becomes the interpolating spline, for which the degrees of freedom is the number of observations,  $n$ .

Thus, intuitively, for values of  $\lambda$  between the extremes of 0 and  $\infty$ , the spline degrees of freedom should lie somewhere between  $\nu$  and  $n$ ; the greater the smoothing, the fewer the *effective* degrees of freedom. How can we capture this notion precisely?

A clue comes from Ordinary Least Squares (OLS) regression, in which the fitted values are given by

$$\hat{\mathbf{y}} = X(X^T X)^{-1} X^T \mathbf{y} = H \mathbf{y},$$

where  $X$  is the  $n \times p$  design matrix and  $p$  is the number of model parameters. Here,

$$H = X(X^T X)^{-1} X^T$$

is called the **hat** matrix, which linearly maps the data  $\mathbf{y}$  onto the fitted values  $\hat{\mathbf{y}}$ . Using the property that  $\text{trace}(QR) = \text{trace}(RQ)$  for matrices  $Q, R$  of conformable dimensions, the trace of the hat matrix is:

$$\begin{aligned} \text{trace}(X(X^T X)^{-1} X^T) &= \text{trace}((X^T X)^{-1} X^T X) \\ &= \text{trace}(I_p) \\ &= p, \end{aligned}$$

where  $I_p$  is the  $p \times p$  identity matrix. Thus, for OLS regression, the trace of the hat matrix equals the number of model parameters.

Now in Equation 13.4 we see that the smoothing matrix  $S_\lambda$  takes the role of a hat matrix, since it linearly maps the data onto the fitted values. This suggests that, for the smoothing spline, we can calculate an effective number of degrees of freedom as:

$$\text{edf}_\lambda = \text{trace } S_\lambda. \quad (13.5)$$

It can be shown from the limiting behaviour of the smoothing splines as  $\lambda \rightarrow \infty$  and  $\lambda \rightarrow 0$  that  $\text{edf}_\infty = \nu$  (the number of parameters in the OLS solution) and  $\text{edf}_0 = n$  (the number of parameters in the interpolating spline).

## 13.6 Generalised Cross-Validation

The cross-validation criterion  $Q_{OCV}(\lambda)$ , defined in Equation 13.3, is used to set the smoothing parameter,  $\lambda$ . However, as noted in Section 13.3, using Equation 13.3 to compute  $Q_{OCV}(\lambda)$  would be impractical for large  $n$ , since it would require fitting a new smoothing spline  $\hat{f}_{\lambda, -j}$  for each *leave-one-out* dataset  $I_{-j}$ .

Fortunately, it is possible to compute  $Q_{OCV}(\lambda)$  directly from the spline  $\hat{f}_\lambda$  fitted to the full dataset. It can be shown that Equation 13.3 can be rewritten:

$$Q_{OCV}(\lambda) = \frac{1}{n} \sum_{j=1}^n \left( \frac{y_j - \hat{f}_\lambda(x_j)}{1 - s_{jj}} \right)^2, \quad (13.6)$$

where  $\hat{f}_\lambda(x_j)$  is the full-data fitted spline value at  $x_j$  given by Equation 13.4, and  $s_{jj}$  is the  $j$ th diagonal element of the smoothing matrix  $S_\lambda$ .

Prior to the discovery of algorithms to compute Equation 13.6 quickly, a computationally efficient approximation to the cross-validation criterion Equation 13.3 was proposed, called the **Generalised Cross-Validation** criterion (GCV):

$$Q_{GCV}(\lambda) = \frac{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{f}_\lambda(x_j))^2}{\left(1 - \frac{1}{n} \text{trace}(S_\lambda)\right)^2}. \quad (13.7)$$

Thus Equation 13.7 replaces  $s_{jj}$  in Equation 13.6 with the average of the diagonal elements of  $S_\lambda$ , which equals  $\frac{1}{n} \text{edf}_\lambda$ . A low value of  $\text{edf}_\lambda$  will deflate  $Q_{GCV}(\lambda)$ , making that value of  $\lambda$  more favourable.

In principle, OCV supplants GCV, but GCV is still used as it is numerically more stable. In particular, GCV is used in the `mgcv` package — this will be discussed in the next chapter.

## 13.7 Exercises

13.1 The file `bmd.csv` (<https://www.richardpmann.com/MATH3701/Datasets/bmd.csv>) contains bone mineral density T-scores for  $n = 66$  women aged between 40 and 70, along with an indicator of whether each woman has been through menopause.

- Read in the data and produce a scatter plot of `Tscore` against `age`. Comment on the shape of the relationship: does it look linear?
- Fit a cubic smoothing spline of `Tscore` on `age` using `smooth.spline()`, first specifying a very small value of the smoothing parameter (e.g. `spar = 0`) and then a very large value (e.g. `spar = 1`). Overlay both fits on your scatter plot and comment on the over- and under-fitting you observe.
- Now fit the smoothing spline letting R choose the smoothing parameter automatically by generalised cross-validation (the default behaviour of `smooth.spline()`). Report the selected value of  $\lambda$  (`fit$lambda`) and the effective degrees of freedom (`fit$df`). Does the fitted curve look reasonable?
- Compare the GCV-selected effective degrees of freedom from (c) to the two extreme cases  $\text{edf} = \nu$  and  $\text{edf} = n$  discussed in Section 13.5. Where does the GCV choice lie, and what does this suggest about the relationship between age and bone density?

Click here to see hints.

Use `read.csv()` to load the data; `age` is the explanatory variable and `Tscore` the response. For (b) and (c), `smooth.spline(data$age, data$Tscore, spar = ...)`, omitting `spar` lets R choose it automatically via GCV.

13.2 The file `iron.csv` (<https://www.richardpmann.com/MATH3701/Datasets/iron.csv>) records an annual measurement, `iron`, for each `Year` from 1950 to 1999 ( $n = 50$ ).

- a. Plot `iron` against `Year`. Does a linear trend look appropriate, or is a smoothing spline needed?
- b. Using the **training/test approach** of Section 13.2, hold out the final 10 years (1990–1999) as the test set  $I_2$  and fit smoothing splines to the remaining  $n - 10$  years (the training set  $I_1$ ) over a grid of  $\lambda$  values, e.g. `10^seq(-4, 4, by = 0.5)`. For each  $\lambda$ , compute the test-set criterion  $Q_{I_1:I_2}(\lambda)$  from Equation 13.2 and plot it against  $\lambda$  on a log scale.
- c. Report the value of  $\lambda$  that minimises  $Q_{I_1:I_2}(\lambda)$ , refit the smoothing spline to the full dataset using this  $\lambda$ , and plot the resulting fit.
- d. Now fit `smooth.spline()` to the full dataset letting R choose  $\lambda$  by GCV. How does the GCV-selected  $\lambda$  compare to the one you found using the training/test approach in (c)? Which approach do you find more reliable here, and why?

Click here to see hints.

Use `read.csv()` to load the data. For (b), fit with `smooth.spline(train$Year, train$iron, lambda = lam)` for each candidate `lam` in your grid, then use `predict()` at the held-out `Year` values to compute the sum of squared errors against the held-out `iron` values. For (d), `smooth.spline(data$Year, data$iron)` with no `lambda` argument selects  $\lambda$  via GCV by default.

# 14 Generalised Additive Models

## 14.1 Overview

So far, we have considered the modelling of a response variable  $y$  in terms of a single explanatory variable  $x$ . In particular, we have assumed that the form of this relationship is unknown but can be written as

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

where the  $\epsilon_i$  are i.i.d. and  $f(x)$  is assumed to be smooth. Given knot positions  $\{x_i, i = 1, \dots, n\}$ , we can estimate  $f(x)$  with a smoothing spline  $\hat{f}_\lambda(x)$  for given smoothing parameter  $\lambda$  and, further, we can estimate  $\lambda$  using ordinary or generalised cross-validation. This approach is in contrast to simple linear regression where  $y$  is expressed as a linear function of the explanatory variable,  $y_i = \alpha + \beta x_i + \epsilon_i$ , which enforces a very inflexible relationship.

In this chapter, we generalise the modelling to describe the dependence of the response variable  $y$  on a set of explanatory variables  $\mathbf{x} = (x_1, x_2, \dots, x_p)$  where, conditionally on  $\mathbf{x}$ , observation  $y$  has a distribution which is not necessarily normal.

Just as with a generalised linear model, the *generalised additive model* relates a continuous or discrete response variable  $Y$  to a set of explanatory variables  $\mathbf{x} = (x_1, x_2, \dots, x_p)$  and the model contains three parts:

**Random part:** The probability (mass or density) function of  $Y$  is assumed to belong to the *two-parameter exponential family* of distributions with parameters  $\theta$  and  $\phi$ .

**Systematic part:** This is a *non-linear predictor equation*:

$$\eta = \sum_{j=1}^p f_j(x_j). \quad (14.1)$$

**Link function:** This is a one-to-one function providing the link between the predictor equation  $\eta$  and the mean  $\mu = E[Y]$ :

$$\eta = g(\mu), \quad \text{and} \quad \mu = g^{-1}(\eta) = h(\eta). \quad (14.2)$$

Here,  $g(\mu)$  is called the *link function*, and  $h(\eta)$  is called the *inverse link function*.

## 14.2 Penalised deviance

The spline theory in the previous chapters has assumed Gaussian (normally distributed) data and the identity link function. For non-Gaussian data and/or a non-identity link function, we replace the penalised least-squares criterion of Equation 12.2 with a *penalised deviance*:

$$R_\nu(f, \lambda, \beta) = D(\mathbf{y}, f, \beta) + \lambda J_\nu(f), \quad (14.3)$$

where  $D(\mathbf{y}, f, \beta)$  is the deviance for the vector  $\mathbf{y}$  of observations modelled by a linear predictor that comprises a spline function  $f(x)$  of order  $\nu$  and possibly also covariate main-effects and interactions. The penalised deviance is minimised with respect to the spline coefficients  $\mathbf{a}$ ,  $\mathbf{b}$  and regression parameters  $\beta$ , if any. Note that the fitted values for  $y$  are obtained by applying the inverse link function to the linear predictor  $f(x)$ , for example the logistic function when the link is logit.

When there are several smooth terms of order  $\nu$  in the model,  $\mathbf{f} = \{f_1, \dots, f_m\}$ , each may be assigned its own roughness penalty  $\lambda_h$ , and Equation 14.3 becomes

$$R_\nu(f_1, \dots, f_m, \lambda_1, \dots, \lambda_m, \beta) = D(\mathbf{y}, f_1, \dots, f_m, \beta) + \sum_{h=1}^m \lambda_h J_\nu(f_h), \quad (14.4)$$

or equivalently

$$R_\nu(\mathbf{f}, \lambda, \beta) = D(\mathbf{y}, \mathbf{f}, \beta) + \sum_{h=1}^m \lambda_h J_\nu(f_h).$$

## 14.3 GAMs in R

Fitting smoothing splines and GAMs is straightforward in practice using **R**. At the beginning of each **R** session, load the package **mgcv**, which makes available a set of routines written by Simon Wood of the University of Bristol.

The main command is **gam**, which fits a smoothing spline (or, more generally, a generalised additive model). The **gam** function is an extension to the **glm** command for fitting generalised linear models, allowing nonparametric functions of explanatory variables.

The syntax of the **gam** command is similar to that of **glm**. Suppose  $\mathbf{y}$  is a vector of length  $\mathbf{n}$  containing observations of a dependent variable and  $\mathbf{x}$  is another vector of length  $\mathbf{n}$  containing the corresponding explanatory variable values. Then each of the commands

fits a cubic smoothing spline to the dependent variable  $\mathbf{y}$ . In the first version above, the user explicitly sets the smoothing parameter  $\lambda$  (denoted **sp**) to the value 3.5. In the second version, the optimal value of  $\lambda$  is chosen by the routine to minimise the Generalised Cross-Validation criterion, GCV. The notation **s(x)** means a smooth function (a cubic smoothing spline in this setting) of the explanatory variable  $\mathbf{x}$ . Setting **fx=FALSE** in function **s** specifies that the dimensionality of the spline should be free. Parameter **k** of function **s** specifies the *maximum* dimensionality of the spline, **and should be set according to the problem**

**and data at hand** since the default value will not be appropriate in general. For example, if `x` contains only 6 distinct values, then it would be appropriate to set `k=6`.

In each of the above examples, output from `gam` is stored in an object called `out`. This object contains several components of interest:

- `out$fitted.values` is a vector of length `n` containing the fitted values of the smoothing spline at the data values.
- `out$gcv.ubre` contains the value of the GCV criterion.
- `out$hat` contains the diagonal values of the smoothing matrix, also called the *hat* matrix. The total degrees of freedom in the model (including the intercept term) is given by `sum(out$hat)`.
- `out$sp` contains the smoothing parameter value.
- `summary.gam(out)` provides a summary of the smoothed model fit.
- `anova.gam(out)` provides an analysis of deviance for the model.

Thus, if `gam` is called without an explicit choice of `sp` (as in the second example above), the output gives the optimal smoothing parameter value and the corresponding value of the GCV criterion.

To plot a smoothing spline:

- define a dense set of values spanning the data, at which to plot the smooth curve;
- use the `predict.gam` function to compute the cubic spline at these values.

For example, suppose `y` is a vector of length 20 containing the responses at times 1, ..., 20.

## 14.4 Coronary heart disease in South Africa

The textbook *Elements of Statistical Learning*, by Hastie, Tibshirani and Friedman (2nd Edn, 2011), refers to a case-control study of coronary heart disease (CHD) in South Africa. The data record the following variables for 462 males in a heart-disease high-risk region:

| Variable             | Description                                       |
|----------------------|---------------------------------------------------|
| <code>tobacco</code> | cumulative tobacco consumption (kg)               |
| <code>famhist</code> | family history of heart disease (Present, Absent) |
| <code>age</code>     | age at onset of the disease (years)               |
| <code>chd</code>     | case-control status (1 = CHD; 0 = no CHD)         |

The dependent variable in our models will be `chd`. We model the CHD status of each individual as the result of a binary experiment (CHD or no CHD), using the Binomial distribution with index 1. The `glm` function allows such binary (0/1) dependent variables to be specified directly in the model formula.

We can examine CHD in relation to tobacco consumption, age and family history of heart disease, with the following **R** commands:

```
Call:
glm(formula = "chd ~ tobacco + age + famhist", family = "binomial")
```

Coefficients:

|                | Estimate  | Std. Error | z value | Pr(> z ) |     |
|----------------|-----------|------------|---------|----------|-----|
| (Intercept)    | -3.620593 | 0.444576   | -8.144  | 3.83e-16 | *** |
| tobacco        | 0.083004  | 0.025712   | 3.228   | 0.00125  | **  |
| age            | 0.048812  | 0.009452   | 5.164   | 2.42e-07 | *** |
| famhistPresent | 0.974791  | 0.220023   | 4.430   | 9.41e-06 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 596.11 on 461 degrees of freedom  
Residual deviance: 495.39 on 458 degrees of freedom  
AIC: 503.39

Number of Fisher Scoring iterations: 4

All variables in this model are statistically significant: disease is positively related to tobacco consumption, age and family history of heart disease. However, this model assumes that the logit of the probability of disease is *linearly* related to both tobacco consumption and age (logit being the default link for Binomial). We can explore more flexible tobacco-consumption and age trends with the following generalised additive model:

Loading required package: nlme

This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.

Family: binomial

Link function: logit

Formula:

chd ~ s(tobacco, k = 20) + s(age, k = 20) + famhist

Parametric coefficients:

|                | Estimate | Std. Error | z value | Pr(> z ) |     |
|----------------|----------|------------|---------|----------|-----|
| (Intercept)    | -1.2379  | 0.1631     | -7.592  | 3.15e-14 | *** |
| famhistPresent | 0.9628   | 0.2233     | 4.311   | 1.62e-05 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```

Approximate significance of smooth terms:
              edf Ref.df Chi.sq  p-value
s(tobacco) 6.080  7.573  17.89   0.0179 *
s(age)      1.002  1.003   24.11 9.53e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.212   Deviance explained = 19.1%
UBRE = 0.083268  Scale est. = 1           n = 462

```

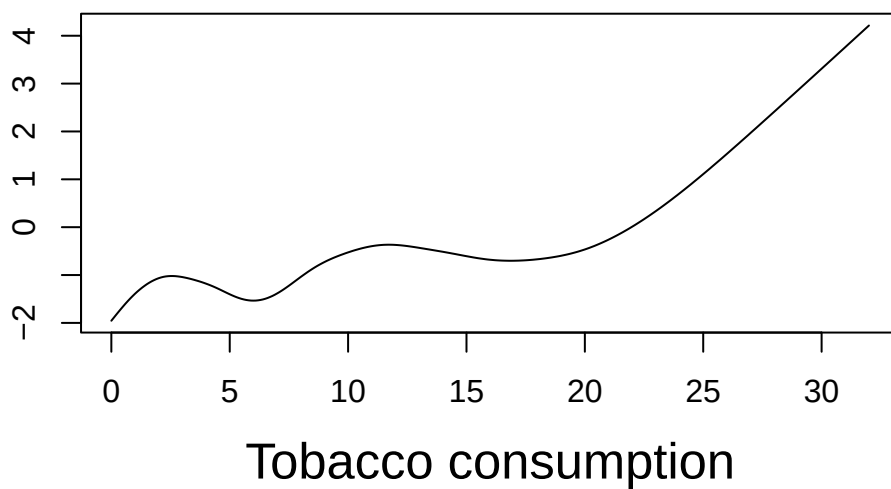
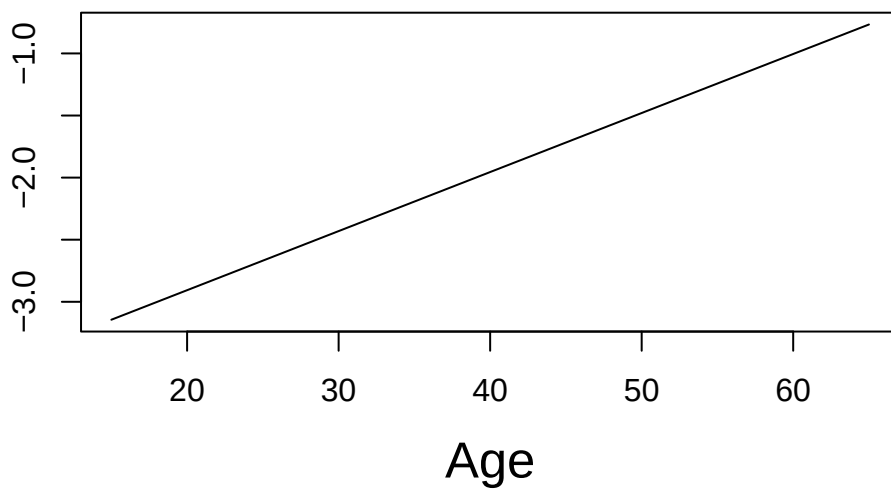
Here, the `s` function specifies a smooth (cubic spline) dependence. Parameter `k` of the `s` function specifies the maximum number of degrees of freedom allocated to this dependence. Setting `k` too small would restrict the set of basis functions used to construct the splines; setting `k` too large would increase the computational burden unnecessarily.

These results show that the fitted smoothing spline for tobacco consumption has an *effective degrees of freedom* (edf) of 6.080, while the dependence on age has edf of only 1.002, implying an almost linear age-dependence (a linear age term would have exactly 1 degree of freedom). See Section 13.5 for further details on the calculation of edf.

The significance of each smooth term is given by the *p*-value column in the table above. These *p*-values are computed from the  $\chi^2$  statistics in the previous column, whose approximate degrees of freedom are given in the column headed `Ref.df`. For example, the *p*-value of 0.0196 for `s(tobacco)` is computed by referring 17.62 to the  $\chi^2$  distribution on 7.573 degrees of freedom. Note that this compares the model above with the model which omits `tobacco` completely.

The following **R** code plots the fitted smooth functions of age and tobacco consumption:

predicted logit probability of CHD



The predicted dependence of the logit probability of CHD on smooth functions of age and tobacco consumption is shown above. We see an almost linear predicted age-dependence, in agreement with its edf. The curious predicted dependence on tobacco consumption may be explained by other factors correlated with tobacco consumption.

## 14.5 Exercises

14.1 Using the South Africa heart disease data (<https://www.richardpmann.com/MATH3701/Datasets/SAheart>), fit a GAM that includes smooth terms for all continuous predictors (tobacco, alcohol, obesity, sbp, ldl, adiposity, typea) along with `famhist` as a parametric term and `chd` as the binary response.

- Report the edf for each smooth term. Which predictors show evidence of non-linearity?
- Use `anova.gam` to assess the significance of each smooth term. Which predictors would you retain?
- Refit the model retaining only the significant smooth terms. Compare the GCV score of this reduced model with that of the full model.

[Click here to see hints.](#)

Fit the full model with `gam(chd ~ s(tobacco,k=10) + s(alcohol,k=10) + ... + famhist, family='binomial')`. An edf close to 1 suggests the smooth term is approximately linear and could be replaced by a parametric linear term.

14.2 Using the `mcycle` data set from the `MASS` package (accelerometer readings from a motorcycle crash test), fit a GAM with `times` as the explanatory variable and `accel` as the response.

- Fit the model with GCV-selected smoothing parameter and report the estimated edf.
- Plot the fitted smooth function with its 95% confidence band.
- Compare the GCV-optimal fit with fits obtained using a fixed `sp` of  $10^{-4}$  and  $10^2$ . Comment on the differences.

[Click here to see hints.](#)

Use `library(MASS); data(mcycle)` to load the data. For part (b), use `plot.gam(out, shade=TRUE)`.

## **Part III**

# **Part III: Survival Analysis**

# 15 Introduction to Survival Analysis

## 15.1 What is survival analysis?

In many statistical problems the response variable of interest is the time taken for a specific event to occur. We call this a **survival time** and denote it by  $T > 0$ . The event is sometimes called the **endpoint**. Although the name “survival analysis” originated in biomedical applications — where the endpoint is literally death — the methods apply far more broadly:

- time from surgery to death or relapse (medicine);
- time to failure of a mechanical or electronic component (reliability);
- time until a criminal reoffends after release from prison (criminology);
- duration of first marriage (sociology);
- time until an insurance claim is made (actuarial science).

Two features distinguish survival data from the more familiar continuous response variables we have seen in Parts I and II.

**Skewness.** Survival times are strictly positive ( $T > 0$ ) and typically right-skewed: most individuals experience the event relatively early, but a small number survive much longer. Normal-linear models are therefore usually inappropriate.

**Censoring.** In many studies it is impossible to observe the exact survival time for every individual. The survival time is then said to be **censored**. Censoring is the central methodological challenge of survival analysis, and handling it correctly is the main focus of Part III.

## 15.2 Motivating example

Example: Aortic valve replacement

Aortic valve replacement (AVR) is a form of open heart surgery in which a faulty valve is replaced with a mechanical or animal valve. A new surgical method has been developed that is easier to perform, but there is concern that it may shorten patients' lives. To investigate, we follow a sample of five patients who underwent AVR ten years ago.

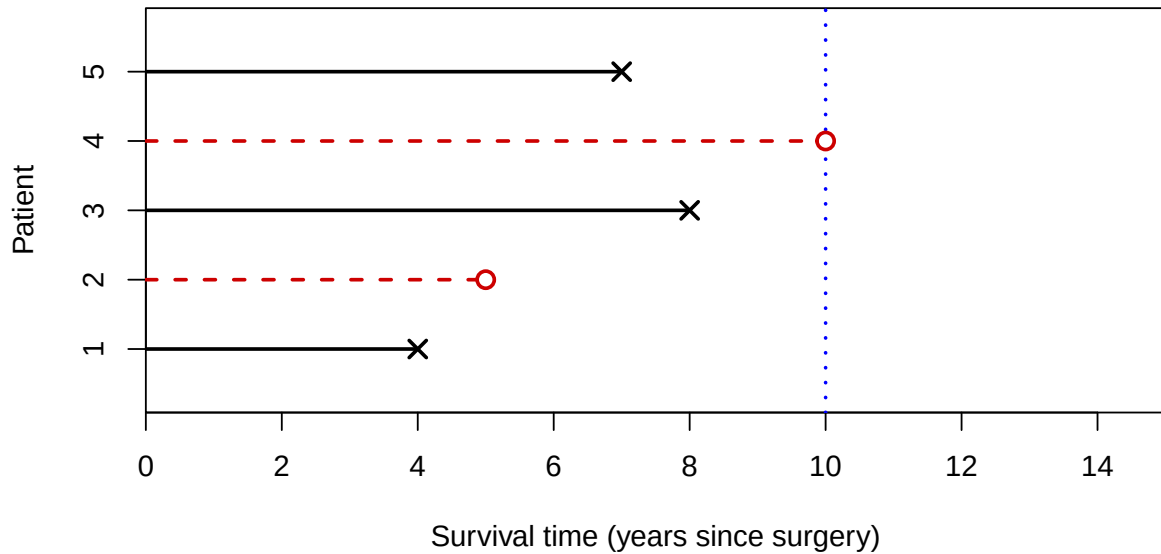


Figure 15.1: Survival times for the 5-patient AVR data set. Deaths ( $\times$ ) and censored observations ( $\circ$ ) are shown. The dashed vertical line marks the end of the 10-year study.

The survival times are summarised in Figure 15.1. Patients 1, 3, and 5 were observed to die at years 4, 8, and 7 after surgery. Patient 2 was **lost to follow-up** at year 5 (perhaps they withdrew from the study), so their exact death time is unknown. Patient 4 was still alive at the end of the 10-year observation period.

For patients 2 and 4, we know only that  $T > t^*$  for some observed time  $t^*$ . These observations are **right-censored**.

Suppose we wish to estimate  $\mathbb{P}(T > 6)$ , the probability that a randomly chosen future patient survives more than 6 years. Three naive approaches all fail:

1. **Ignore censoring, treat as deaths:** Only 1 of 5 patients died within 6 years, so  $\hat{p} = 4/5$ . But this assumes censored patients did not die after leaving the study — optimistic, and tends to **overestimate** survival.
2. **Treat censored observations as deaths:** 2 of 5 patients “died” within 6 years, so  $\hat{p} = 3/5$ . This is pessimistic and tends to **underestimate** survival.
3. **Exclude censored observations:** 1 of 3 observed deaths was before year 6, so  $\hat{p} = 2/3$ . But this discards the information that patient 4 definitely survived beyond 6 years, and would give the nonsensical estimate  $\mathbb{P}(T > 9) \approx 0$ .

All three approaches are unsatisfactory. The methods developed in this part of the course are designed to make proper use of censored data.

## 15.3 Types of censoring

Definition: Right censoring

A survival time  $T$  is **right-censored** if the observed time  $t^*$  is less than the actual (unknown) survival time:  $T > t^*$ .

Right censoring is the most common type and is the focus of this course. Other types include:

- **Left censoring:** the event occurred before a certain time, but the exact time is unknown ( $T < t^*$ ). For example, a patient whose cancer was present before their first screening but the exact onset is unknown.
- **Interval censoring:** it is known only that  $T$  falls in some interval  $(a, b)$ . For example, if annual screenings reveal a disease first at time  $b$  but it was absent at time  $a$ .

## 15.4 Assumptions

Throughout this part of the course we make the following assumptions.

! Important

**Assumption 1 (Uninformative censoring).** Censoring times are independent of survival times. There is no systematic tendency for individuals who would die sooner (or later) to be censored sooner (or later).

! Important

**Assumption 2 (Eventual event).**  $\mathbb{P}(T < \infty) = 1$ . Every individual will eventually experience the event of interest.

! Important

**Assumption 3 (i.i.d. sample).** The survival times  $T_1, T_2, \dots, T_n$  form an independent and identically distributed (i.i.d.) random sample from a common distribution.

In many analyses we will additionally observe covariates (explanatory variables) for each individual, and the survival distribution will depend on those covariates. Regression models for this are introduced in Chapter [22](#).

## 15.5 Observed data structure

In practice, for each individual  $i = 1, \dots, n$  we observe a pair  $(t_i, \delta_i)$  where:

$$t_i = \min(T_i, C_i), \quad \delta_i = \mathbf{1}(T_i \leq C_i)$$

and  $C_i$  is the censoring time. That is,  $t_i$  is the observed time (either the true event time if uncensored, or the censoring time if censored), and  $\delta_i$  is the **event indicator**:  $\delta_i = 1$  if the event was observed,  $\delta_i = 0$  if censored.

For the AVR example:

| Patient | $t_i$ | $\delta_i$   |
|---------|-------|--------------|
| 1       | 4     | 1 (death)    |
| 2       | 5     | 0 (censored) |
| 3       | 8     | 1 (death)    |
| 4       | 10    | 0 (censored) |
| 5       | 7     | 1 (death)    |

In R, this structure is represented using `Surv()` from the `survival` package:

The `+` symbol in the output of `Surv()` denotes a censored observation.

## Exercises

**15.1.** Twenty bedridden patients are placed on a new pressure-relieving mattress and checked daily for 12 days for the development of a pressure ulcer. Ten patients developed an ulcer, discovered at daily checks on days 1, 2, 2, 2, 5, 5, 8, 8, 8, and 8. Five patients were removed from the study between the day-6 and day-7 check (they had not yet developed an ulcer). The remaining five patients lasted the full 12 days and were gradually discharged over the following week without having developed an ulcer.

- What type of censoring applies to the **five patients removed between day 6 and day 7**?
- The five patients removed between day 6 and day 7 are subject to:
  - (A) Right censoring only
  - (B) Right censoring (and the ulcer times themselves are interval-censored)
  - (C) Interval censoring only
  - (D) Left censoring

- b. What type of censoring applies to the **ulcer event times themselves** (e.g., an ulcer discovered at the day-8 check)?
- c. An ulcer discovered at a daily check is an example of:
  - (A) Right censoring
  - (B) Left censoring
  - (C) Interval censoring
  - (D) No censoring — the exact time is known
- c. Given the combination of censoring types present, which non-parametric survival estimator is most appropriate?
- d. The most appropriate estimator for these data is:
  - (A) Kaplan–Meier estimator
  - (B) Actuarial (life-table) estimator
  - (C) Nelson–Aalen estimator
  - (D) Empirical distribution function

**15.2.** Using the pressure-ulcer data from Exercise 15.1, compute the actuarial estimate of the survival function  $S(t)$  by hand.

[Click here to see hints.](#)

For each interval  $[t_{j-1}, t_j)$ : let  $n_j$  be the number at risk at the start of the interval,  $d_j$  the number of events (ulcers discovered), and  $c_j$  the number of censored observations *within* the interval. The **effective risk set** is  $n'_j = n_j - c_j/2$ . The actuarial factor for that interval is  $(n'_j - d_j)/n'_j$  (set to 1 if  $d_j = 0$ ). The survival estimate is the product of all factors up to and including interval  $j$ .

- a. Construct the actuarial life table, showing columns for interval,  $n_j$ ,  $d_j$ ,  $c_j$ ,  $n'_j$ , the interval factor, and the cumulative  $\hat{S}(t)$ .
- b. State the estimated probability of *not* developing an ulcer over the 12-day study period.

# 16 Key Functions in Survival Analysis

## 16.1 The survival function

Let  $T > 0$  be a continuous random survival time with distribution function  $F(t) = \mathbb{P}(T \leq t)$  and density  $f(t) = F'(t)$ .

Definition: Survival function

The **survival function** is

$$S(t) = \mathbb{P}(T > t) = 1 - F(t), \quad t \geq 0.$$

$S(t)$  is a non-increasing function with  $S(0) = 1$  and  $\lim_{t \rightarrow \infty} S(t) = 0$ . For continuous  $T$ ,

$$S'(t) = -f(t) \quad \Rightarrow \quad f(t) = -S'(t).$$

## 16.2 The hazard function

Definition: Hazard function

The **hazard function** (or **hazard rate**) is

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{\mathbb{P}(T < t + \delta t \mid T \geq t)}{\delta t}.$$

Intuitively,  $h(t) \delta t \approx \mathbb{P}(\text{event in } [t, t + \delta t) \mid \text{survived to } t)$  for small  $\delta t$ .

Using conditional probability and the definition of the density:

$$h(t) = \frac{f(t)}{S(t)} = -\frac{S'(t)}{S(t)} = -\frac{d}{dt} \log S(t).$$

The hazard can take any non-negative value. Unlike the density  $f(t)$ , it is **not** a probability density and need not integrate to 1.

## 16.3 The cumulative hazard

Definition: Cumulative hazard

The **cumulative hazard function** is

$$H(t) = \int_0^t h(s) ds.$$

Since  $h(t) = -d \log S(t)/dt$ , integrating gives

$$H(t) = -\log S(t) \quad \Longleftrightarrow \quad S(t) = e^{-H(t)}.$$

This is one of the most important relationships in survival analysis. The three functions  $S(t)$ ,  $h(t)$ , and  $H(t)$  each completely characterise the distribution of  $T$ : knowing any one determines the others.

Example: Exponential distribution

For the exponential distribution with rate  $\lambda > 0$ :

$$f(t) = \lambda e^{-\lambda t}, \quad S(t) = e^{-\lambda t}, \quad h(t) = \lambda, \quad H(t) = \lambda t.$$

The hazard is **constant**: the risk of death in the next instant does not depend on how long the individual has already survived. This is the **memoryless property** of the exponential distribution.

## 16.4 Relationship between the three functions

| Function | From $S(t)$   | From $h(t)$        |
|----------|---------------|--------------------|
| $S(t)$   | —             | $e^{-H(t)}$        |
| $f(t)$   | $-S'(t)$      | $h(t)e^{-H(t)}$    |
| $h(t)$   | $-S'(t)/S(t)$ | —                  |
| $H(t)$   | $-\log S(t)$  | $\int_0^t h(s) ds$ |

## 16.5 Mean and median survival time

The **mean survival time** is

$$E[T] = \int_0^\infty t f(t) dt = \int_0^\infty S(t) dt,$$

where the second equality follows from integration by parts (under Assumption 2 of Chapter 15,  $\mathbb{P}(T < \infty) = 1$ ).

The **median survival time**  $\mu_{1/2}$  satisfies  $S(\mu_{1/2}) = 0.5$ , i.e., it is the time by which half the population has experienced the event. The median is often preferred over the mean in survival analysis because it is less sensitive to the (often unknown) tail behaviour of  $S(t)$ .

For the exponential distribution:  $E[T] = 1/\lambda$  and  $\mu_{1/2} = \log(2)/\lambda$ .

More generally, the  $p$ th quantile satisfies  $S(t_p) = 1 - p$ .

## 16.6 Empirical estimation (no censoring)

When data are uncensored,  $S(t)$  can be estimated by the **empirical survival function**

$$\hat{S}(t) = \frac{\#\{i : t_i > t\}}{n}.$$

This is an unbiased estimator:  $E[\hat{S}(t)] = S(t)$ .

### Empirical survival function

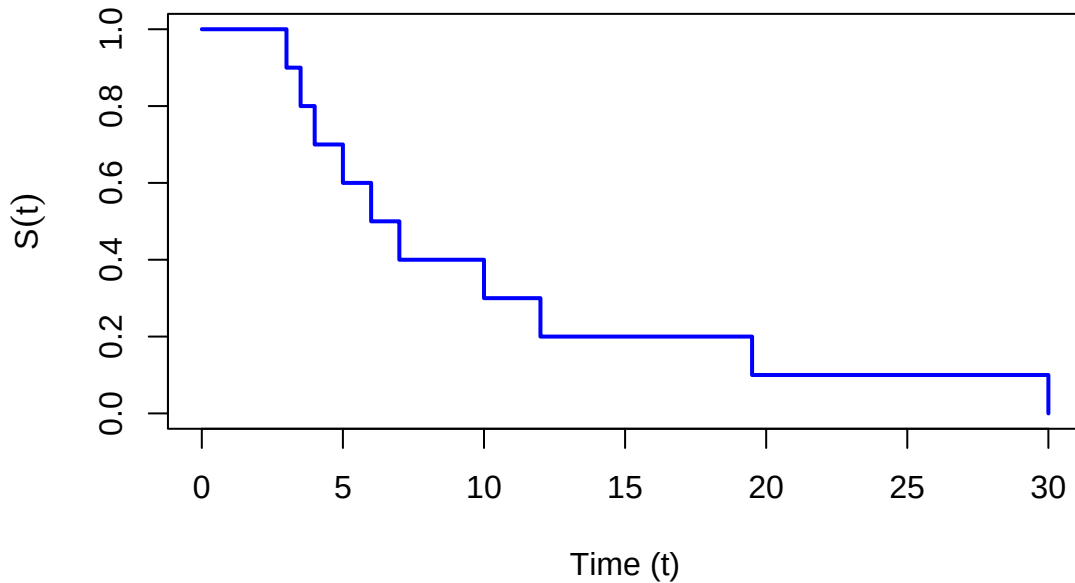


Figure 16.1: Empirical survival function for 10 uncensored survival times.

#### Example: Empirical estimation

For the 10 uncensored observations  $t = 3, 3.5, 4, 5, 6, 7, 10, 12, 19.5, 30$ , the empirical survival function takes the value  $\hat{S}(4.5) = 7/10$  because 7 of the 10 observations exceed 4.5. The estimated median is the smallest  $t$  such that  $\hat{S}(t) < 0.5$ , which is  $t = 7$ .

In R:

When data include **censored** observations, the empirical survival function is biased. The Kaplan–Meier estimator, introduced in Chapter 17, handles this correctly.

## Exercises

**16.1.** Suppose a survival time  $T$  has hazard function

$$h(t) = \frac{2}{t+8}, \quad t \geq 0.$$

- Find the cumulative hazard  $H(t)$ .
- Hence write down the survival function  $S(t)$  and the probability density function  $f(t)$ .

[Click here to see hints.](#)

Recall  $H(t) = \int_0^t h(u) du$ ,  $S(t) = e^{-H(t)}$ , and  $f(t) = h(t)S(t)$ .

**16.2.** Continuing from Exercise 16.1 with  $S(t) = 64/(8+t)^2$ .

- Find the mean survival time  $E[T]$  using the formula  $E[T] = \int_0^\infty S(t) dt$ .
- Find the median survival time  $t_{0.5}$ , defined by  $S(t_{0.5}) = 0.5$ .

[Click here to see hints.](#)

For part (a), integrate  $64/(8+t)^2$  from 0 to  $\infty$ . For part (b), solve  $64/(8+t_{0.5})^2 = 1/2$  for  $t_{0.5}$ .

# 17 Kaplan–Meier Estimator

## 17.1 Overview

When data include censored observations, the empirical survival function of Chapter 16 is biased. The **Kaplan–Meier (KM) estimator** is the non-parametric maximum likelihood estimator of  $S(t)$  that correctly accounts for right censoring. It is the standard starting point for any survival analysis.

## 17.2 The product-limit formula

Let  $t_{(1)} < t_{(2)} < \dots < t_{(k)}$  denote the ordered, distinct **event times** (ignoring censored observations). For each event time  $t_{(j)}$ , define:

- $n_j$  = number of individuals at risk (alive and uncensored) immediately before  $t_{(j)}$ ,
- $d_j$  = number of events (deaths) at  $t_{(j)}$ .

Definition: Kaplan–Meier estimator

The Kaplan–Meier estimator of  $S(t)$  is

$$\widehat{S}(t) = \prod_{j: t_{(j)} \leq t} \frac{n_j - d_j}{n_j}.$$

This is a step function that drops at each observed event time. Between event times,  $\widehat{S}(t)$  is constant. Censored observations contribute to the **risk set**  $n_j$  before they are censored but do not cause the estimator to drop.

Example: AVR data

For the AVR data  $(t, \delta) = (4, 1), (5, 0), (7, 1), (8, 1), (10, 0)$ , the event times are 4, 7, 8. Constructing the KM table:

| $t_{(j)}$ | $n_j$ | $d_j$ | $(n_j - d_j)/n_j$ | $\widehat{S}(t_{(j)})$  |
|-----------|-------|-------|-------------------|-------------------------|
| 4         | 5     | 1     | 4/5               | 4/5 = 0.800             |
| 7         | 3     | 1     | 2/3               | (4/5)(2/3) = 0.533      |
| 8         | 2     | 1     | 1/2               | (4/5)(2/3)(1/2) = 0.267 |

At  $t_{(1)} = 4$ : all 5 are at risk. At  $t_{(2)} = 7$ : patient 2 was censored at time 5, so only 3 remain at risk. At  $t_{(3)} = 8$ : patient 4 was censored at time 10 (still at risk at 8), so 2 are at risk.

Therefore  $\hat{S}(6) = 0.800$ ,  $\hat{S}(9) = 0.267$ , and the estimated median is  $\min\{t_{(j)} : \hat{S}(t_{(j)}) < 0.5\} = 7$  years.

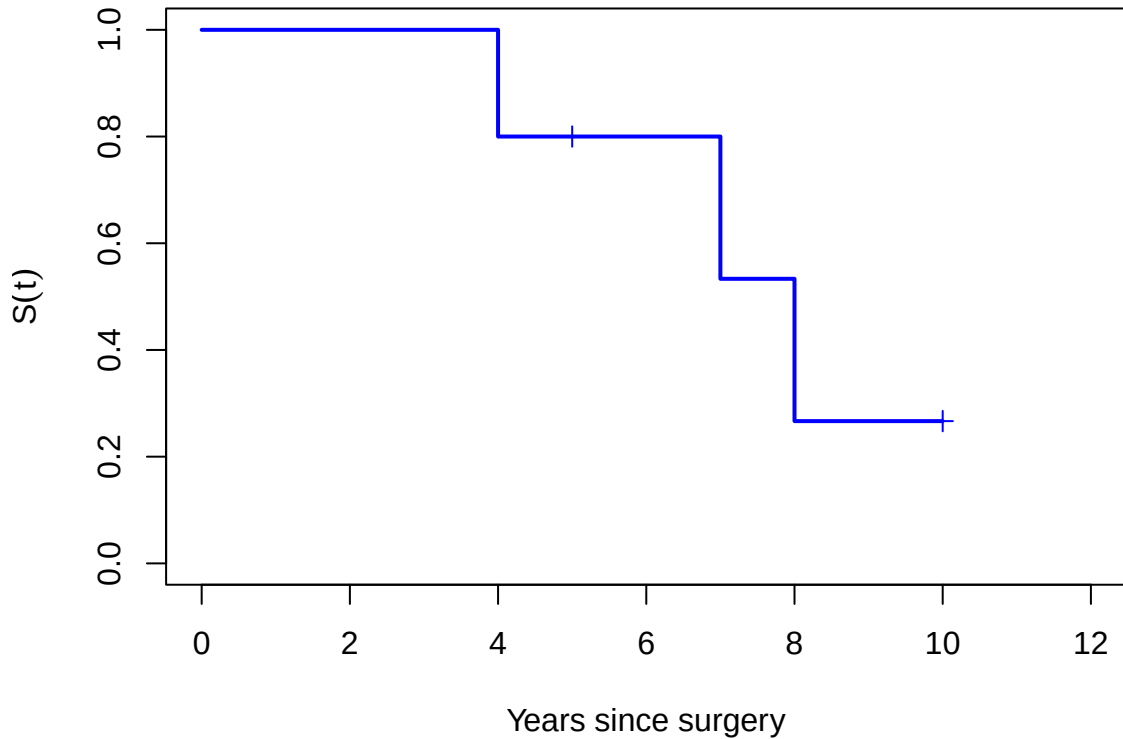


Figure 17.1: Kaplan–Meier estimate for the AVR data. Tick marks on the curve indicate censored observations.

### 17.3 Standard error: Greenwood's formula

Since the KM estimator is random, we need a measure of its precision. **Greenwood's formula** gives the approximate variance:

$$\widehat{\text{Var}}(\hat{S}(t)) = \hat{S}(t)^2 \sum_{j: t_{(j)} \leq t} \frac{d_j}{n_j(n_j - d_j)}.$$

An approximate 95% pointwise confidence interval for  $S(t)$  is

$$\hat{S}(t) \pm 1.96 \sqrt{\widehat{\text{Var}}(\hat{S}(t))}.$$

In practice, confidence intervals based on the **log-log transformation** of  $S(t)$  are preferred because they automatically respect the constraint  $0 < S(t) < 1$  (the plain linear interval can exceed these bounds). These are computed automatically by `survfit()` in R.

## 17.4 The delta method

Constructing a confidence interval on a transformed scale, as above, requires the variance of the *transformed* quantity, not just of  $\hat{S}(t)$  itself. The **delta method** provides a general way to approximate the variance of a function of an estimator, given the variance of the estimator itself.

### Definition: Delta method

Let  $\hat{\theta}$  be an estimator of  $\theta$  with approximate variance  $\text{Var}(\hat{\theta})$ , and let  $g$  be a differentiable function. A first-order Taylor expansion of  $g(\hat{\theta})$  about  $\theta$  gives

$$g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta),$$

so that, taking variances,

$$\text{Var}(g(\hat{\theta})) \approx [g'(\theta)]^2 \text{Var}(\hat{\theta}).$$

In practice  $g'(\theta)$  is evaluated at  $\hat{\theta}$ , since  $\theta$  itself is unknown.

This is exactly how the log-log and logit transformations of  $\hat{S}(t)$  are handled: apply the delta method with  $g$  equal to the chosen transformation, substitute Greenwood's formula for  $\text{Var}(\hat{S}(t))$ , construct the interval on the transformed scale, and back-transform via  $g^{-1}$  to recover a confidence interval for  $S(t)$  that automatically respects  $0 < S(t) < 1$ .

## 17.5 Kaplan–Meier in R

## 17.6 Two-group comparison

When we have two groups (e.g., treatment vs control), we can compare their KM curves visually and test for equality using the **log-rank test**. In R:

### Note: KM as descriptive tool

The KM estimator is a **non-parametric, descriptive** method. It makes no assumptions about the shape of  $S(t)$  and does not allow for adjustment of covariates. For modelling, we use parametric regression (Chapters 17–18) or the Cox model (Chapter 23).

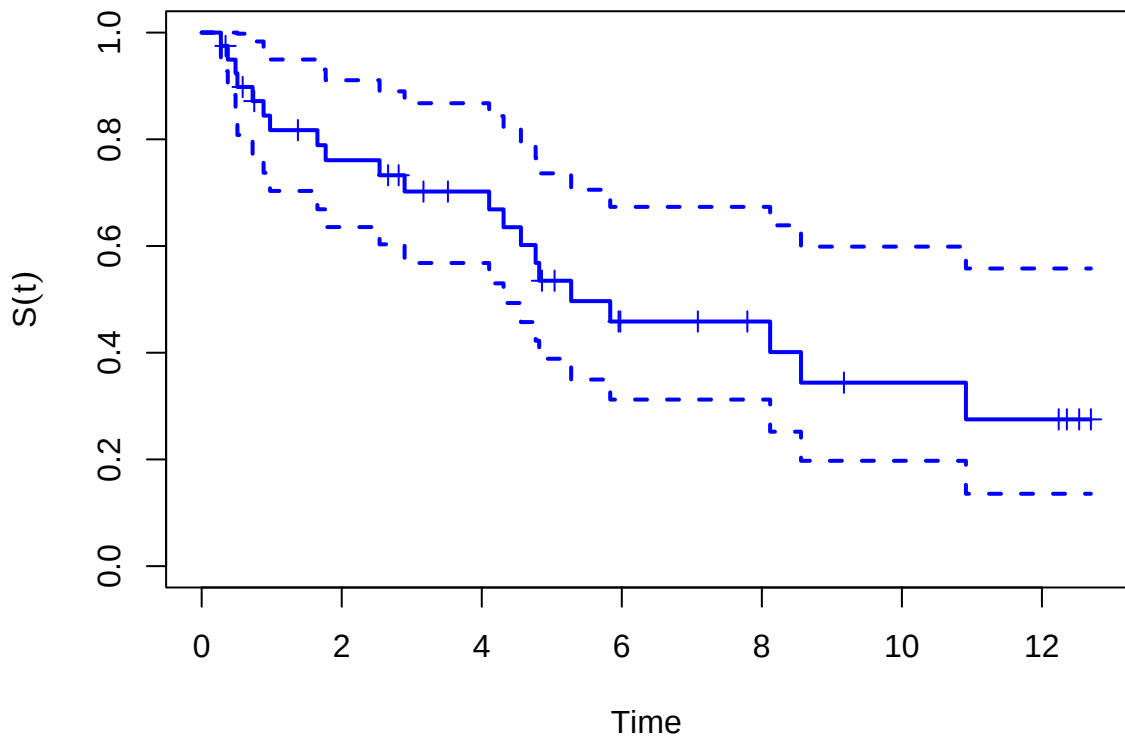


Figure 17.2: Kaplan–Meier estimate with 95% pointwise confidence band (log-log transformation) for a larger simulated dataset.

## Exercises

**17.1.** The following are survival times (in months) for ten colorectal cancer patients:

2, 10, 14, 27, 27, 37, 44, 46, 54, 60.

All ten times are exact (no censoring in this part).

- Compute the empirical survival function  $\hat{S}(t)$  and sketch it. Estimate the median survival time from the plot.
- Now suppose the observation at  $t = 44$  months is instead a **right-censored** time (the patient was still alive at 44 months). Re-compute the survival estimate using the Kaplan–Meier method, showing all working. State how the estimates  $\hat{S}(46)$  and  $\hat{S}(54)$  change compared with part (a).

[Click here to see hints.](#)

For the KM estimator, at each event time  $t_{(j)}$  with  $d_j$  events and risk set  $n_j$ : multiply the running product by  $(n_j - d_j)/n_j$ . Censored observations reduce the risk set at the next event time but do not contribute a multiplicative factor of their own.

**17.2.** Continuing with the censored colorectal cancer data from Exercise 17.1(b).

[Click here to see hints.](#)

For part (a), use `library(survival)` and `survfit(Surv(t, d) ~ 1)`. For part (b), Greenwood's formula is  $\widehat{\text{Var}}(\hat{S}(t)) = [\hat{S}(t)]^2 \sum_{t_{(j)} \leq t} d_j / (n_j(n_j - d_j))$ .

- a. Reproduce the KM analysis from Exercise 17.1(b) in R. Plot the KM curve with 95% confidence bands.
- b. Using Greenwood's formula, compute an approximate 95% confidence interval for  $S(11)$  (the probability of surviving beyond 11 months). Work by hand.

**17.3.** Let  $\hat{S}(t)$  be the Kaplan–Meier estimator and define the **logit transform**

$$\hat{S}_{\text{logit}}(t) = \log \left( \frac{\hat{S}(t)}{1 - \hat{S}(t)} \right).$$

- a. Using the delta method and Greenwood's formula, derive an expression for the approximate variance of  $\hat{S}_{\text{logit}}(t)$ .
- b. Explain how this variance can be used to construct a confidence interval for  $S(t)$  that is guaranteed to lie in  $(0, 1)$ .

[Click here to see hints.](#)

Let  $g(x) = \log(x/(1 - x))$ . Compute  $g'(x)$  and apply the delta method:  $\text{Var}(g(\hat{S}(t))) \approx [g'(S(t))]^2 \text{Var}(\hat{S}(t))$ . Then substitute Greenwood's formula. For part (b), construct the CI on the logit scale and back-transform via the inverse logit.

# 18 Parametric Survival Distributions

## 18.1 Overview

A **parametric** survival model specifies a functional form for  $S(t)$  (or equivalently  $h(t)$ ) up to a finite number of unknown parameters. The parameters are then estimated from data by maximum likelihood. This chapter introduces the most important parametric families.

## 18.2 Exponential distribution

Definition: Exponential distribution

$T \sim \text{Exp}(\lambda)$  if

$$h(t) = \lambda, \quad S(t) = e^{-\lambda t}, \quad f(t) = \lambda e^{-\lambda t}$$

for  $t > 0, \lambda > 0$ .

Key properties: - **Constant hazard**: the risk of dying does not increase or decrease with time (memoryless property). - Mean:  $E[T] = 1/\lambda$ ; Median:  $\mu_{1/2} = \log 2/\lambda$ . - Simple to fit and interpret, but often too restrictive in practice.

## 18.3 Weibull distribution

Definition: Weibull distribution

$T \sim \text{Weibull}(\lambda, \gamma)$  if

$$h(t) = \lambda \gamma t^{\gamma-1}, \quad S(t) = e^{-\lambda t^\gamma}, \quad f(t) = \lambda \gamma t^{\gamma-1} e^{-\lambda t^\gamma}$$

for  $t > 0, \lambda > 0$  (scale),  $\gamma > 0$  (shape).

The shape parameter  $\gamma$  controls how the hazard changes with time:

| $\gamma$ | Hazard     | Interpretation                                       |
|----------|------------|------------------------------------------------------|
| $< 1$    | Decreasing | Early high risk, then improving (“infant mortality”) |
| $= 1$    | Constant   | Exponential model                                    |

| $\gamma$ | Hazard     | Interpretation    |
|----------|------------|-------------------|
| $> 1$    | Increasing | Ageing / wear-out |

The Weibull distribution is the most widely used parametric model in survival analysis because of its flexibility and its **proportional hazards** property (see Chapter 21).

## 18.4 Other parametric families

**Gompertz distribution:**

$$h(t) = \lambda e^{\theta t}, \quad \lambda > 0, \theta \in \mathbb{R}.$$

The hazard grows exponentially with  $t$ . Used extensively in actuarial science to model human mortality.

**Gamma distribution** with shape  $\alpha > 0$  and rate  $\lambda > 0$ :

$$f(t) = \frac{\lambda^\alpha t^{\alpha-1}}{\Gamma(\alpha)} e^{-\lambda t}.$$

The hazard starts at 0 if  $\alpha > 1$ , at  $\infty$  if  $\alpha < 1$ , and at  $\lambda$  if  $\alpha = 1$  (exponential). Unlike the Weibull, the gamma does **not** have the proportional hazards property.

**Log-normal distribution:**  $\log T \sim N(\mu, \sigma^2)$ . The hazard first increases then decreases (hump-shaped). Common in engineering reliability.

## 18.5 Graphical assessment of parametric fit

Before fitting a model, it is useful to inspect the data graphically to see which family is plausible.

**Exponential check.** The Kaplan–Meier estimate of  $H(t) = -\log \hat{S}(t)$  should be approximately linear through the origin:

$$H(t) = \lambda t \quad \Rightarrow \quad \hat{H}(t) \approx \lambda t.$$

**Weibull check.** Taking logs of  $H(t) = \lambda t^\gamma$ :

$$\log H(t) = \log \lambda + \gamma \log t.$$

A plot of  $\log \hat{H}(t)$  against  $\log t$  (the **log-cumulative hazard plot**) should be approximately linear. The slope estimates  $\gamma$  and the intercept estimates  $\log \lambda$ .

## 18.6 Simulation and fitting in R

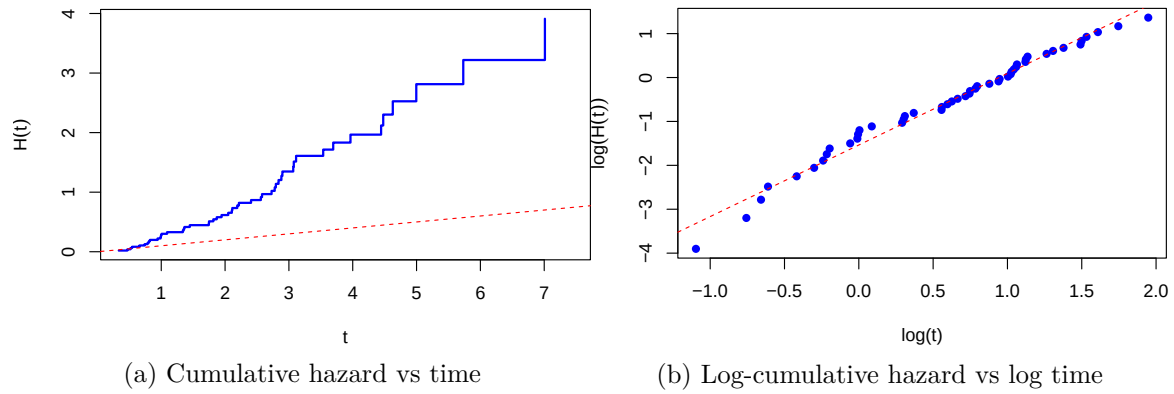


Figure 18.1: Log-cumulative hazard plots for simulated Weibull data. The left panel shows  $\widehat{H}(t)$  vs  $t$  (exponential check); the right panel shows  $\log \widehat{H}(t)$  vs  $\log t$  (Weibull check). The linear trend on the right confirms the Weibull fit.

Note: R's parametrisation

The `survreg()` function uses an accelerated failure time (AFT) parametrisation. For a Weibull model, the output reports an intercept  $\mu$  and a scale  $\sigma$ . To convert to the hazard parametrisation  $h(t) = \lambda\gamma t^{\gamma-1}$ :

$$\hat{\gamma} = 1/\hat{\sigma}, \quad \hat{\lambda} = e^{-\hat{\gamma}\hat{\mu}}.$$

## Exercises

**18.1.** Two groups of patients have the following survival times (in days), with right-censored times marked \*:

- **Group I:** 25, 30, 31, 44, 54, 59, 70\*, 88, 111, 139, 149.
- **Group II:** 25\*, 54, 86, 111, 152, 187\*, 195, 212, 301, 357.

The resulting Kaplan–Meier estimates are:

**Group I KM:**

| $t$ | $\widehat{S}(t)$ |
|-----|------------------|
| 25  | 0.909            |
| 30  | 0.818            |
| 31  | 0.727            |
| 44  | 0.636            |
| 54  | 0.545            |
| 59  | 0.455            |
| 88  | 0.341            |

| $t$ | $\hat{S}(t)$ |
|-----|--------------|
| 111 | 0.227        |
| 139 | 0.114        |
| 149 | 0.000        |

**Group II KM:**

| $t$ | $\hat{S}(t)$ |
|-----|--------------|
| 54  | 0.889        |
| 86  | 0.778        |
| 111 | 0.667        |
| 152 | 0.556        |
| 195 | 0.417        |
| 212 | 0.278        |
| 301 | 0.139        |
| 357 | 0.000        |

- a. For each group, estimate the percentiles  $t(P)$  for  $P = 10, 20, \dots, 90$  from the KM table. Arrange your results in a two-column table.

[Click here to see hints.](#)

The  $P$ th percentile satisfies  $S(t(P)) = 1 - P/100$ . Because  $\hat{S}(t)$  is a step function (not a smooth curve), do not interpolate:  $t(P)$  is simply the smallest observed event time at which  $\hat{S}(t)$  first drops to or below  $1 - P/100$ .

- b. For each group, produce the graphical Weibull check: plot  $\log \hat{H}(t)$  against  $\log t$ , where  $\hat{H}(t) = -\log \hat{S}(t)$ , using the KM jump points from the tables above. Do the plots support a Weibull model for each group's survival times?
- c. From your plots in (b), read off rough graphical estimates of the shape  $\gamma$  and (via the intercept) the scale  $\lambda$  for each group.

[Click here to see hints.](#)

Recall  $\hat{H}(t) = -\log \hat{S}(t)$ , and under a Weibull model  $\log H(t) = \log \lambda + \gamma \log t$ , so plot  $\log \hat{H}(t)$  against  $\log t$ : the slope of the best-fit line estimates  $\gamma$ , and the intercept estimates  $\log \lambda$ .

**18.2.** Using the raw survival data (times and censoring indicators) from Exercise 18.1, fit a Weibull model separately to each group in R.

- a. Write the R code to fit each model and report  $\hat{\gamma}$  and  $\hat{\lambda}$  for each group, using the parametrisation conversion given earlier in this chapter. How do these compare with your graphical estimates from 18.1(c)?

- b. The two groups have similar estimated shape parameters  $\hat{\gamma}$ . Since the hazards are  $h(t) = \lambda\gamma t^{\gamma-1}$ , when  $\gamma$  is (approximately) the same for both groups the ratio of hazards is approximately  $\hat{\lambda}_I/\hat{\lambda}_{II}$ . Compute this ratio and interpret it: which group has the worse survival experience, and by roughly what factor?

[Click here to see hints.](#)

Use `survreg(Surv(t, d) ~ 1, dist = "weibull")` separately on each group's data, with `d = 1` for an event and `d = 0` for a censored time (matching the \* superscripts in Exercise 18.1). Convert to the hazard parametrisation via  $\hat{\gamma} = 1/\hat{\sigma}$ ,  $\hat{\lambda} = e^{-\hat{\gamma}\hat{\mu}}$ , as in the “R's parametrisation” note above.

# 19 Maximum Likelihood with Censored Data

## 19.1 Likelihood contributions

Recall from Chapter 15 that the observed data for each individual consist of a pair  $(t_i, \delta_i)$  where  $t_i$  is the observed time and  $\delta_i = 1$  if the event was observed,  $\delta_i = 0$  if censored.

For an uncensored observation ( $\delta_i = 1$ ), the event occurred at exactly time  $t_i$ , contributing the density  $f(t_i)$  to the likelihood.

For a right-censored observation ( $\delta_i = 0$ ), the individual survived beyond  $t_i$ , contributing  $\mathbb{P}(T > t_i) = S(t_i)$ .

Under the uninformative censoring assumption (Chapter 15), the full likelihood is

$$L(\theta) = \prod_{i=1}^n f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}$$

Using  $f(t) = h(t)S(t)$ , this can be written equivalently as

$$L(\theta) = \prod_{i=1}^n h(t_i)^{\delta_i} S(t_i). \quad (19.1)$$

The second form has a natural interpretation: the survival function  $S(t_i)$  represents the probability of surviving until the observation time, and  $h(t_i)^{\delta_i}$  captures the (conditional) probability of the event occurring at that time if observed.

## 19.2 Exponential model: MLE derivation

For the exponential model with  $h(t) = \lambda$  and  $S(t) = e^{-\lambda t}$ , the log-likelihood is

$$\ell(\lambda) = \log L(\lambda) = \sum_{i=1}^n [\delta_i \log \lambda - \lambda t_i] = r \log \lambda - \lambda v,$$

where  $r = \sum_i \delta_i$  is the **number of observed events** and  $v = \sum_i t_i$  is the **total person-time**.

Setting  $\ell'(\lambda) = 0$ :

$$\frac{d\ell}{d\lambda} = \frac{r}{\lambda} - v = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{r}{v} = \frac{\text{events}}{\text{total time}}.$$

This has an intuitive interpretation: the estimated hazard rate is the number of events per unit of total follow-up time.

Example: AVR data

For the AVR data ( $n = 5$ , observed times 4, 5, 8, 10, 7; deaths at times 4, 8, 7):

$$r = 3, \quad v = 4 + 5 + 8 + 10 + 7 = 34.$$

Therefore  $\hat{\lambda} = 3/34 \approx 0.088$  deaths per year. The estimated mean survival is  $1/\hat{\lambda} \approx 11.3$  years.

## 19.3 Weibull model: MLE

For the Weibull model with  $h(t) = \lambda\gamma t^{\gamma-1}$  and  $S(t) = e^{-\lambda t^\gamma}$ , the log-likelihood is

$$\ell(\lambda, \gamma) = r \log(\lambda\gamma) + (\gamma - 1) \sum_{i=1}^n \delta_i \log t_i - \lambda \sum_{i=1}^n t_i^\gamma.$$

There is no closed form for  $\hat{\gamma}$ . The score equation  $\partial\ell/\partial\gamma = 0$  gives

$$\frac{1}{\gamma} + \frac{1}{r} \sum_{i=1}^n \delta_i \log t_i - \frac{\sum_{i=1}^n t_i^\gamma \log t_i}{\sum_{i=1}^n t_i^\gamma} = 0.$$

This is solved numerically. Given  $\hat{\gamma}$ , the profile MLE for  $\lambda$  is

$$\hat{\lambda} = \frac{r}{\sum_{i=1}^n t_i^{\hat{\gamma}}}.$$

## 19.4 Fitting parametric models in R

## 19.5 Model assessment via graphical checks

Having estimated the parameters, we should check the fit graphically by overlaying the fitted survival curve on the Kaplan–Meier estimate (see Chapter 17).

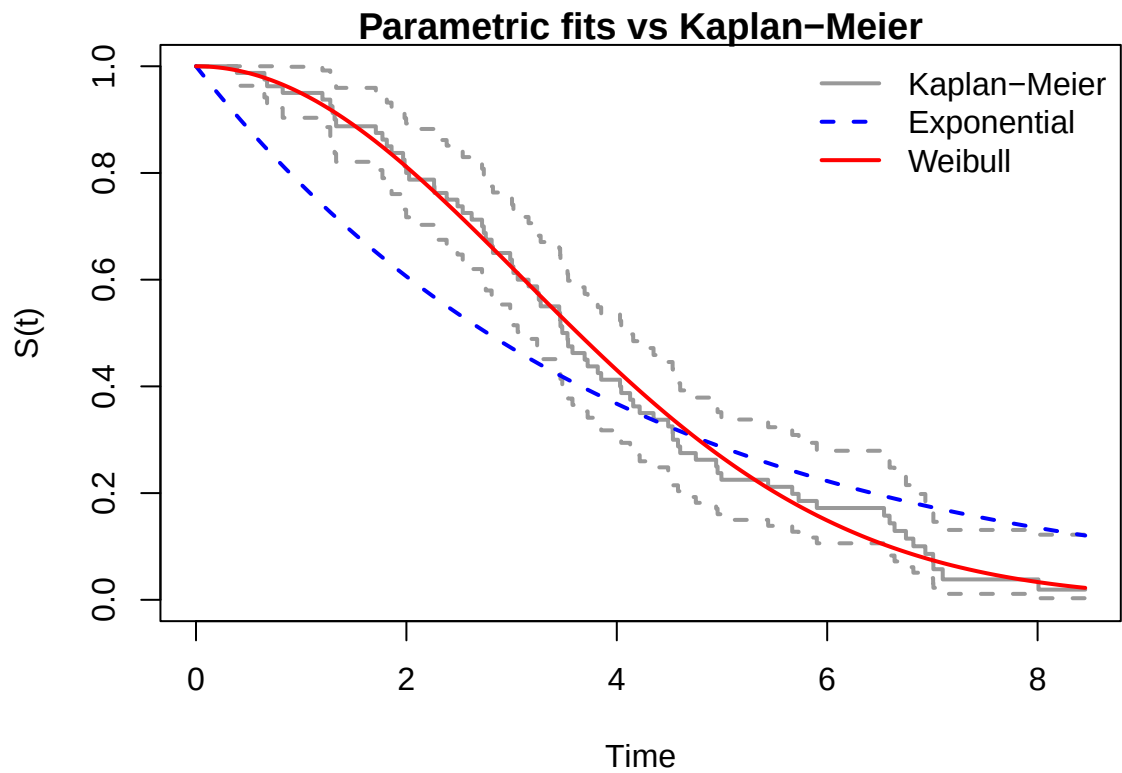


Figure 19.1: Kaplan-Meier estimate (step function) with fitted exponential (dashed) and Weibull (solid) curves for simulated data.

## Exercises

**19.1.** Consider a survival model with hazard function

$$h(t) = \lambda e^t, \quad t \geq 0, \lambda > 0.$$

Find the survival function  $S(t)$ .

[Click here to see hints.](#)

Compute  $H(t) = \int_0^t h(u) du$  first, then use  $S(t) = e^{-H(t)}$ .

**19.2.** Continuing from Exercise 19.1. Suppose we observe  $n$  independent individuals with survival times  $t_1, \dots, t_n$ , of which  $r$  are uncensored events ( $\delta_i = 1$ ) and the rest are right-censored ( $\delta_i = 0$ ).

[Click here to see hints.](#)

Write the log-likelihood as  $\ell(\lambda) = \sum_i \delta_i \log f(t_i) + \sum_i (1 - \delta_i) \log S(t_i)$ . Substitute  $f(t) = h(t)S(t) = \lambda e^t e^{-\lambda(e^t - 1)}$ . Differentiate with respect to  $\lambda$  and set to zero.

- Write down the log-likelihood  $\ell(\lambda)$  in terms of  $r$ ,  $\{t_i\}$ , and  $\lambda$ .
- Derive the maximum likelihood estimator  $\hat{\lambda}$ .

## 20 Parametric Two-Sample Comparison

### 20.1 Comparing two groups

Suppose individuals belong to one of two groups (e.g., two treatment arms, male/female, exposed/unexposed). We wish to test whether the survival distributions in the two groups are the same.

Let  $h_j(t)$  denote the hazard in group  $j$  ( $j = 1, 2$ ). For each group, observe  $(t_{ji}, \delta_{ji})$  for  $i = 1, \dots, n_j$ .

Define: -  $r_j = \sum_i \delta_{ji}$ : number of events in group  $j$ , -  $v_j = \sum_i t_{ji}$ : total person-time in group  $j$ .

### 20.2 Exponential two-sample model

Under the exponential model,  $h_j(t) = \lambda_j$  (constant), and the MLEs are  $\hat{\lambda}_j = r_j/v_j$  (from Chapter 19).

To compare the two groups, we use the **hazard ratio**

$$\psi = \frac{\lambda_2}{\lambda_1},$$

which is estimated by  $\hat{\psi} = \hat{\lambda}_2/\hat{\lambda}_1 = (r_2 v_1)/(r_1 v_2)$ .

The log hazard ratio has an approximately normal distribution:

$$\log \hat{\psi} \sim N\left(\log \psi, \frac{1}{r_1} + \frac{1}{r_2}\right),$$

giving a 95% confidence interval for  $\psi$ :

$$\exp\left(\log \hat{\psi} \pm 1.96 \sqrt{\frac{1}{r_1} + \frac{1}{r_2}}\right).$$

A likelihood ratio test of  $H_0 : \psi = 1$  (equal hazards) uses

$$\Lambda = -2(\hat{\ell}_0 - \hat{\ell}_1) \sim \chi_1^2$$

under  $H_0$ , where  $\hat{\ell}_0$  is the maximised log-likelihood under  $\lambda_1 = \lambda_2$  and  $\hat{\ell}_1$  is the maximised log-likelihood with separate  $\lambda_j$ .

#### Example: Breast cancer data

In a study of breast cancer, tumours were classified by HPA staining (negative: group 1,  $n_1 = 13$ ; positive: group 2,  $n_2 = 32$ ). The observed events and total person-months are:

$$r_1 = 5, \quad v_1 = 1652, \quad r_2 = 21, \quad v_2 = 2679.$$

Estimated hazard rates:

$$\hat{\lambda}_1 = 5/1652 \approx 0.00303, \quad \hat{\lambda}_2 = 21/2679 \approx 0.00784.$$

Estimated hazard ratio:

$$\hat{\psi} = \frac{21/2679}{5/1652} = \frac{21 \times 1652}{5 \times 2679} \approx 2.59.$$

95% CI for  $\psi$ :

$$\exp(\log 2.59 \pm 1.96\sqrt{1/5 + 1/21}) = [0.98, 6.87].$$

Interpretation: patients with positive staining have approximately 2.6 times the hazard of patients with negative staining, though the confidence interval includes 1 (just).

## 20.3 Graphical check: the log-cumulative hazard

Before fitting any parametric model, it is important to check whether the chosen family is appropriate. For the exponential model:

$$H_j(t) = \lambda_j t \quad \Rightarrow \quad \log H_j(t) = \log \lambda_j + \log t.$$

A plot of  $\log \hat{H}(t)$  against  $\log t$  should be linear with slope 1 for both groups. If the slopes differ from 1 but are equal to each other, a Weibull model with a common shape  $\gamma$  may be more appropriate.

## 20.4 Weibull two-sample model

For a common shape  $\gamma$  but different scales  $\lambda_1, \lambda_2$ :

$$h_j(t) = \lambda_j \gamma t^{\gamma-1}.$$

The hazard ratio  $\psi = \lambda_2/\lambda_1$  does not depend on  $t$  — this is the **proportional hazards** (PH) property, which is explored in detail in Chapter 21.

The MLEs of  $\lambda_j$  (given  $\hat{\gamma}$ ) are

$$\hat{\lambda}_j = \frac{r_j}{\sum_i t_{ji}^{\hat{\gamma}}},$$

and  $\hat{\gamma}$  is obtained numerically by maximising the joint log-likelihood.

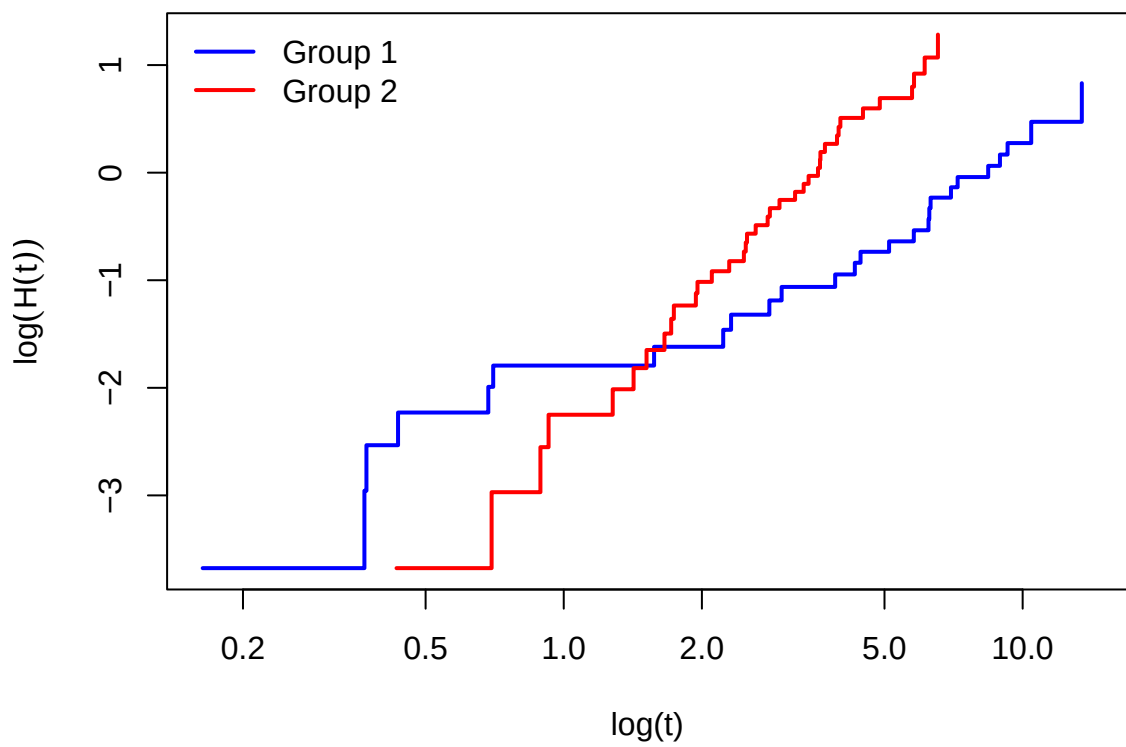


Figure 20.1: Log-cumulative hazard plots for two simulated groups. Parallel lines with approximately the same slope confirm the proportional hazards assumption.

## 20.5 Fitting in R

### Exercises

**20.1.** A randomised trial compared two surgical approaches for a procedure, measuring operating time (minutes) until the surgeon achieved an adequate result. Times marked \* are right-censored at 120 minutes (procedure stopped without achieving the result):

- **Group 1 (robot-assisted):** 50, 51, 66\*, 82, 92, 120\*, 120\*, 120\*
  - **Group 2 (traditional):** 63, 65, 69, 69, 79, 82, 82, 120\*
- a. Write R code to enter these data and produce a single plot showing the Kaplan–Meier survival curves for both groups, with appropriate labels, a legend, and 95% confidence intervals.
  - b. Based on the KM curves, which surgical approach achieves the adequate result sooner? Comment on the degree of overlap of the confidence intervals.

[Click here to see hints.](#)

Use `survfit(Surv(t, d) ~ group)` and then `plot()` with `col`, `lty`, and `legend()`. A lower KM curve means events occur sooner — in this context that is the favourable outcome.

**20.2.** Using the robot surgery data from Exercise 20.1, perform a formal two-sample comparison of the survival curves.

[Click here to see hints.](#)

For the log-rank test, at each event time  $t_{(j)}$  compute  $E_{1j} = n_{1j}d_j/n_j$ . The test statistic is  $U_L = \sum_j (O_{1j} - E_{1j})$  with variance  $\text{Var}(U_L) = \sum_j n_{1j}n_{2j}d_j(n_j - d_j)/(n_j^2(n_j - 1))$ .

- a. Carry out the **normal-form log-rank test** at the 10% significance level by hand, showing the calculation table.
- b. State the chi-squared form of the test statistic and compare your conclusion with part (a). Verify your result using R.

# 21 The Proportional Hazards Property

## 21.1 Definition

Definition: Proportional hazards

Two hazard functions  $h_1(t)$  and  $h_2(t)$  are **proportional** if

$$h_2(t) = \psi h_1(t) \quad \text{for all } t \geq 0,$$

for some constant  $\psi > 0$  called the **hazard ratio**.

Under proportional hazards, the relative risk  $h_2(t)/h_1(t) = \psi$  is constant over time. Group 2 is always  $\psi$  times as likely to experience the event as group 1, regardless of how long they have survived.

Equivalent conditions:

$$H_2(t) = \psi H_1(t) \quad \Leftrightarrow \quad S_2(t) = S_1(t)^\psi.$$

## 21.2 Weibull and exponential models satisfy PH

For a Weibull model with common shape  $\gamma$  and rates  $\lambda_1, \lambda_2$ :

$$h_j(t) = \lambda_j \gamma t^{\gamma-1}, \quad \frac{h_2(t)}{h_1(t)} = \frac{\lambda_2}{\lambda_1} = \psi.$$

The hazard ratio is  $\psi = \lambda_2/\lambda_1$  and does not depend on  $t$ . The exponential model is the special case  $\gamma = 1$ .

Not all distributions satisfy PH. For example, the log-normal and gamma distributions generally do **not**.

## 21.3 Graphical assessment

Taking log of  $H_j(t) = \lambda_j t^\gamma$ :

$$\log H_j(t) = \log \lambda_j + \gamma \log t.$$

If the PH property holds, the log-cumulative hazard plots for the two groups should be **parallel lines**. The vertical distance between them is  $\log \psi = \log \lambda_2 - \log \lambda_1$ .

Equivalently, plotting  $\log \widehat{H}(t)$  (computed from the Kaplan–Meier estimator) against  $\log t$  for each group provides both:

1. A **Weibull fit check**: the curves should be linear.
2. A **PH assumption check**: the curves should be parallel.

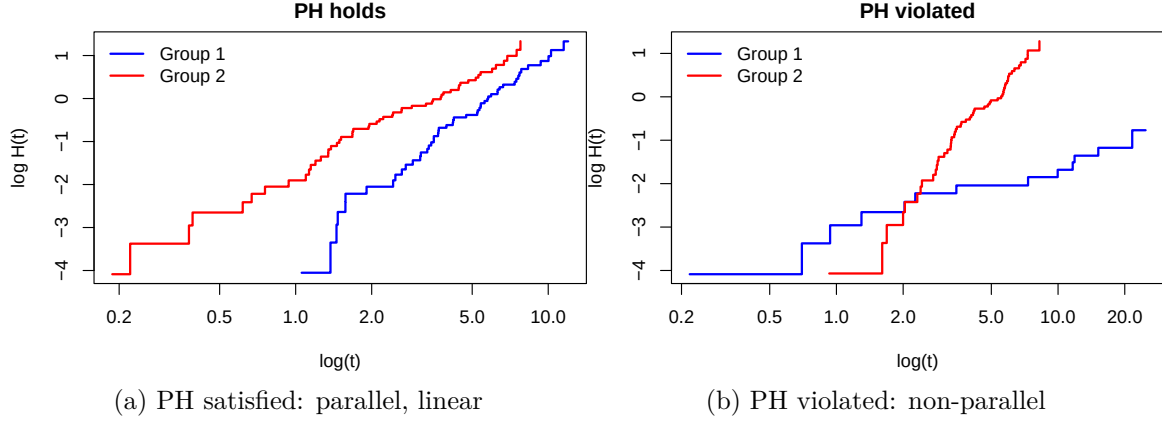


Figure 21.1: Log-cumulative hazard plots for three scenarios. Left: PH holds (parallel lines). Right: PH violated (non-parallel).

## 21.4 Prognostic index

In a multiple regression setting (several covariates  $\mathbf{x} = (x_1, \dots, x_p)$ ), the PH model takes the form

$$h(t | \mathbf{x}) = h_0(t) \exp(\beta^T \mathbf{x}),$$

where  $h_0(t)$  is the **baseline hazard** (the hazard when all covariates are zero) and  $\beta = (\beta_1, \dots, \beta_p)^T$  is the parameter vector.

The quantity  $\beta^T \mathbf{x} = \beta_1 x_1 + \dots + \beta_p x_p$  is called the **prognostic index** or **linear predictor**. Individuals with larger prognostic indices have higher hazard.

Hazard ratio for two individuals with covariate vectors  $\mathbf{x}$  and  $\mathbf{x}^*$ :

$$\frac{h(t | \mathbf{x})}{h(t | \mathbf{x}^*)} = \exp(\beta^T (\mathbf{x} - \mathbf{x}^*)),$$

which is constant in  $t$  — the PH property holds for any combination of covariates.

For a single binary covariate ( $x = 0$  or  $1$ ), the hazard ratio is  $\psi = e^\beta$ .

## Exercises

**21.1.** For each of the following hazard functions, determine whether the **proportional hazards (PH)** property holds. Recall that a family  $\{h(t; \beta)\}$  has the PH property if multiplying the hazard by any positive constant  $\psi$  yields another hazard in the same family.

**(i) Gompertz:**  $h(t) = \lambda e^{\theta t}$ ,  $\lambda > 0$ ,  $\theta \in \mathbb{R}$ .

(i) Does the Gompertz distribution have the PH property?

- (A) Yes — multiplying by just replaces with
- (B) No — the ratio  $h(t)/h(t)$  depends on  $t$

**(ii) Makeham:**  $h(t) = \gamma + \lambda e^{-\theta t}$ ,  $\gamma > 0$ ,  $\lambda \geq -\gamma$ ,  $\theta \geq 0$ .

(ii) Does the Makeham distribution have the PH property?

- (A) Yes — multiplying by scales both and simultaneously
- (B) No — the ratio  $h(t)/h(t)$  depends on  $t$

**(iii) Bathtub:**  $h(t) = \alpha t + \beta/(1 + \gamma t)$ ,  $\gamma \geq 0$ ,  $\alpha \geq 0$ ,  $\beta > 0$ .

(iii) Does the Bathtub distribution have the PH property?

- (A) Yes — multiplying by scales both and simultaneously
- (B) No — the ratio  $h(t)/h(t)$  depends on  $t$

For each case that has the PH property, explicitly show how the reparametrisation works: write  $\psi h(t)$  in the original functional form with new parameter values.

## 22 Parametric Proportional Hazards Regression

### 22.1 The model

The parametric PH regression model specifies both the baseline hazard  $h_0(t)$  and the effect of covariates:

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\beta^T \mathbf{x}_i),$$

where  $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^T$  is the covariate vector for individual  $i$ .

We use a **Weibull baseline hazard**  $h_0(t) = \lambda \gamma t^{\gamma-1}$ , giving

$$h(t \mid \mathbf{x}_i) = \lambda \gamma t^{\gamma-1} \exp(\beta^T \mathbf{x}_i).$$

The survival function is

$$S(t \mid \mathbf{x}_i) = \exp(-\lambda t^\gamma \exp(\beta^T \mathbf{x}_i)).$$

### 22.2 Log-likelihood

Using the censored likelihood from Chapter 19, with  $h(t_i \mid \mathbf{x}_i) = \lambda \gamma t_i^{\gamma-1} \exp(\beta^T \mathbf{x}_i)$  and  $S(t_i \mid \mathbf{x}_i)$  as above:

$$\ell(\lambda, \gamma, \beta) = \sum_{i=1}^n \left[ \delta_i \{ \beta^T \mathbf{x}_i + \log(\lambda \gamma) + (\gamma - 1) \log t_i \} - \lambda t_i^\gamma \exp(\beta^T \mathbf{x}_i) \right]. \quad (22.1)$$

The MLEs  $\hat{\lambda}, \hat{\gamma}, \hat{\beta}$  are obtained numerically by maximising Equation 22.1.

### 22.3 Interpretation of coefficients

The **hazard ratio** for a one-unit increase in covariate  $x_k$ , with all other covariates fixed, is

$$\psi_k = e^{\beta_k}.$$

- $\beta_k > 0$ : higher  $x_k$  increases the hazard (worse prognosis).
- $\beta_k < 0$ : higher  $x_k$  decreases the hazard (better prognosis).

- $\beta_k = 0$ :  $x_k$  has no effect on survival.

For a binary covariate (e.g., group indicator  $x_k \in \{0, 1\}$ ):  $\psi_k = e^{\beta_k}$  is the hazard ratio between groups.

## 22.4 Fitting Weibull PH models in R

The `survreg()` function uses the accelerated failure time (AFT) parametrisation. For a Weibull model, we must convert from the AFT output to the PH parametrisation.

Note: Sign convention in `survreg()`

In `survreg()`, the covariate effects  $\tilde{\beta}_k$  are in the AFT parametrisation:  $\log T = \mu + \tilde{\beta}^T \mathbf{x} + \sigma \varepsilon$ . The relationship to the PH coefficient is  $\beta_k^{PH} = -\hat{\gamma} \cdot \tilde{\beta}_k$ , so the **signs are reversed** when converting.

Example: Breast cancer data

For the breast cancer data from Chapter 20, fitting a Weibull model with a group indicator ( $x = 0$ : negative staining,  $x = 1$ : positive):

$$\hat{\gamma} \approx 1.0 \quad (\text{essentially exponential}), \quad \hat{\psi} = e^{\hat{\beta}} \approx 2.59.$$

The estimated hazard ratio matches the two-sample exponential result, confirming consistency. The 95% CI [0.98, 6.87] is just consistent with  $\psi = 1$ .

## 22.5 Multiple covariates: Weibull regression

With multiple continuous or categorical covariates:

## 22.6 Model checking

After fitting, check:

1. **Log-cumulative hazard plot**: should be linear in  $\log t$  for each group/covariate pattern (Weibull check).
2. **Parallel log-cumulative hazard curves**: should be parallel for different groups (PH check).
3. **Residual plots**: deviance or Cox–Snell residuals should be approximately exponentially distributed if the model fits.

## Exercises

**22.1.** Consider the survival model with hazard  $h(t) = \lambda e^t$  (from Exercise 19.1) extended to a proportional hazards regression model with a single binary covariate  $x_i \in \{0, 1\}$ :

$$h_i(t) = e^{\beta x_i} \cdot \lambda e^t, \quad i = 1, \dots, n.$$

There are  $r$  uncensored event times and  $n - r$  right-censored observations.

[Click here to see hints.](#)

Write the full log-likelihood  $\ell(\lambda, \beta) = \sum_i \delta_i \log h_i(t_i) + \sum_i \log S_i(t_i)$ . Note that  $S_i(t) = \exp(-e^{\beta x_i} \lambda (e^t - 1))$ . Differentiate with respect to  $\lambda$  and  $\beta$  separately and set both partial derivatives to zero.

- Write down the log-likelihood  $\ell(\lambda, \beta)$ .
- Derive the two score equations  $\partial \ell / \partial \lambda = 0$  and  $\partial \ell / \partial \beta = 0$ .
- Show that the two equations can be reduced to a single equation in  $\beta$  alone, plus a formula expressing  $\hat{\lambda}$  in terms of  $\hat{\beta}$ .

# 23 The Cox Proportional Hazards Model

## 23.1 Motivation

Parametric models (Chapters 17–22) require us to specify the shape of the baseline hazard  $h_0(t)$ . In practice, we may not be confident in this choice, and a misspecified baseline can bias the estimated covariate effects  $\hat{\beta}$ .

In 1972, Sir David Cox proposed a **semi-parametric** approach that estimates  $\beta$  without specifying  $h_0(t)$  at all. The baseline hazard is treated as an unspecified nuisance function, and inference on  $\beta$  is based on a **partial likelihood** that eliminates  $h_0(t)$ .

## 23.2 The Cox model

Definition: Cox proportional hazards model

$$h(t | \mathbf{x}_i) = h_0(t) \exp(\beta^T \mathbf{x}_i),$$

where  $h_0(t) \geq 0$  is an **unspecified** baseline hazard function and  $\beta = (\beta_1, \dots, \beta_p)^T$  are regression coefficients.

The covariate effects are the same as in the parametric PH model (Chapter 22). What is new is that  $h_0(t)$  is left completely unspecified.

## 23.3 Partial likelihood

Cox's key insight was to construct a likelihood for  $\beta$  that does not involve  $h_0(t)$ .

Let  $t_{(1)} < t_{(2)} < \dots < t_{(r)}$  be the ordered event times (ties handled separately). At each event time  $t_{(j)}$ , define the **risk set**

$$\mathcal{R}(t_{(j)}) = \{\text{individuals still at risk immediately before } t_{(j)}\}.$$

Given that an event occurs at  $t_{(j)}$ , the conditional probability that it is individual  $i_{(j)}$  who experiences the event (rather than any other at-risk individual) is

$$\frac{h_0(t_{(j)}) \exp(\beta^T \mathbf{x}_{i_{(j)}})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} h_0(t_{(j)}) \exp(\beta^T \mathbf{x}_\ell)} = \frac{\exp(\beta^T \mathbf{x}_{i_{(j)}})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\beta^T \mathbf{x}_\ell)}.$$

The baseline hazard  $h_0(t_{(j)})$  cancels in the numerator and denominator.

Definition: Partial likelihood

$$L(\beta) = \prod_{j=1}^r \frac{\exp(\beta^T \mathbf{x}_{(j)})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\beta^T \mathbf{x}_{\ell})},$$

where  $\mathbf{x}_{(j)}$  is the covariate vector of the individual who dies at time  $t_{(j)}$ .

This can be written compactly as

$$L(\beta) = \prod_{i=1}^n \left[ \frac{\exp(\beta^T \mathbf{x}_i)}{\sum_{\ell \in \mathcal{R}(t_i)} \exp(\beta^T \mathbf{x}_{\ell})} \right]^{\delta_i}.$$

Only the  $\delta_i = 1$  terms contribute to the product.

## 23.4 Partial log-likelihood

$$\ell(\beta) = \sum_{i=1}^n \delta_i \left\{ \beta^T \mathbf{x}_i - \log \sum_{\ell \in \mathcal{R}(t_i)} \exp(\beta^T \mathbf{x}_{\ell}) \right\}. \quad (23.1)$$

The **maximum partial likelihood estimator (MPLE)**  $\hat{\beta}$  maximises Equation 23.1. Under regularity conditions,  $\hat{\beta}$  has the same asymptotic properties as an MLE:

$$\hat{\beta} \sim N_p(\beta, \mathcal{I}(\beta)^{-1}),$$

where  $\mathcal{I}(\beta) = -\partial^2 \ell / \partial \beta \partial \beta^T$  is the observed information matrix.

## 23.5 Handling tied event times

The partial likelihood above assumes no ties. With ties, the most common approximation is **Breslow's approximation**:

$$L(\beta) \approx \prod_{j=1}^r \frac{\exp(\beta^T \mathbf{s}_{(j)})}{\left( \sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\beta^T \mathbf{x}_{\ell}) \right)^{d_j}},$$

where  $d_j$  is the number of deaths at  $t_{(j)}$  and  $\mathbf{s}_{(j)}$  is the sum of covariate vectors of those who die.

## 23.6 Manual calculation: coronavirus example

Example: Recovery from coronavirus

Six patients from two groups (Group I: lung condition; Group II: no lung condition) with event indicator and ordered recovery times:

| Individual | Group | Time | Event        |
|------------|-------|------|--------------|
| 1          | I     | 16   | 1            |
| 2          | I     | 24   | 0 (censored) |
| 3          | I     | 28   | 0 (censored) |
| 4          | I     | 30   | 1            |
| 5          | I     | 42   | 1            |
| 6          | I     | 42   | 0 (censored) |
| 7          | II    | 6    | 1            |
| 8          | II    | 10   | 0 (censored) |
| 9          | II    | 12   | 0 (censored) |
| 10         | II    | 30   | 0 (censored) |
| 11         | II    | 32   | 1            |
| 12         | II    | 41   | 1            |

Let  $x_i = 0$  for Group I,  $x_i = 1$  for Group II. There are  $r = 6$  events at ordered times 6, 16, 30, 32, 41, 42.

The partial log-likelihood (ignoring the intercept, which does not identify  $\beta$ ) is constructed term by term. By symmetry of the data, the MPLE turns out to be  $\hat{\beta} = \log 2 \approx 0.693$ , giving hazard ratio  $\hat{\psi} = e^{\hat{\beta}} = 2$  for Group II relative to Group I: patients without a lung condition (Group II) recover at twice the rate of those with one (Group I) – equivalently, patients with a lung condition recover at half the rate of those without.

## 23.7 Cox model in R

### Exercises

**23.1.** A small clinical trial compares Drug A ( $x = 0$ ) and Drug B ( $x = 1$ ). Survival times (days) are:

- **Drug A:** 2 (event), 4 (event), 9\* (censored), 12 (event)
- **Drug B:** 3\* (censored), 5\* (censored), 8 (event), 14 (event)

The Cox proportional hazards model is  $h(t, x) = e^{\beta x} h_0(t)$ .

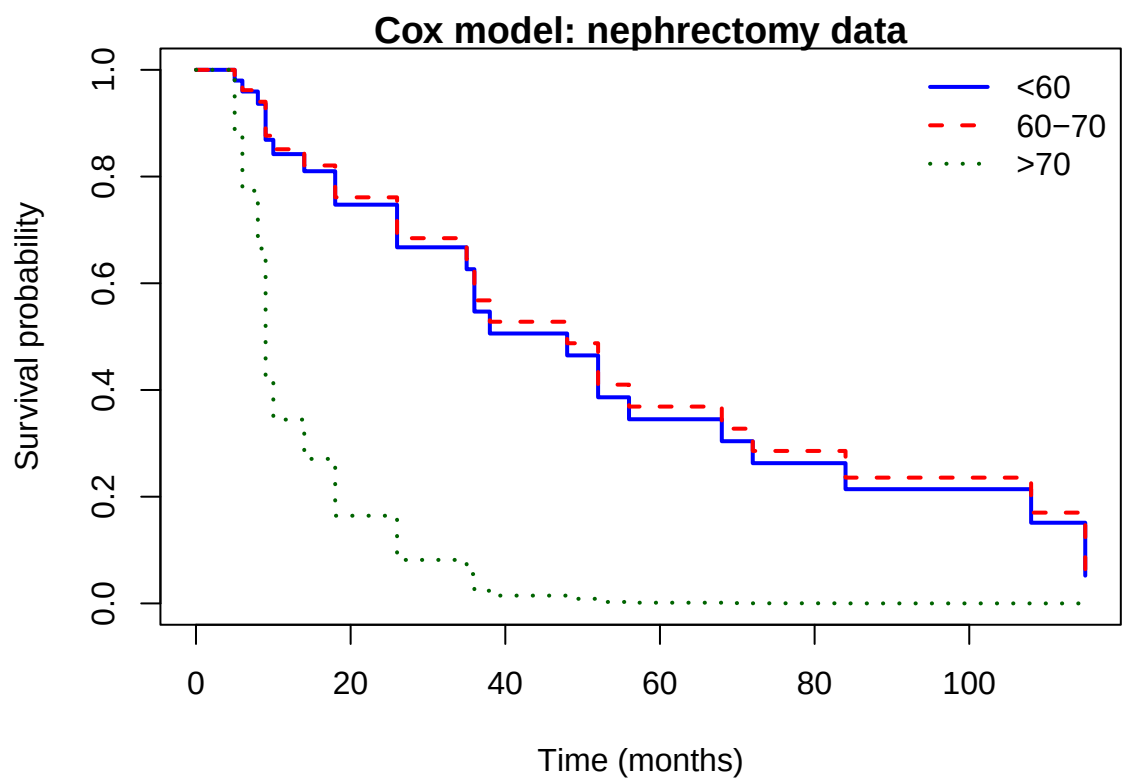


Figure 23.1: Baseline survival function estimated from the Cox model for the nephrectomy data. Separate curves are shown for patients under 60 (solid), 60–70 (dashed), and over 70 (dotted).

- a. Identify all event times and the corresponding risk sets. Hence write down the Cox partial likelihood  $L(\beta)$  and show that

$$L(\beta) = C \cdot \frac{e^\beta}{(1 + e^\beta)^4},$$

stating the value of the constant  $C$ .

[Click here to see hints.](#)

At each event time  $t_{(j)}$ , the partial likelihood contribution is  $e^{\beta x_{(j)}} / \sum_{k \in R(t_{(j)})} e^{\beta x_k}$ , where  $x_{(j)}$  is the covariate of the subject who had the event. The risk set  $R(t_{(j)})$  contains all subjects with  $t_k \geq t_{(j)}$ . Multiply contributions over all event times.

- b. Find the maximum partial likelihood estimator  $\hat{\beta}$ .

[Click here to see hints.](#)

Differentiate  $\ell(\beta) = \log L(\beta)$  with respect to  $\beta$  and set to zero.

**23.2.** Continuing from Exercise 23.1 with  $\hat{\beta} = -\log 3$ .

- a. Compute the approximate variance of  $\hat{\beta}$  using the observed information  $-\ell''(\hat{\beta})$ .
- b. Test  $H_0 : \beta = 0$  (drugs equally effective) against a two-sided alternative at the 5% significance level using a Wald test.

[Click here to see hints.](#)

Compute  $\ell''(\beta) = d^2\ell/d\beta^2$  from  $\ell'(\beta) = 1 - 4e^\beta/(1 + e^\beta)$ . Evaluate at  $\hat{\beta}$  and take the negative reciprocal for the variance. The Wald statistic is  $\hat{\beta}/\widehat{\text{se}}(\hat{\beta})$ .

## 24 Cox Model Diagnostics

### 24.1 Overview

After fitting a Cox model, we should check two key assumptions:

1. **Proportional hazards:** the hazard ratio between any two individuals is constant over time.
2. **Linearity:** continuous covariates enter the log-hazard linearly.

Several diagnostic tools are available.

### 24.2 Graphical check: log-cumulative hazard plots

The simplest PH check for a categorical covariate is the **log-cumulative hazard plot** (introduced in Chapter 21): plot  $\log \hat{H}(t)$  from the KM estimator against  $\log t$  for each group. Under PH, the curves should be parallel.

### 24.3 Schoenfeld residuals

Definition: Schoenfeld residuals

At each event time  $t_{(j)}$ , the Schoenfeld residual for covariate  $k$  is the observed covariate value minus the risk-set-weighted expected value:

$$r_{jk} = x_{(j)k} - \frac{\sum_{\ell \in \mathcal{R}(t_{(j)})} x_{\ell k} \exp(\hat{\beta}^T \mathbf{x}_{\ell})}{\sum_{\ell \in \mathcal{R}(t_{(j)})} \exp(\hat{\beta}^T \mathbf{x}_{\ell})}.$$

Under the PH assumption, the Schoenfeld residuals should have mean zero and no trend with time. A **plot of  $r_{jk}$  against event time  $t_{(j)}$**  that shows a systematic trend (increasing, decreasing, or non-linear) suggests that the PH assumption is violated for covariate  $k$ .

The `cox.zph()` function in R tests formally for a time trend in each covariate's Schoenfeld residuals.

A small  $p$ -value from `cox.zph()` suggests the PH assumption is violated for that covariate.

## 24.4 Martingale residuals

Definition: Martingale residuals

$$M_i = \delta_i - \widehat{H}_0(t_i) \exp(\widehat{\beta}^T \mathbf{x}_i),$$

where  $\widehat{H}_0(t)$  is the Breslow estimate of the baseline cumulative hazard.

Martingale residuals lie in  $(-\infty, 1]$  and sum to zero. Plotting them against a continuous covariate that was assumed to enter linearly can reveal non-linearity: a non-linear trend in the residuals suggests that a transformation of the covariate (or a smooth term) would improve the fit.

## 24.5 Deviance residuals

Deviance residuals are a transformation of the martingale residuals designed to be more symmetric around zero for a well-fitting model:

$$d_i = \text{sign}(M_i) \sqrt{-2(M_i + \delta_i \log(\delta_i - M_i))}.$$

Large absolute deviance residuals (say,  $|d_i| > 3$ ) may indicate influential observations or outliers.

## 24.6 Complete diagnostics workflow

### Exercises

**24.1.** After fitting the Cox proportional hazards model from Exercise 23.1 to the Drug A/B data, a researcher wishes to check the model assumptions. Without using R, describe the diagnostic approach for each of the following.

[Click here to see hints.](#)

Think about log-log survival plots ( $\log(-\log \widehat{S}(t))$  vs  $\log t$  for each group), Schoenfeld residuals plotted against time, and dfbeta or deviance residuals for influence assessment.

- Checking the proportional hazards assumption.** What plot(s) would you produce, what would you look for, and what would departures from PH look like?
- Identifying influential observations.** What residual-based diagnostics would you use, and what pattern in the plot would flag an observation as influential?

**24.2.** Using the robot surgery data from Exercise 20.1, perform a full set of Cox model diagnostics in R.

[Click here to see hints.](#)

Use `coxph()` to fit the model, `cox.zph()` to test PH, `plot(cox.zph(fit))` to visualise Schoenfeld residuals, and `residuals(fit, type="deviance")` for deviance residuals.

- a. Fit a Cox model with group as the covariate. Run `cox.zph()` and interpret the output.
- b. Plot the Schoenfeld residuals against time. What pattern would confirm that the PH assumption is satisfied?
- c. Plot the deviance residuals against the linear predictor. Identify any potential outliers.

## 25 Model Selection and Comparison

### 25.1 Nested models and the change in deviance

Recall from Part I (Section 4.3) that nested models can be compared via the **change in deviance**. The same idea applies directly to survival models.

Two models  $M_0 \subset M_1$  are **nested** if  $M_0$  is obtained from  $M_1$  by fixing one or more parameters at specified values — not necessarily zero. (For example, the exponential model is nested within the Weibull model by fixing the shape parameter  $\gamma = 1$ .)

For a fitted model  $M$  with maximised log-likelihood  $\hat{\ell}$ , its deviance is  $D = -2\hat{\ell}$  (up to an additive constant that cancels whenever we compare two nested models). Then, for nested models  $M_0$  (with  $p$  parameters) and  $M_1$  (with  $q > p$  parameters),

$$\Delta D = D_0 - D_1 \sim \chi^2_{q-p} \quad \text{under } H_0,$$

exactly as in Section 4.3. We reject  $H_0$  (prefer  $M_1$ ) if  $\Delta D > \chi^2_{q-p, 1-\alpha}$ .

For **Cox models**, the same test applies using the deviance based on the maximised **partial** log-likelihood. The asymptotic  $\chi^2$  distribution holds under the same conditions.

**No separate dispersion step needed.** Unlike the Normal/Gamma case in Section 4.3, where an unknown dispersion  $\phi$  required an  $F$ -test with  $\hat{\phi}$  estimated from a larger model, no such step is needed here. A parametric survival model's shape parameter (e.g.  $\gamma$  for the Weibull) is a genuine parameter of the likelihood, not a separate dispersion term scaling an otherwise  $\phi$ -free deviance — it is already fully accounted for in  $\hat{\ell}_0$  and  $\hat{\ell}_1$  once each model is fitted. The Cox partial likelihood has no dispersion parameter at all. In both cases the plain asymptotic  $\chi^2$  result applies directly, exactly as it does for the known- $\phi$  GLM families (Poisson, Binomial).

### 25.2 Step-by-step model selection strategy

A practical approach for building a survival regression model:

1. **Univariate screening:** fit each covariate separately; identify those that are significant at a liberal threshold (e.g.,  $p < 0.10$ ).
2. **Multivariable model:** include all significant covariates from step 1. Drop non-significant ones using backward elimination.

3. **Add borderline variables:** consider adding covariates not significant alone but that may be confounders; use LRT to decide.
4. **Final check:** ensure no term can be significantly added or removed.
5. **Check interactions:** for key covariates, test interaction terms.

## 25.3 Case study: Nephrectomy data

The `hypernephroma.dat` dataset contains survival data for 36 patients with kidney cancer. Covariates are age group (three levels:  $< 60$ ,  $60-70$ ,  $> 70$ ) and whether nephrectomy (kidney removal) was performed.

The key comparisons:

| Comparison                         | $\Delta$              | df | $p$ -value |
|------------------------------------|-----------------------|----|------------|
| $M_1$ vs $M_0$ (age)               | $177.7 - 172.2 = 5.5$ | 2  | 0.064      |
| $M_2$ vs $M_0$ (nephrectomy)       | $177.7 - 170.2 = 7.4$ | 1  | 0.006      |
| $M_3$ vs $M_1$ (nephrectomy   age) | $172.2 - 165.5 = 6.7$ | 1  | 0.010      |
| $M_4$ vs $M_3$ (interaction)       | $165.5 - 162.5 = 3.0$ | 2  | 0.220      |

Nephrectomy is significant, both alone and after adjusting for age (nephrectomy | age,  $p = 0.010$ ). Age on its own is only marginal at the 5% level ( $p = 0.064$  for the joint, 2 df test across all three groups) – this joint test is diluted by the middle (60–70) group, which is not clearly different from the youngest group, even though the oldest group ( $> 70$ ) does show a clearly elevated hazard (see the confidence interval in the next section). The interaction is clearly non-significant ( $p = 0.220$ ); we retain the main-effects model  $M_3$  as the final model. This illustrates that even when a covariate’s overall significance is marginal, individual category effects can still be clinically important, and that we should always verify the added value of an interaction term.

## 25.4 Parametric vs Cox: which to use?

| Criterion       | Parametric (Weibull)    | Cox                                                 |
|-----------------|-------------------------|-----------------------------------------------------|
| Baseline hazard | Specified               | Unspecified                                         |
| Efficiency      | Higher if model correct | Lower                                               |
| Robustness      | Biased if $h_0$ wrong   | Robust                                              |
| Prediction      | Full $\hat{S}(t)$ easy  | $\hat{S}(t)$<br>requires<br>estimate of<br>$H_0(t)$ |

| Criterion        | Parametric (Weibull)  | Cox                |
|------------------|-----------------------|--------------------|
| Interpretability | Hazard ratios + shape | Hazard ratios only |

**Recommendation:** start with a graphical check (log-cumulative hazard plot). If the parametric form looks appropriate, use the parametric model for efficiency. When in doubt, use the Cox model.

## Exercises

**25.1.** Using the Drug A/B Cox partial likelihood from Exercise 23.1, perform a **likelihood ratio test** of  $H_0 : \beta = 0$  against  $H_1 : \beta \neq 0$ .

Click here to see hints.

Simplify  $\ell(\hat{\beta}) - \ell(0)$  by cancelling the  $\log(1/24)$  terms. Use  $\log(4/3) = \log 4 - \log 3$  and collect terms.

The partial log-likelihoods are:

$$\ell(0) = \log\left(\frac{1}{24}\right) - 4 \log 2, \quad \ell(\hat{\beta}) = \log\left(\frac{1}{24}\right) - \log 3 - 4 \log\left(\frac{4}{3}\right).$$

- Compute the likelihood ratio test statistic  $T = -2(\ell(0) - \ell(\hat{\beta}))$ .
- State the null distribution of  $T$  and the critical value at the 5% level.
- State your conclusion. Is it consistent with the Wald test in Exercise 23.2?

**25.2.** When analysing survival data, one must often choose between a fully parametric model (e.g., Weibull PH) and the semi-parametric Cox model.

- Which of the following is an **advantage of the Cox model** over a Weibull PH model?
- Advantage of the Cox model:
  - (A) No assumption about the shape of the baseline hazard is needed
  - (B) The Cox model always gives more precise parameter estimates
  - (C) The Cox model can be fitted without any censored observations
  - (D) The Cox model gives explicit estimates of the survival function at arbitrary future times without additional estimation
- Which of the following is an **advantage of the Weibull PH model** over the Cox model?
- Advantage of the Weibull model:

- (A) The Weibull model makes no distributional assumptions
  - (B) When the model is correctly specified, parametric estimates are more efficient and allow direct survival prediction at any future time
  - (C) The Weibull model is always preferred when the sample size is large
  - (D) The Weibull model cannot be used when there are tied event times
- c. Suppose you fit both a Weibull PH model and an exponential PH model to the same data. Since the exponential model is the Weibull with shape  $\gamma = 1$ , the two models are nested. The deviances ( $-2 \times \log\text{-likelihood}$ ) are: Weibull = 83.4, Exponential = 89.2. Use a likelihood ratio test (i.e. the change in deviance) to decide whether the extra shape parameter is justified.

## 26 Survival Analysis: Extended Case Study

### 26.1 Dataset: Hypernephroma

We carry out a complete survival analysis of the **hypernephroma (kidney cancer)** dataset, illustrating the full pipeline from exploratory analysis through to final model selection.

The dataset contains  $n = 36$  patients. Variables:

- **time**: survival time in months.
- **status**: event indicator (\$1 = \$ death, \$0 = \$ censored).
- **age**: age group (1 = < 60, 2 = 60–70, 3 = > 70).
- **nephrectomy**: whether nephrectomy was performed (\$0 = \$ no, \$1 = \$ yes).

### 26.2 Step 1: Exploratory analysis

We begin with Kaplan–Meier estimates for each covariate separately.

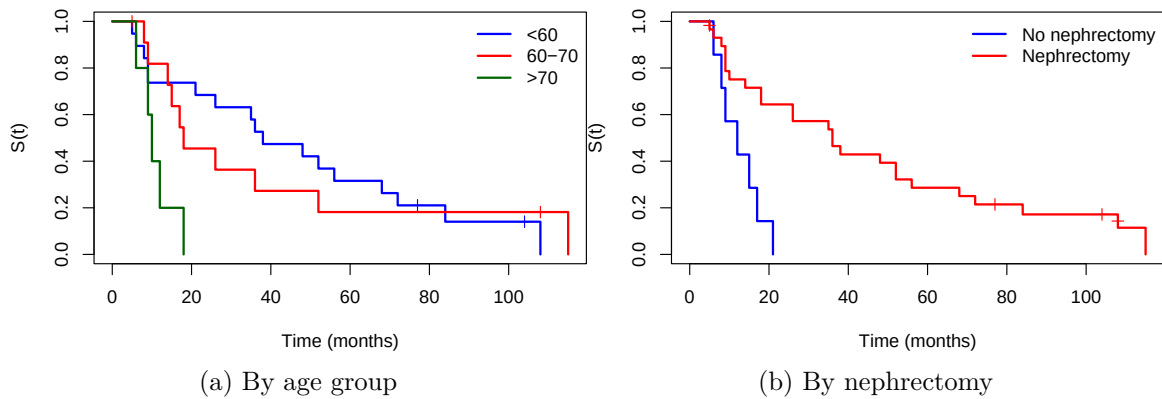


Figure 26.1: Kaplan–Meier estimates by age group (left) and nephrectomy status (right).

**Observations:** - Age group > 70 has clearly worse survival. - Patients who had nephrectomy appear to survive longer on average.

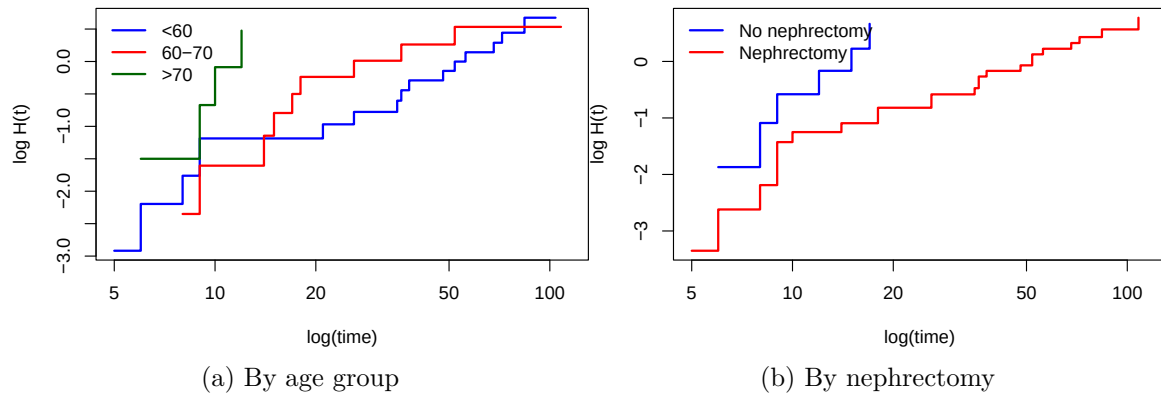


Figure 26.2: Log-cumulative hazard plots. Approximately parallel lines support the proportional hazards assumption.

## 26.3 Step 2: Graphical PH checks

Both plots show approximately parallel lines, supporting the PH assumption for both age group and nephrectomy.

## 26.4 Step 3: Fit the Cox model

We fit a series of nested Cox models and select the best using the LRT.

| Model                     | Parameters | $-2\hat{\ell}$ | LRT vs previous | $p$ -value |
|---------------------------|------------|----------------|-----------------|------------|
| $M_0$ : null              | 0          | 177.67         | —               | —          |
| $M_1$ : age               | 2          | 172.17         | 5.50 (df = 2)   | 0.064      |
| $M_3$ : age + nephrectomy | 3          | 165.51         | 6.66 (df = 1)   | 0.010      |

Nephrectomy is significant after adjusting for age ( $p = 0.010$ ); age on its own is only marginal at the 5% level ( $p = 0.064$ ), though it is retained as an established prognostic factor and, as seen below, the oldest age group specifically does show a clearly elevated hazard. We adopt  $M_3$  as the final model.

## 26.5 Step 4: Interpret the final model

|                | exp(coef) | exp(-coef) | lower .95  | upper .95  |
|----------------|-----------|------------|------------|------------|
| age60-70       | 1.0126102 | 0.9875468  | 0.44058221 | 2.3273282  |
| age>70         | 3.8250311 | 0.2614358  | 1.19928960 | 12.1996083 |
| nephrectomyyes | 0.2437888 | 4.1019116  | 0.08880618 | 0.6692436  |

Key results from the final model:

- **Age 60–70 vs  $< 60$ :**  $\hat{\psi} \approx 1.01$  (95% CI: [0.44, 2.33]) — not significantly different.
- **Age  $> 70$  vs  $< 60$ :**  $\hat{\psi} \approx 3.83$  (95% CI: [1.20, 12.20]) — patients over 70 have nearly 4 times the hazard.
- **Nephrectomy (yes vs no):**  $\hat{\psi} \approx 0.24$  (95% CI: [0.09, 0.67]) — nephrectomy reduces hazard by about 76%.

## 26.6 Step 5: Diagnostics

The `cox.zph()` test should show no significant time trends for age or nephrectomy, confirming the PH assumption. Most deviance residuals should lie within  $\pm 2$ .

## 26.7 Step 6: Comparison with parametric model

As a sensitivity analysis, we compare with a Weibull PH model.

The Weibull model provides additional shape information ( $\hat{\gamma}$ ) and should yield similar hazard ratio estimates to the Cox model if the Weibull fit is reasonable.

## 26.8 Summary

This case study has demonstrated the complete survival analysis workflow:

1. **Exploratory analysis:** KM curves by group reveal patterns.
2. **PH assumption check:** log-cumulative hazard plots support PH.
3. **Model selection:** LRT identifies which covariates to include.
4. **Interpretation:** hazard ratios quantify covariate effects.
5. **Diagnostics:** Schoenfeld residuals and deviance residuals confirm model adequacy.
6. **Sensitivity analysis:** comparison with parametric model confirms robustness.

The finding that nephrectomy substantially reduces hazard (by ~76%) is clinically meaningful and consistent with established medical knowledge. The strong age effect ( $> 70$  group) highlights that patient age is an important prognostic factor in kidney cancer.

## Exercises

**26.1.** This exercise guides you through a complete survival analysis of the **robot surgery data** introduced in Exercise 20.1. Work through each part sequentially, using R where indicated.

Recall the data: operating time (minutes) until the surgeon achieved an adequate result, censored at 120 minutes.

- **Group 1 (robot-assisted):** 50, 51, 66\*, 82, 92, 120\*, 120\*, 120\*
  - **Group 2 (traditional):** 63, 65, 69, 69, 79, 82, 82, 120\*
- a. **Define the endpoint and censoring.** State the event of interest, identify the censoring mechanism, and write down the observed data pairs  $(t_i, \delta_i)$  for all 16 patients.
  - b. **Kaplan–Meier analysis.** Plot the KM survival curves for both groups. Then use the log-log transformation (`fun="cloglog"`) to assess whether the proportional hazards assumption is plausible.
  - c. **Cox model.** Fit a Cox PH model with group as the covariate. Write down the estimated hazard ratio with a 95% confidence interval and interpret it in context.
  - d. **PH diagnostics.** Run `cox.zph()` and plot the Schoenfeld residuals. Comment on whether the PH assumption appears to be satisfied.
  - e. **Parametric comparison.** Using `survreg()`, fit a Weibull model and an exponential model. Since the exponential is the Weibull with shape = 1, compare these two nested parametric models using a likelihood ratio test (change in deviance), and state which you would recommend overall.

[Click here to see hints.](#)

For (b), `plot(km_fit, fun="cloglog")` — parallel lines support PH. For (c), `exp(coef(cox_fit))` gives the hazard ratio. For (d), `cox.zph(cox_fit)` and `plot(cox.zph(cox_fit))`. For (e), fit `survreg(Surv(t,d)~g, dist="weibull")` and `survreg(Surv(t,d)~g, dist="exponential")`, then compute the deviance of each as  $-2 \times$  its log-likelihood (`logLik()`), and compare via a  $\chi^2_1$  test exactly as in the change-in-deviance method of Chapter 25.